Ex. No:1 **STUDY OF UML**

Date:

AIM:

General study of UML

DESCRIPTION:

The heart of Software Engineering problem solving is the construction of a model. The model abstracts the essential details of the underlying problem from its usually complicated real world. Several modeling tools are wrapped under the heading of the **UML**, which stands for Unified Modeling Language. The purpose of this course is to present important highlights of the UML.

At the center of the UML are its nine kinds of modeling diagrams, which we describe here.

- ✓ Use case diagrams
- ✓ Class diagrams
- ✓ Object diagrams
- ✓ Sequence diagrams
- ✓ Collaboration diagrams
- ✓ State chart diagrams
- ✓ Activity diagrams
- ✓ Component diagrams
- ✓ Deployment diagrams

Some of the sections of this course contain links to pages with more detailed information. And every section has short questions. Use them to test your understanding of the section topic.

Why is UML important?

Let's look at this question from the point of view of the construction trade. Architects design buildings. Builders use the designs to create buildings. The more complicated the building, the more critical the communication between architect and builder. Blueprints are the standard graphical language that both architects and builders must learn as part of their trade.

Writing software is not unlike constructing a building. The more complicated the underlying system, the more critical the communication among everyone involved in creating and deploying the software. In the past decade, the UML has emerged as the software blueprint language for analysts, designers, and programmers alike. It is now part of the software trade. The UML gives everyone from business analyst to designer to programmer a common vocabulary to talk about software design.

The UML is applicable to Software Engineering problem solving. Anyone interested in learning UML must be familiar with the underlying tenet of Software Engineering problem solving -- it all begins with the construction of a model. A model is an abstraction of the underlying problem. The domain is the actual world from which the problem comes.

Models consist of objects that interact by sending each other message. Think of an object as "alive." Objects have things they know (attributes) and things they can do (behaviors or operations). The values of an object's attributes determine their state. Classes are the "blueprints" for objects. A class wraps attributes (data) and behaviors (methods or functions) into a single distinct entity. Objects are instances of classes.

Use case diagrams:

Use case diagrams describe what a system does from the standpoint of an external observer. The emphasis is on *what* a system does rather than *how.*

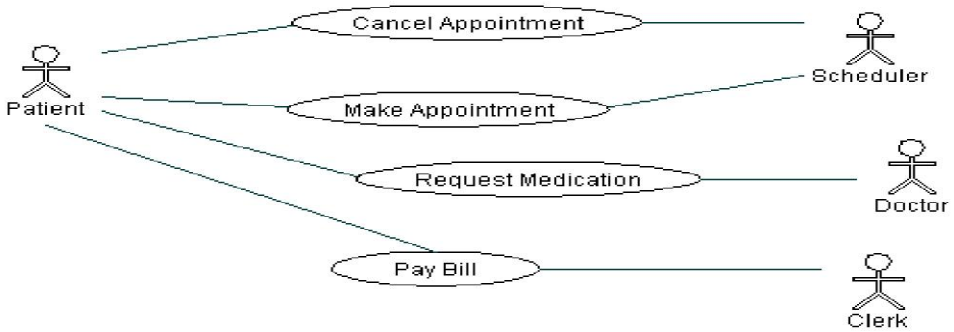
Use case diagrams are closely connected to scenarios. A scenario is an example of what happens when someone interacts with the system. Here is a scenario for a medical clinic."A patient calls the clinic to make an appointment for a yearly checkup. The receptionist finds the nearest empty time slot in the appointment book and schedules the appointment for that time slot. "

A use case is a summary of scenarios for a single task or goal. An actor is who or what initiates the events involved in that task. Actors are simply roles that people or objects play. The picture below is a Make Appointment use case for the medical clinic. The actor is a Patient. The connection between actor and use case is a communication association (or communication for short).



Actors are stick figures. Use cases are ovals. Communications are lines that link actors to use cases.

A use case diagram is a collection of actors, use cases, and their communications. We've put **Make Appointment** as part of a diagram with four actors and four use cases. Notice that a single use case can have multiple actors.



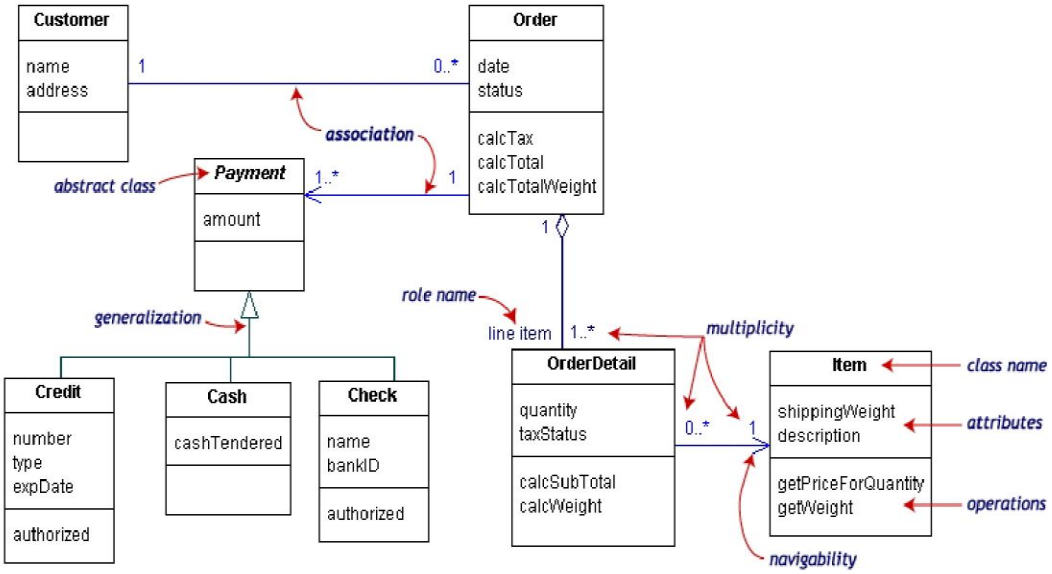
Use case diagrams are helpful in three areas.

- **Determining features (requirements)-** New use cases often generate new requirements as the system is analyzed and the design takes shape.

- **Communicating with clients**- Their notational simplicity makes use case diagrams a good way for developers to communicate with clients.
- **Generating test cases**- The collection of scenarios for a use case may suggest a suite of test cases for those scenarios.

Class diagrams:

A Class diagram gives an overview of a system by showing its classes and the relationships among them. Class diagrams are static -- they display what interacts but not what happens when they do interact. The class diagram below models a customer order from a retail catalog. The central class is the Order. Associated with it are the Customer making the purchase and the Payment. A Payment is one of three kinds: Cash, Check, or Credit. The order contains OrderDetails (line items), each with its associated Item.



UML class notation is a rectangle divided into three parts: class name, attributes, and operations. Names of abstract classes, such as *Payment,* are in italics. Relationships between classes are the connecting links.

Our class diagram has three kinds of relationships.

Association -- a relationship between instances of the two classes. There is an association between two classes if an instance of one class must know about the other in order to perform its work. In a diagram, an association is a link connecting two classes.

Aggregation -- an association in which one class belongs to a collection. An aggregation has a diamond end pointing to the part containing the whole. In our diagram, Order has a collection of Order Details.

Generalization -- an inheritance link indicating one class is a superclass of the other. A generalization has a triangle pointing to the superclass. *Payment* is a superclass of Cash, Check, and Credit.

An association has two ends. An end may have a role name to clarify the nature of the association. For example, an OrderDetail is a line item of each Order.

A navigability arrow on an association shows which direction the association can be traversed or queried. An OrderDetail can be queried about its Item, but not the other way around. The arrow also lets you know who "owns" the association's implementation; in this case, OrderDetail has an Item. Associations with no navigability arrows are bi-directional.

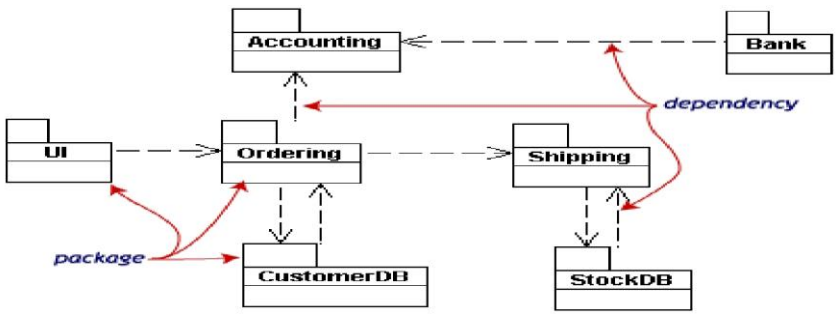
The multiplicity of an association end is the number of possible instances of the class associated with a single instance of the other end. Multiplicities are single numbers or ranges of numbers. In our example, there can be only one Customer for each Order, but a Customer can have any number of **Orders**.
This table gives the most common multiplicities.

Multiplicities	Meaning
0..1	zero or one instance. The notation ***n . . m*** indicates ***n*** to ***m*** instances.
0..* *or* *	no limit on the number of instances (including none).
1	exactly one instance
1..*	at least one instance

Every class diagram has classes, associations, and multiplicities. Navigability and roles are optional items placed in a diagram to provide clarity.

Packages and object diagrams

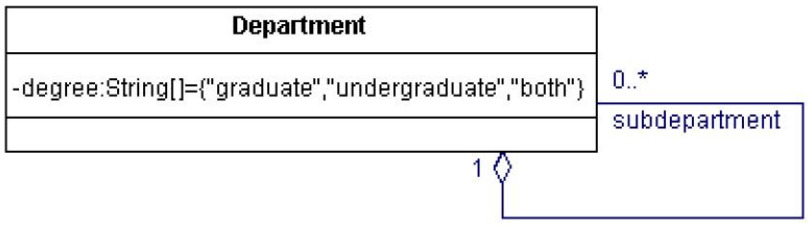
To simplify complex class diagrams, you can group classes into packages. A package is a collection of logically related UML elements. The diagram below is a business model in which the classes are grouped into packages.



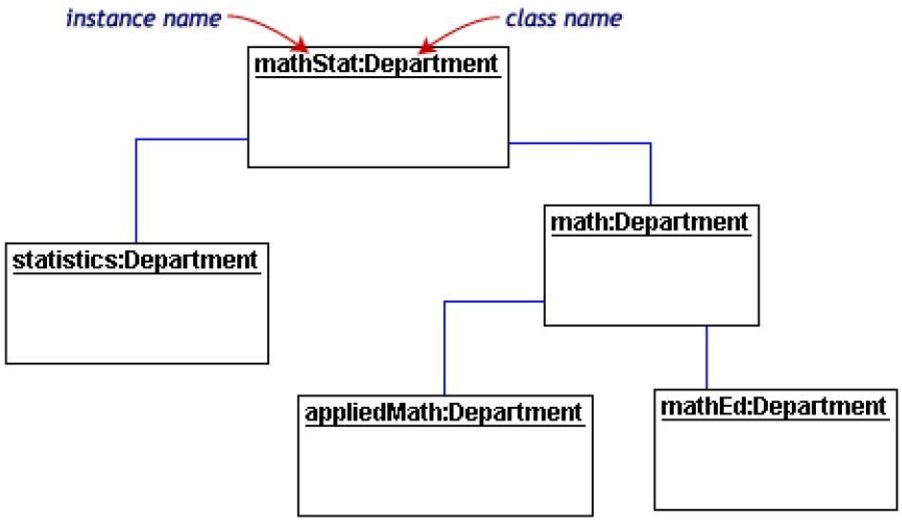
Packages appear as rectangles with small tabs at the top. The package name is on the tab or inside the rectangle. The dotted arrows are dependencies. One package depends on another if changes in the other could possibly force changes in the first.

Object diagrams show instances instead of classes. They are useful for explaining small pieces with complicated relationships, especially recursive relationships.

This small class diagram shows that a university Department can contain lots of other Departments.



The object diagram below instantiates the class diagram, replacing it by a concrete example.



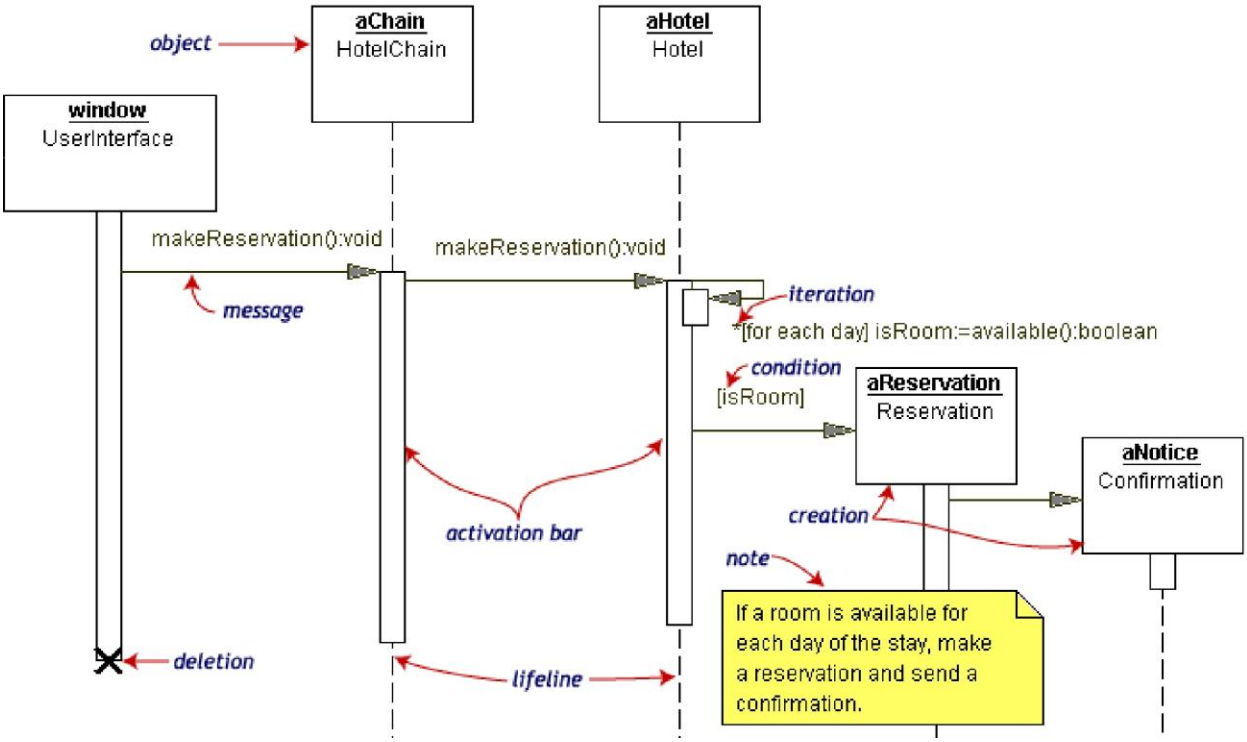
Each rectangle in the object diagram corresponds to a single instance. Instance names are underlined in UML diagrams. Class or instance names may be omitted from object diagrams as long as the diagram meaning is still clear.

Sequence diagrams:

Class and object diagrams are static model views. Interaction diagrams are dynamic. They describe how objects collaborate.

A sequence diagram is an interaction diagram that details how operations are carried out -- what messages are sent and when. Sequence diagrams are organized according to time. The time progresses as you go down the page. The objects involved in the operation are listed from left to right according to when they take part in the message sequence.

Below is a sequence diagram for making a hotel reservation. The object initiating the sequence of messages is a Reservation window.



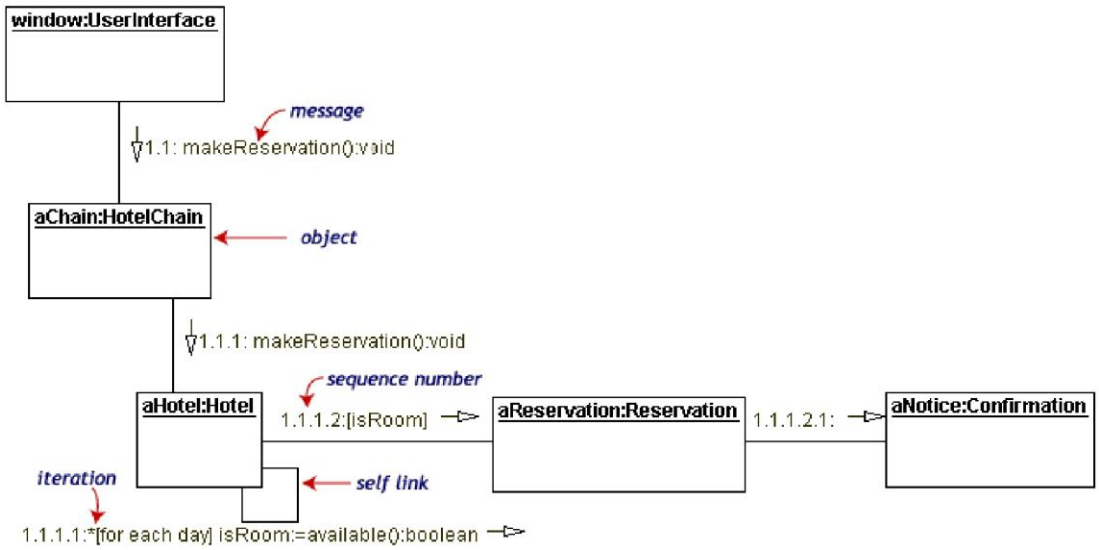
The Reservation window sends a makeReservation() message to a HotelChain. The HotelChain then sends a makeReservation() message to a Hotel. If the Hotel has available rooms, then it makes a Reservation and a Confirmation.

Each vertical dotted line is a lifeline, representing the time that an object exists. Each arrow is a message call Each vertical dotted line is a lifeline, representing the time that an object exists. The activation bar represents the duration of execution of the message.

In our diagram, the Hotel issues a self call to determine if a room is available. If so, then the Hotel creates a Reservation and a Confirmation. The asterisk on the self call means iteration (to make sure there is available room for each day of the stay in the hotel). The expression in square brackets, [ ], is a condition.

Collaboration diagrams:

Collaboration diagrams are also interaction diagrams. They convey the same information as sequence diagrams, but they focus on object roles instead of the times that messages are sent. In a sequence diagram, object roles are the vertices and messages are the connecting links.



The object-role rectangles are labeled with either class or object names (or both). Class names are preceded by colons (:).

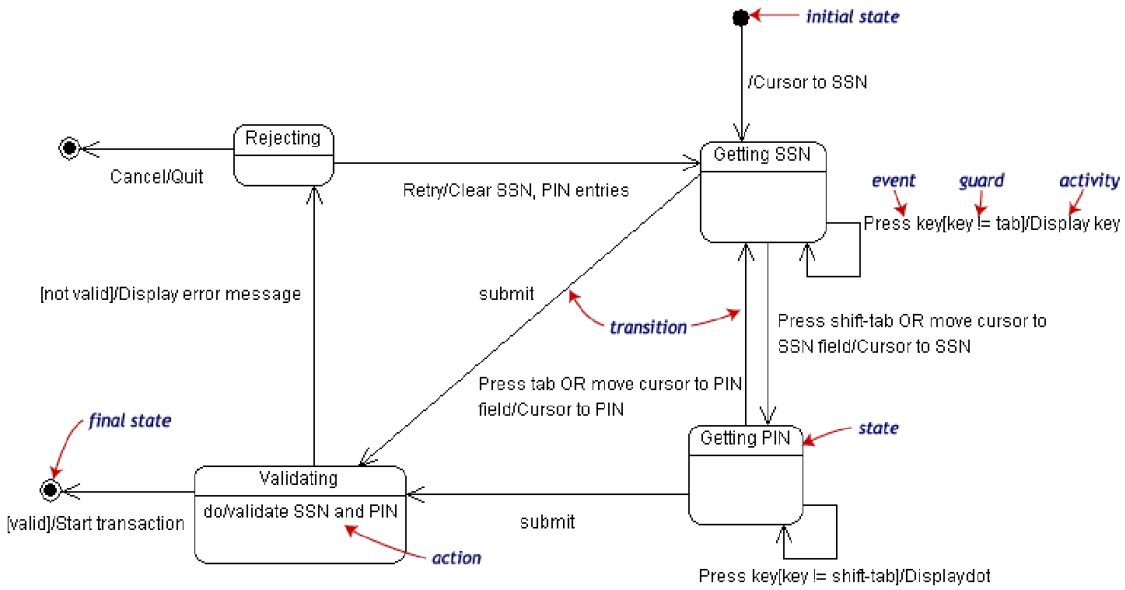
Each message in a collaboration diagram has a **sequence number**. The top-level message is numbered 1. Messages at the same level (sent during the same call) have the same decimal prefix but suffixes of 1, 2, etc. according to when they occur.

Statechart diagrams:

Objects have behaviors and state. The state of an object depends on its current activity or condition. A **statechart diagram** shows the possible states of the object and the transitions that cause a change in state.

Our example diagram models the login part of an online banking system. Logging in consists of entering a valid social security number and personal id number, then submitting the information for validation.

Logging in can be factored into four non-overlapping states: **Getting SSN**, **Getting PIN**, **Validating**, and **Rejecting**. From each state comes a complete set of **transitions** that determine the subsequent state.



States are rounded rectangles. Transitions are arrows from one state to another. Events or conditions that trigger transitions are written beside the arrows. Our diagram has two self-transition, one on Getting SSN and another on Getting PIN.

The initial state (black circle) is a dummy to start the action. Final states are also dummy states that terminate the action.

The action that occurs as a result of an event or condition is expressed in the form /action. While in its Validating state, the object does not wait for an outside event to trigger a transition. Instead, it performs an activity. The result of that activity determines its subsequent state.

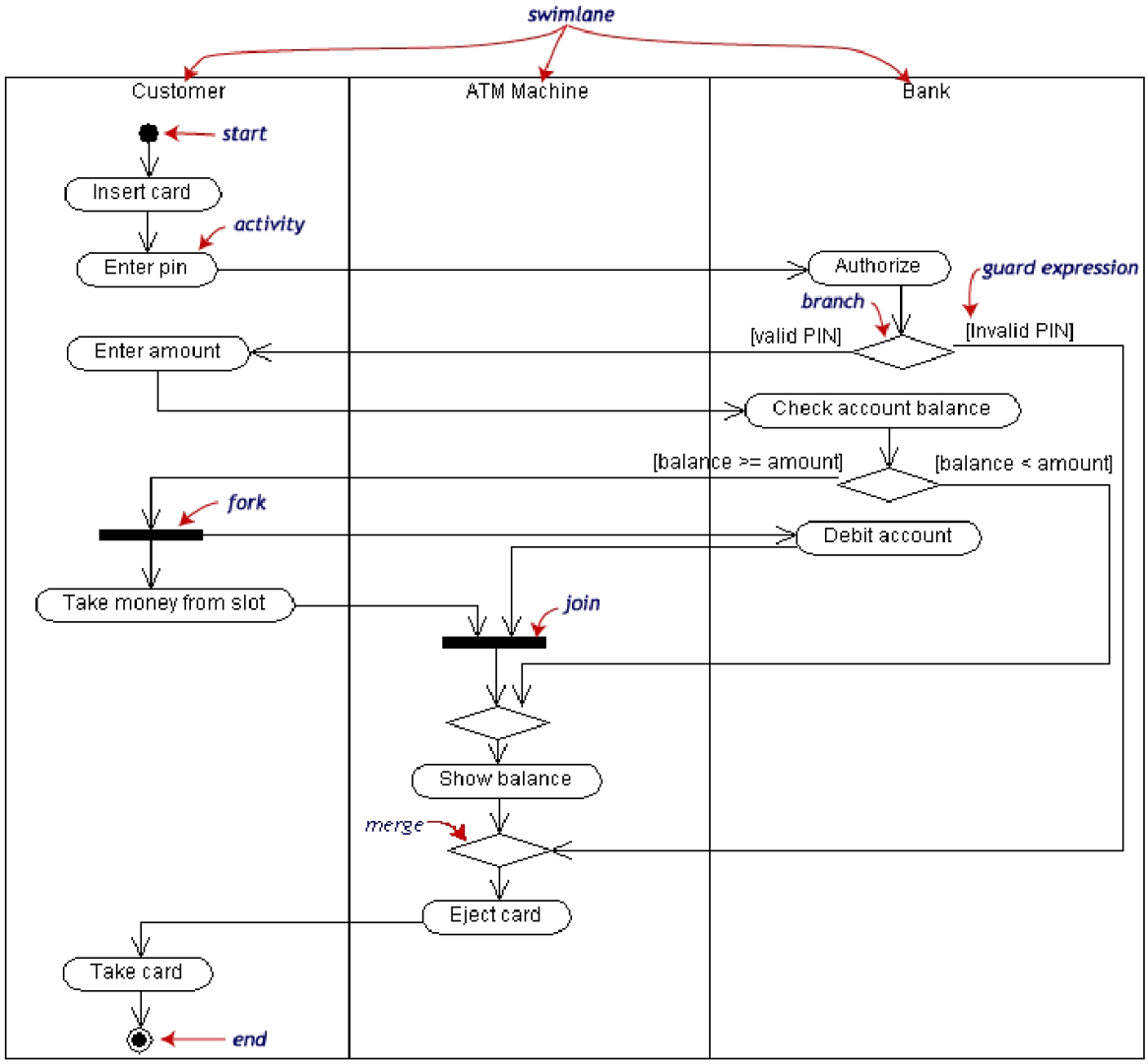
Activity diagrams:

An activity diagram is essentially a fancy flowchart. Activity diagrams and statechart diagrams are related. While a statechart diagram focuses attention on an object undergoing a process (or on a process as an object), an activity diagram focuses on the flow of activities involved in a single process. The activity diagram shows the how those activities depend on one another.

For our example, we used the following process.

"Withdraw money from a bank account through an ATM."

The three involved classes (people, etc.) of the activity are Customer, ATM, and Bank. The process begins at the black start circle at the top and ends at the concentric white/black stop circles at the bottom. The activities are rounded rectangles.



Activity diagrams can be divided into object swimlanes that determine which object is responsible for which activity. A single transition comes out of each activity, connecting it to the next activity.

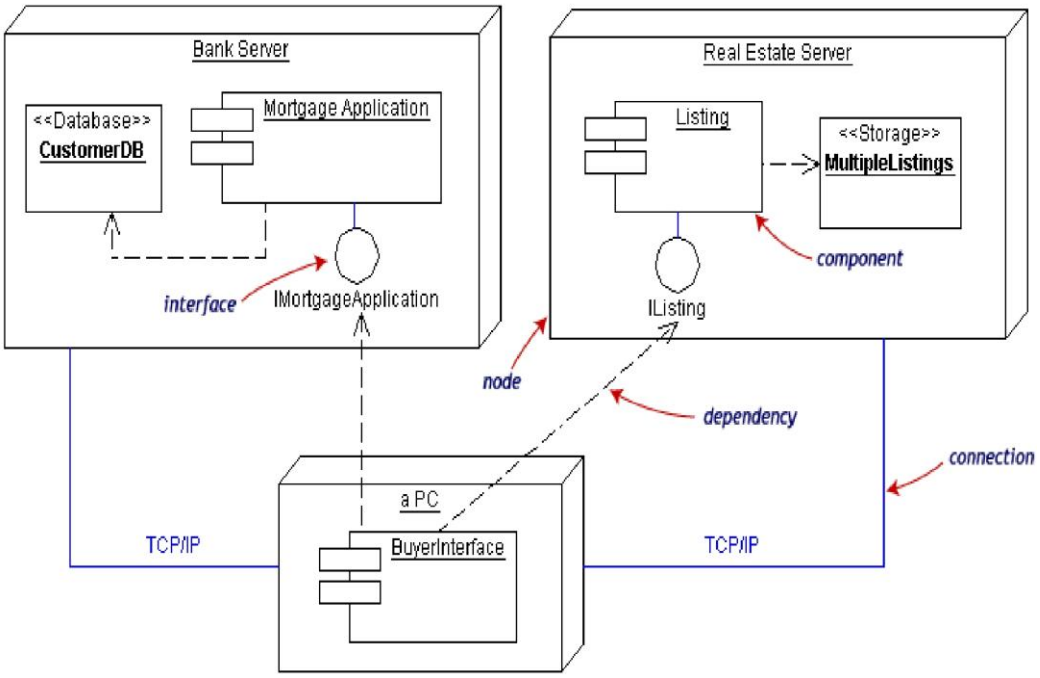
A transition may branch into two or more mutually exclusive transitions. Guard expressions (inside [ ]) label the transitions coming out of a branch. A branch and its subsequent merge marking the end of the branch appear in the diagram as hollow diamonds.

A transition may fork into two or more parallel activities. The fork and the subsequent join of the threads coming out of the fork appear in the diagram as solid bars.

Component and deployment diagrams:

A component is a code module. Component diagrams are physical analogs of class diagram. Deployment diagrams show the physical configurations of software and hardware.

The following deployment diagram shows the relationships among software and hardware components involved in real estate transactions.



The physical hardware is made up of nodes. Each component belongs on a node. Components are shown as rectangles with two tabs at the upper left.

RESULT:

Thus the study of various UML diagram has been analyzed.

Ex No: 2 PASSPORT AUTOMATION SYSTEM

Date:

Aim:

To analyze and design passport automation system using Umbrello Software.

Procedure:

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

The passport Automation system includes actors such as user and passport agent. Also it includes the use cases such as User details. In the Passport automation system Applicant can easily get the passport without difficulties in order to use the passport, applicant needs the bio data and sufficient certificates. At first the passport can be obtained by giving the applicant details, after that the given details is verified by the agent and request for payment process, after the payment corresponding authorities, will send passport through mail.

Project scope

The passport automation system deals with the process of sending passport applicant when payment has been pay. The customer needs their bio data and deposit.

Project objectives

The aim of the passport automation system is to makes the process of sending passport to the applicant.

Software requirement analysis

It deals with following actors

- Request for passport
- Certificate verification
- Issuing passport

Request for passport: Applicant is the person who is applying for passport, and giving the sufficient details and payment.

Certificate verification: The government agent is the person who has accepting the applicant details and verifying their certificate details and payment thereafter providing the passport.

Issuing passport: Government officer is the person who is accepting the passport, verifying it and issuing the passport to the respective applicant.

SOFTWARE REQUIREMENT SPECIFICATION:

1.0 INTRODUCTION

Passport Automation System is an interface between the Applicant and the Authority responsible for the Issue of Passport. It aims at improving the efficiency in the Issue of Passport and reduces the complexities involved in it to the maximum possible extent.

1.1 PURPOSE

If the entire process of 'Issue of Passport' is done in a manual manner then it would takes several months for the passport to reach the applicant. Considering the fact that the number of applicants for passport is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process. As this is a matter of National Security, the system has been carefully verified and validated in order to satisfy it.

1.2 SCOPE

- The System provides an online interface to the user where they can fill in their personal details and submit the necessary documents (may be by scanning).
- The authority concerned with the issue of passport can use this system to reduce his workload and process the application in a speedy manner.
- Provide a communication platform between the applicant and the administrator.
- Transfer of data between the Passport Issuing Authority and the Local Police for verification of applicant's information.
- Users/Applicants will come to know their status of application and the date in which they must subject themselves for manual document verification.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

- **Administrator**
  Refers to the super user who is the Central Authority with the privilege to manage the entire system. It can be any higher official in the Regional Passport Office of Ministry of External Affairs.
- **Applicant**
  One who wishes to obtain the Passport.
- **PAS**
  Refers to this Passport Automation System.
- **HTML**
  Markup Language used for creating web pages.
- **J2EE**
  Java 2 Enterprise Edition is a programming platform java platform for developing and running distributed java applications.
- **HTTP**
  Hyper Text Transfer Protocol.
- **TCP/IP**
  Transmission Control Protocol/Internet Protocol is the communication protocol used to connect hosts on the Internet.

1.4 REFERENCES

IEEE Software Requirement Specification format.

1.5 TECHNOLOGIES TO BE USED

• HTML
• JSP
• Javascript
• Java

1.6 TOOLS TO BE USED

• Eclipse IDE (Integrated Development Environment)
• Umbrello tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements

Overall Description will describe major role of the system components and inter-connections.
Specific Requirements will describe roles & functions of the actors.

2.0 OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

The PAS acts as an interface between the 'applicant' and the 'administrator'. This system tries to make the interface as simple as possible and at the same time not risking the security of data stored in. This minimizes the time duration in which the user receives the passport.

2.2 SOFTWARE INTERFACE

• **Front End Client -** The applicant and Administrator online interface is built using JSP and HTML. The Administrator's local interface is built using Java.
• **Web Server** – Apache Tomcat application server (Oracle Corporation).
• **Back End** – Oracle11g database.

2.3 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

2.4 SYSTEM FUNCTIONS

- Secure Registration of information by the Applicants.
- Schedule the applicants an appointment for manual verification of original documents.
- Panel for Passport Application Status Display by the Administrator.
- SMS and Mail updates to the applicants by the administrator.
- Administrator can generate reports from the information and is the only authorized personnel to add the eligible application information to the database.

2.5 USER CHARACTERISTICS

- **Applicant**

  These are the person who desires to obtain the passport and submit the information to the database.

- **Administrator**

  He has the certain privileges to add the passport status and to approve the issue of passport. He may contain a group of persons under him to verify the documents and give suggestion whether or not to approve the dispatch of passport.

- **Police**

  He is the person who upon receiving intimation from the PAS, perform a personal verification of the applicant and see if he has any criminal case against him before or at present. He has been vetoed with the power to decline an application by suggesting it to the Administrator if he finds any discrepancy with the applicant. He communicates via this PAS.

2.6 CONSTRAINTS

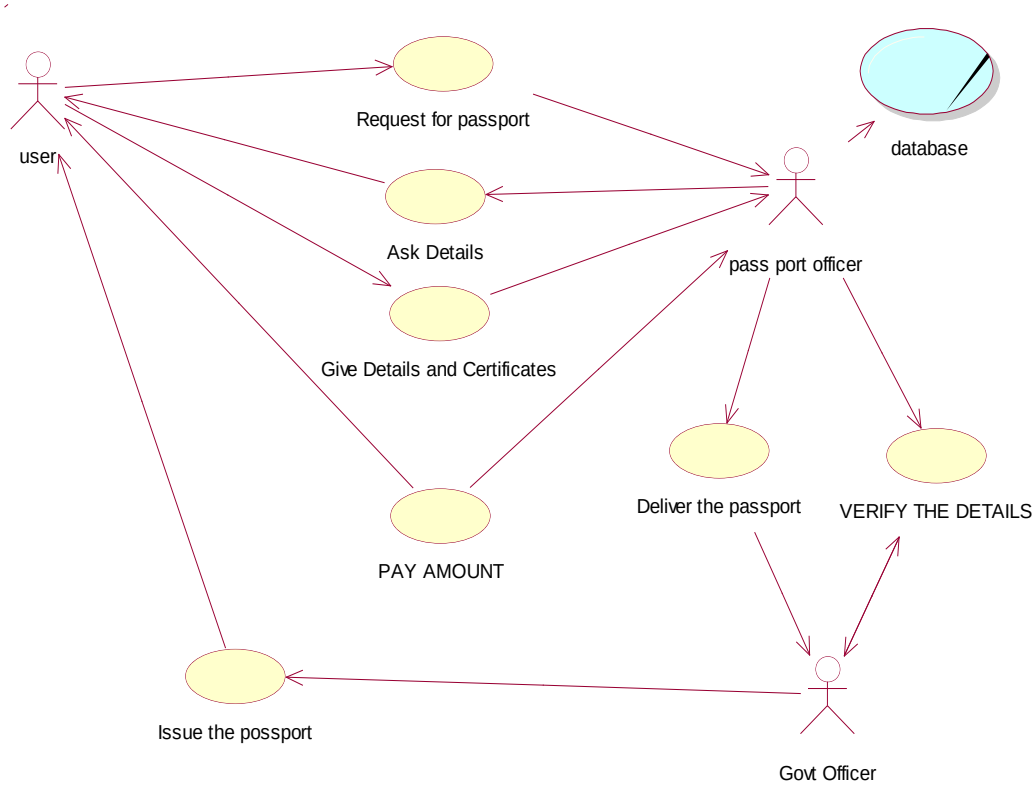
- The applicants require a computer to submit their information.
- Although the security is given high importance, there is always a chance of intrusion in the web world which requires constant monitoring.
- The user has to be careful while submitting the information. Much care is required.

2.7 ASSUMPTIONS AND DEPENDENCIES

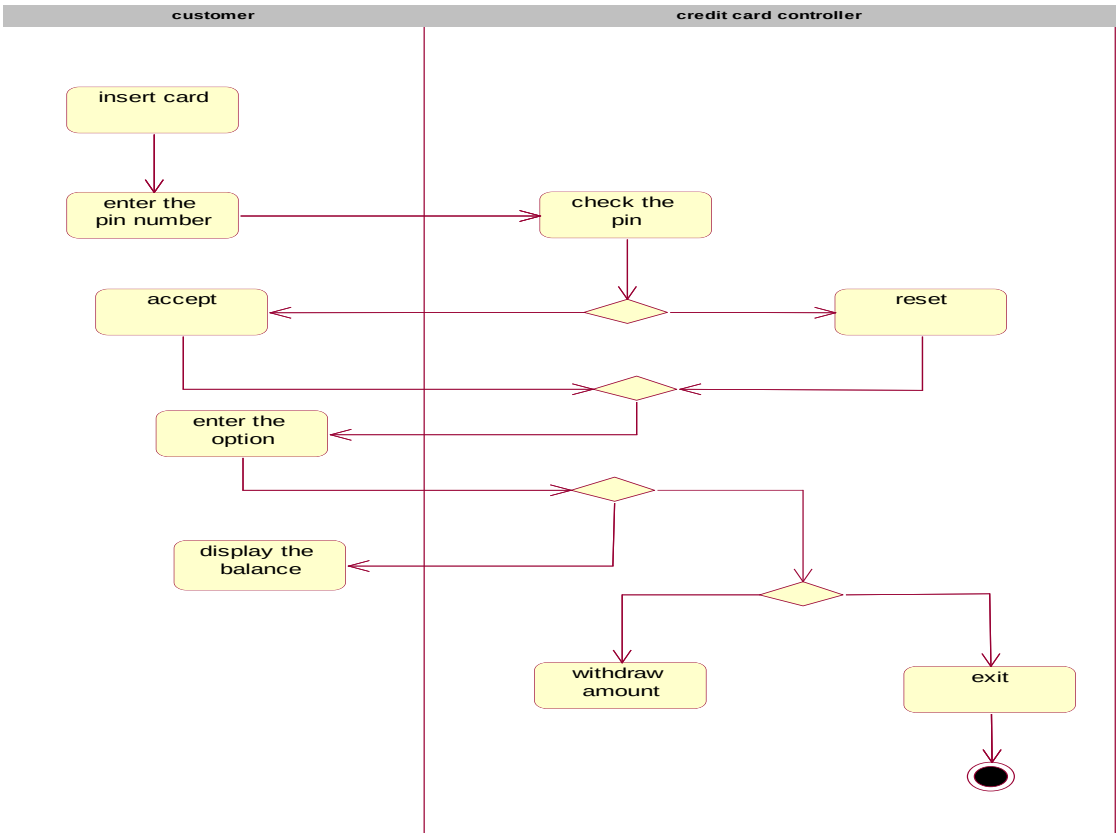
- The Applicants and Administrator must have basic knowledge of computers and English Language.
- The applicants may be required to scan the documents and send.

UML diagrams:

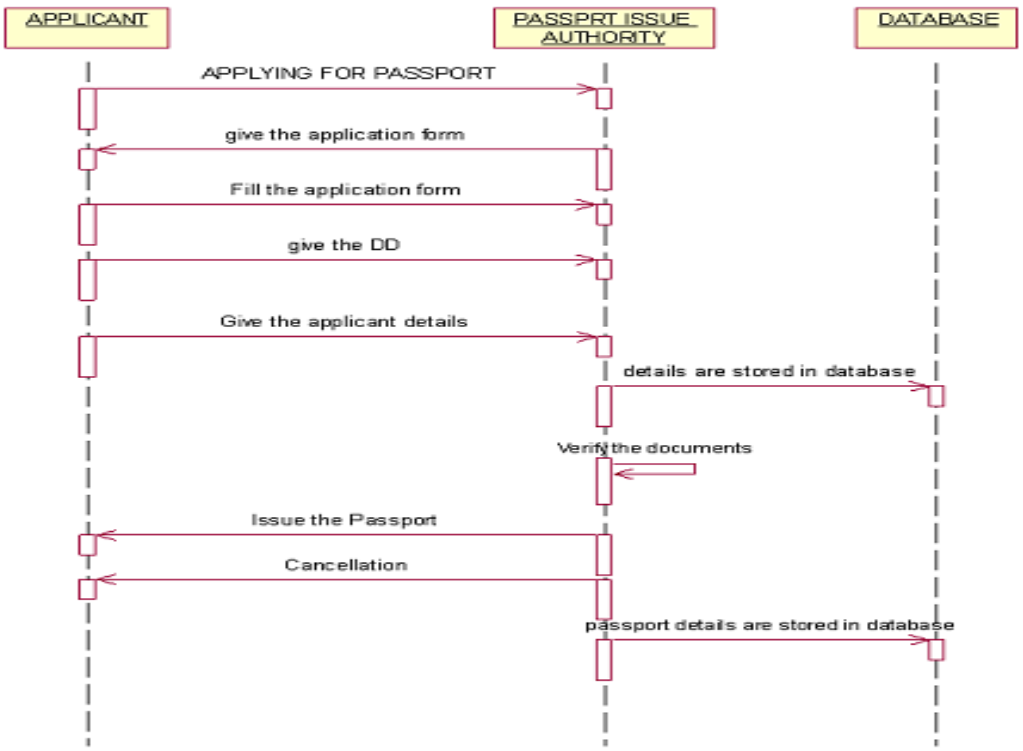
Use case diagram:



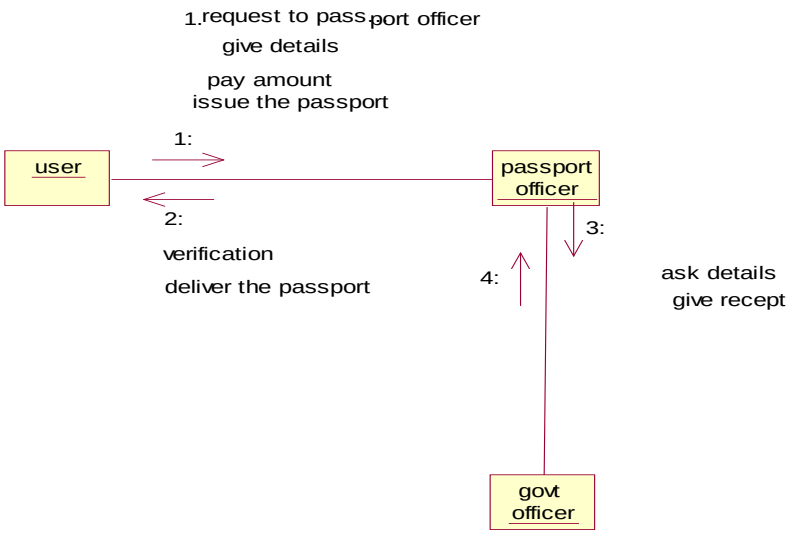
Activity diagram:



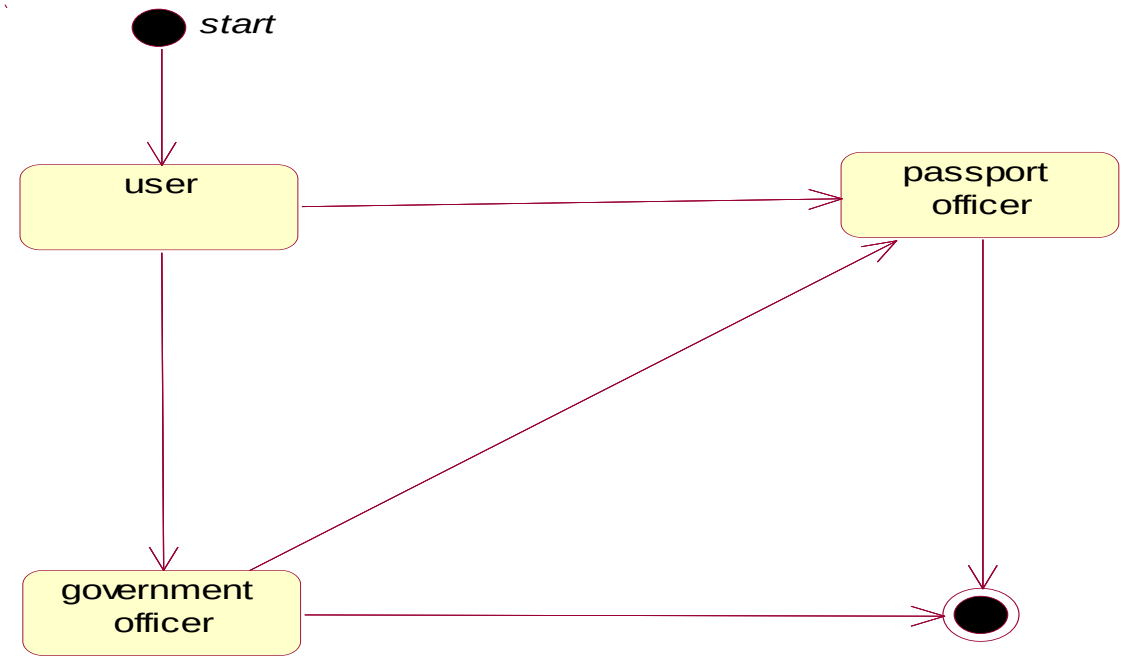
Sequence diagram



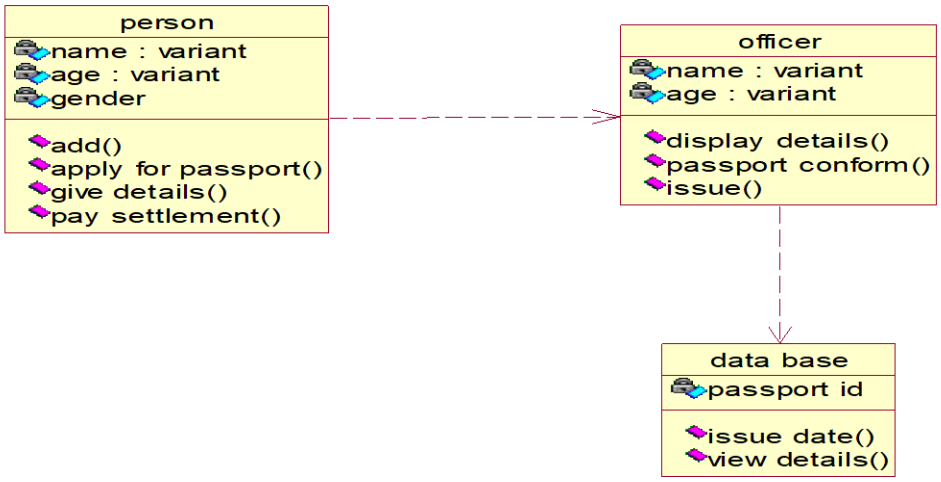
Collaboration diagram:



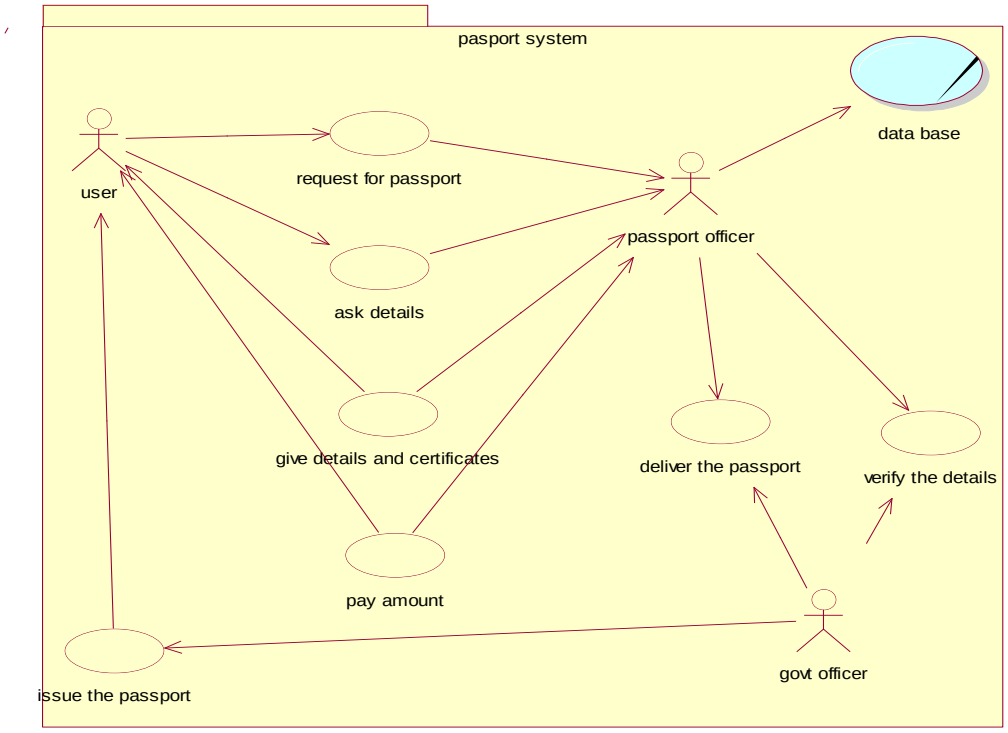
State diagram



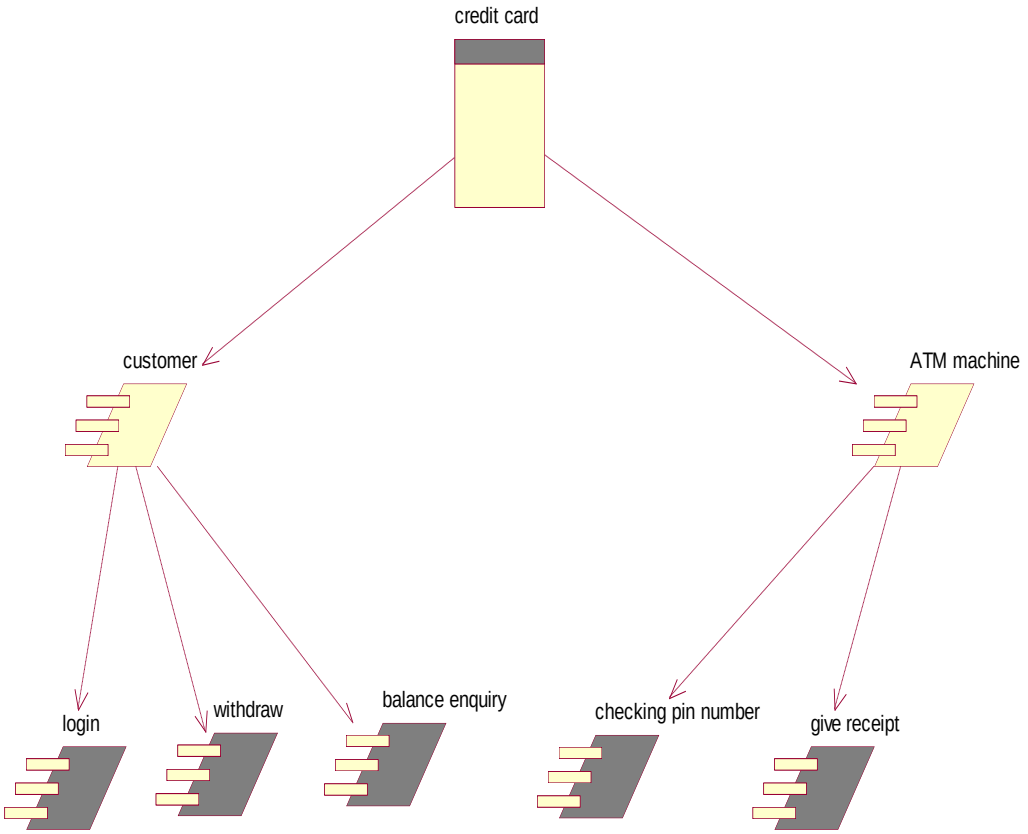
Class diagram:



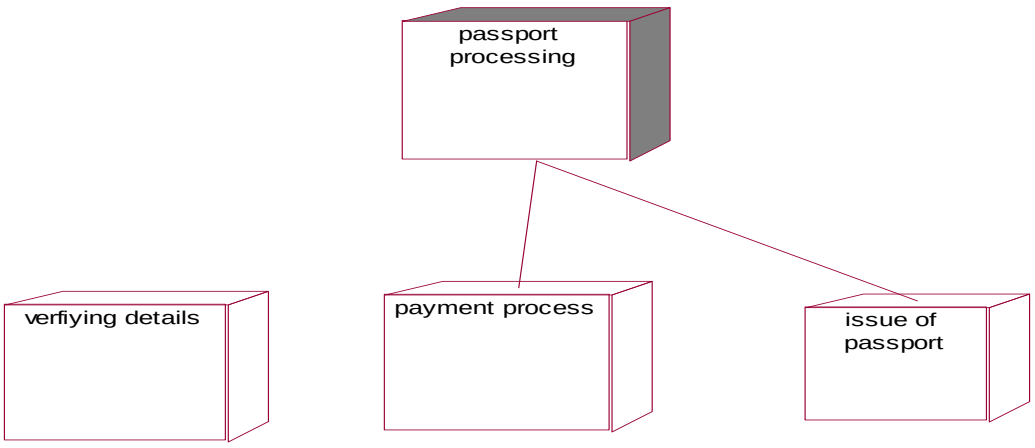
Package diagram:



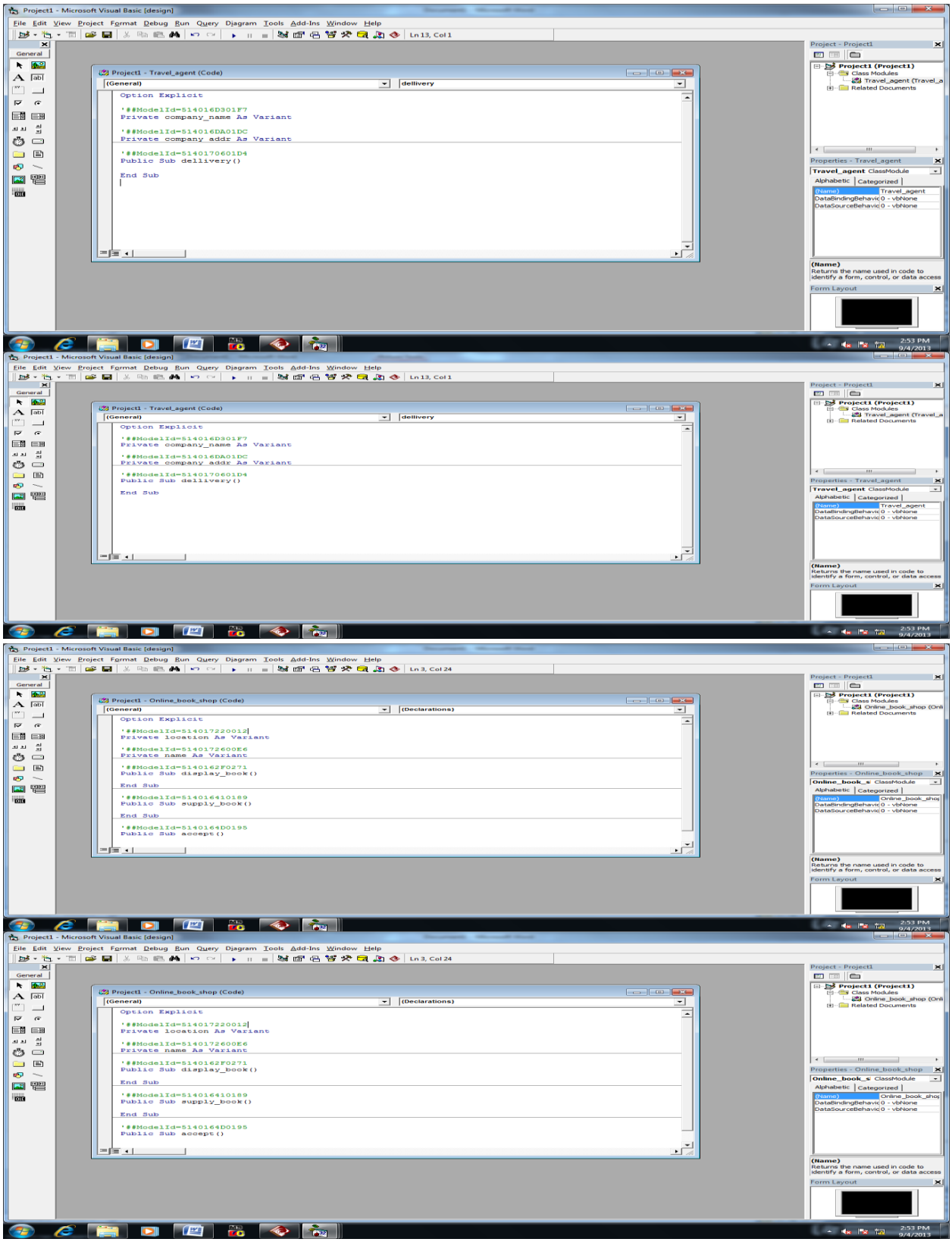
Component diagram:



Deployment diagram



Output



Result:

Thus the passport automation system was designed and the code was generated successfully.

EX. No: 3 BOOK BANK

Date :

Aim

To analyze and design book bank using Umbrello Software.

Procedure

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

To promote effective sharing of books among students, book bank is used. Books purchased under the book bank scheme are the property of that particular book bank. The users are entitled to borrow books from the faculty. The faculty will determine duration of a loan for a book but that should in no case exceed duration period. A user must return a borrowed book within the prescribed loan period. With the exception of certain categories of books in the book bank, all books may be borrowed from the faculty book bank by filling in the borrowing form. The date and the user's name will be entered on the data card. The book card will be removed from the book packet. The signing of the book card and borrowing form constitute the loan. A user remains responsible for a book as long as borrowing records in the faculty remain un-cancelled. A user is not authorized to loan a book to another reader. Borrowed books must be returned to the faculty book bank. A user who loses a book or fails to return book will be required to pay the fine.

Project scope

The book bank deals with sharing of books among user like students and staffs. The faculty book bank also provides membership to the user and it allows the member to take the book from the book bank. Mainly, the students and staffs in colleges utilize these facilities.

Project objectives

The aim of the book bank is to help in the selection of books for the university's departments, schools and Institutes. This book bank should provide limited funds allocated for text books.

Software requirement analysis:

The modules of the book bank are:

- Book details
- Issue borrowing form and book card
- Issue book
- Return book
- Pay fine
- Issuing receipt

Book details:

The module of book details consists of certain book details such as Engineering books, medical books, business management books etc,.

Issue borrowing form and book card:

The faculty will issue the borrowing form which consists of data and the user's name. The faculty will also issue book card to get a sign from a user and these two constitute the loan.

Issue book:

This module is used by the faculty of the book bank to issue books to the user.

Return book:

This module is used by the user to return the books to the faculty of the book bank on or before duration period.

Pay fine:

The user will pay fine to the accountant through this module.

Issuing Receipt:

After paying the fine, the accountant will give receipt to the user through this module.

SOFTWARE REQUIREMENTS SPECIFICATION

1.0 INTRODUCTION

Book Bank is the interface between the students and Librarian. It aims at improving the efficiency in the Issue of books or magazines and reduce the complexities involved in it to the maximum possible extent.

1.1 PURPOSE

If the entire process of 'Issue of Books or Magazines' is done in a manual manner then it would take several months for the books or magazines to reach the applicant. Considering the fact that the number of students for Book Bank is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process. The system has been carefully verified and validated in order to satisfy it.

1.2 SCOPE

The System provides an online interface to the user where they can fill in their personal details and submit the necessary documents (may be by scanning). The authority concerned with the issue of books can use this system to reduce his workload and process the application in a speedy manner.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

- **Librarian** - Refers to the super user who is the Central Authority who has been vested with the privilege to manage the entire system.
- **Student** - One who wishes to obtain the Books or Magazines.

- **HTML** - Markup Language used for creating web pages.
- **J2EE** - Java 2 Enterprise Edition is a programming platform and it is the part of the java platform for developing and running distributed java applications.
- **HTTP** -Hyper Text Transfer Protocol
- **TCP/IP** - Transmission Control Protocol/Internet Protocol is the communication protocol used to connect hosts on the Internet.

1.4 REFERENCES

IEEE Software Requirement Specification format

1.5 TECHNOLOGIES TO BE USED

- Visual Basic
- Oracle 11g

1.6 TOOLS TO BE USED

- Visual Basic Tools
- Umbrello tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements.

Overall description will describe major role of the system components and inter-connections.

Specific requirements will describe roles & functions of the actors.

2.0 OVERALL DESCRIPTION

It will describe major role of the system components and inter-connections.

2.1 PRODUCT PERSPECTIVE

The SRS acts as an interface between the 'Students' and the 'Librarian'. This system tries to make the interface as simple as possible and at the same time not risking the security of data stored in. This minimizes the time duration in which the user receives the books or magazines.

2.2 SOFTWARE INTERFACE

- **Front End Client** - The Student and Librarian online interface is built using Visual studio.
- **Back End** - Oracle 11 g database

2.3 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

2.4 SYSTEM FUNCTIONS

- Secure Registration of information by the Students.
- Librarian can generate reports from the information and is the only authorized personnel to add the eligible application information to the database.

2.5 USER CHARACTERISTICS

- **Student** - They are the people who desire to obtain the books and submit the information to the database.
- **Librarian** - He has the certain privileges to add the books and to approval of the reservation of books.

2.6 CONSTRAINTS

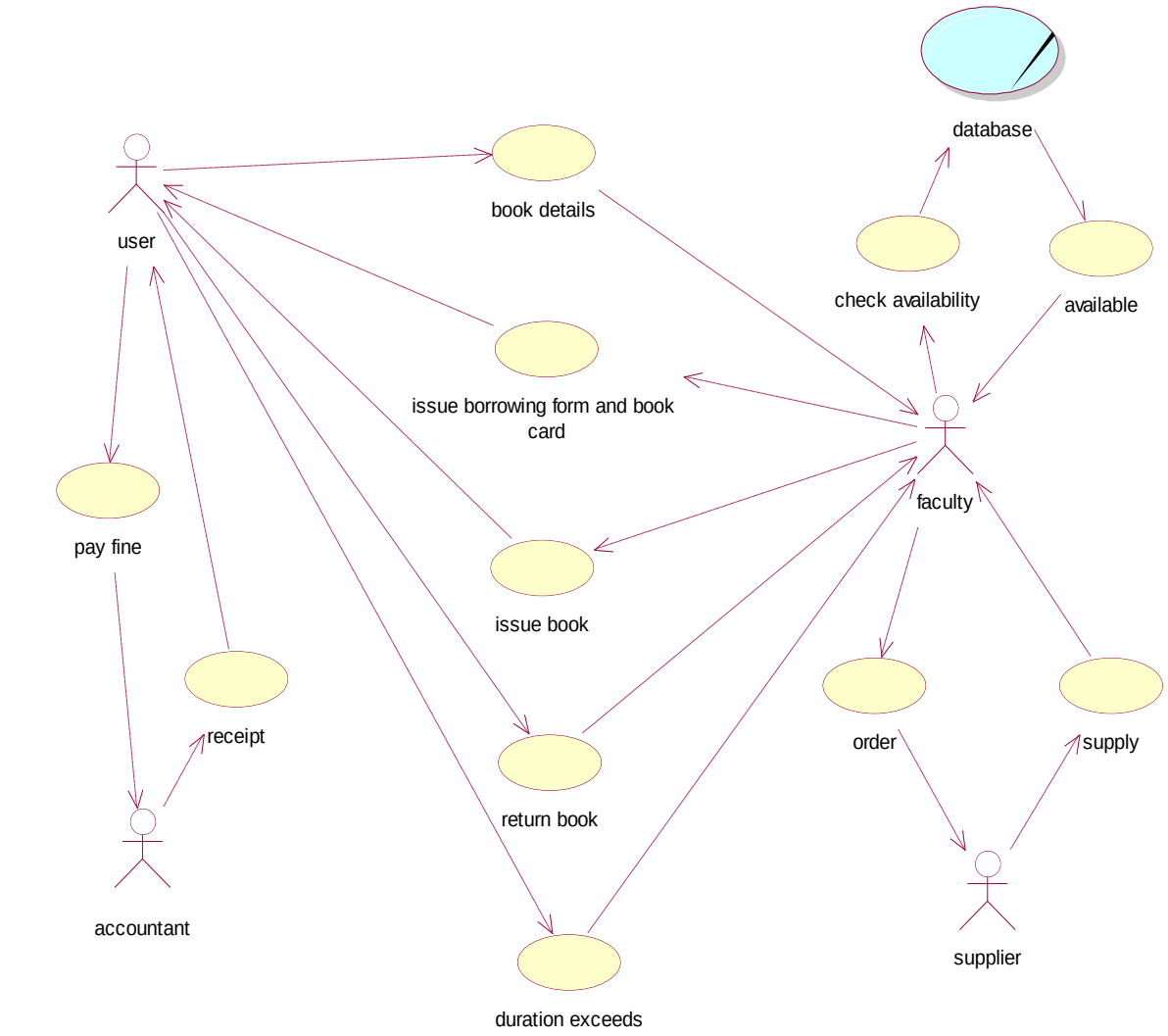
- The Students require a computer to submit their information.
- Although the security is given high importance, there is always a chance of intrusion in the web world which requires constant monitoring.
- The Students has to be careful while submitting the information. Much care is required.

2.7 ASSUMPTIONS AND DEPENDENCIES

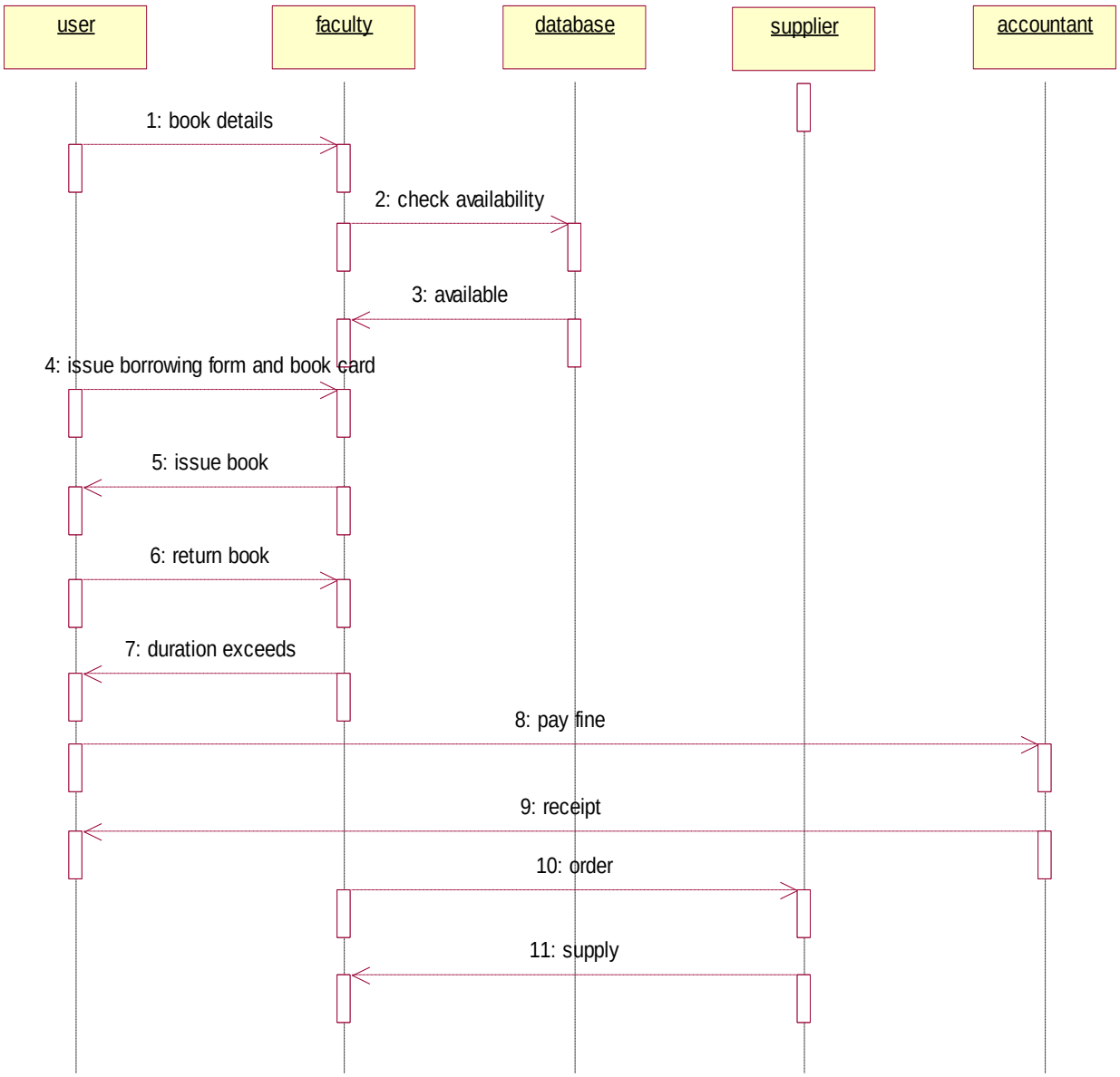
- The Student and Librarian must have basic knowledge of computers and English Language.
- The Students may be required to scan the documents and send.

UML diagrams

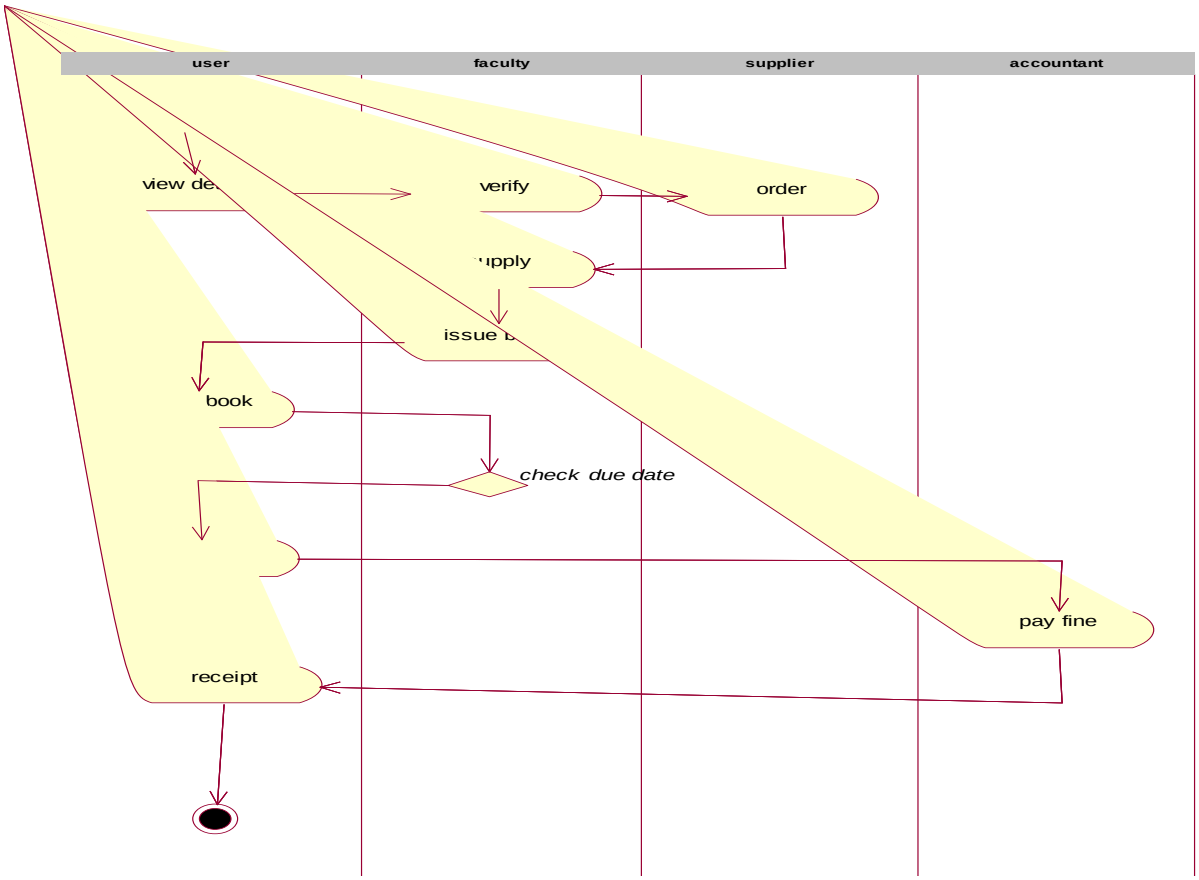
Use case diagram



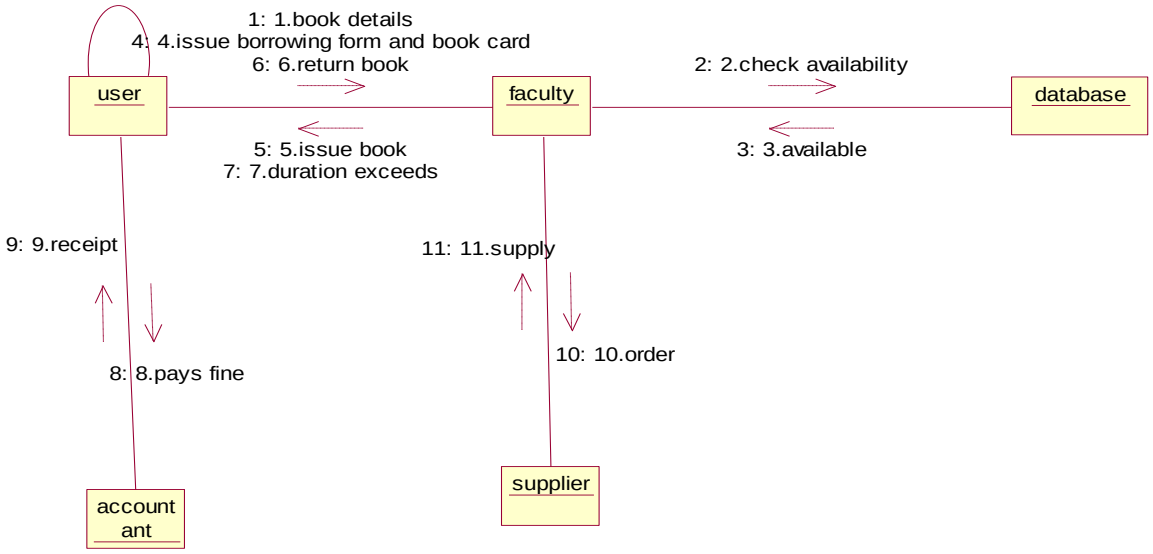
Sequence diagram:



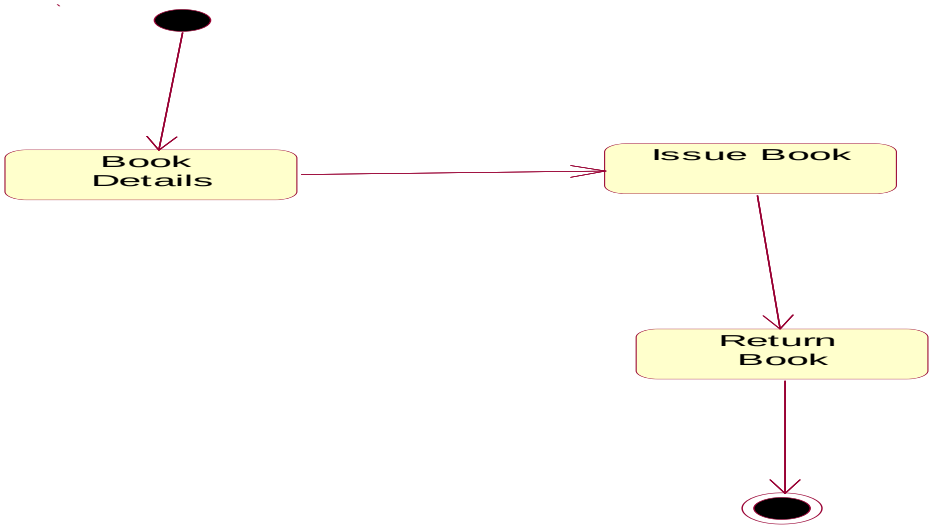
Activity diagram:



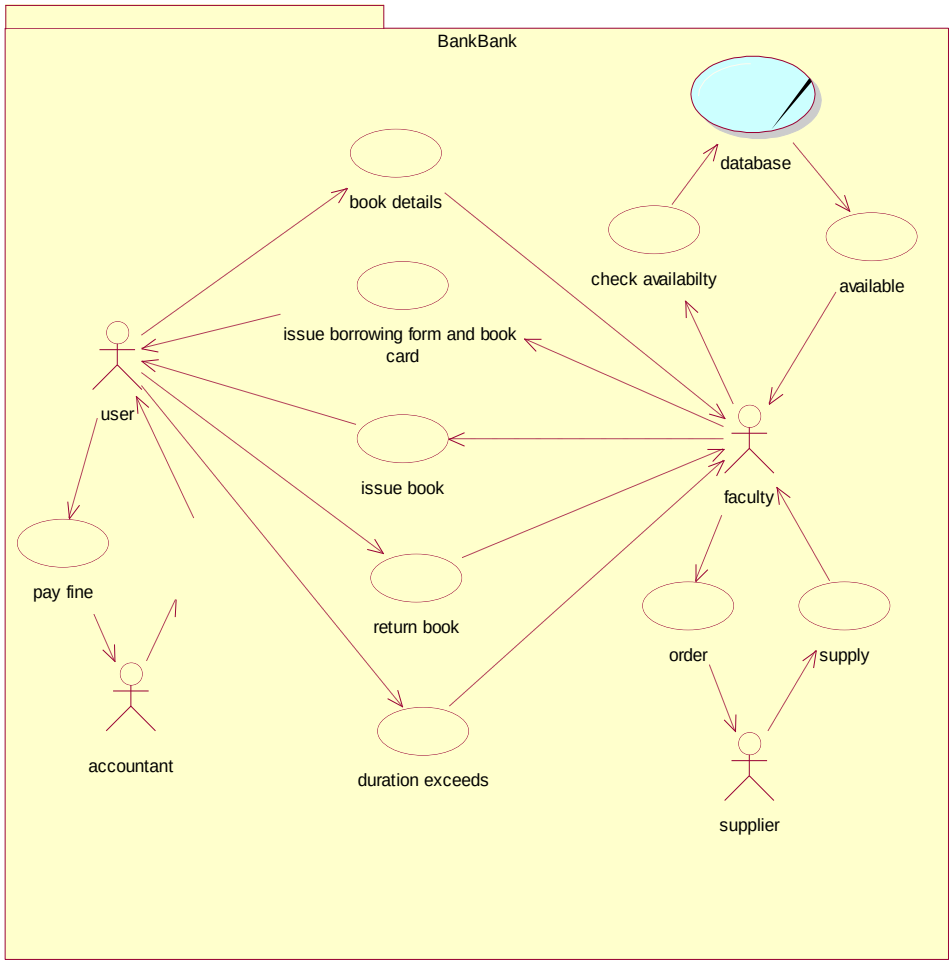
Collaboration Diagram:



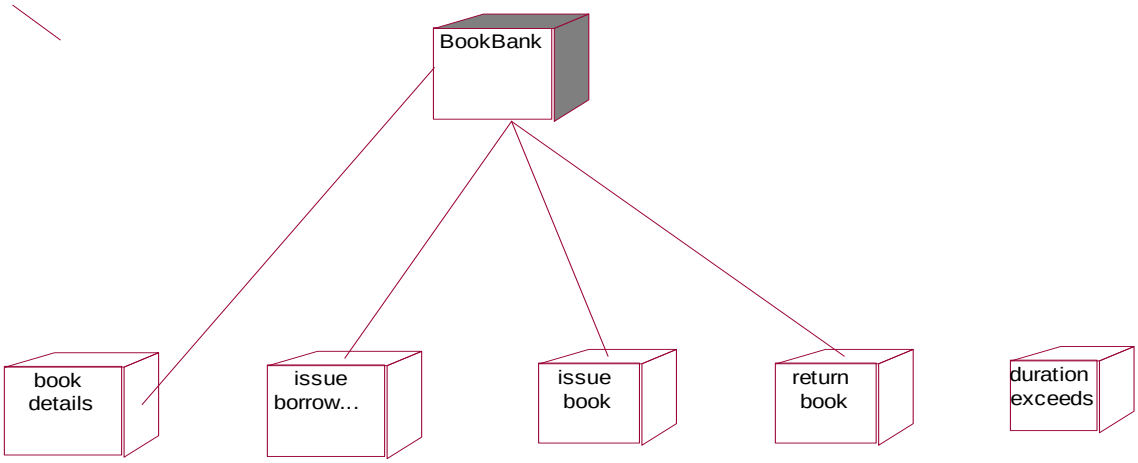
State Diagram



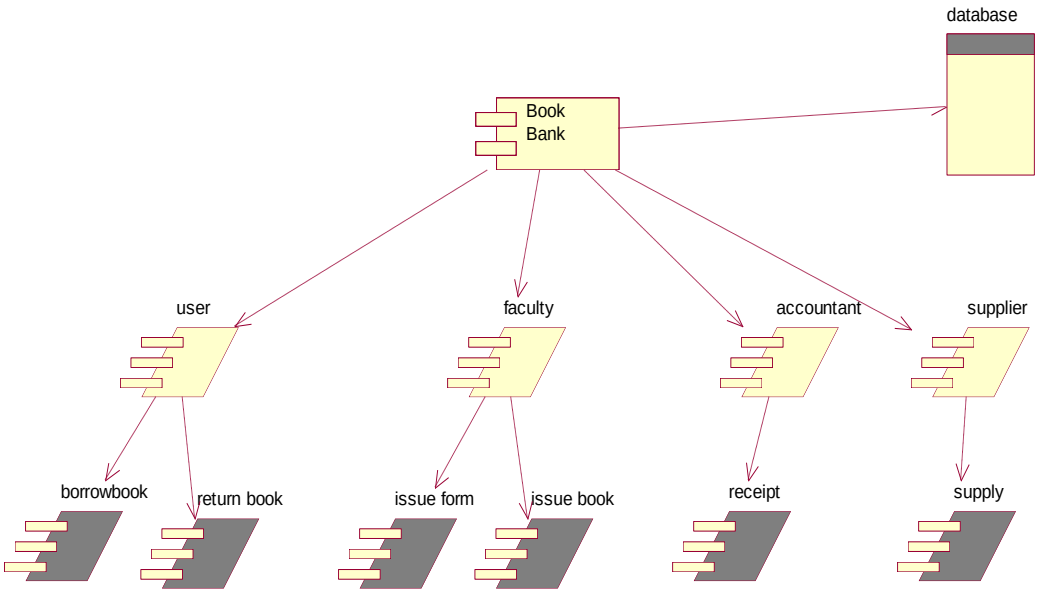
PackageDiagram:



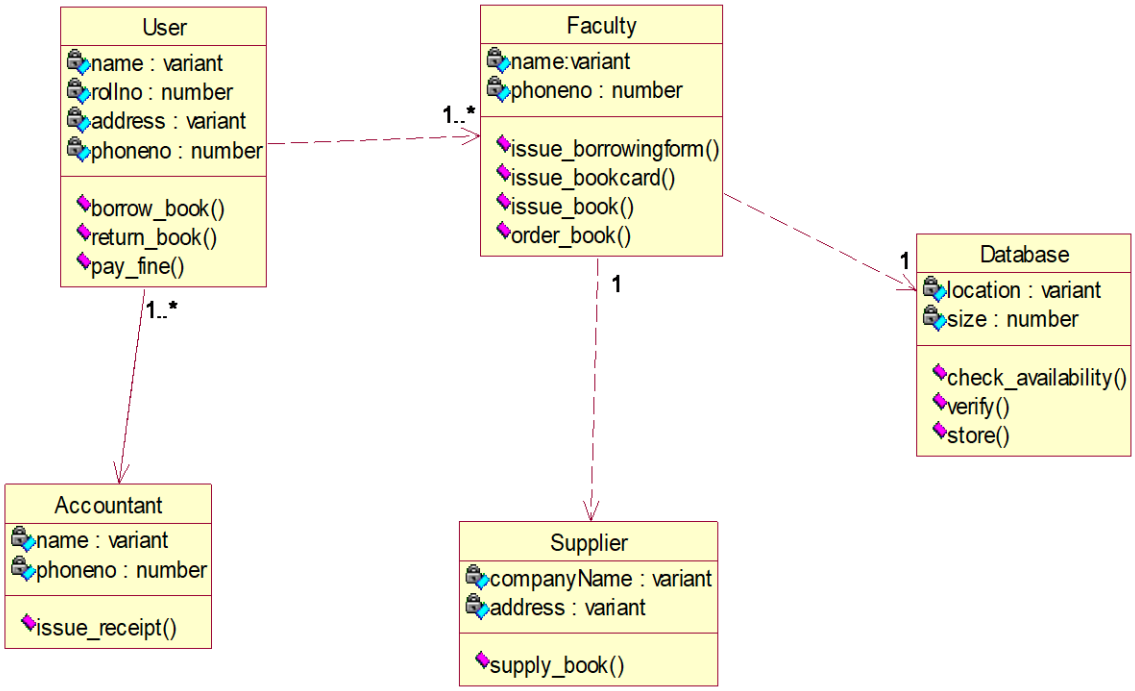
Deployment Diagram:



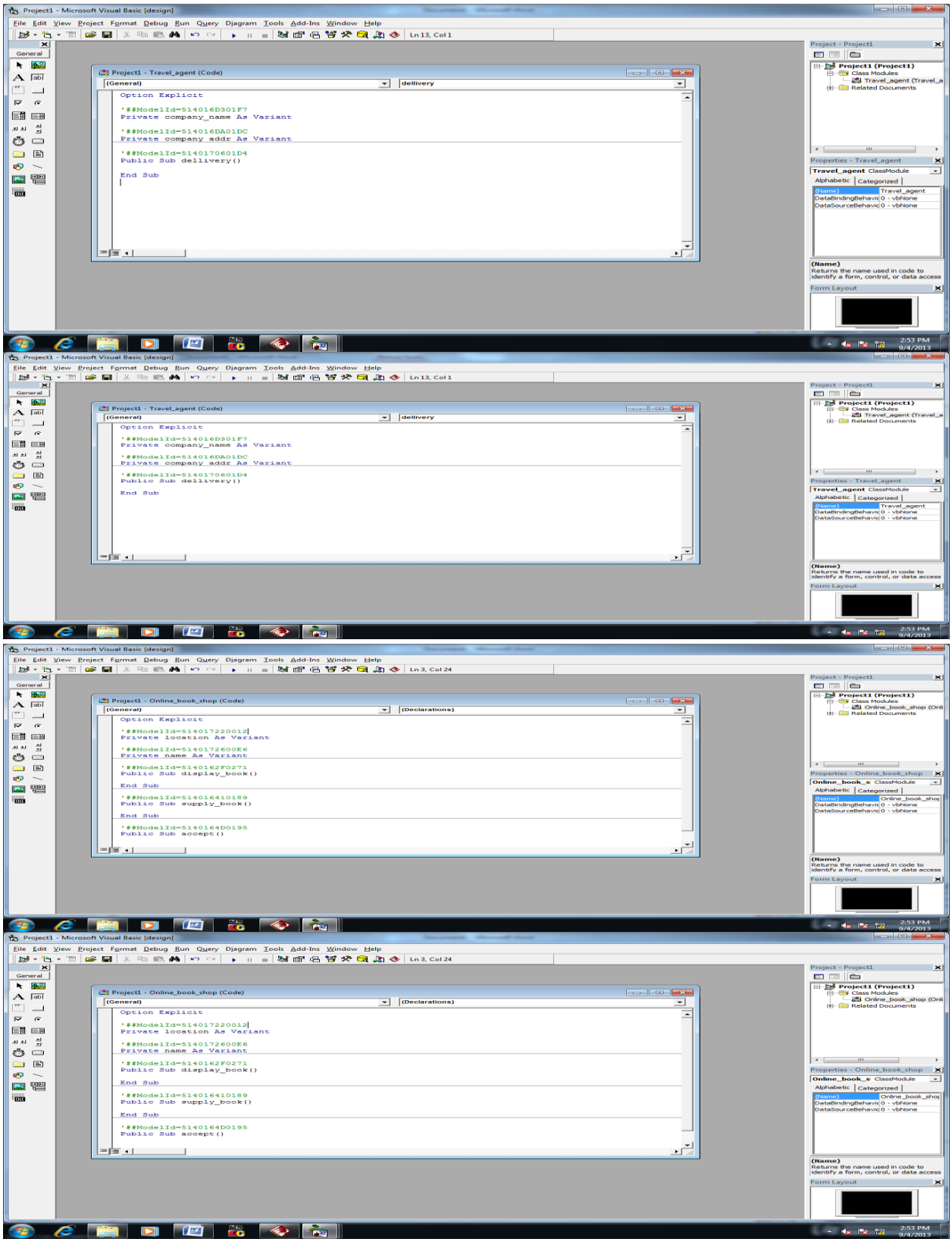
Component Diagram:



Class diagram:



Output:



Result:

Thus the book bank was designed and the code was generated successfully.

Ex. No: 4 EXAM REGISTRATION SYSTEM

Date:

Aim:

To analyze and design Exam Registration system using Umbrello Software.

Procedure:

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

Exam Registration includes the following functions view the exam details, apply to get the form fill the application form, submit the form. The user can view the exam details through the online. The user can choose the corresponding exam which they need. If the user selects the corresponding exam then user can apply for application form. The application form contains the user details. The application form has been filled up by the user. If the application form was fill up then the user can submit the application form to the exam coordinator.

Project scope

In the Exam registration the user can apply for corresponding exam through online and get the application form from the Exam coordinator. Fill that application form and send it to the exam coordinator. They check that form and issue Hall Ticket for the exam.

Project objectives

The aim of exam registration is to help for the students to know, view and apply the exam.

Software requirement analysis

The modules of the Exam registration are:

- View the exam details
- Select the corresponding exam
- Apply for Application form
- Fill and summit the Application form
- Submit the form
- Issue the Hall Ticket

View the exam details:

This module is used to view the corresponding exam through the online.

Select the corresponding exam:

This module is used to select the corresponding exam from list of exam details.

Apply for Application Form:

This module is used for apply the application form for corresponding exam.

Fill and summit the Application Form

This module is used to fill the Name, mark details, DOB and user details. After filling the form then submit the form to exam coordinator.

Issuing Hall Ticket

The exam coordinator checks the application form and sends the Hall ticket to the user. The user issues the Hall Ticket.

SOFTWARE REQUIREMENT SPECIFICATION:

1.0 INTRODUCTION

Exam Registration System is an interface between the Student and the Exam Controller responsible for the Issue of Hall Ticket. It aims at improving the efficiency in the Issue of Hall ticket and reduces the complexities involved in it to the maximum possible extent.

1.1 PURPOSE

If the entire process of 'Issue of Hall ticket' is done in a manual manner then it would takes several days for the hall ticket to reach the student. Considering the fact that the number of students for hall ticket is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process. As this is a matter of National Security, the system has been carefully verified and validated in order to satisfy it.

1.2 SCOPE

• The System provides an online interface to the user where they can fill in their personal details and submit the necessary documents (may be by scanning).
• The controller concerned with the issue of hall ticket can use this system to reduce his workload and process the application in a speedy manner.
• Provide a communication platform between the student and the controller.
• Students will come to know their status of application and the date in which they must subject themselves for manual document verification.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

• **Exam Controller** - Refers to the super user who is the Central Authority who has been vested with the privilege to manage the entire system.
• **Student** - One who wishes to obtain the Hall Ticket.
• **ERS** - Refers to this Examination Registration System.
• **HTML** - Markup Language used for creating web pages.
• **J2EE** – Java 2 Enterprise Edition is a programming platform java platform for developing and running distributed java applications.
• **HTTP** - Hyper Text Transfer Protocol.
• **TCP/IP** – Transmission Control Protocol/Internet Protocol is the communication protocol used to connect hosts on the Internet.

1.4 REFERENCES

IEEE Software Requirement Specification format.

1.5 TECHNOLOGIES TO BE USED

- ✓ HTML
- ✓ JSP
- ✓ Javascript
- ✓ Java

1.6 TOOLS TO BE USED

- o Eclipse IDE (Integrated Development Environment)
- o Umbrello tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements.

Overall Description will describe major role of the system components and inter-connections.
Specific Requirements will describe roles & functions of the actors.

2.0. OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

The ERS acts as an interface between the 'student' and the 'exam controller'. This system tries to make the interface as simple as possible and at the same time not risking the security of data stored in. This minimizes the time duration in which the user receives the hall ticket.

2.2 SOFTWARE INTERFACE

- ✓ **Front End Client** - The exporter online interface is built using JSP and HTML.
- ✓ **Web Server** – Apache Tomcat Server (Oracle Corporation)
- ✓ **Back End** - Oracle 11g database

2.3 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

2.4 SYSTEM FUNCTIONS

• Secure Registration of information by the Students.
• SMS and Mail updates to the students by the controller.
• Controller can generate reports from the information and is the only authorized personnel to add the eligible application information to the database.

2.5 USER CHARACTERISTICS

• **Student** - They are the people who desire to obtain the hall ticket and submit the information to the database.
• **Exam controller** - He has the certain privileges to add the registration status and to approve the issue of hall ticket. He may contain a group of persons under him to verify the documents and give suggestion whether or not to approve the dispatch of hall ticket.

2.6 CONSTRAINTS

• The applicants require a computer to submit their information.
• Although the security is given high importance, there is always a chance of intrusion in the web world which requires constant monitoring.
• The user has to be careful while submitting the information. Much care is required.

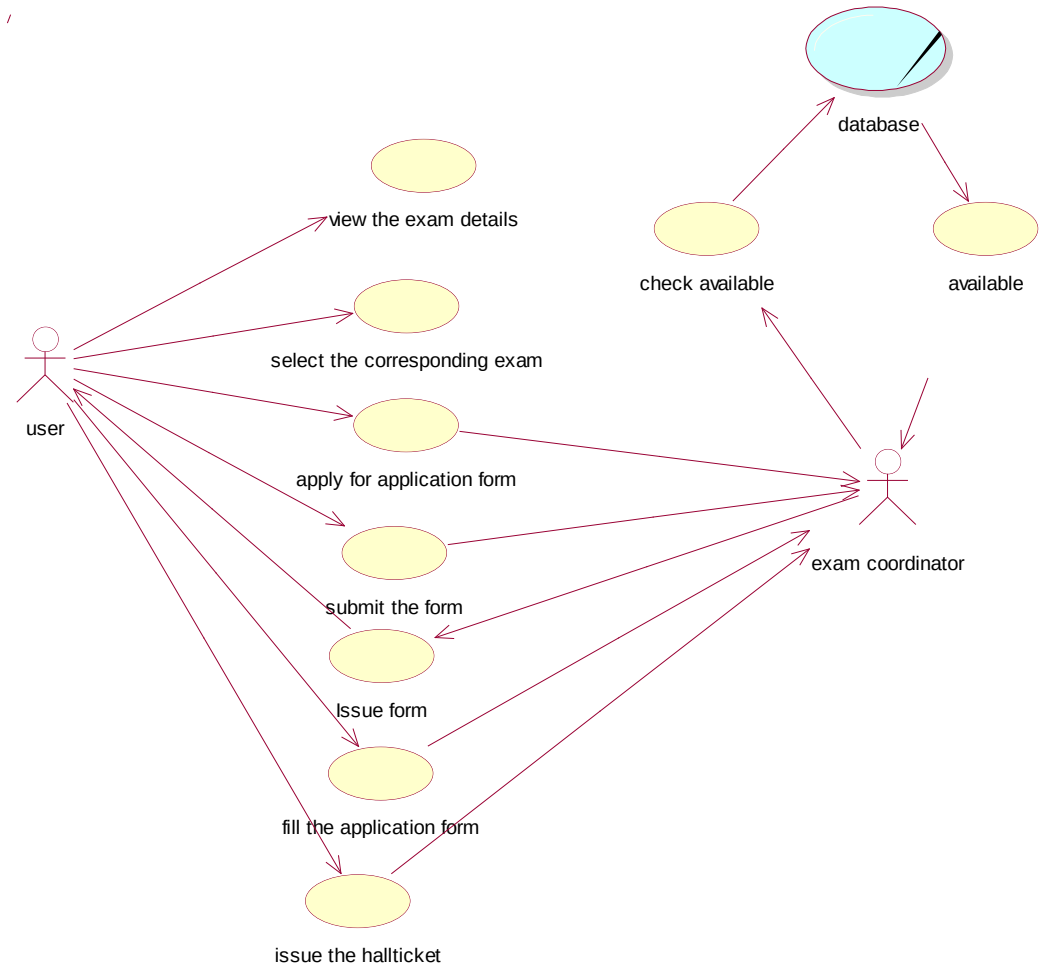
2.7 ASSUMPTIONS AND DEPENDENCIES

• The Students and Exam Controller must have basic knowledge of computers and English Language.

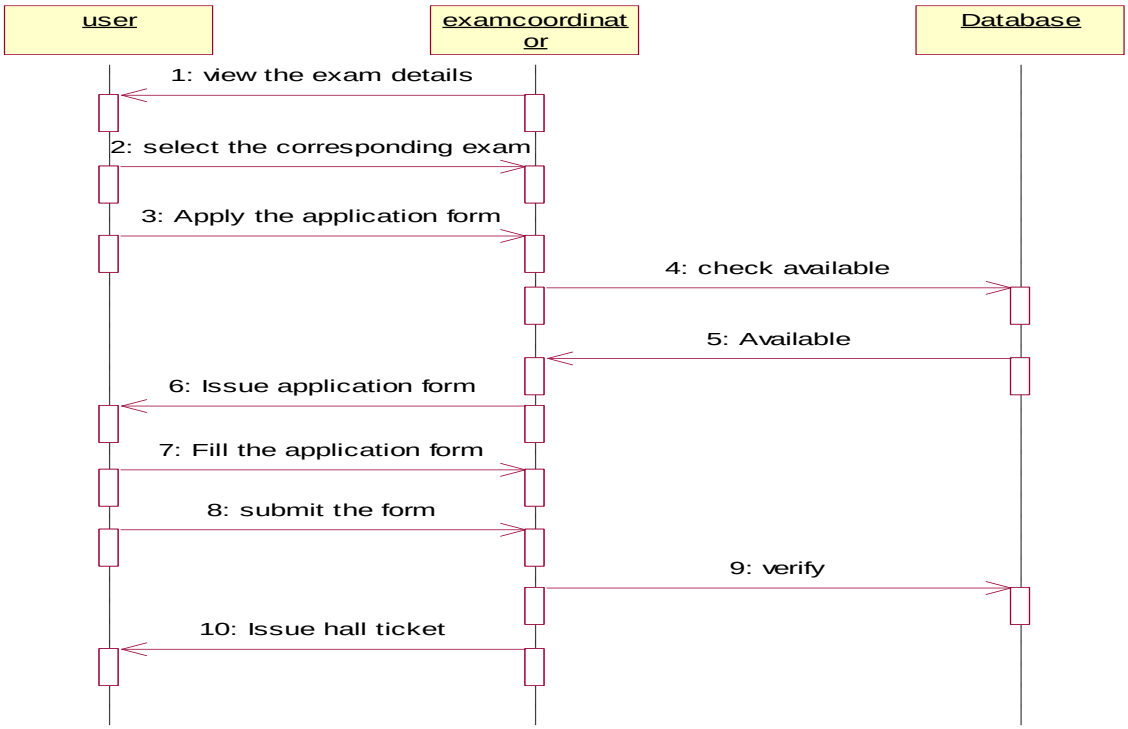
• The student may be required to scan the documents and send.

UML Diagrams:

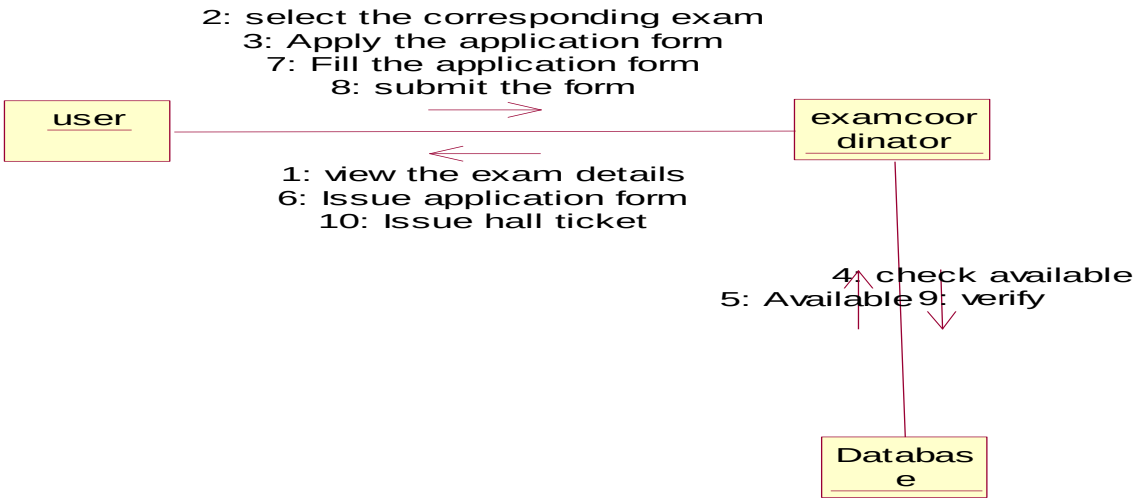
UsecaseDiagram:



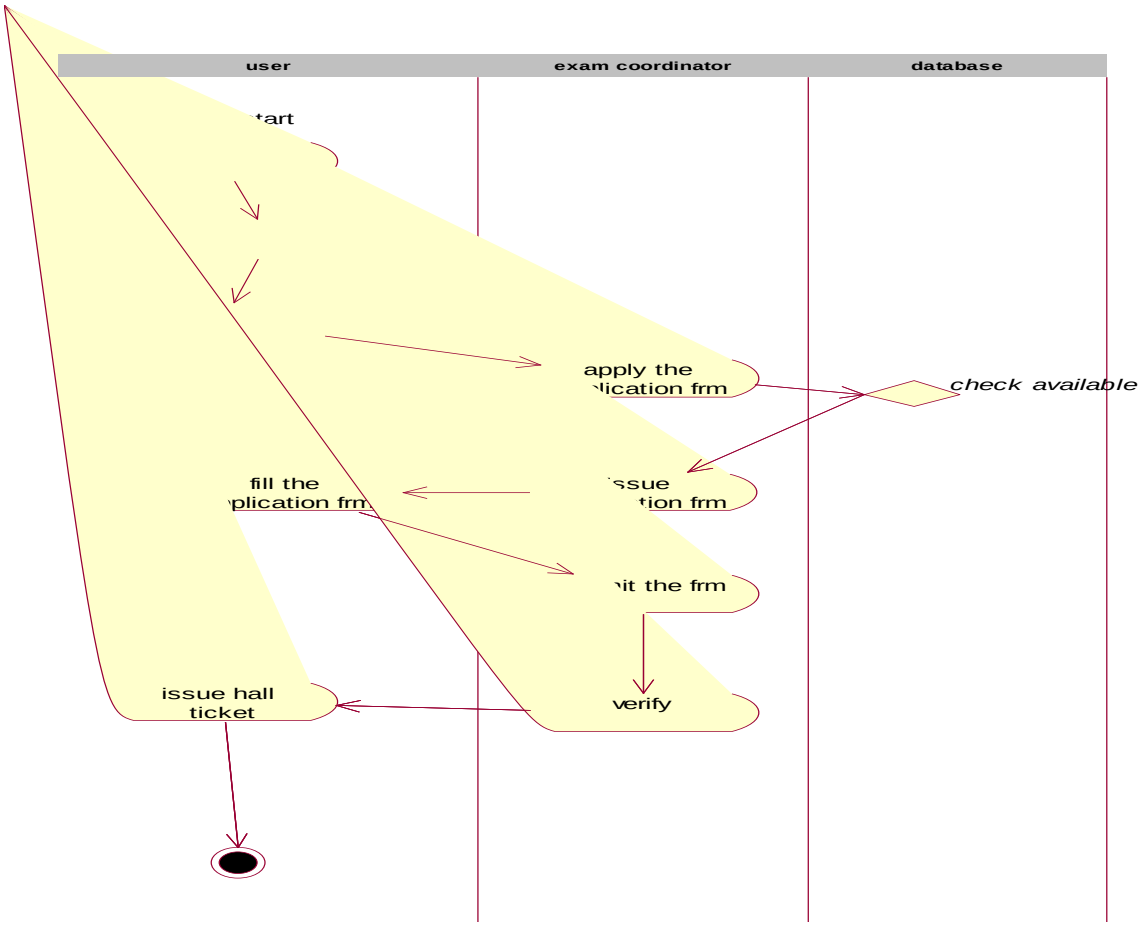
Sequence diagram:



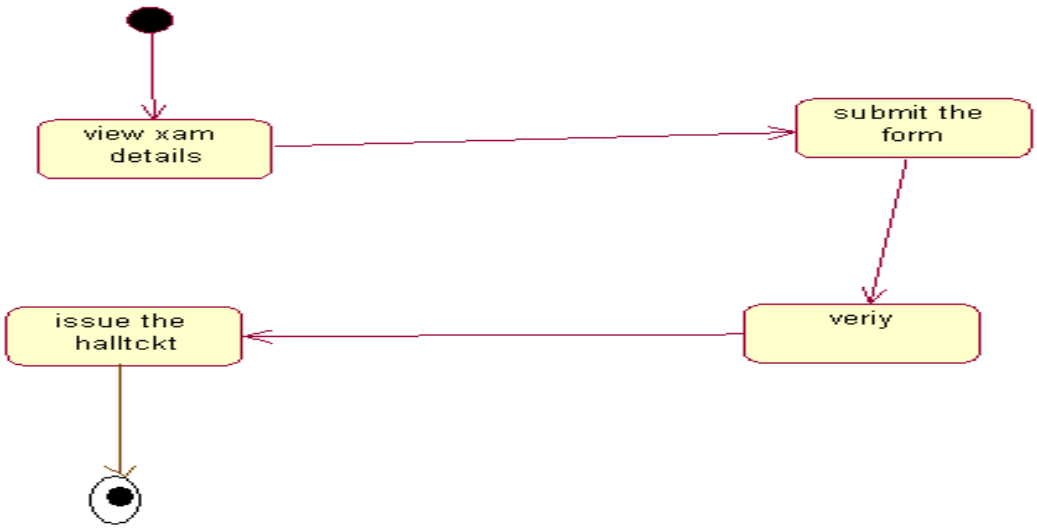
CollaborationDiagram:



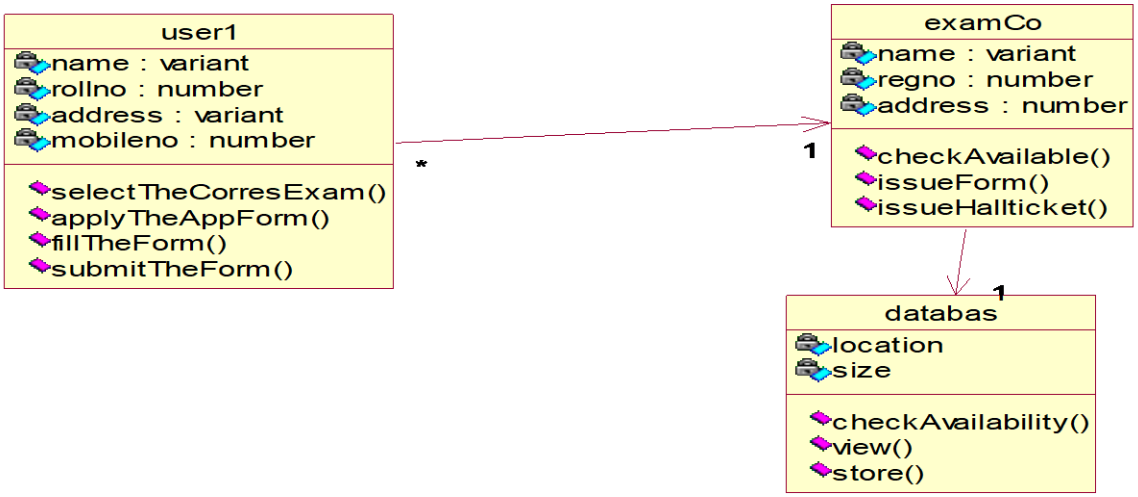
Activity diagram



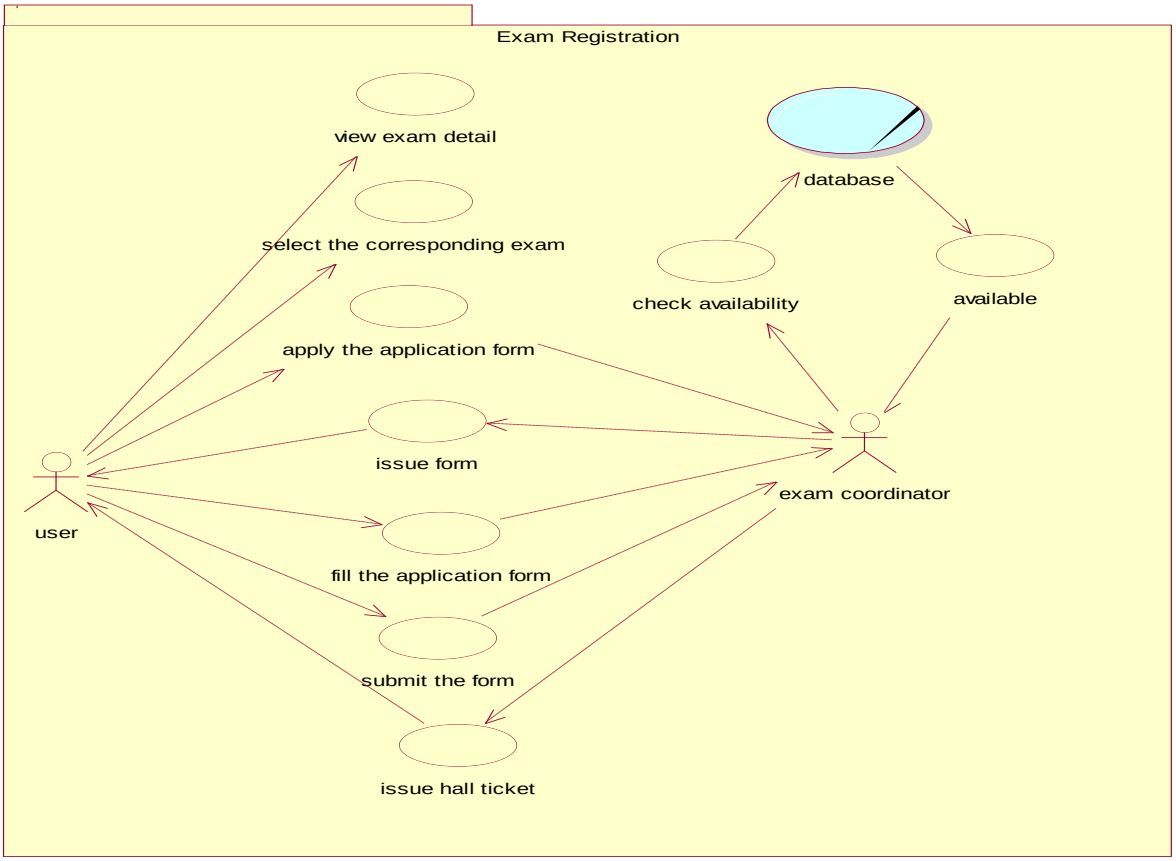
State Diagram:



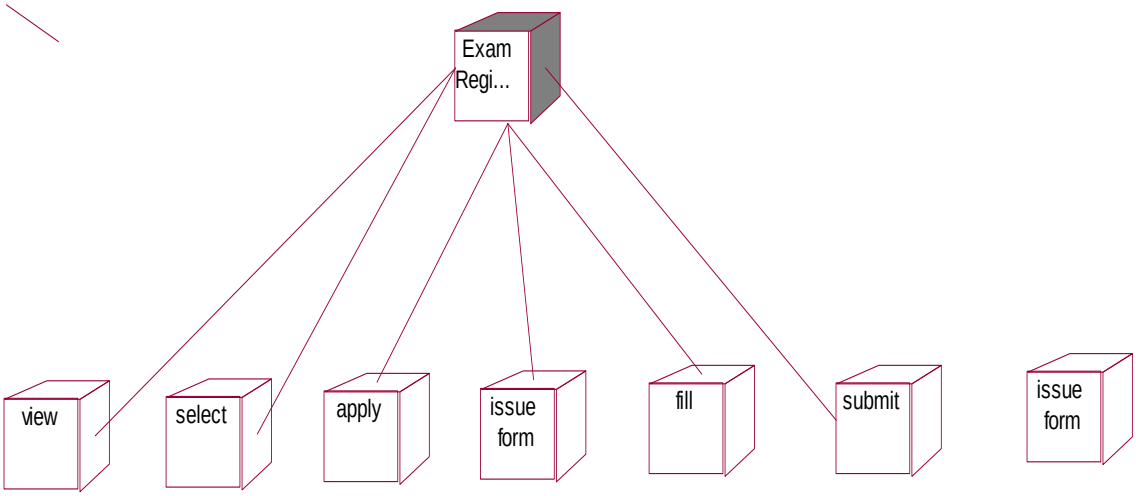
Class diagram:



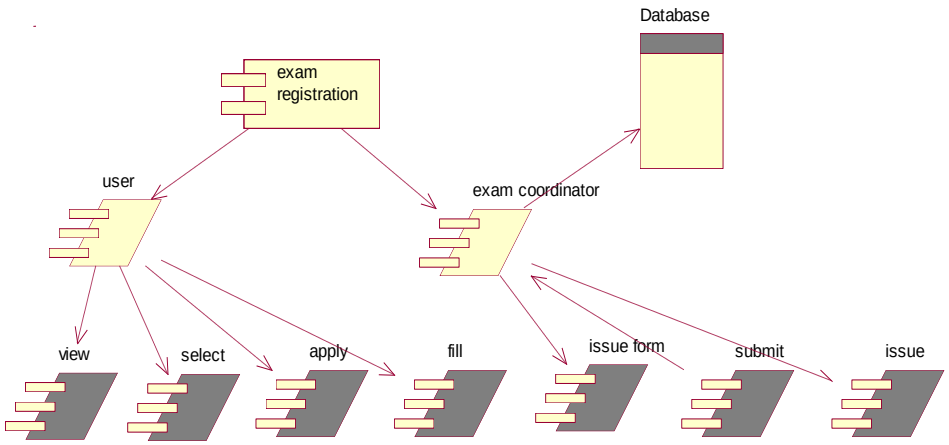
Package Diagram:



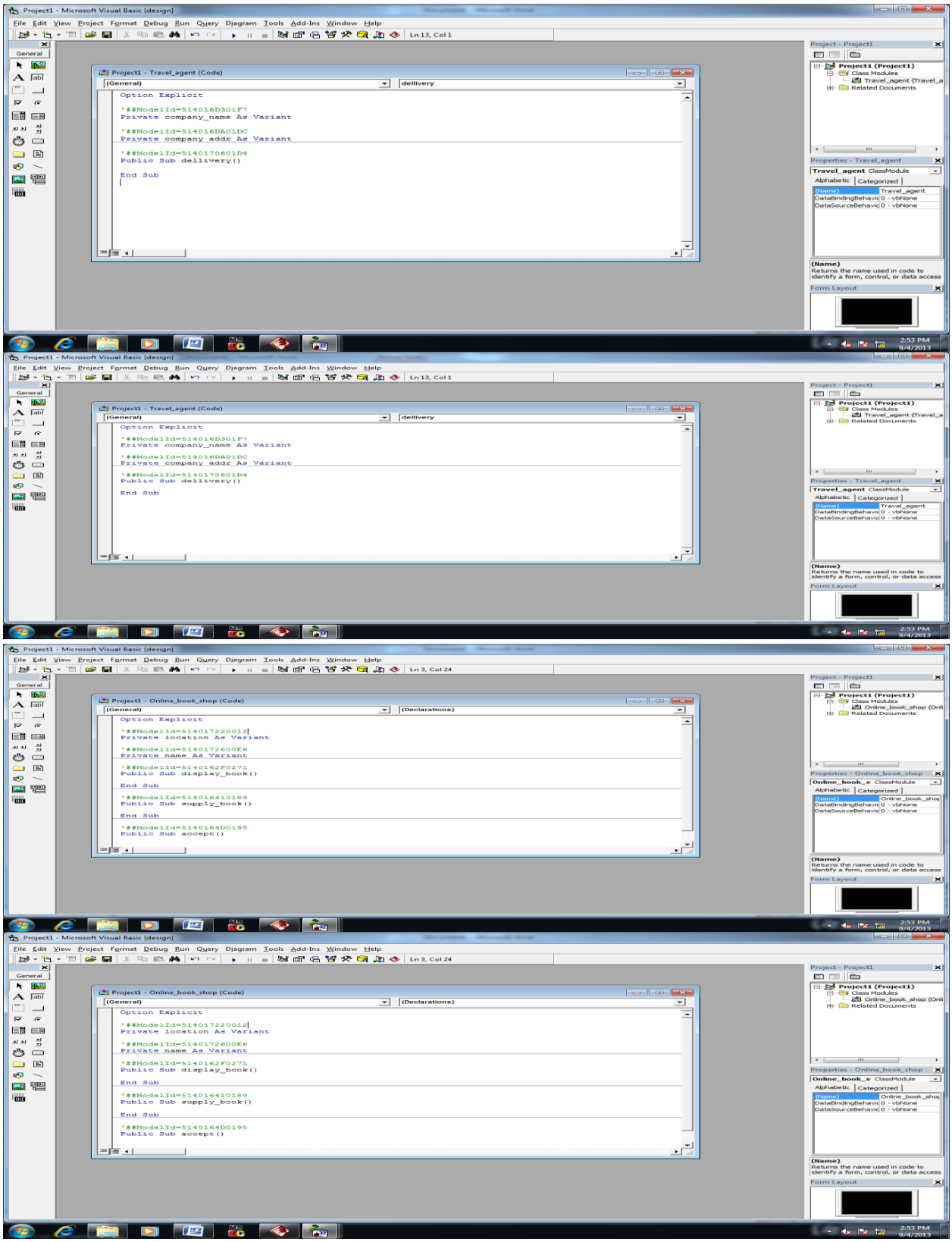
Deployment Diagram:



Component Diagram:



Output:



Result:

Thus the Exam Registration system was designed and the code was generated successfully.

EX.NO: 5 STOCK MAINTENANCE SYSTEM

DATE:

Aim:

To analyze and design Stock maintenance system using Rational Rose Software.

Procedure:

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

The new system must allow the customer to select the products purchase the product, and pay the bill. The main function of the stock according to the necessary of the market .The customer will select the product, purchase the selected product and pay the bill. The shopkeeper verifies the purchased product according to the bill. The shopkeeper issues the bill and receives the payment. The administrator must update the stock's price and introduce new offer for the customers.

Project scope

The stock maintenance system deals with maintaining the stock, which means maintaining the availability of the product on the market.

Project objectives

The aim of the stock maintenance is to maintain the availability of the product in the market.

Software requirement analysis:

The stock maintenance system includes the following modules:

- Selecting a product.
- Purchasing the product.
- Payment.
- Issuing Bill.
- Verify Item.
- Updating price and stock.
- Updating stock.
- Introducing offers.

Selecting a Product:

The customer will select a product according to its quality and price.

Purchasing the Product:

The customer will select the product and then purchase it by giving money.

Payment:

The customer will pay for the product and receiver the bill.

Issuing Bill:

While purchasing the shopkeeper will issue the bill for the purchased product

Verify Item:

When the customer leaves the shop the shopkeeper will verify the item with the bill

Updating Price:

The administrator will update the price of the product from time to time. The administrator must replace the stock so that the product is available in the market.

Introducing Offers:

The administrator will introduce new offers for product so that more customers can be attracted.

SOFTWARE REQUIREMENT SPECIFICATION

1.0 INTRODUCTION

Stock maintenance is an interface between the customer and the sales person. It aims at improving the efficiency in maintaining the stocks.

1.1 PURPOSE

The entire process of Stock maintenance is done in a manual manner Considering the fact that the number of customers for purchase is increasing every year, a maintenance system is essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process.

1.2 SCOPE

• The System provides an interface to the customer where they can fill in orders for the item needed.
• The sales person is concerned with the issue of items and can use this system.
• Provide a communication platform between the customer and the sales person.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

- **Market Data provider**: One who analyze the product and distribute the news.
- **Customer:** One who takes order of product
- **Sales person:** One who maintains the stock details

1.4 REFERENCES

IEEE Software Requirement Specification format

1.5 TECHNOLOGIES TO BE USED

- Visual Studio
- VB Script

1.6 TOOLS TO BE USED

- Eclipse IDE (Integrated Development Environment)
- Rational Rose tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements

Overall Description will describe major role of the system components and inter-Connections

Specific Requirements will describe roles & functions of the actors.

2 OVERALL DESCRIPTIONS

2.1 PRODUCT PERSPECTIVE

The Stock maintenance acts as an interface between the 'customer' and the 'sales person'. This system tries to make the interface as simple as possible and at the same time not risking the work of data stored.

2.2 SYSTEM FUNCTIONS

- Secure order of information by the customer
- Schedule the customer an appointment for manual delivery of the product.

2.3 USER CHARACTERISTICS

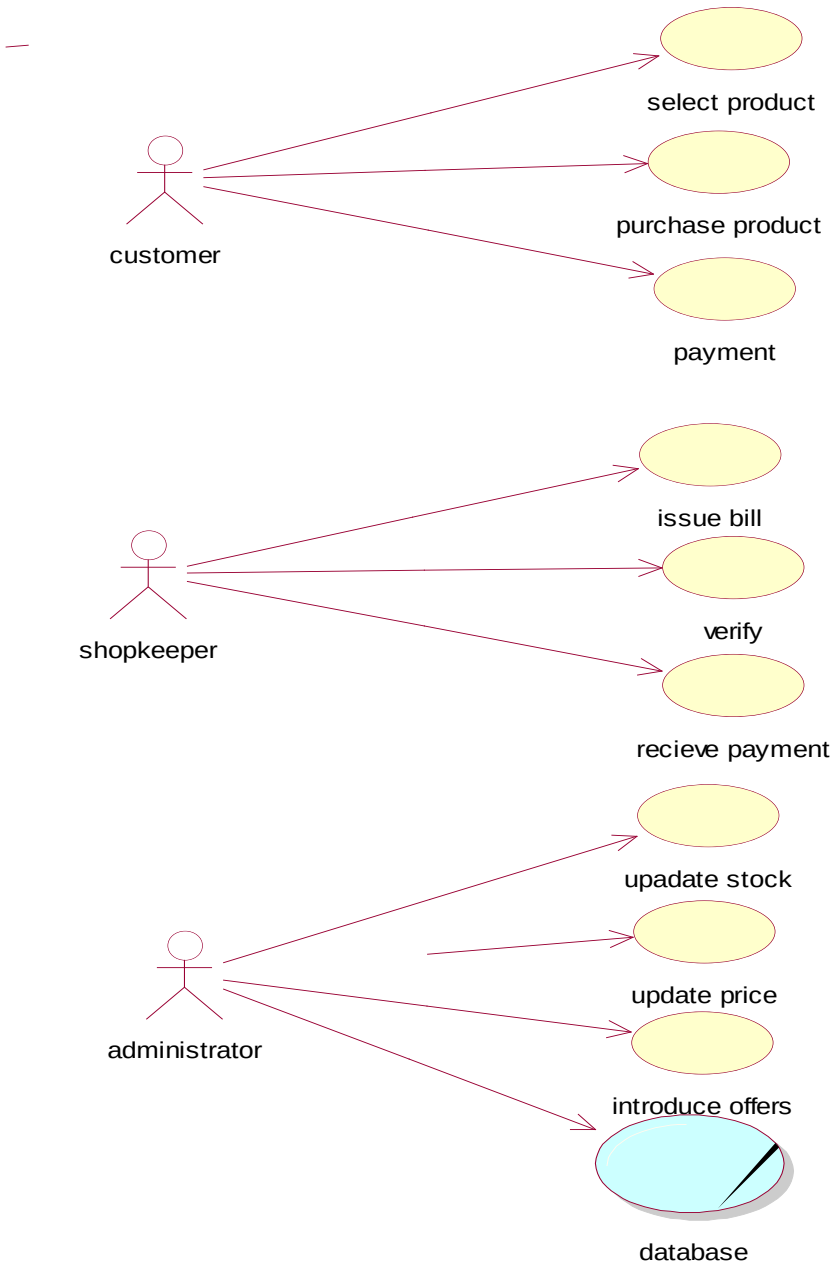
1. **Customer:** The person who orders for the item.
2. **Validate customer:** The items ordered by the customer are validated.
3. **Sales Detail:** Maintains the stock details after delivering the items to the customer.

2.4 CONSTRAINTS

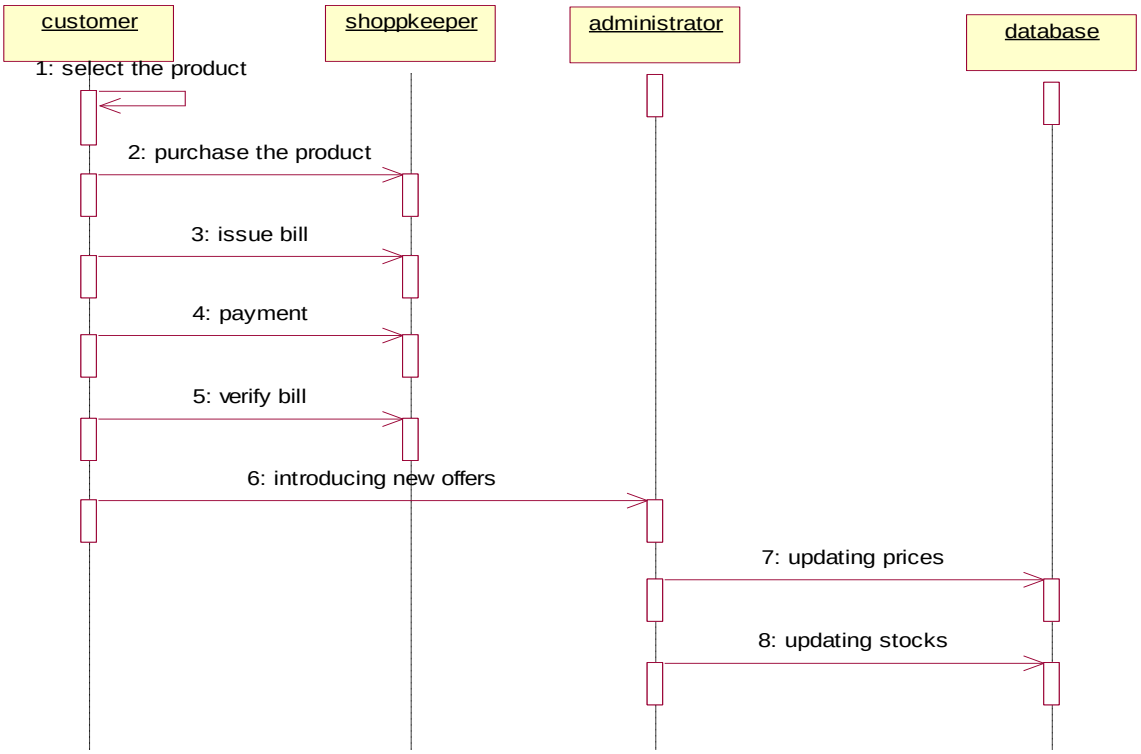
1. The customer should wait until the trade contractor and other to analyze the product.
2. After the distribution of the news about the product. The customer can take order and request of sales person to fill it.
3. Finally the sales person delivers the order.

UML Diagrams:

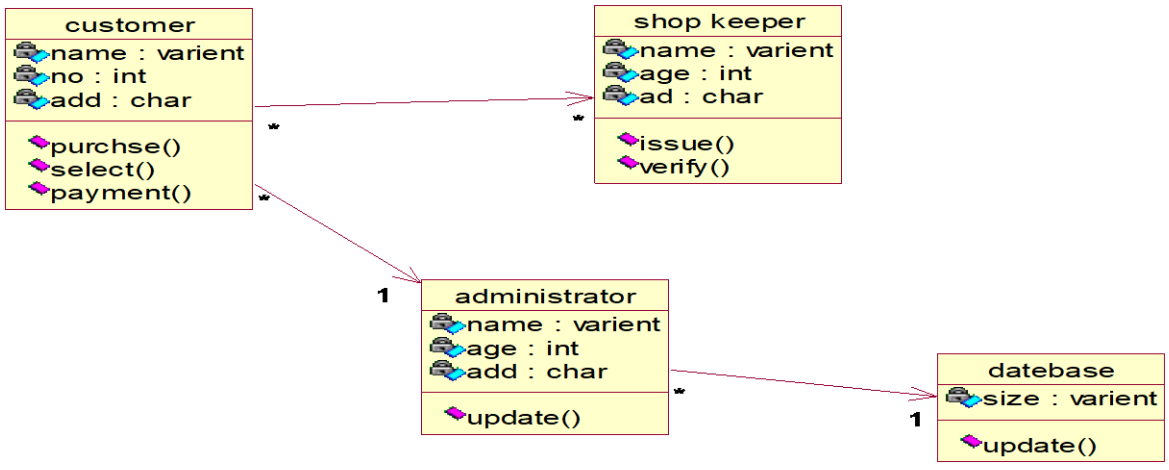
Use case Diagram:



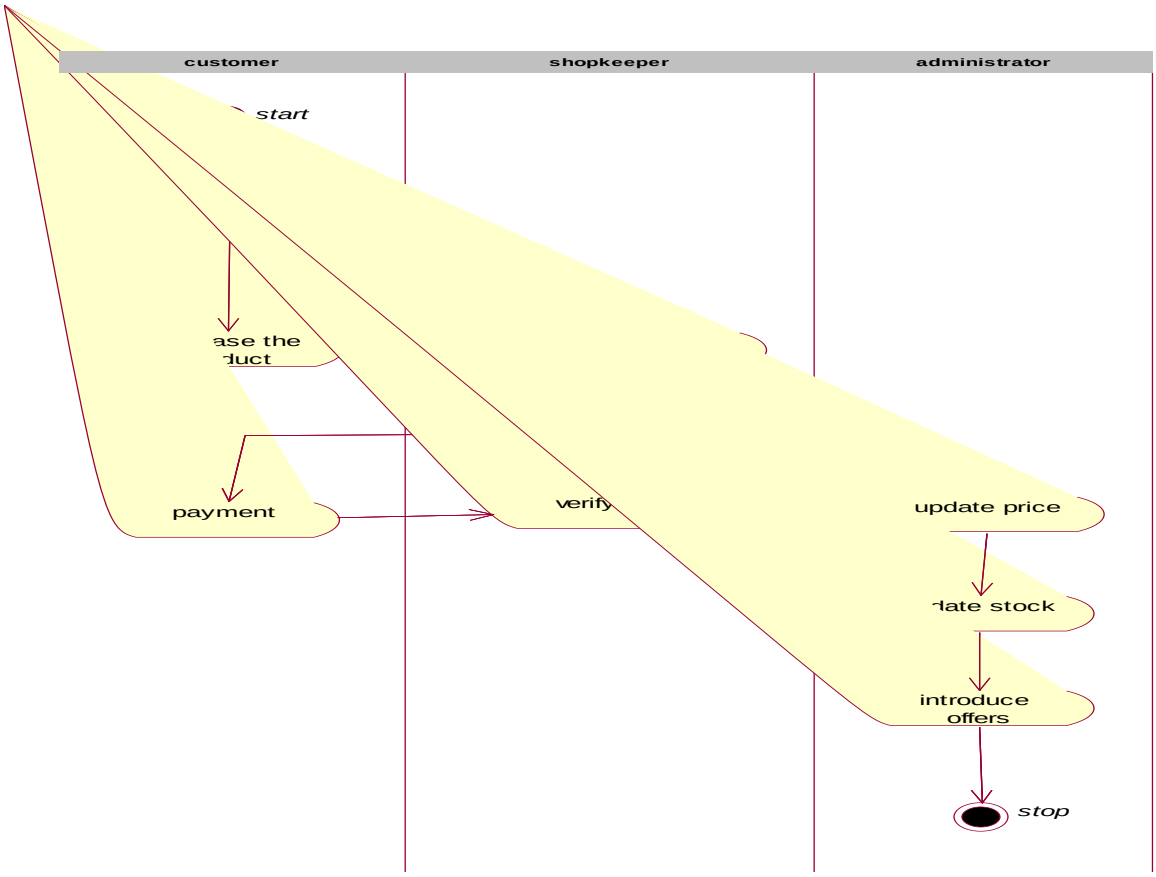
SequenceDiagram:



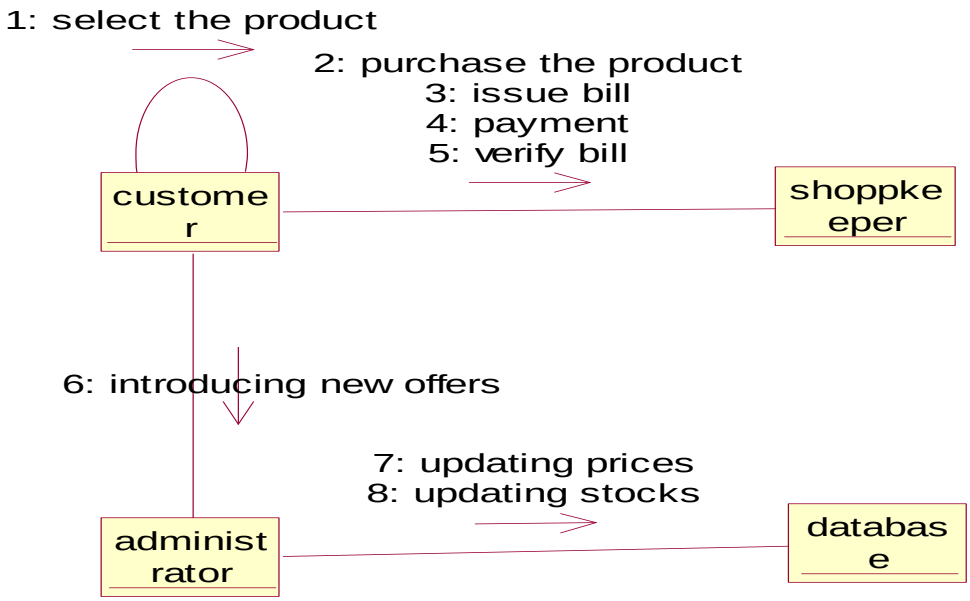
Class Diagram:



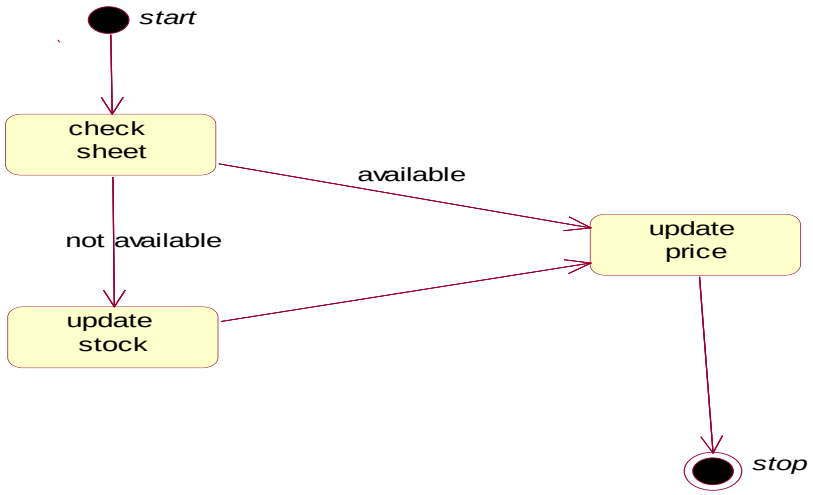
Activity diagram



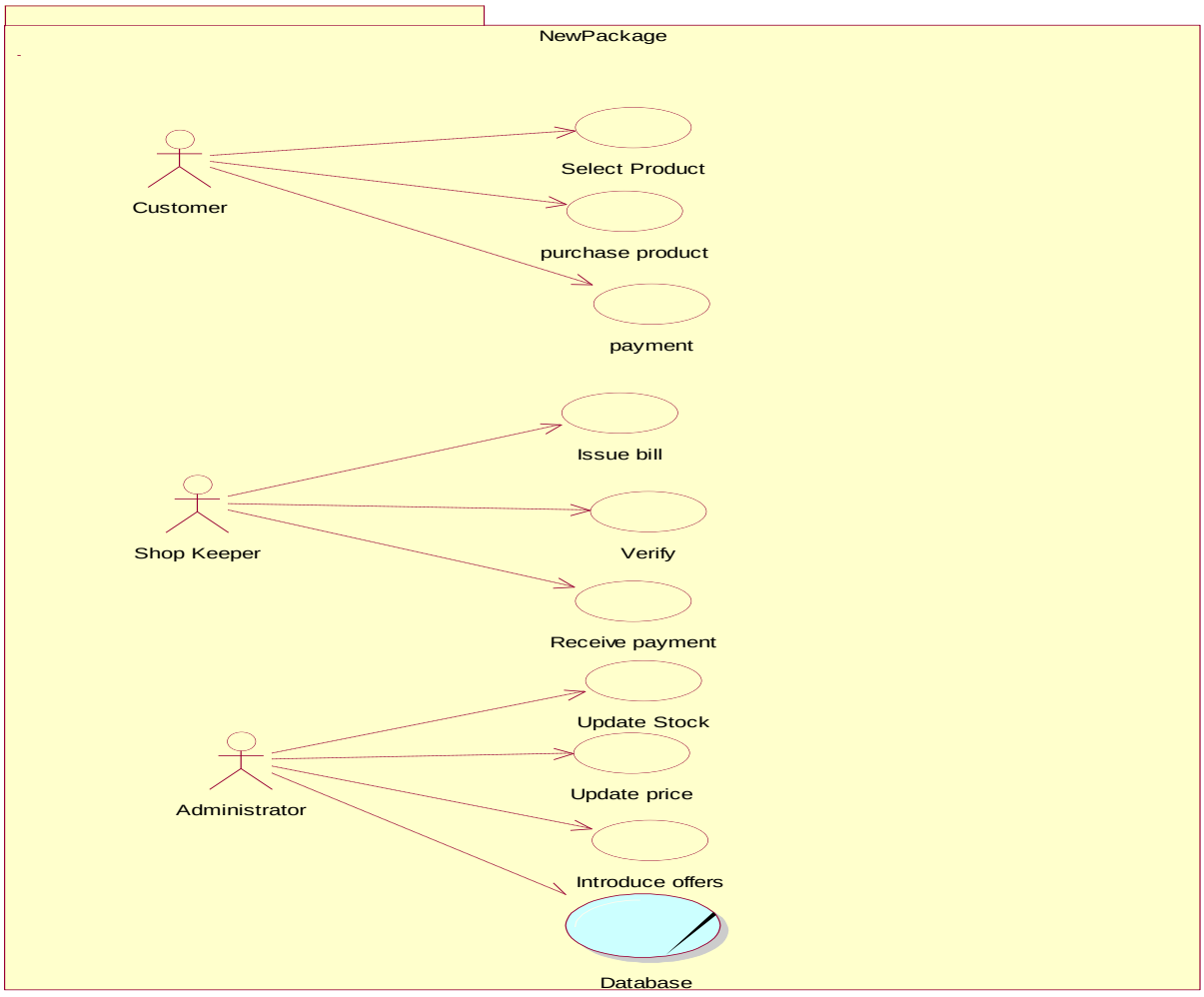
Collaboration Diagram:



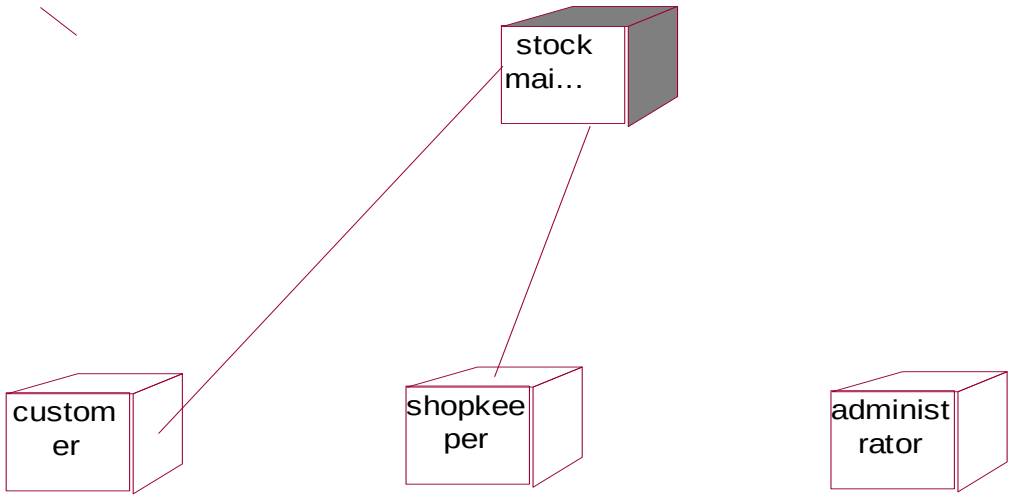
State Diagram:



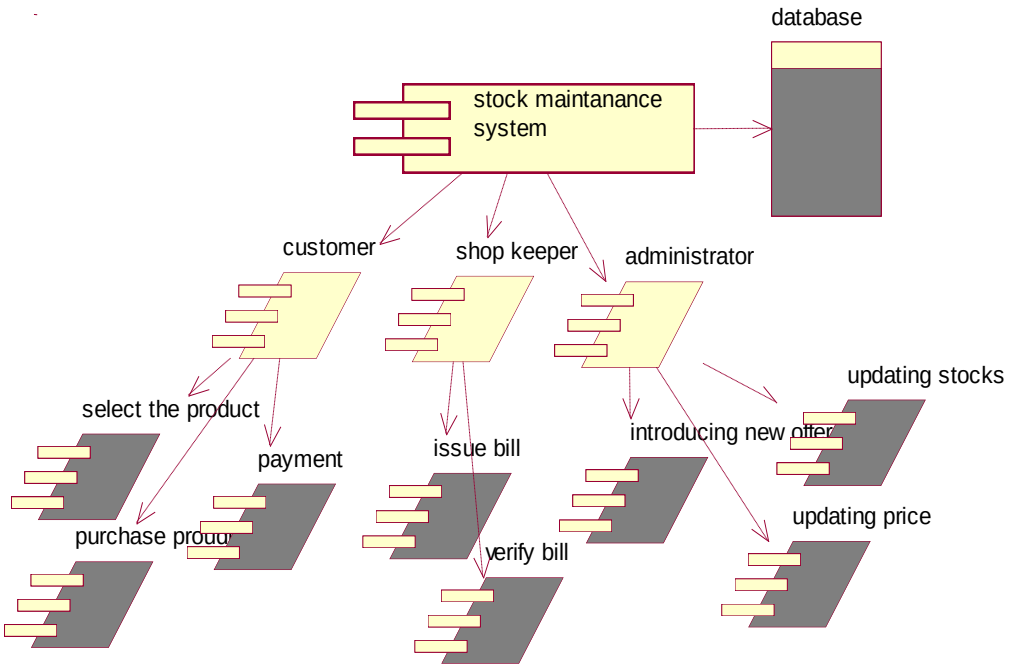
Package Diagram:



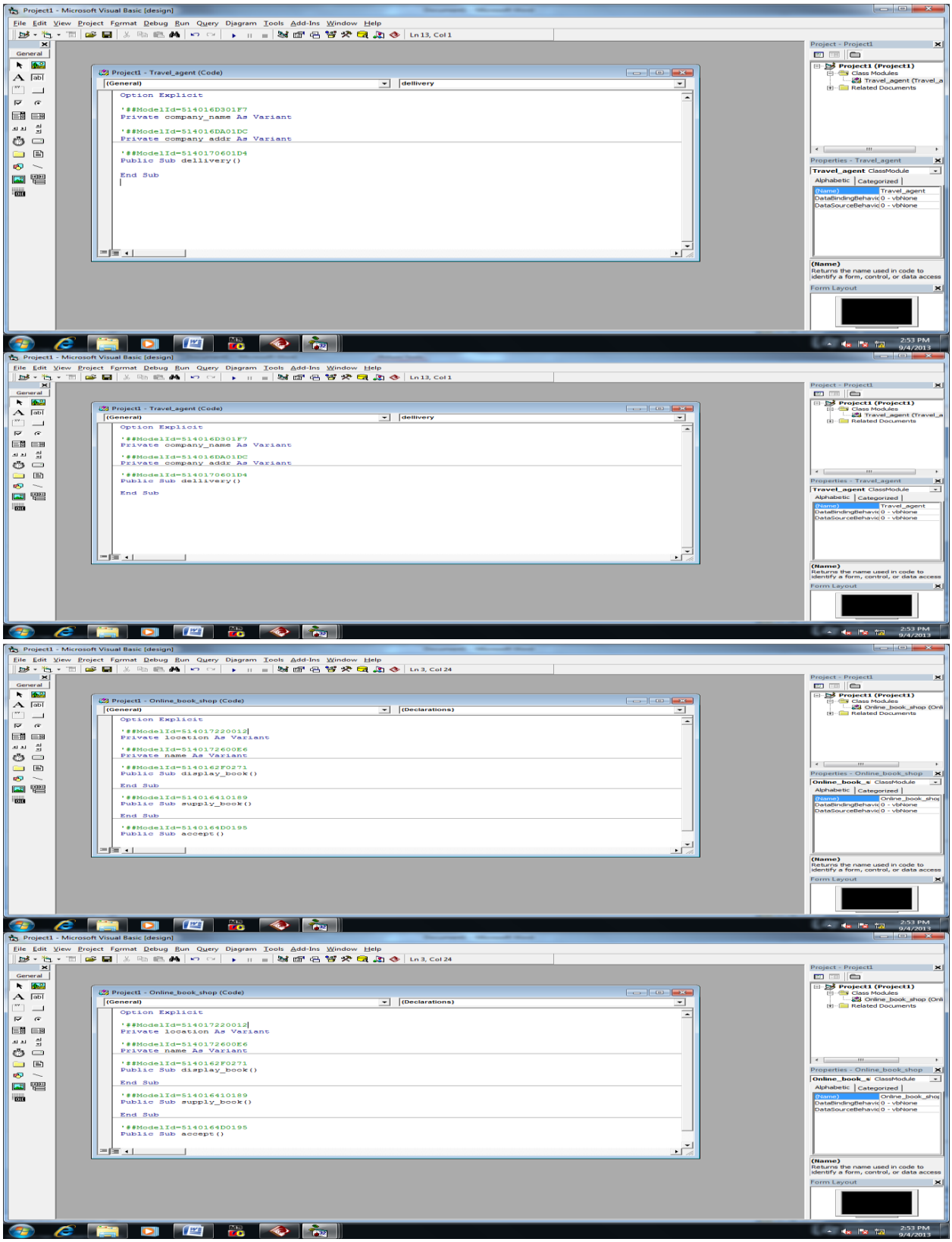
Deployment Diagram:



Component Diagram:



Output:



Result:

Thus the Stock Maintenance system was designed and the code was generated successfully.

EX.NO: 6 ONLINE COURSE RESERVATION SYSTEM

DATE:

Aim:

To analyze and design online course reservation system using Umbrello Software.

Procedure:

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

Students can search the courses in a particular university. Then the course we wanted in that university will be searched. Then we can find out how many colleges will have that particular condition for marks and age. If the percentage was satisfied we will check out the fees. After that we can register that particular course either in website or in forms given from the college. For registration they wanted the name, DOB, gender, address, phone no, the percentage for 10th and 12th with some condition in marks. Finally submit the registration form.

Project scope

The course registration system deals with the registration of course through websites or directly. In this students can search the different courses available and know details about their registration.

Project objectives

The course registration system deals with the registration of course through websites or directly. In this students can search the different courses available and know details about their registration.

Software requirement analysis

The modules of the online course registration system are:

- Search the course
- Register the course
- Cancel the registration

Search the course: The course could be searched directly to the college or through website. The course can be divided into UG and PG.

Register the course: The course must be registered through website or from the college itself. The registration must have some condition of marks. Staff must handle the registration.

Cancel the registration: The registered course must be cancelled in the website or directly. The staffs also handle the cancellation of course.

SOFTWARE REQUIREMENT SPECIFICATION

1.0. INTRODUCTION

Course Reservation System is an interface between the Student and the Registrar responsible for the issue of Course. It aims at improving the efficiency in the issue of Course and reduces the complexities involved in it to the maximum possible extent.

1.1 PURPOSE

If the entire process of 'Issue of Course' is done in a manual manner then it would takes several months for the course to reach the applicant. Considering the fact that the number of applicants for course is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process.

1.2 SCOPE

- ✓ The System provides an online interface to the user where they can fill in their personal details and submit the necessary documents (may be by scanning).
- ✓ The Registrar concerned with the issue of course can use this system to reduce his workload and process the application in a speedy manner.
- ✓ Provide a communication platform between the Student and the Registrar.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

- **Registrar**
  Refers to the super user with the privilege to manage the entire system.
- **Applicant**
  One who wishes to register the Course.
- **OCRS**
  Refers to online Course Reservation System.
- **HTML**
  Markup Language used for creating web pages.
- **J2EE**
  Java 2 Enterprise Edition is a programming platform java platform for developing and running distributed java applications.
- **HTTP**
  Hyper Text Transfer Protocol.
- **TCP/IP**
  Transmission Control Protocol/Internet Protocol is the communication protocol used to connect hosts on the Internet.

1.4 REFERENCES

IEEE Software Requirement Specification format.

1.5 TECHNOLOGIES TO BE USED

- HTML
- JSP
- Javascript
- Java

1.6 TOOLS TO BE USED

• Eclipse IDE (Integrated Development Environment)
• Umbrello tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements.
Overall Description will describe major role of the system components and inter-connections.
Specific Requirements will describe roles & functions of the actors.

2.0 OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

The OCRS acts as an interface between the 'Student' and the 'Registrar'. This system tries to make the interface as simple as possible and at the same time not risking the security of data stored in. This minimizes the time duration in which the user receives the course.

2.2 SOFTWARE INTERFACE

• **Front End Client** - The Student and Registrar online interface is built using JSP and HTML. The Administrators's local interface is built using Java.
• **Web Server** – Tomcat Apache application server (Oracle Corporation).
• **Back End** – Oracle 11g database.

2.3 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

2.4 SYSTEM FUNCTIONS

• Secure Reservation of information by the Students.
• SMS and Mail updates to the students by the Registrar
• Registrar can generate reports from the information and is the only authorized personnel to add the eligible application information to the database.

2.5 USER CHARACTERISTICS

• **Applicant** - They are the person who desires to obtain the course and submit the information to the database.
• **Administrator** - He has the certain privileges to add the course status and to approve the issue of course. He may contain a group of persons under him to verify the documents and give suggestion whether or not to approve the dispatch of course.

2.6 CONSTRAINTS

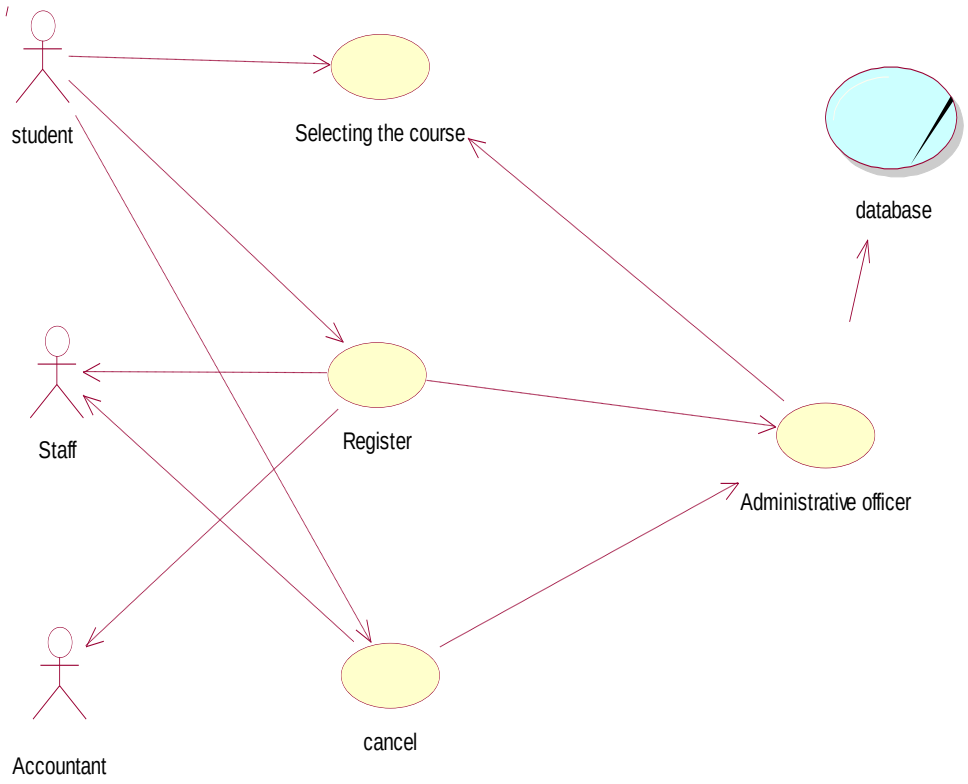
• The applicants require a computer to submit their information.
• Although the security is given high importance, there is always a chance of intrusion in the web world which requires constant monitoring.
• The user has to be careful while submitting the information. Much care is required.

2.7 ASSUMPTIONS AND DEPENDENCIES

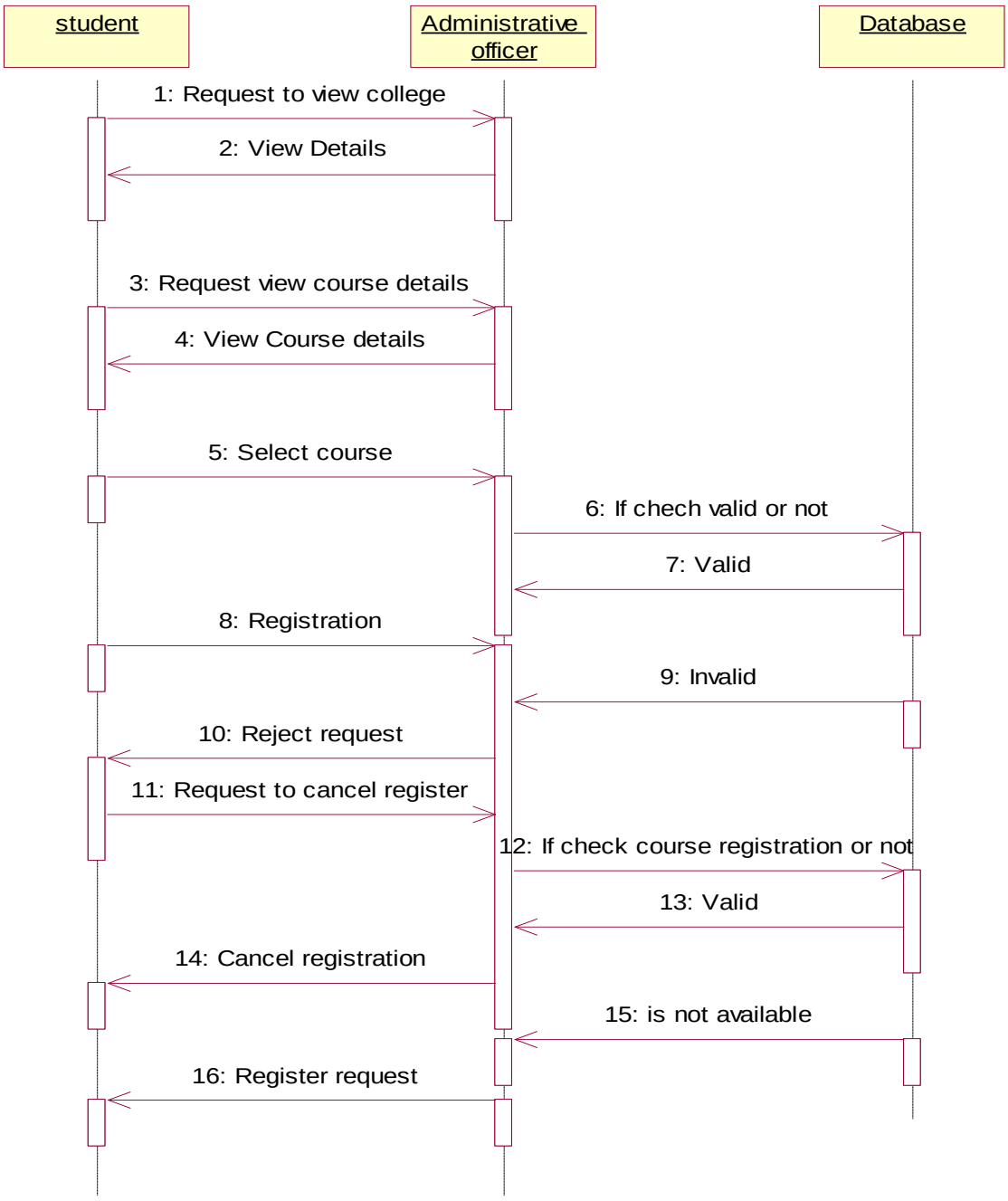
• The Applicants and Administrator must have basic knowledge of computers and English Language.
• The applicants may be required to scan the documents and send

UML Diagrams:

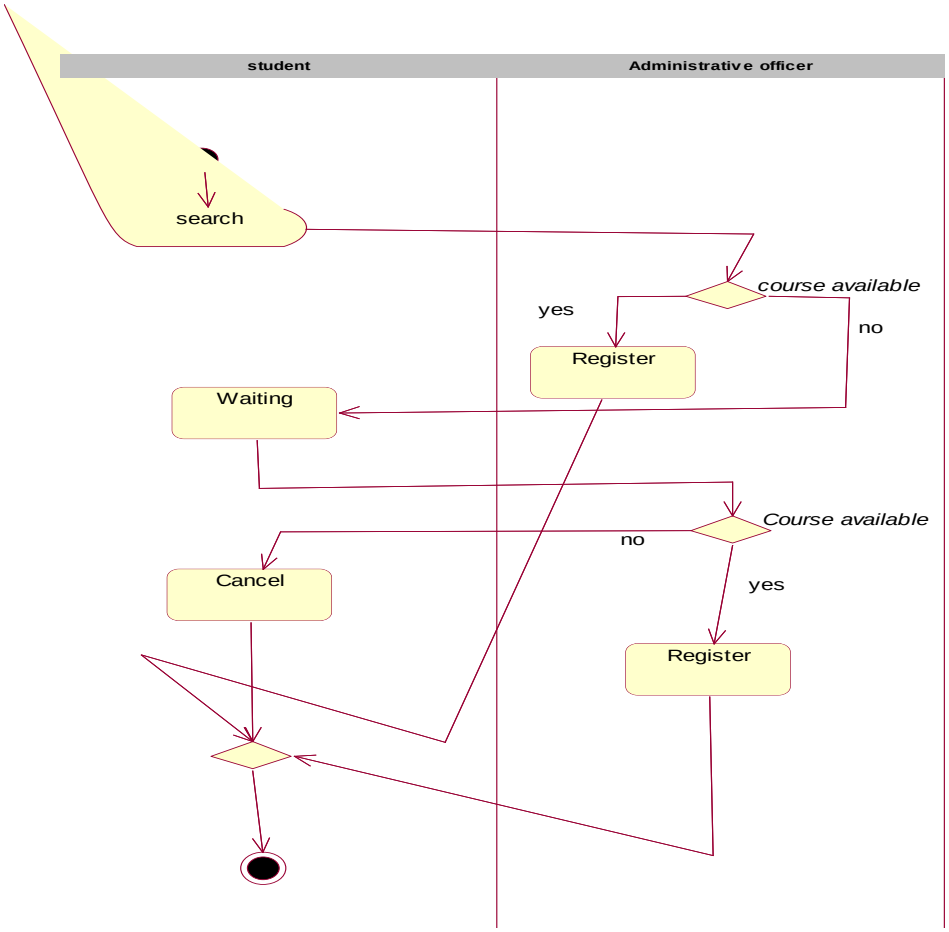
Use case Diagram:



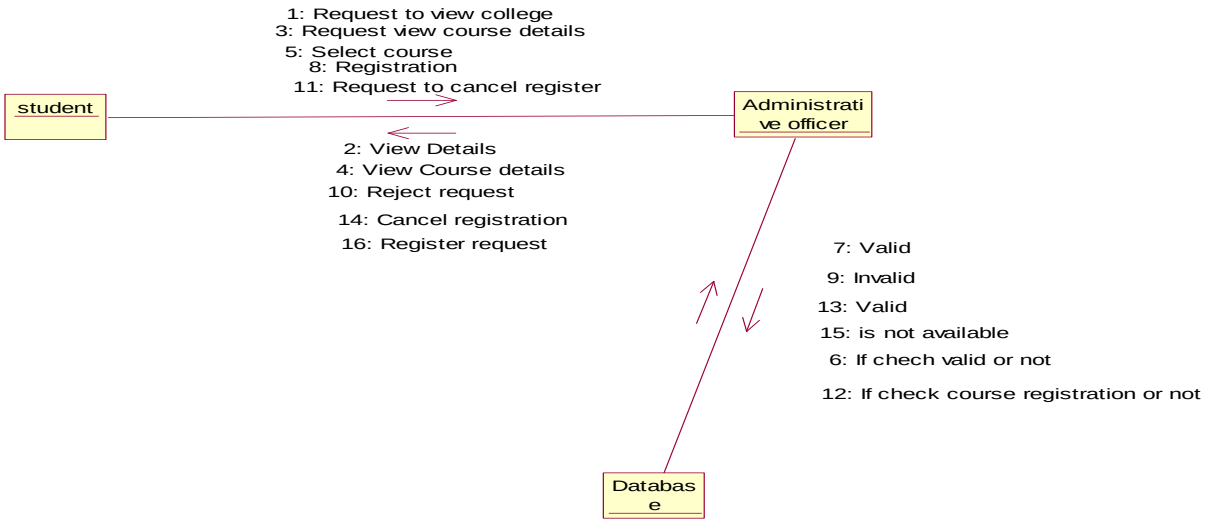
Sequence diagram:



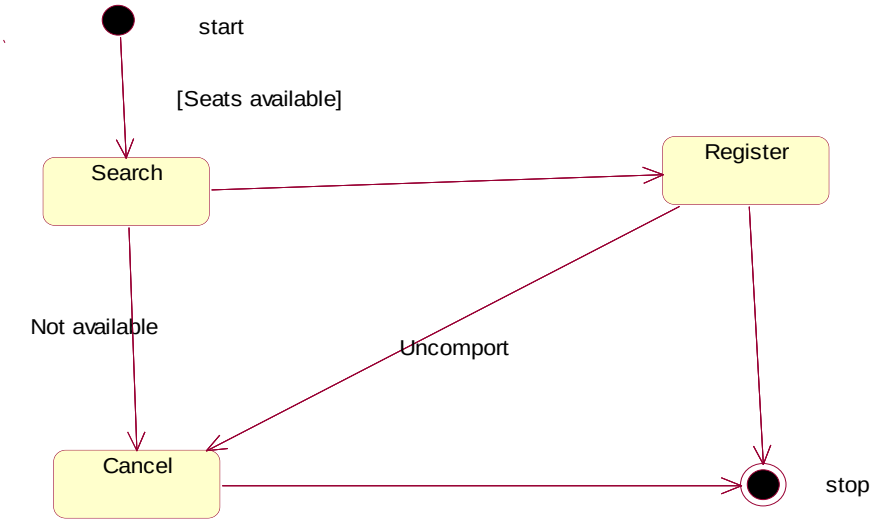
Activity diagram:



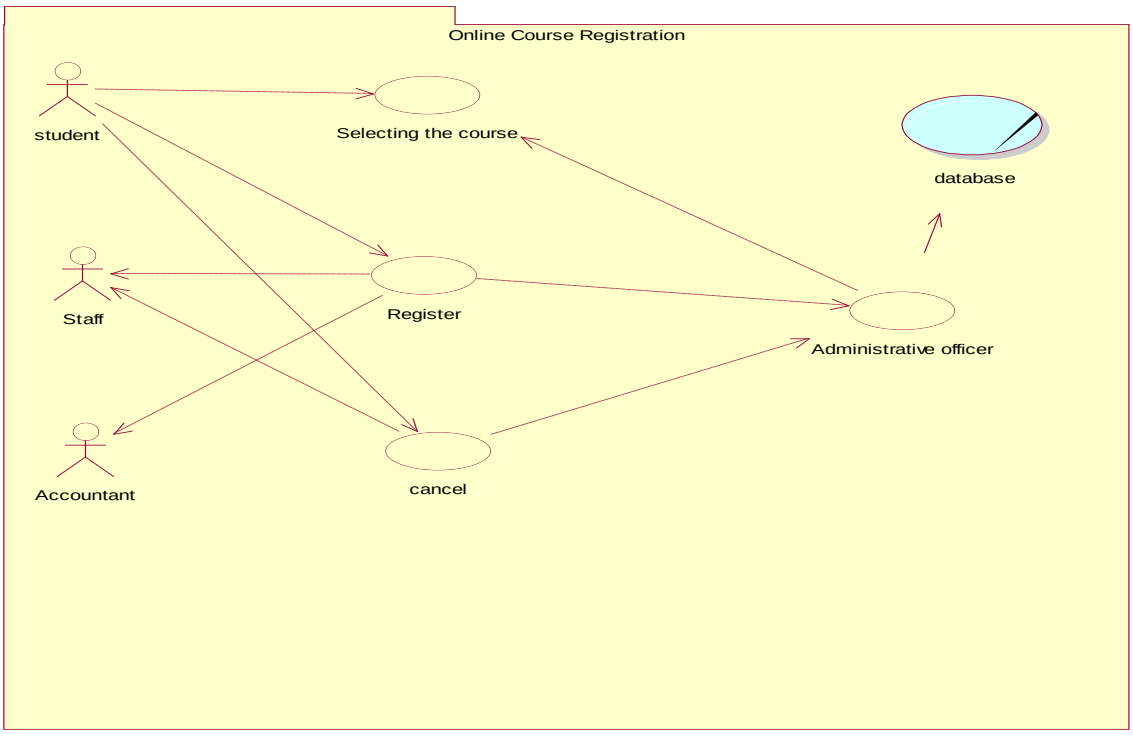
Collaboration Diagram:



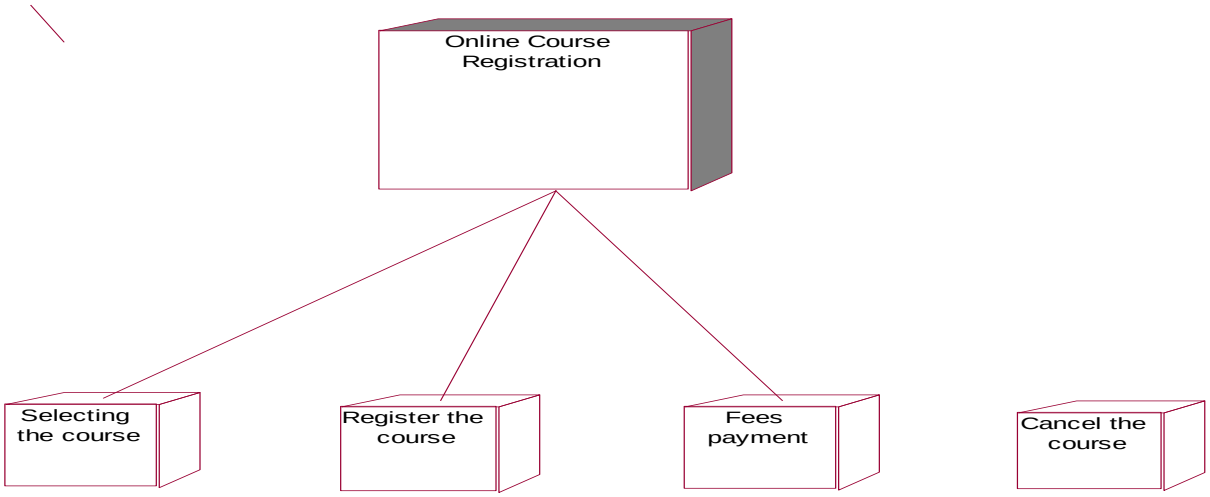
State Diagram:



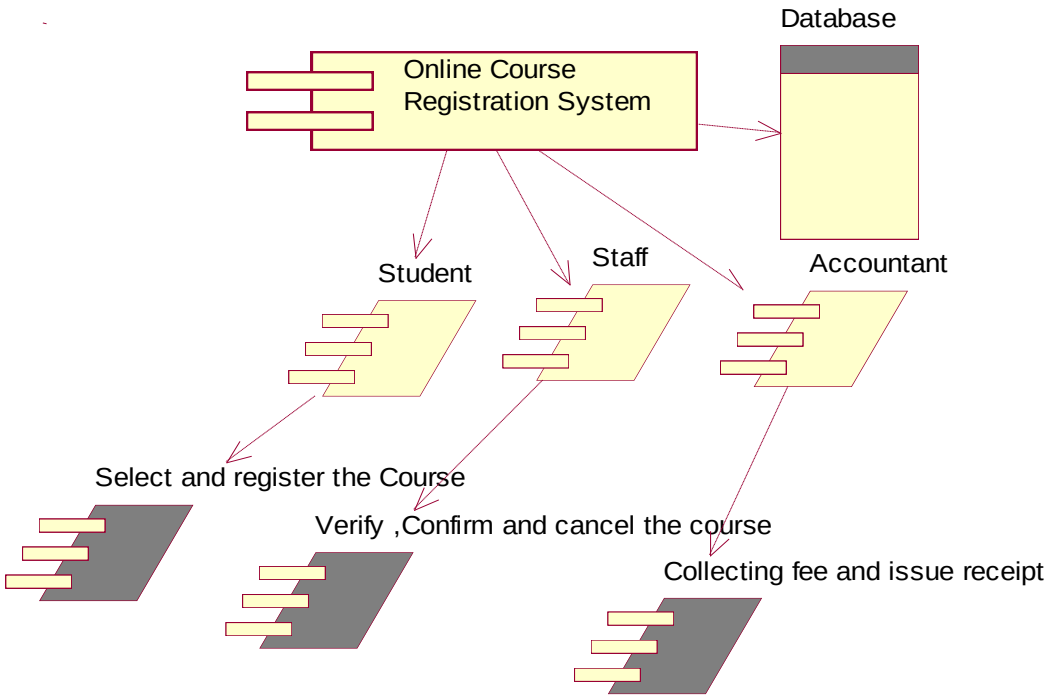
PackageDiagram:



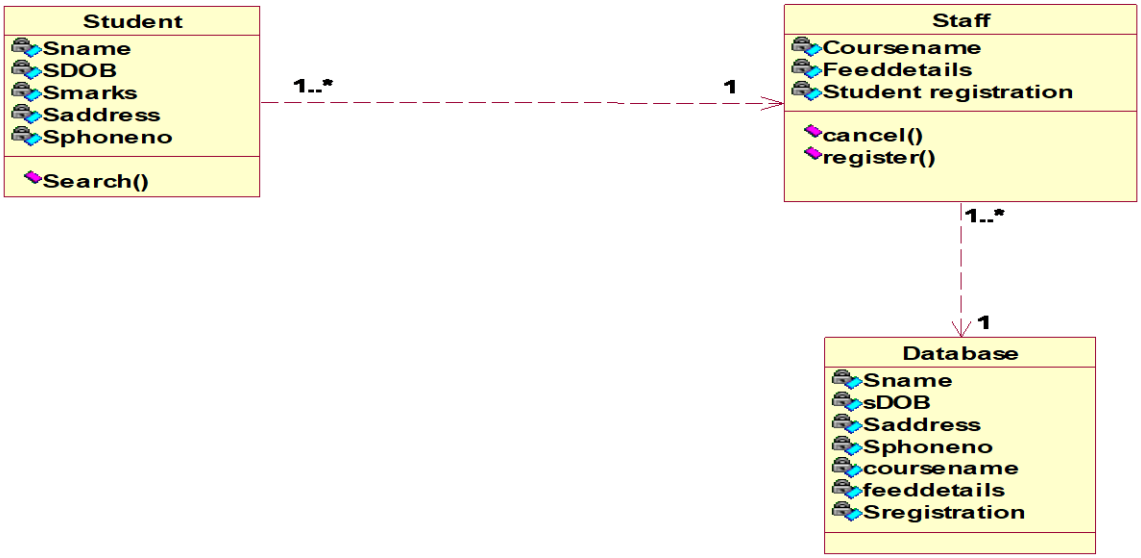
Deployment Diagram:



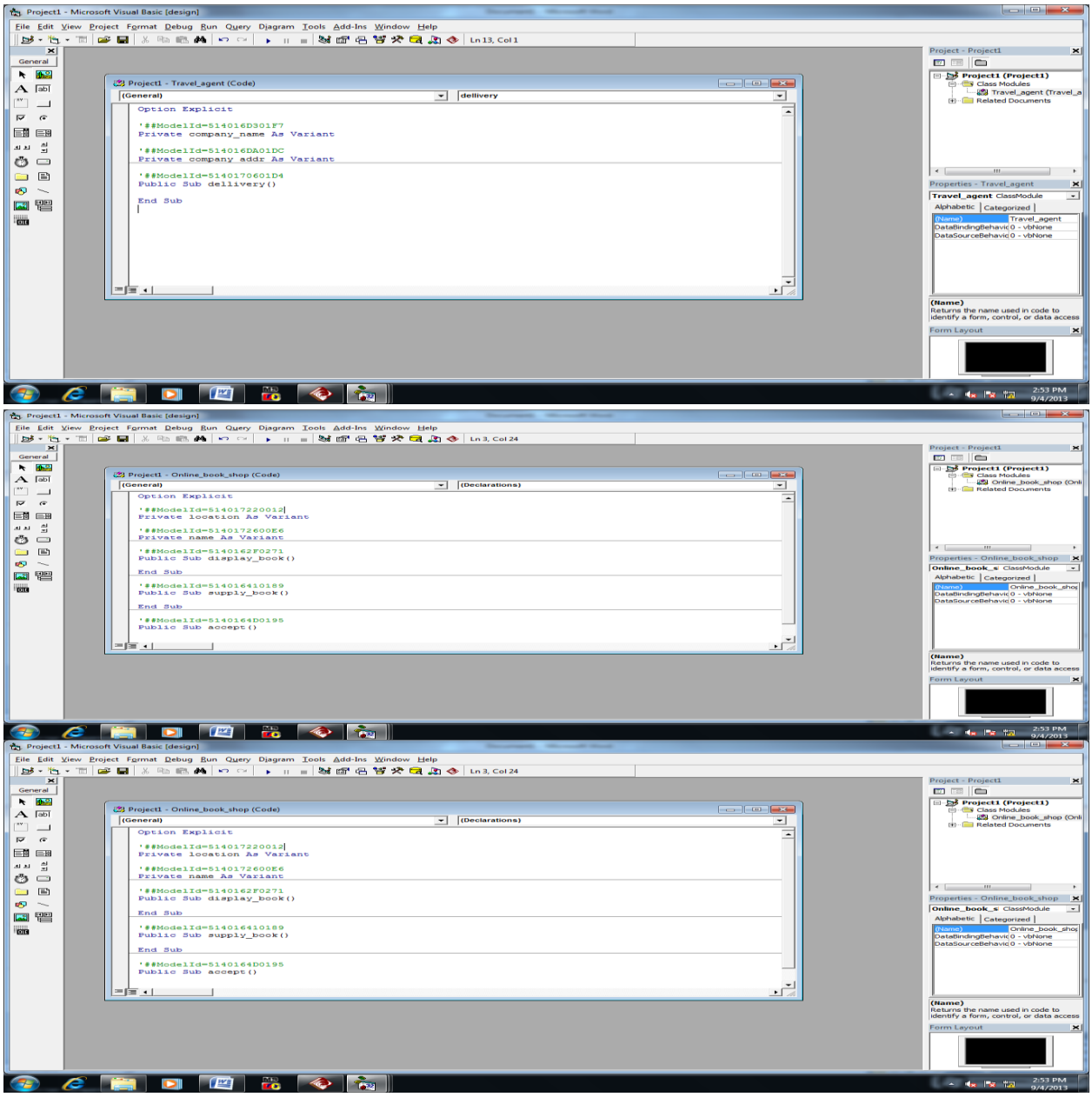
Component Diagram:



Class diagram:



Output:



Result:

Thus the online course registration system was designed and the code was generated.

EX.NO:7 E-TICKETING

DATE:

Aim:

To analyze and design E-Ticketing system using Umbrello Software.

Procedure:

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

The function of this system is we can reserve ticket through online. So for using this system we can reserve from our house without going the reservation counter. For using online reservation system we can reserve the railway tickets. In this reservation system we can reserve the starting station and destination. For using this online reservation system we can reserve the tickets very easily.

Project scope:

This e-ticketing system deals with reserving the tickets and views the details. If we want to cancel the tickets we can cancel. The passenger and the station master utilize these facilities.

Project objectives

The aim of this e-ticketing system is to get the ticket through online reservation. This should provide the required features to be passenger.

Software requirement analysis

This e-ticketing system includes the following modules:

- Reservation
- Cancel ticket
- Update ticket
- View train details

Reservation:-

The passenger will request to reserve ticket. The administrator checks whether seats is available or not. Then the administrator conformed if seat is available otherwise administration gives the information that there is no seat is available.

Cancel ticket:-

If the passenger wants to cancel the ticket, the administrator will check whether the ticket is reserved or not. Then he cancel the ticket.

Update ticket:-

The passenger asks the administrator to show the update details. Then he show the update details to the passenger.

View train details:-

The passenger will request to view the train details. The administrator will show the details to the passenger.

SOFTWARE REQUIREMENT SPECIFICATION

1.0 INTRODUCTION

The manual system of ticket reservation takes more time and the number of reservations per day is limited. To increase the efficiency of the process, we go for online ticket reservation system. This system supports online ticket booking.

1.1 PURPOSE

If the entire process of reservation is done in a manual manner then it would takes several months for reservation to reach the applicant. Considering the fact that the number of passenger is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process. As this is a matter of National Security, the system has been carefully verified and validated in order to satisfy it.

1.2 SCOPE

• The System provides an online interface to the user where they can fill in their personal details and submit the necessary documents (may be by scanning).
• The authority concerned with the issue of railway can use this system to reduce his workload and process the application in a speedy manner.
• Provide a communication platform between the passenger and the administrator.
• Passenger will come to know their status of application and the date in which they must subject themselves for manual document verification.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

• **Passenger** – The person that who wishes to obtain the railway ticket.
• **PNR** – Passenger Name Records
• **HTML** - Markup Language used for creating web pages.
• **J2EE** – Java 2 Enterprise Edition is a programming platform java platform for developing and running distributed java applications.
• **HTTP** - Hyper Text Transfer Protocol.
•**TCP/IP** – Transmission Control Protocol/Internet Protocol is the communication protocol used to connect hosts on the Internet.

1.4 REFERENCES

IEEE Software Requirement Specification format.

1.5 TECHNOLOGIES TO BE USED

• HTML
• JSP
• Javascript
• Java

1.6 TOOLS TO BE USED

• Eclipse IDE (Integrated Development Environment)
• Umbrello tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements –
Overall Description will describe major role of the system components and inter-connections.
Specific Requirements will describe roles & functions of the actors.

2.0 OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

This system tries to make the interface as simple as possible and at the same time not risking the security of data stored in. This minimizes the time duration in which the user receives the ticket.

2.2 SOFTWARE INTERFACE

• **Front End Client** - The passenger and System online interface is built using JSP and HTML. The Administrators's local interface is built using Java.

- **Web Server** – Apache Tomcat Server (Oracle Corporation)
- **Back End** - Oracle 11g database

2.3 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

2.4 SYSTEM FUNCTIONS

• Secure Registration of information by the Passengers.
• System can generate reports from the information and is the only authorized personnel to add the eligible application information to the database.
• Display the requested pages to the user.
• Update the database after every successful process.

2.5 USER CHARACTERISTICS

• **Passenger** - They are the people who desire to obtain the ticket and submit the information to the database.

2.6 CONSTRAINTS

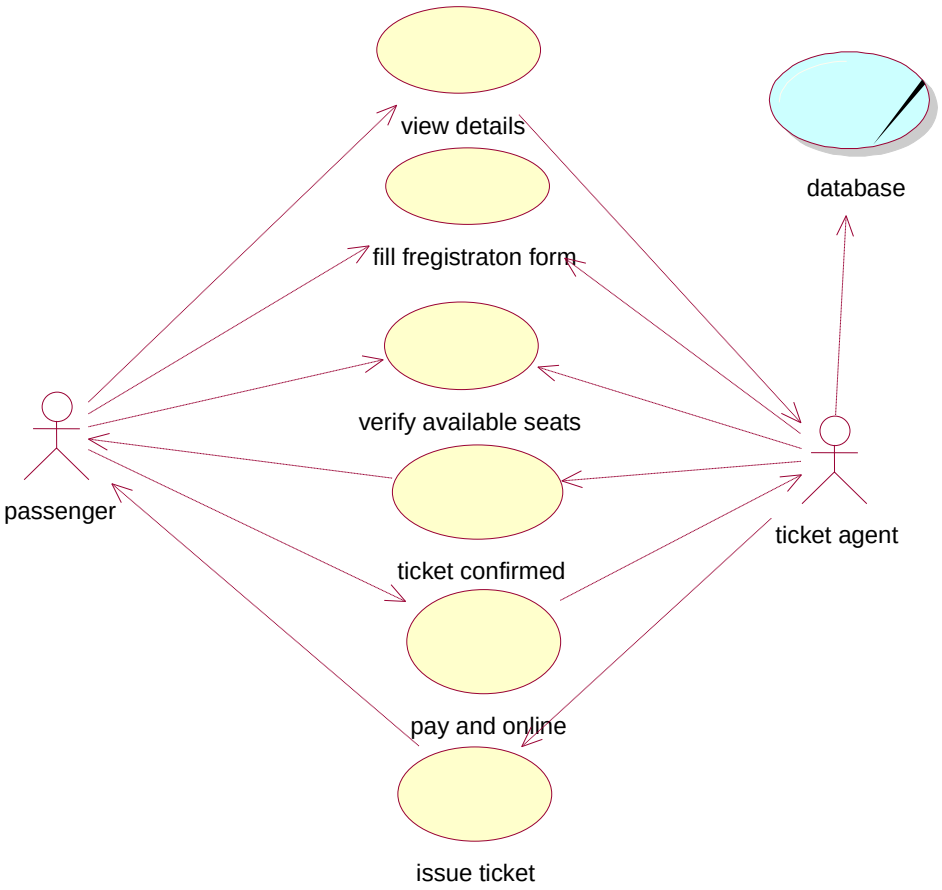
• The passengers require a computer to submit their information.
• Although the security is given high importance, there is always a chance of intrusion in the web world which requires constant monitoring.
• The user has to be careful while submitting the information. Much care is required.

2.7 ASSUMPTIONS AND DEPENDENCIES

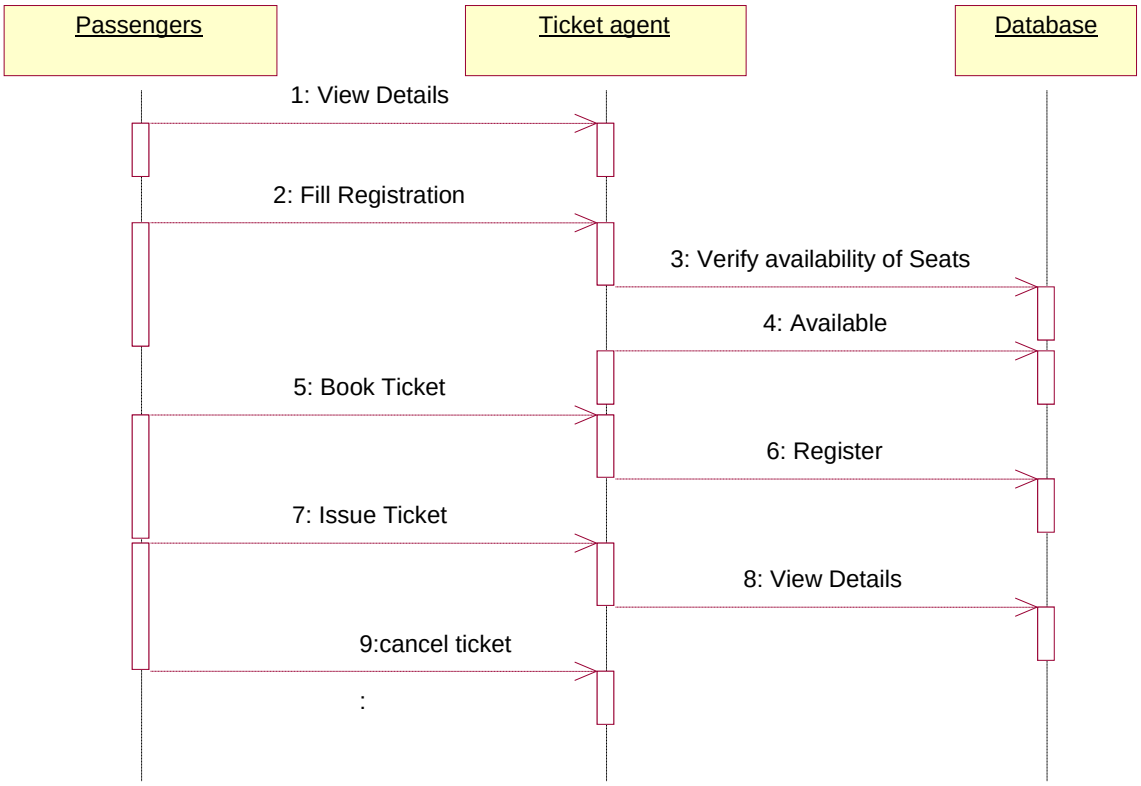
• The Passengers must have basic knowledge of computers and English Language.
• The passengers may be required to scan the documents and send.

UML Diagrams

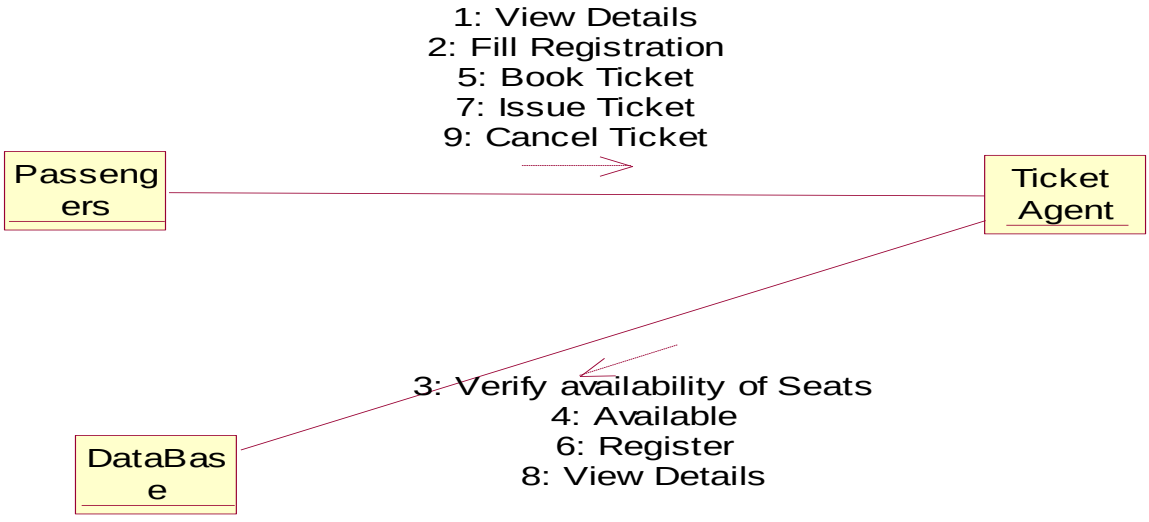
Use case Diagram



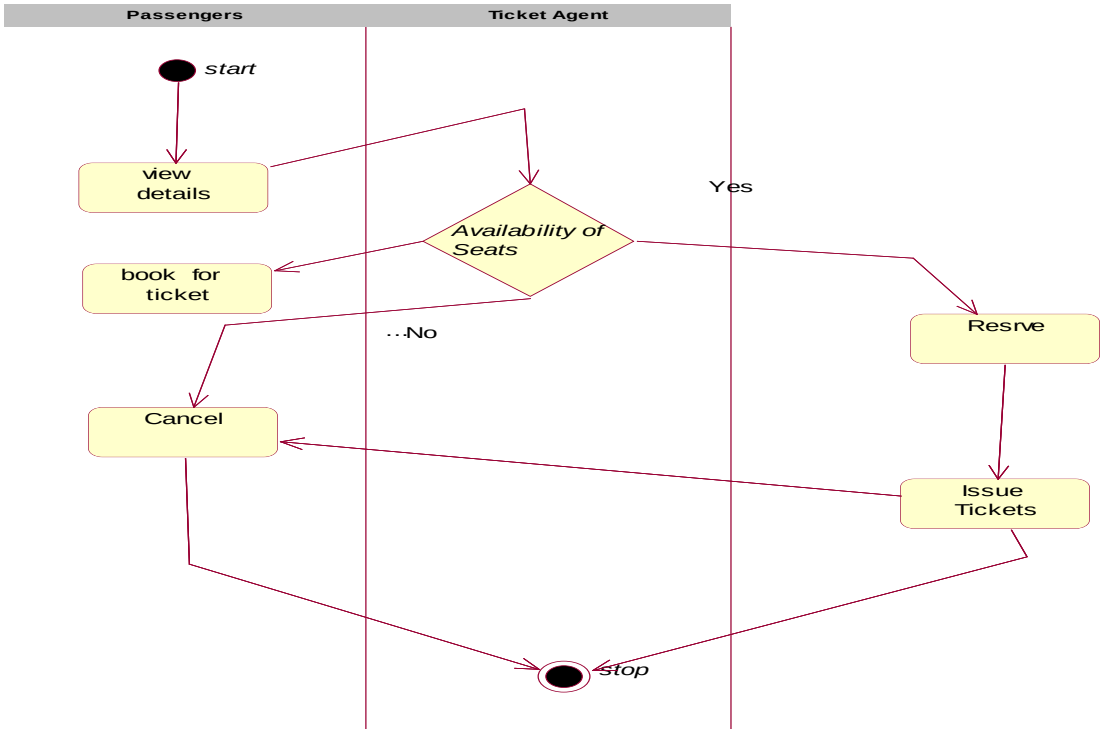
Sequence diagram:



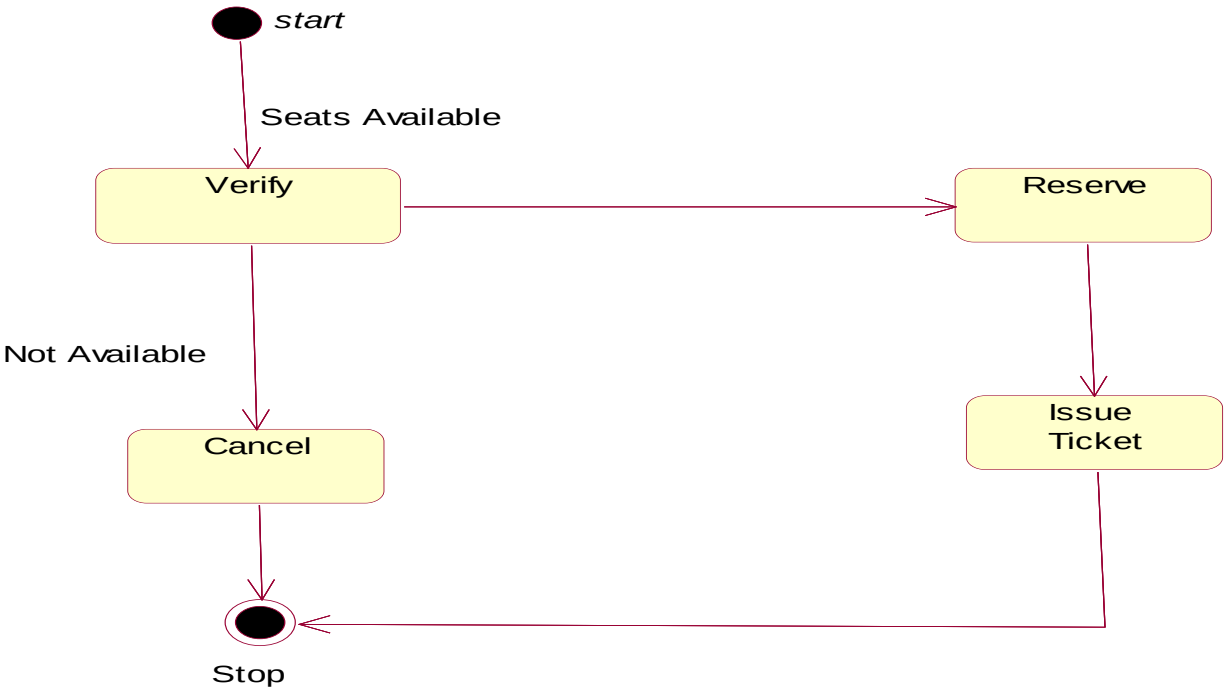
Collaboration Diagram:



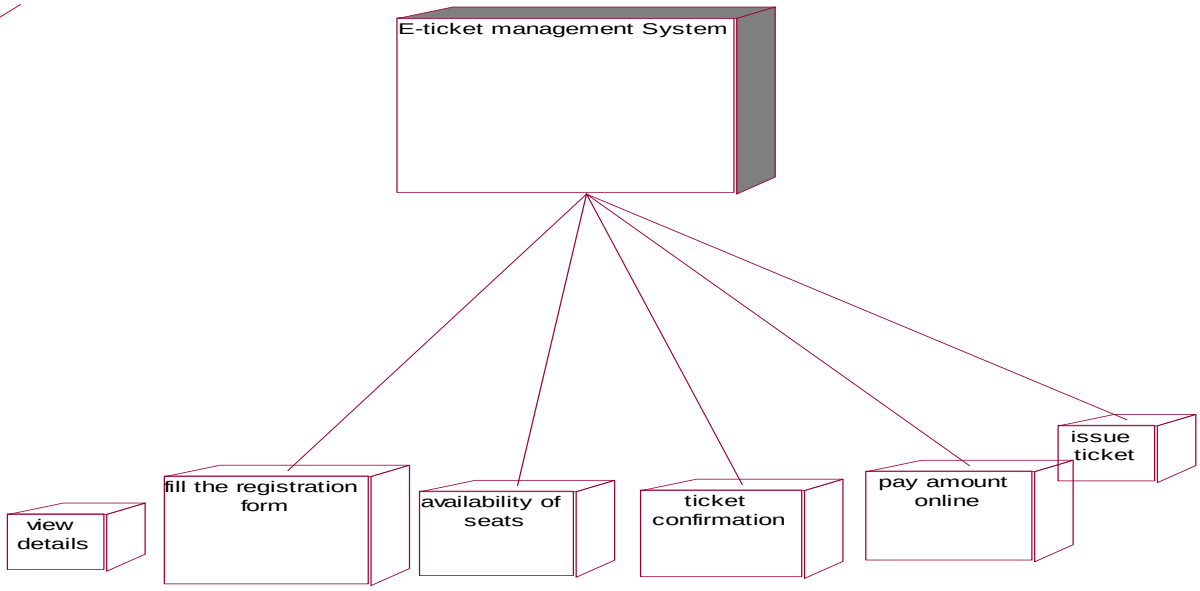
Activity diagram



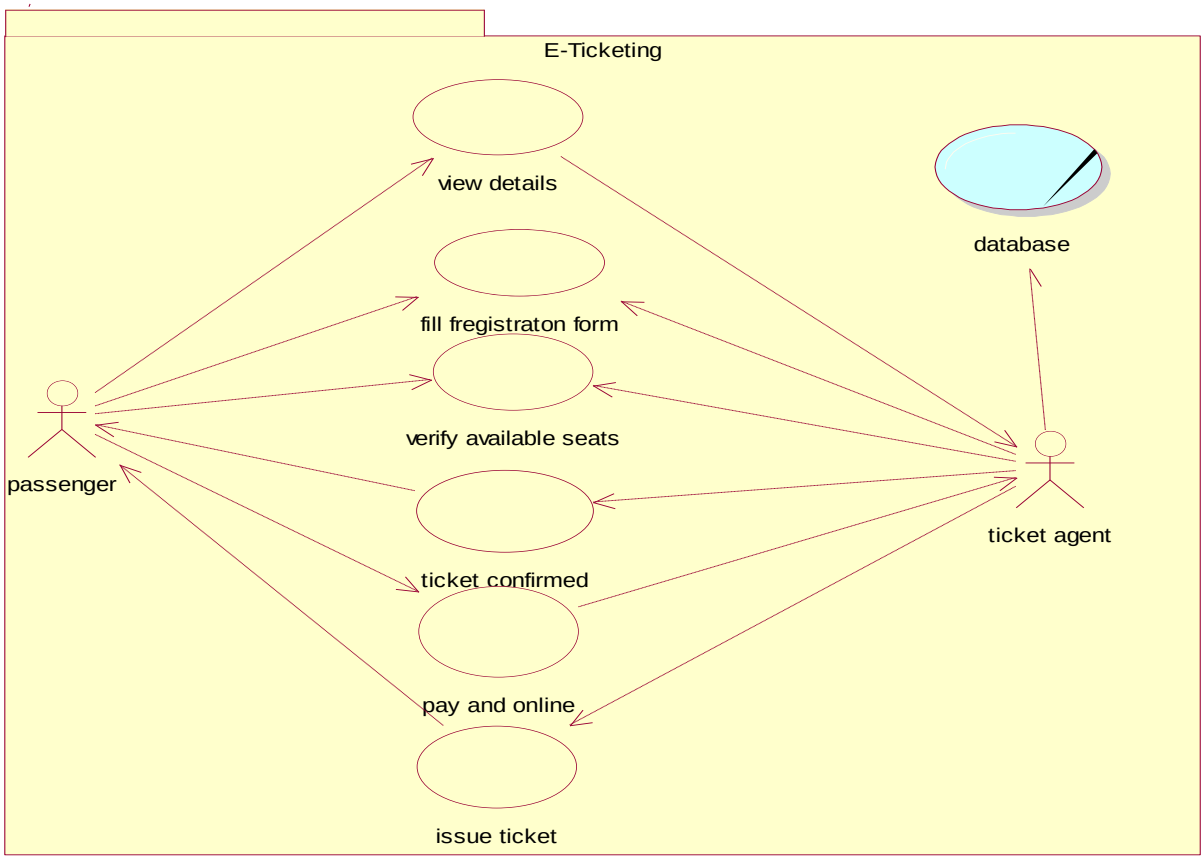
State Diagram:



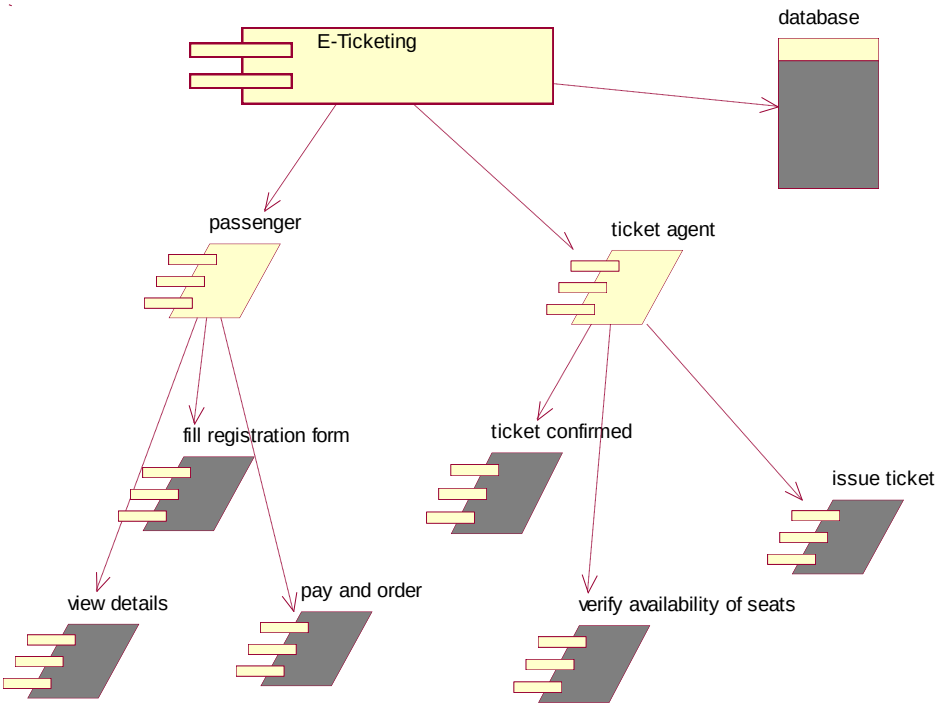
Deployment Diagram:



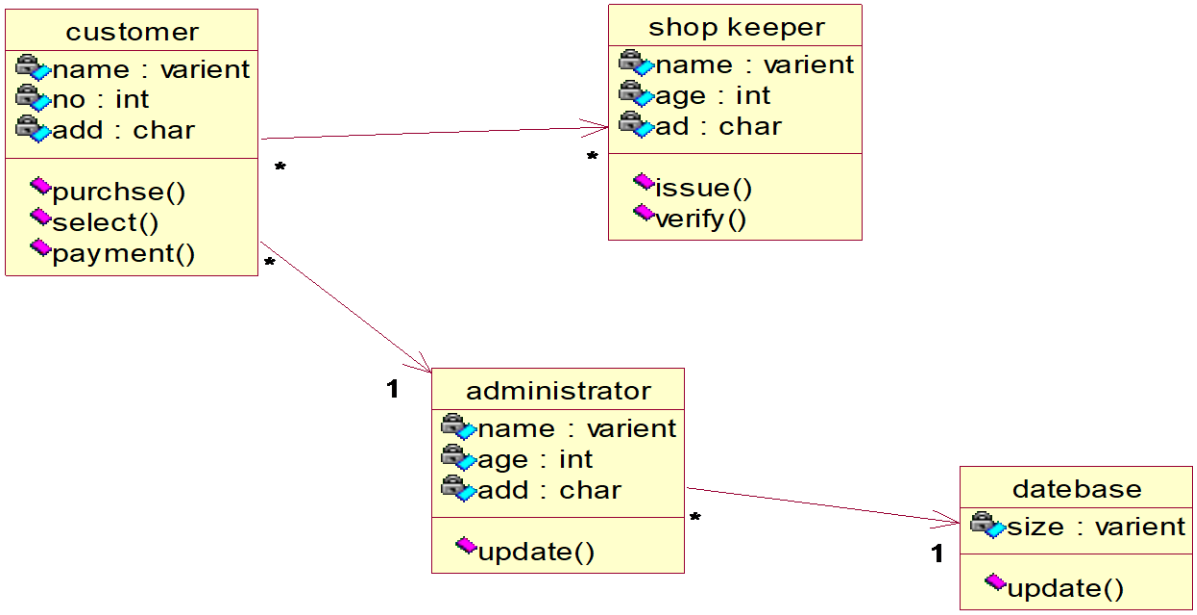
Package Diagram:



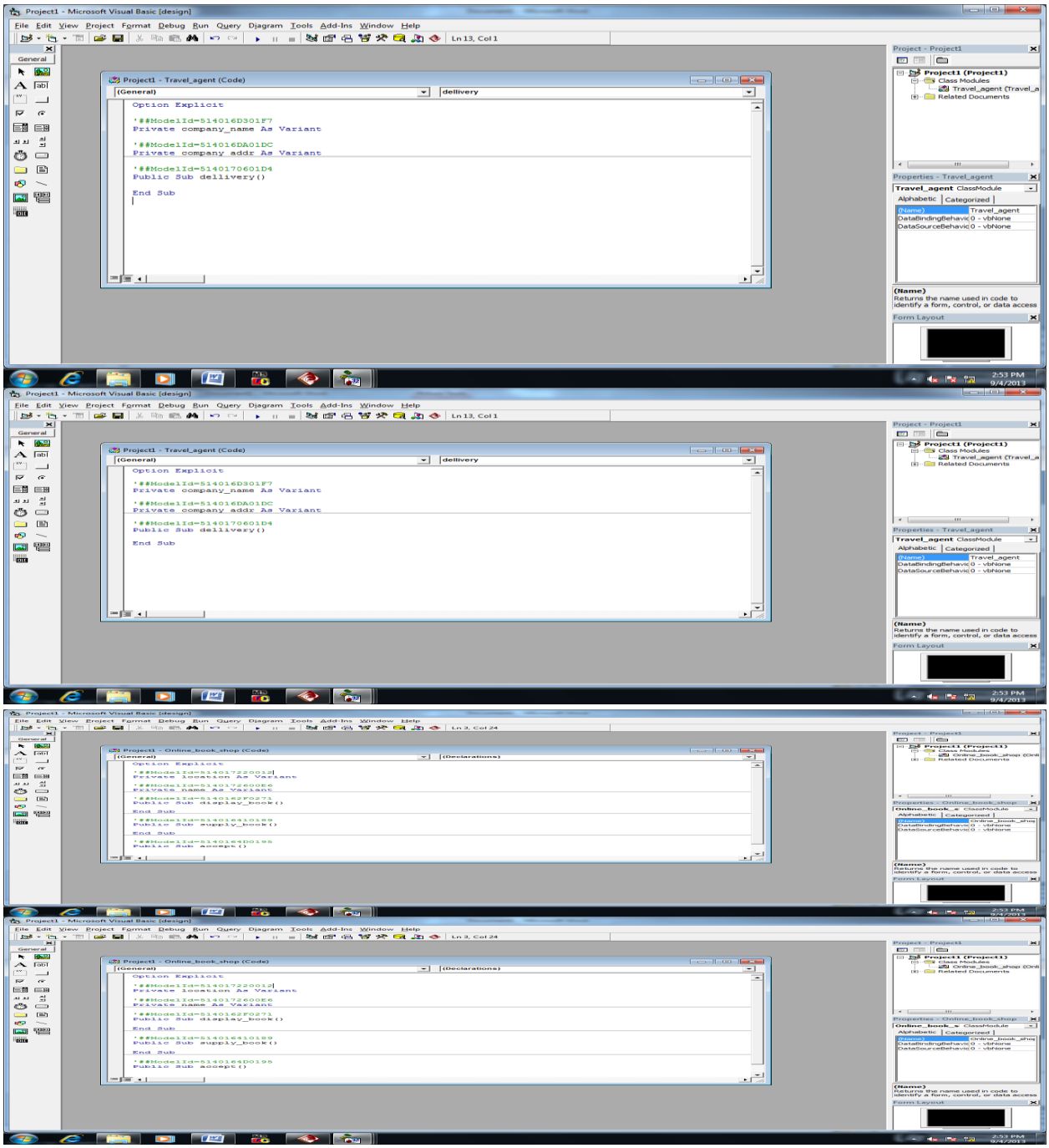
Component Diagram:



Class diagram:



Output:



Result:

Thus the E-ticketing was designed and the code was generated successfully.

EX.NO: 8 SOFTWARE PERSONNEL MANAGEMENT SYSTEM

DATE:

AIM:

To implement a software for software personnel management system.

(I)PROBLEM STATEMENT:

Human Resource management system project involves new and/or system upgrades of software of send to capture information relating to the hiring termination payment and management of employee. He uses system to plan and analyze all components and performance of metrics driven human resource functions, including recruitment, attendance, compensation, benefits and education. Human resources management systems should align for maximum operating efficiency with financial accounting operations customer relationship management,security and business lines as organization.

(II)SOFTWARE REQUIREMENT SPECIFICATION:

2.1 SOFTWARE INTERFACE

- **Front End Client** - The applicant and Administrator online interface is built using

  JSP and HTML. The HR's local interface is built using Java.

- **Server** - Glassfish application server(SQL Corporation).

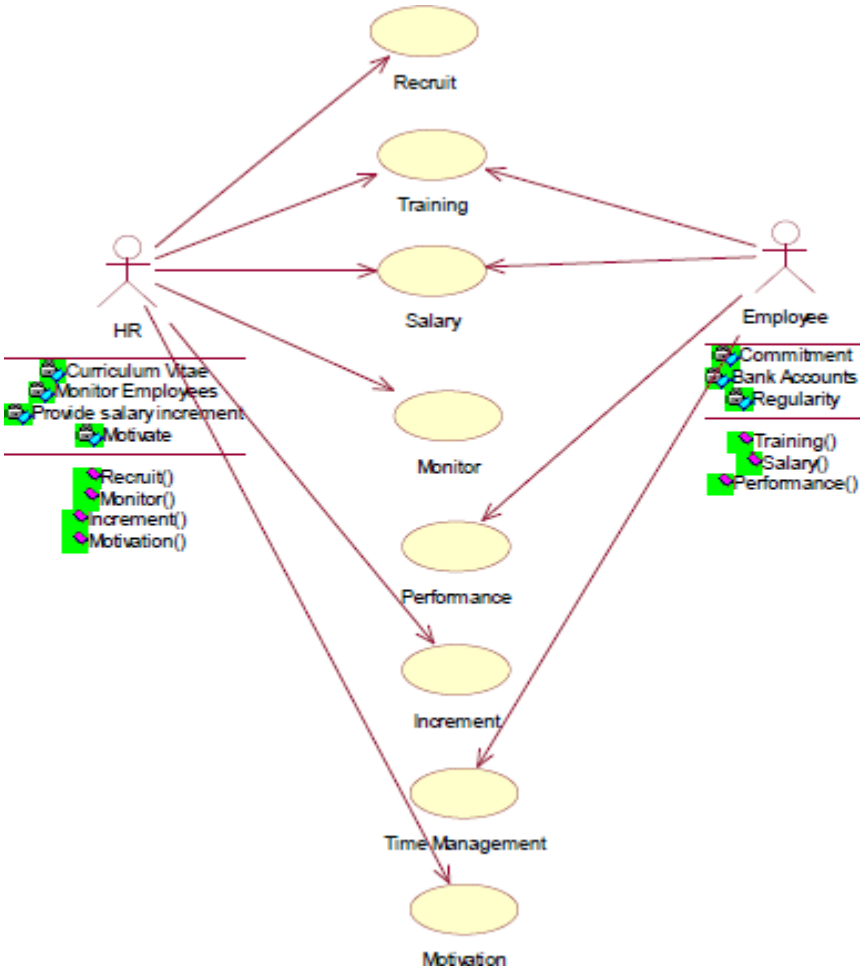
- **Back End** - SQL database.

2.2 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

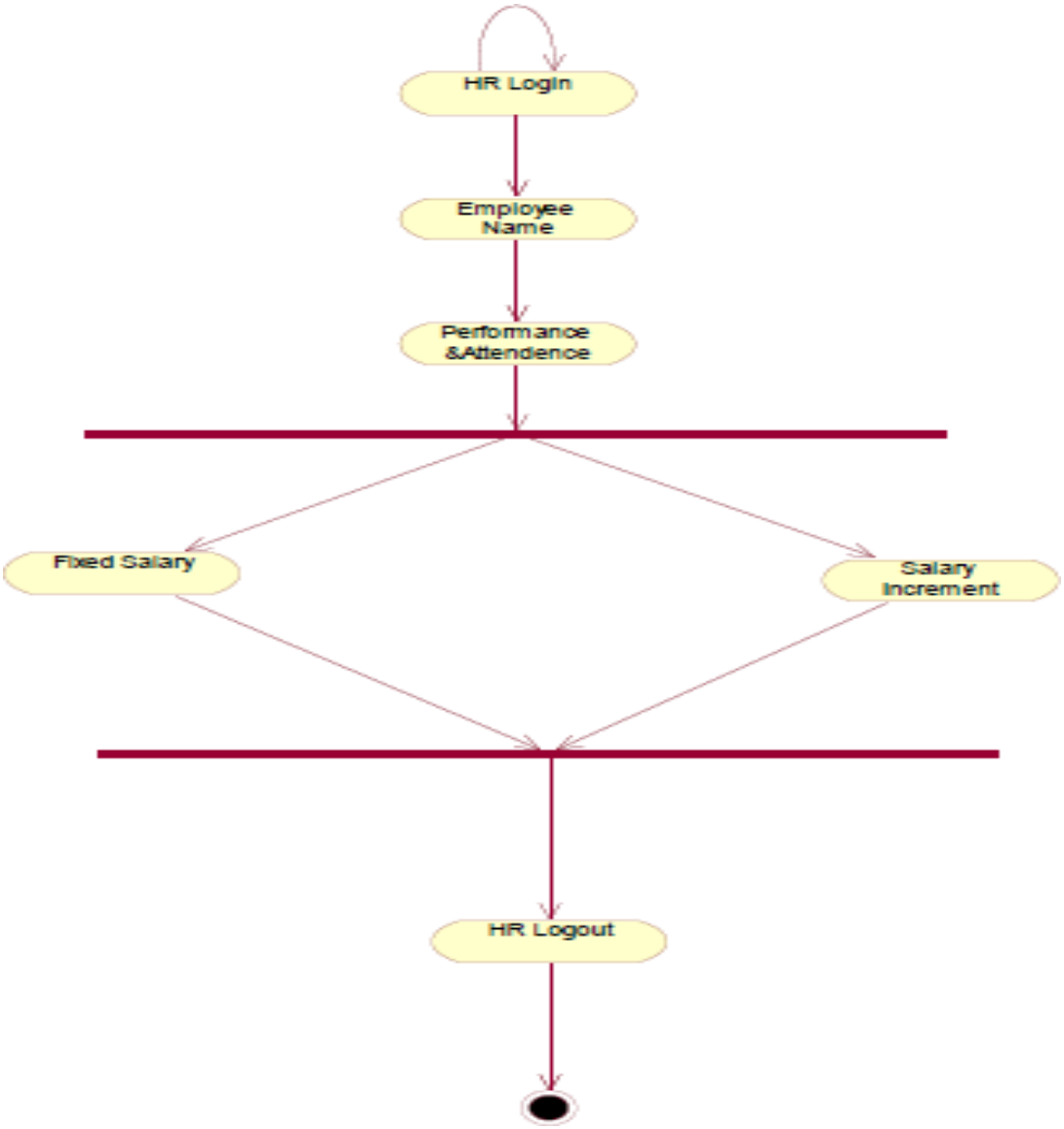
(III)USECASE DIAGRAM:

The HR of an organization involves recruitment training, monitoring and motivation of an employee. The HR also involves gives salary as observed in the payroll sheet. The employee undergoes training, receives the salary , gives the expected performance and manages time in order to complete a given task within the required period.



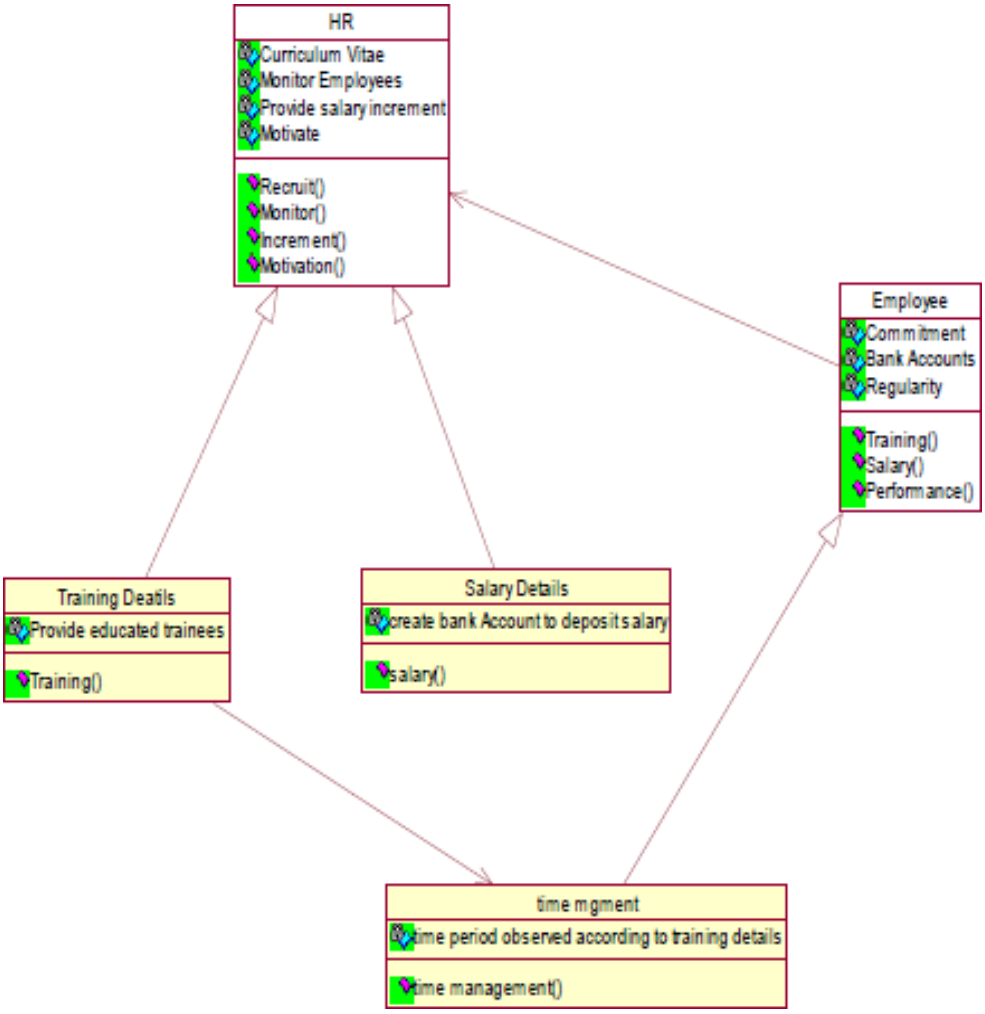
(IV) ACTIVITY DIAGRAM:

The activity diagram notation is an action, partition, fork join and object node. Most of the notation is self explanatory, two subtle points. Once an action finished, there is an automatic outgoing transaction. The diagram can show both control flow and data flow.



(V) CLASS DIAGRAM:

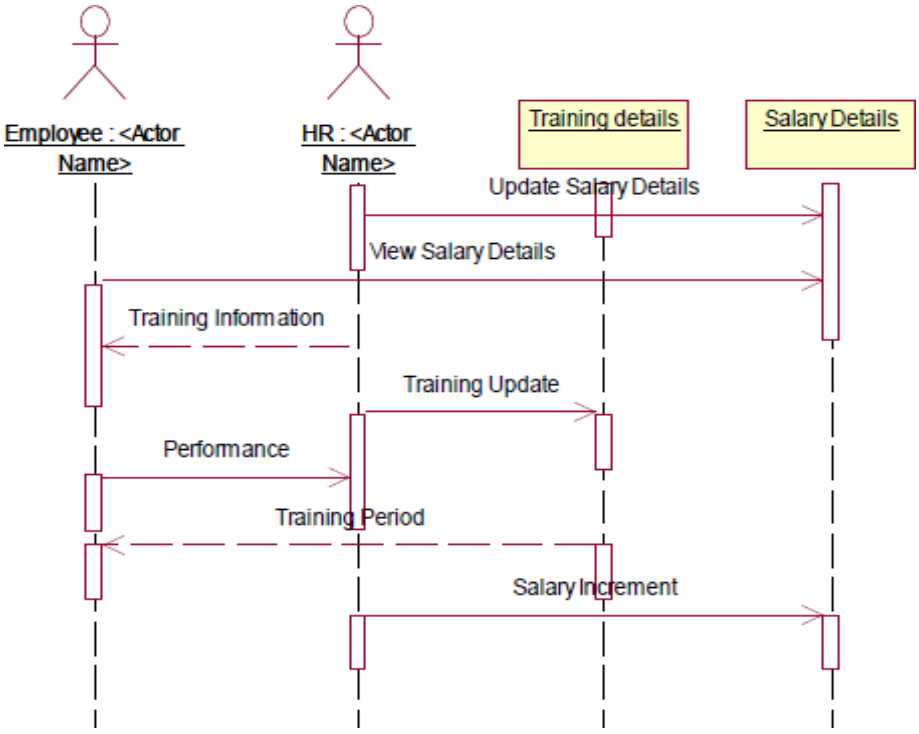
The class diagram, also referred to as object modeling is the main static analysis diagram. The main task of object modeling is to graphically show what each object will do in the problem domain. The problem domain describes the structure and the relationships among objects.



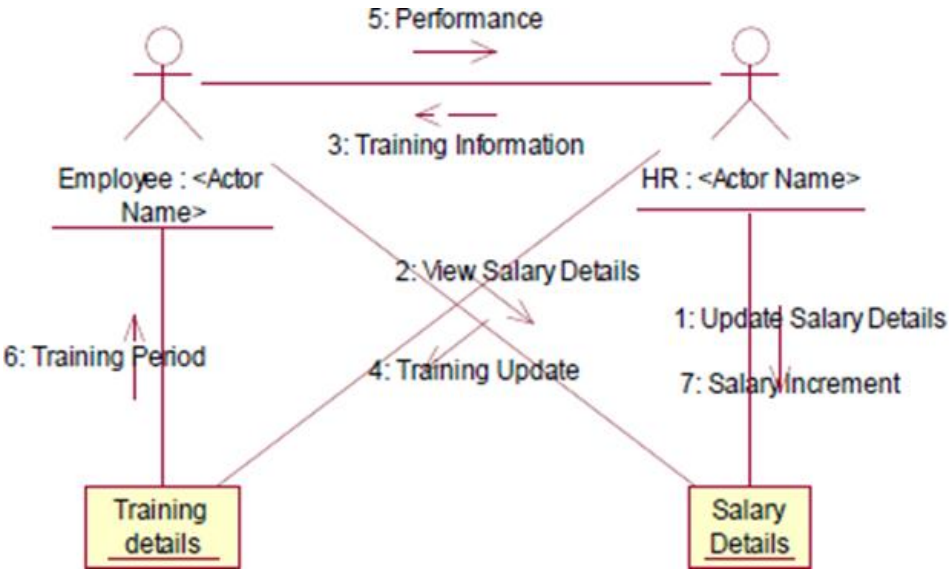
(VI) INTERACTION DIAGRAM:

A sequence diagram represents the sequence and interactions of a given USE-CASE or scenario. Sequence diagrams can capture most of the information about the system. Most object to object interactions and operations are considered events and events include signals, inputs, decisions, interrupts, transitions and actions to or from users or external devices.

SEQUENCE DIAGRAM

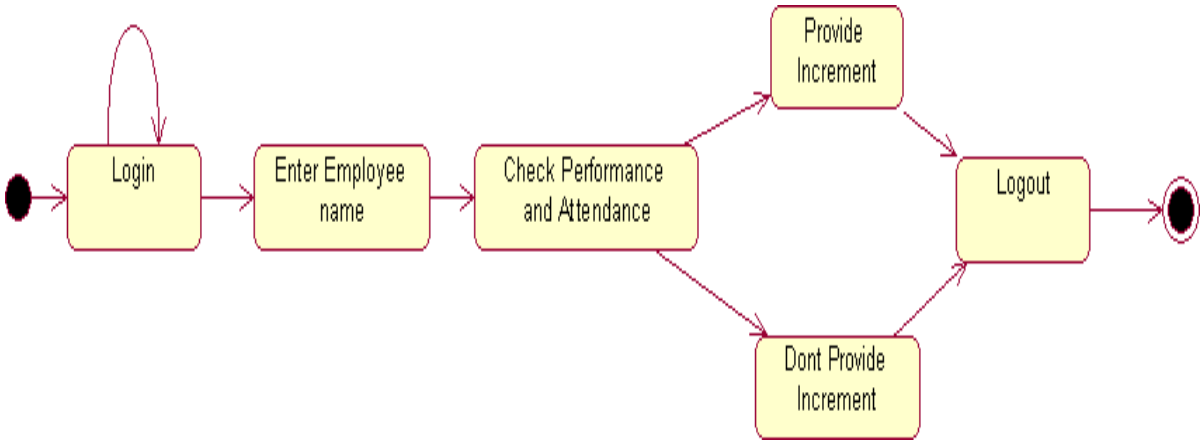


COLLABORATION DIAGRAM



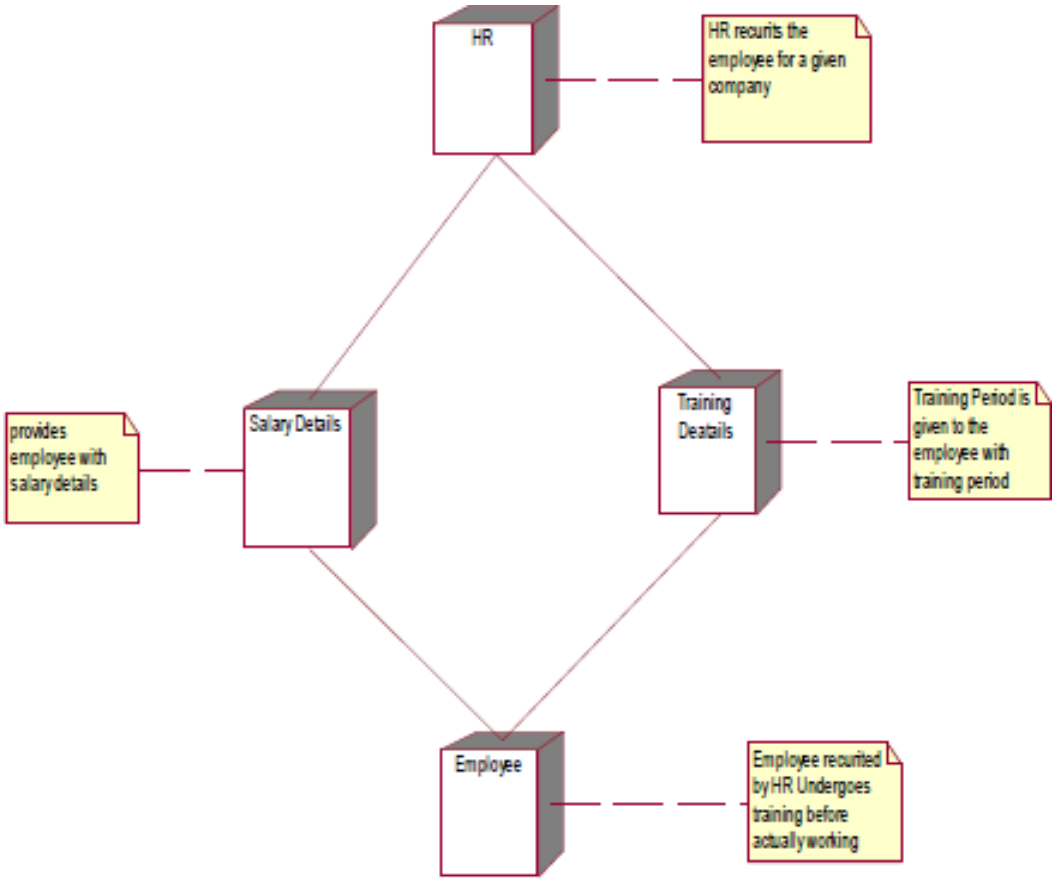
(VII) State Transition Diagram

States of object are represented as rectangle with round corner, the transaction between the different states. A transition is a relationship between two state that indicates that when an event occur the object moves from the prior state to the subsequent.



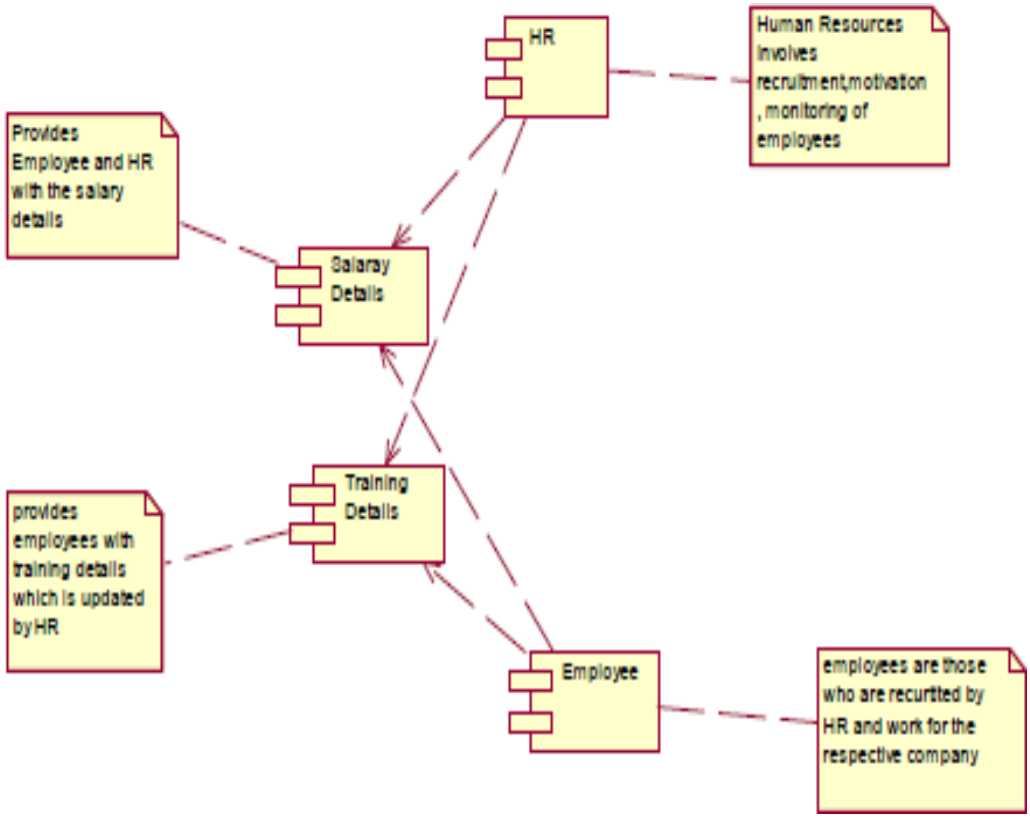
(VIII) DEPLOYMENT DIAGRAM AND COMPONENT DIAGRAM

HR recruits employee for a company employee recruited by HR goes under training before actually working. Training period is given to the employee with the training details. The salary details for the employee are provided.

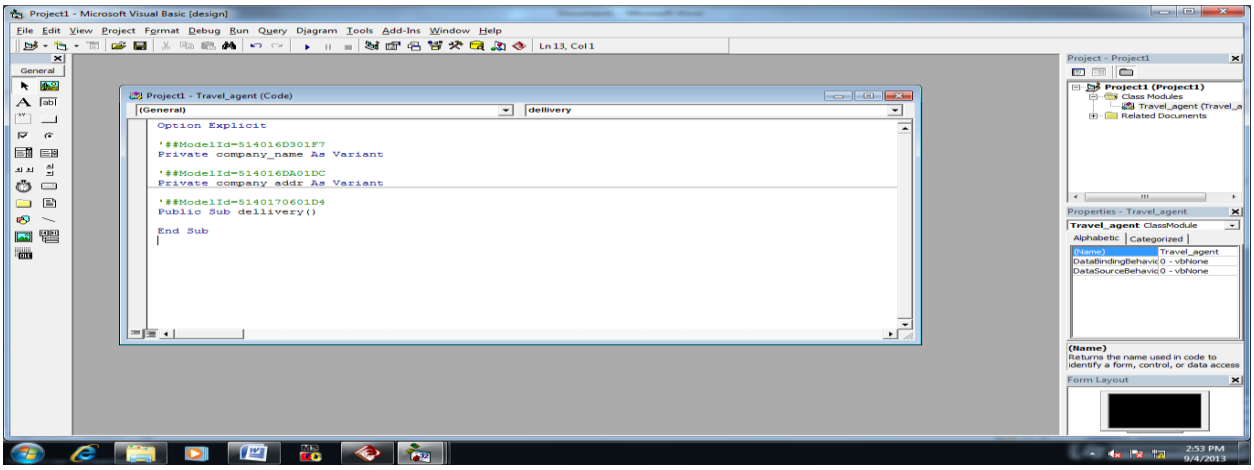


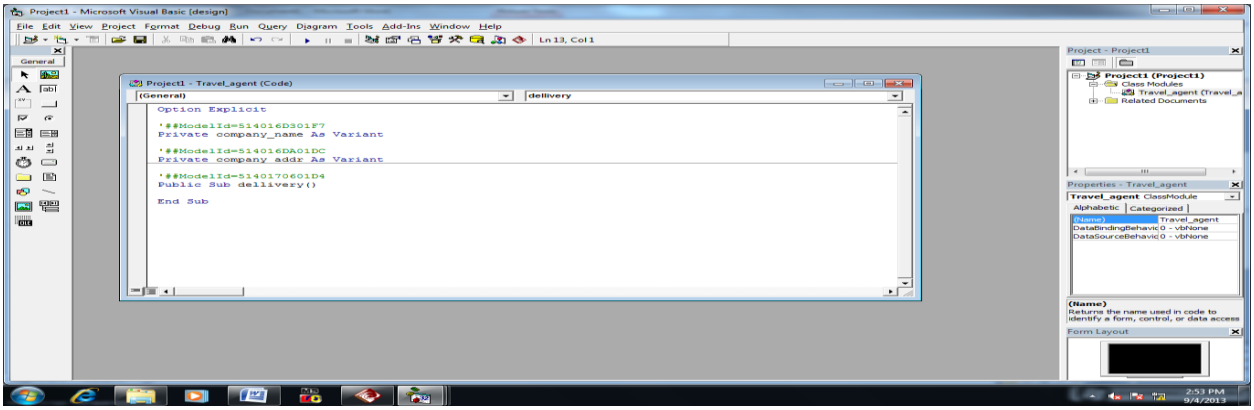
COMPONENT DIAGRAM

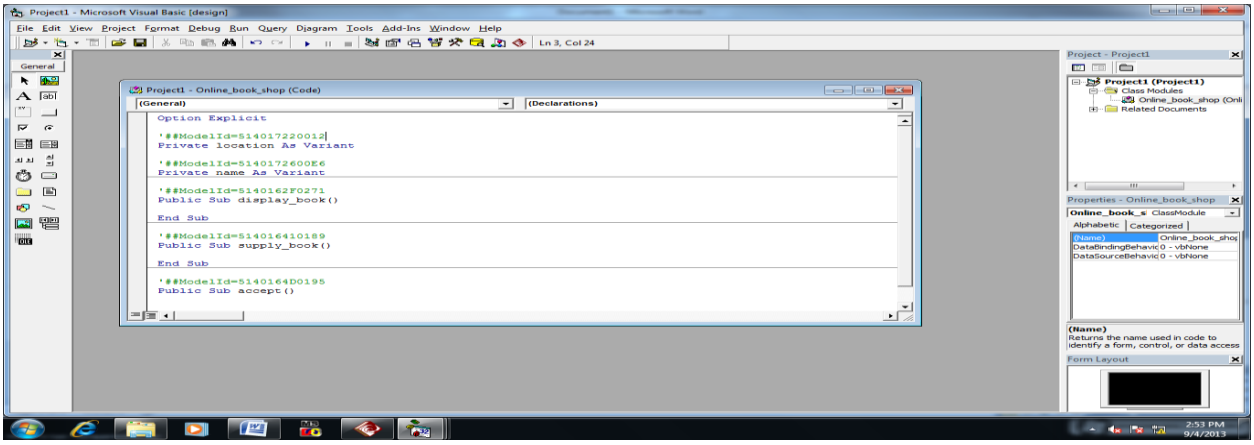
The HR recruits, motivate and monitor the employee, HR also update the salary details and training details for reference. The employee are those who are recruited by HR and work for the company. The training details provide employees with training details which is updated by HR

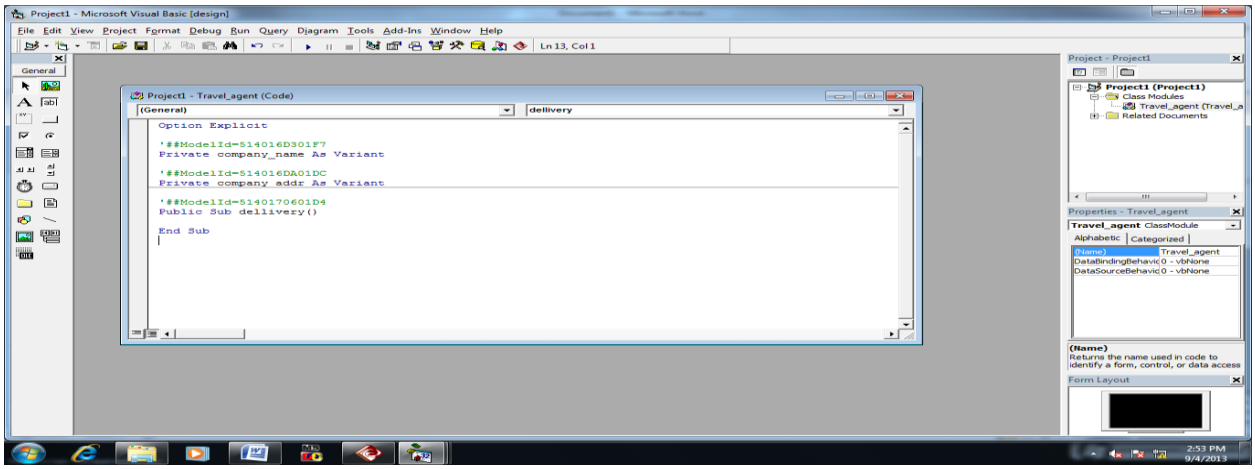


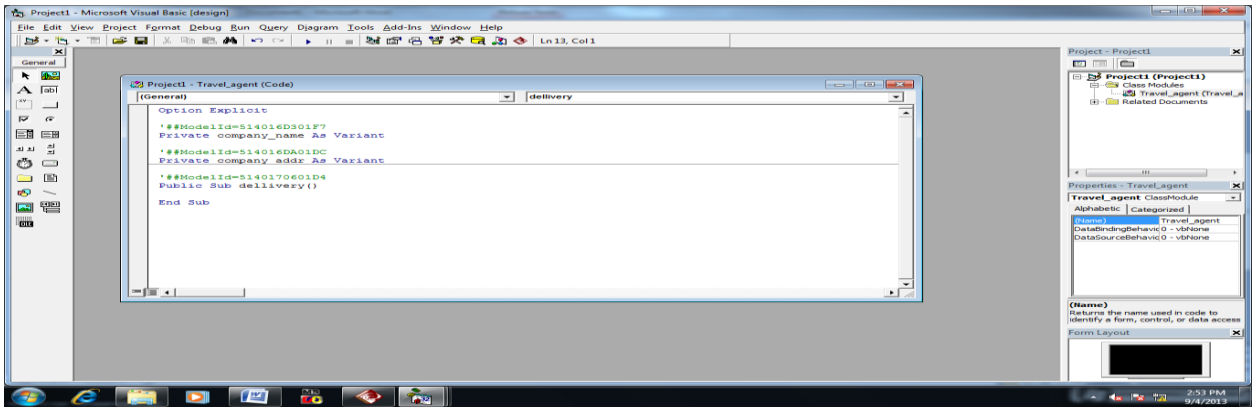
OUTPUT:


Project1 - Microsoft Visual Basic [design]
Project1 - Travel_agent (Code)
(General)
delivery
Option Explicit
'##ModelId=514016D301F7
Private company_name As Variant
'##ModelId=514016DA01DC
Private company_addr As Variant
'##ModelId=5140170601D4
Public Sub delivery()
End Sub


Project1 - Microsoft Visual Basic [design]
Project1 - Travel_agent (Code)
(General)
delivery
Option Explicit
'##ModelId=514016D301F7
Private company_name As Variant
'##ModelId=514016DA01DC
Private company_addr As Variant
'##ModelId=5140170601D4
Public Sub delivery()
End Sub


Project1 - Microsoft Visual Basic [design]
Project1 - Online_book_shop (Code)
(General)
(Declarations)
Option Explicit
'##ModelId=514017220012
Private location As Variant
'##ModelId=5140172600E6
Private name As Variant
'##ModelId=5140162F0271
Public Sub display_book()
End Sub
'##ModelId=514016410189
Public Sub supply_book()
End Sub
'##ModelId=5140164D0195
Public Sub accept()





RESULT:

Thus the software personnel management system has been successfully executed and codes are generated.

EX.NO:9 CREDIT CARD PROCESSING SYSTEM

DATE:

Aim:

To analyze and design credit card processing system using Umbrello Software.

Procedure:

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

The credit card system includes actors such as customer and credit card controller. Also it includes the use cases such as withdraw, membership ID. In the credit card system customer can easily withdraw the money without any difficulties in order to use the credit card system, customer needs the credit card. At first the card can be obtained by opening the account in any one of the bank, after 15 days the credit card send to the house by corresponding authorities, the customer can withdraw the money by inserting the credit card system into the credit machine. Then select the option language, then it will ask the pin number. Then it will ask to enter the amount. After entering the touch yes option. Then the corresponding amount will obtained from credit card machine after getting the leave the credit card controller.

Project scope:

The credit card system deals with the process of transaction the money to the customer and money is to be deposited. The customer needs the credit card for withdrawing the money. The customers who have the credit card utilize these facilities.

Software requirement analysis

It deals with following operations

- Withdraw money
- Verification and Updating

Withdraw money:

Customer gets the membership card from the bank. Then the customer will use the membership card to take money from the credit machine. The customer can perform withdraw of money in account.

Verification and updating:

The credit controller accesses the id and after accessing it checks the pin number whether it is correct or not. If it is correct means then the requirement amount is transacted from the customer account.

SOFTWARE REQUIREMENT SPECIFICATION

1.0 INTRODUCTION

A **credit card** is a small plastic card issued to users as a system of payment. It allows its holder to buy goods and services based on the holder's promise to pay for these goods and services. The issuer of the card creates a revolving account and grants a line of credit to the consumer (or the user) from which the user can borrow money for payment to a merchant or as a cash advance to the user.

When a purchase is made the merchant swipes the card. The information goes to a gateway processor, which either accepts or rejects the transaction. If it is accepted, the transaction is held until the end of the business day. The merchant then reenters the transaction via the gateway processor, the data is logged, and the debt is transferred to the account. The use of an ATM for cash advance is a similar process.

If you are selling to consumers, merchant services will allow you to expand your customer base and provide a more convenient method of payment than cash or checks. And if you are interested in selling over the Internet, accepting credit card processing is a must. Accepting credit cards allows funds to be transferred to your bank account in less than a week. This can be a welcome relief for businesses that experience a tight cash flow.

The two purchase options for Credit Card Processing facility are:

- Validation only
- Credit card processing (which secures deposits at the time of booking)

With either option, credit card accounts entered during booking are validated to assure that the account is active and in good standing. The credit card processing option also allows properties to process credit card deposits.

1.1 PURPOSE

When customers complete their shopping cart, their credit card is preauthorized and the order is entered into Sales Order. Credit Card Processing dials out and obtains a credit card payment. Within five minutes the customer receives an e-mail receipt.

1.2 SCOPE

- Automatically connects to your financial network for credit card authorizations and settlements
- Integrates with Sales Order, Accounts Receivable, and e-Business Manager
- Support for dial-up (modem) connections or secure Internet connections through TCP/IP and SSL
- Compliant with Visa and MasterCard Electronic Commerce Indicator (ECI) regulations
    - Multiple address verification options available .

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

• **Authorization service** - The issuer of the card creates a revolving account and grants a line of credit to the consumer (or the user) from which the user can borrow money for payment to a merchant or as a cash advance to the user.

•**User**- One who wishes to uses the credit card.

• **CCP** - Refers to this Credit Card Processing.

• **HTML** - Markup Language used for creating web pages.

•**J2EE** – Java 2 Enterprise Edition is a programming platform java platform for developing and running distributed java applications.

• **HTTP** - Hyper Text Transfer Protocol.

•**TCP/IP** – Transmission Control Protocol/Internet Protocol is the communication protocol used to connect hosts on the Internet.

1.4 REFERENCES

IEEE Software Requirement Specification format.

1.5 TECHNOLOGIES TO BE USED

• HTML
• JSP
• Javascript
• Java

1.6 TOOLS TO BE USED

• Eclipse IDE (Integrated Development Environment)
• Umbrello tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements –
Overall Description will describe major role of the system components and inter-connections.
Specific Requirements will describe roles & functions of the actors.

2.0 OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

This solution involves signing up for a free Business Account. Once this is done and the e-commerce site is properly configured, you can accept payments from Visa, MasterCard, Amex, and Discover cards payments.

2.2 SOFTWARE INTERFACE

• **Front End Client** - The applicant and Administrator online interface is built using JSP and HTML. The Administrators's local interface is built using Java.
• **Web Server** - Glassfish application server(SQL Corporation).
• **Back End** - SQL database.

2.3 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

2.4 SYSTEM FUNCTIONS

1. Accept credit card numbers on the web, store them in a database, then process them off-line

2. Credit card processing with CCP
3. Credit card processing with a third-party credit card processing company

2.5 USER CHARACTERISTICS

1) **User/Customer** - They are the people who desires to purchase the goods using credit card.
2) **Authorization Service**

- Validate the credit card payments to ensure that the card number is valid and the card has not expired
- Deposit processing to apply the deposit payment to the card
- Prepare Credit card transaction reports that show authorization codes, amounts, and error/success messages

2.6 CONSTRAINTS

- Trusted if using a well known third-party processor
- Must suite for higher-volume sites
- Cheaper transaction rates

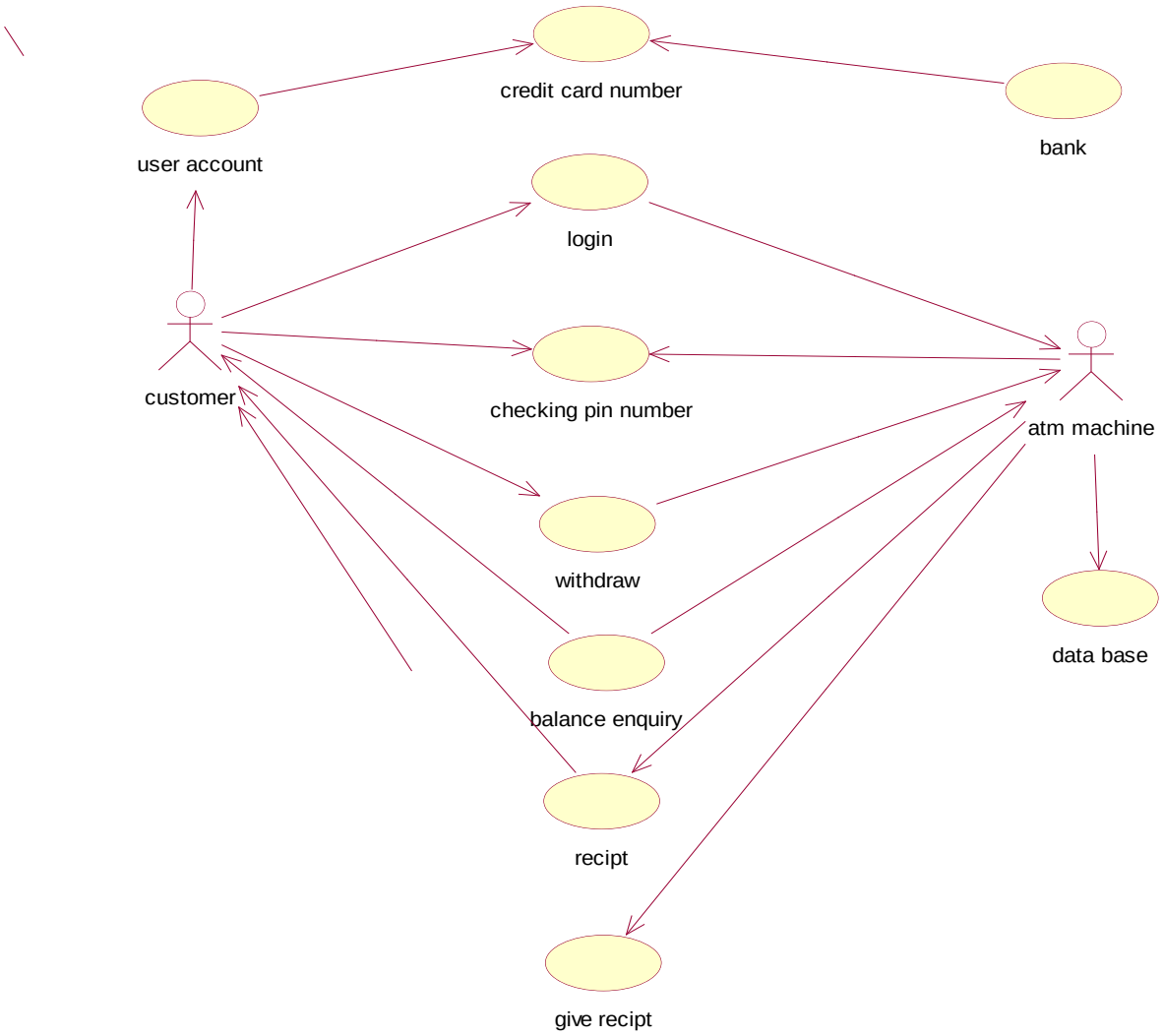
- Getting money transferred may be very fast
- Must provide fraud prevention measures and fraud protection programs

2.7 ASSUMPTIONS AND DEPENDENCIES

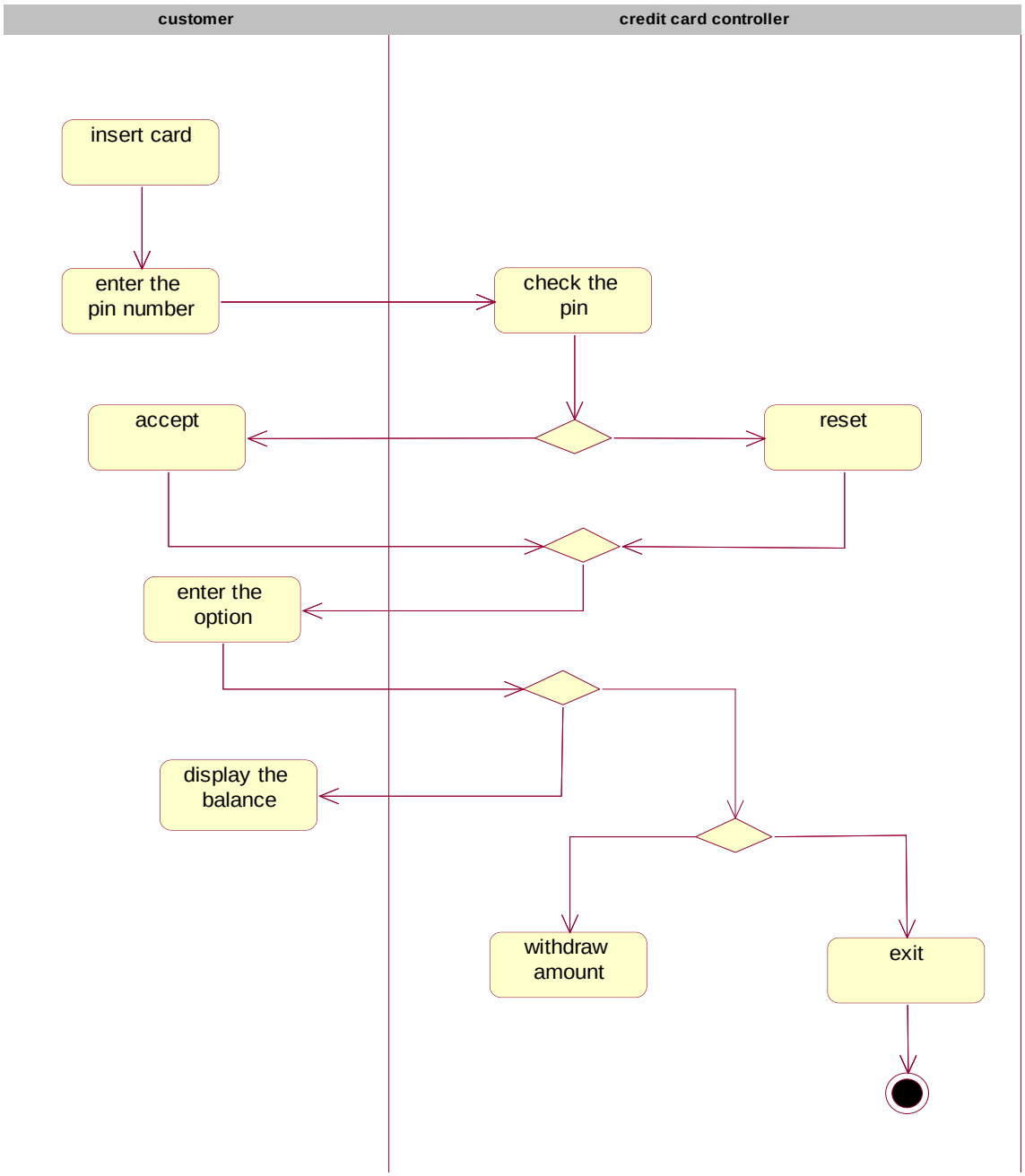
- The Applicants and Administrator must have basic knowledge of computers and English Language.
    - The applicants may be required to scan the documents and send.

UML Diagrams:

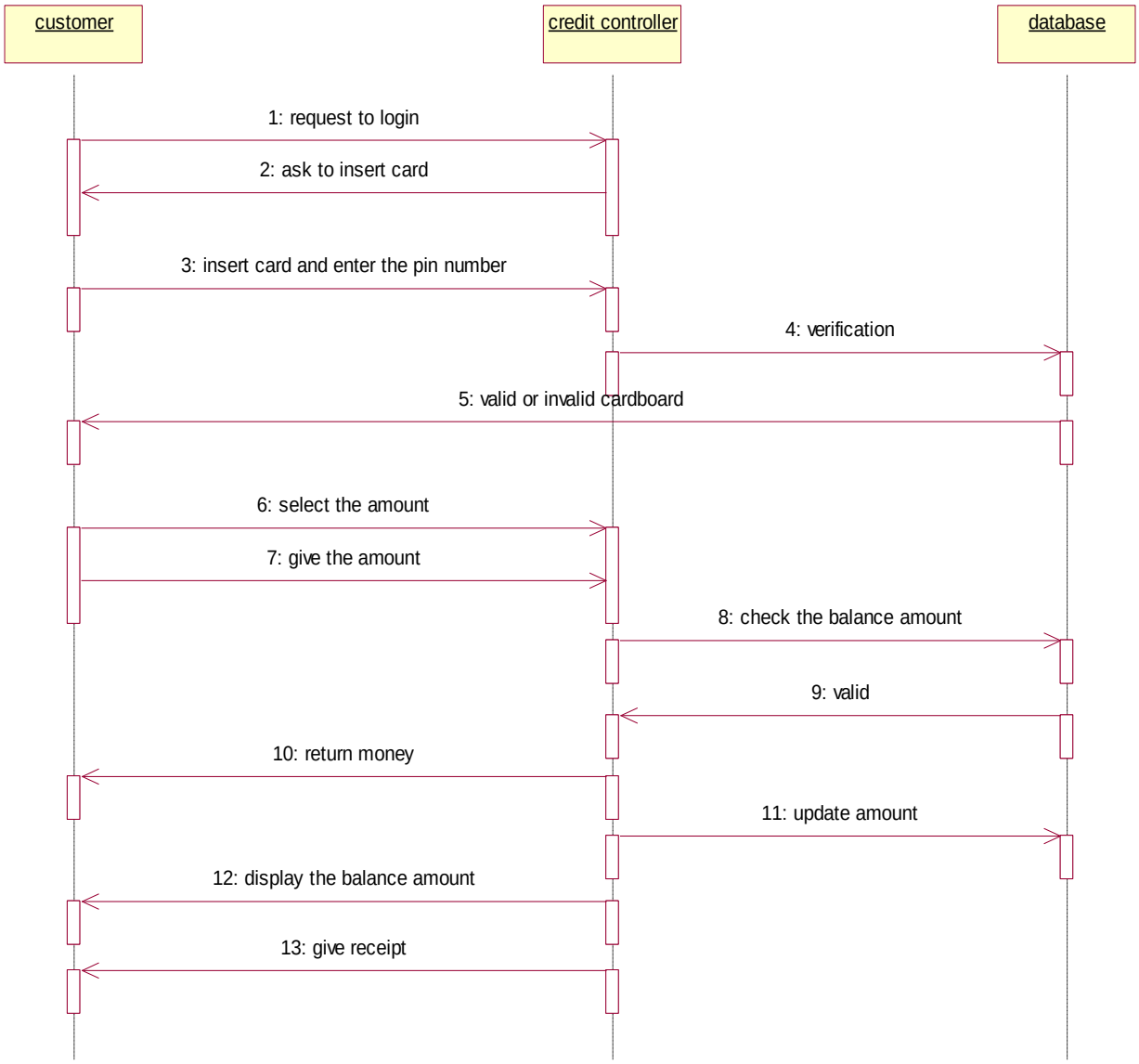
Use case Diagram:



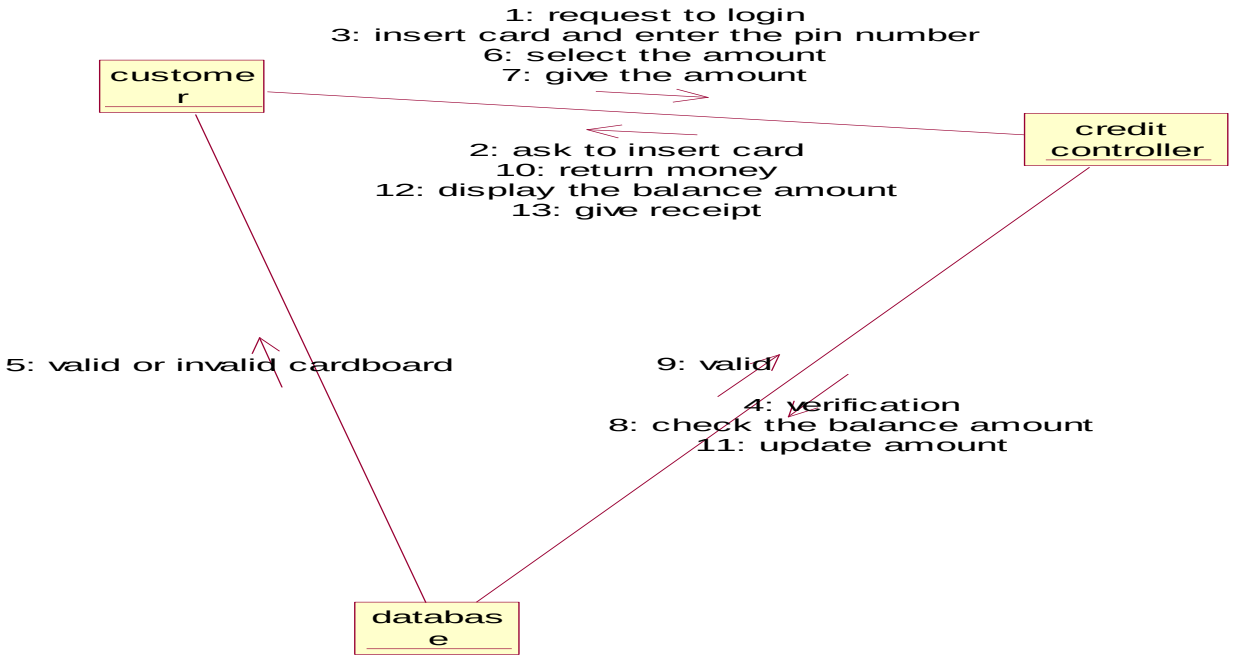
Activity Diagram:



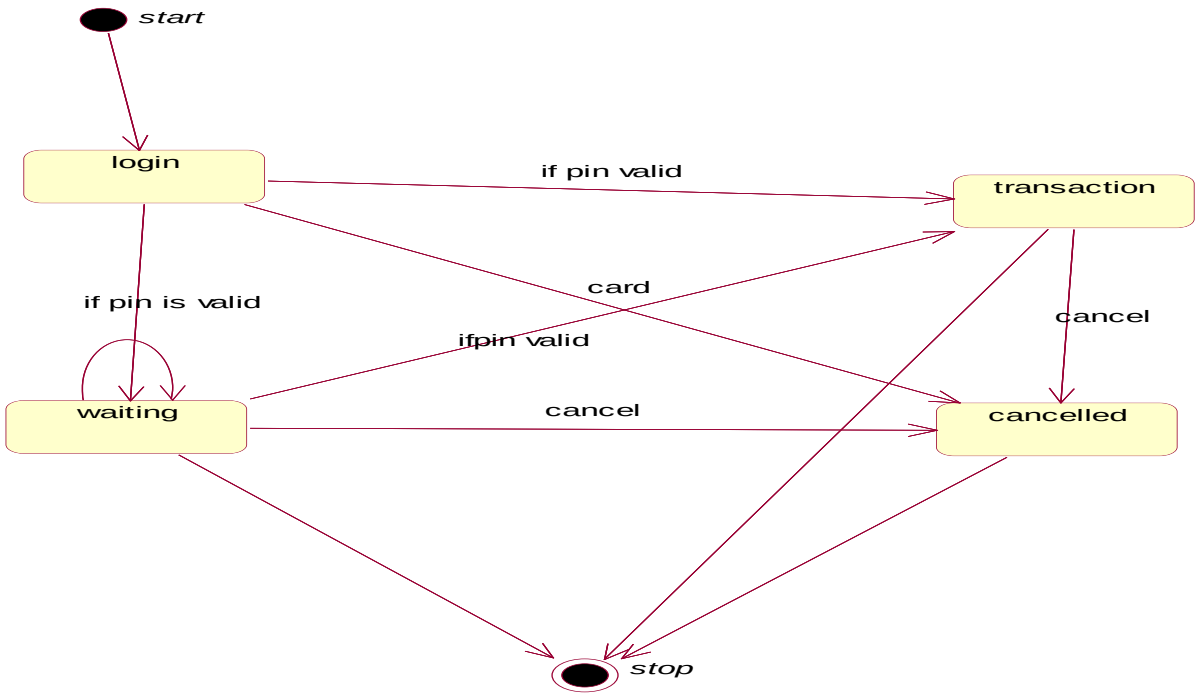
Sequence diagram:



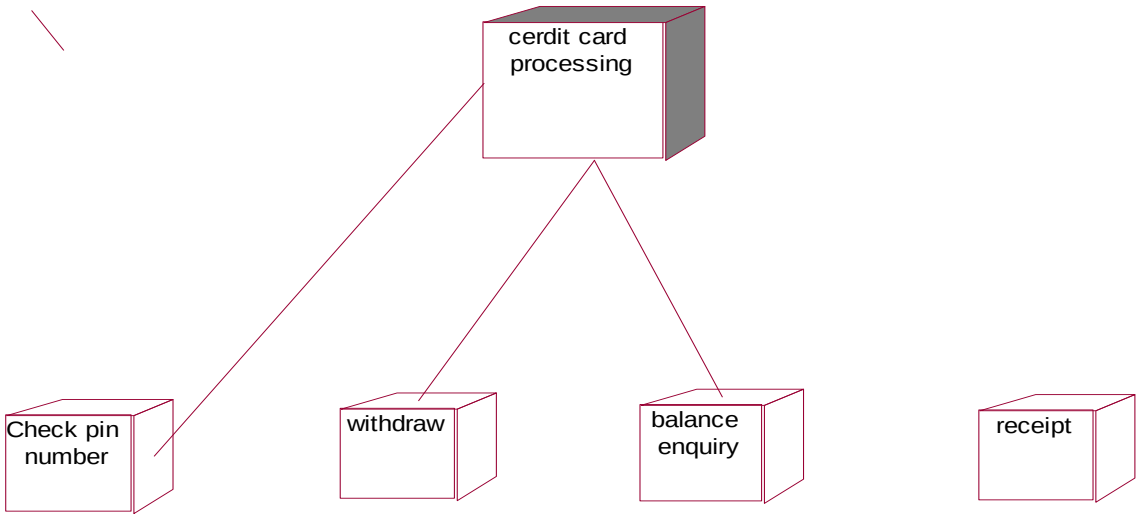
Collaboration Diagram:



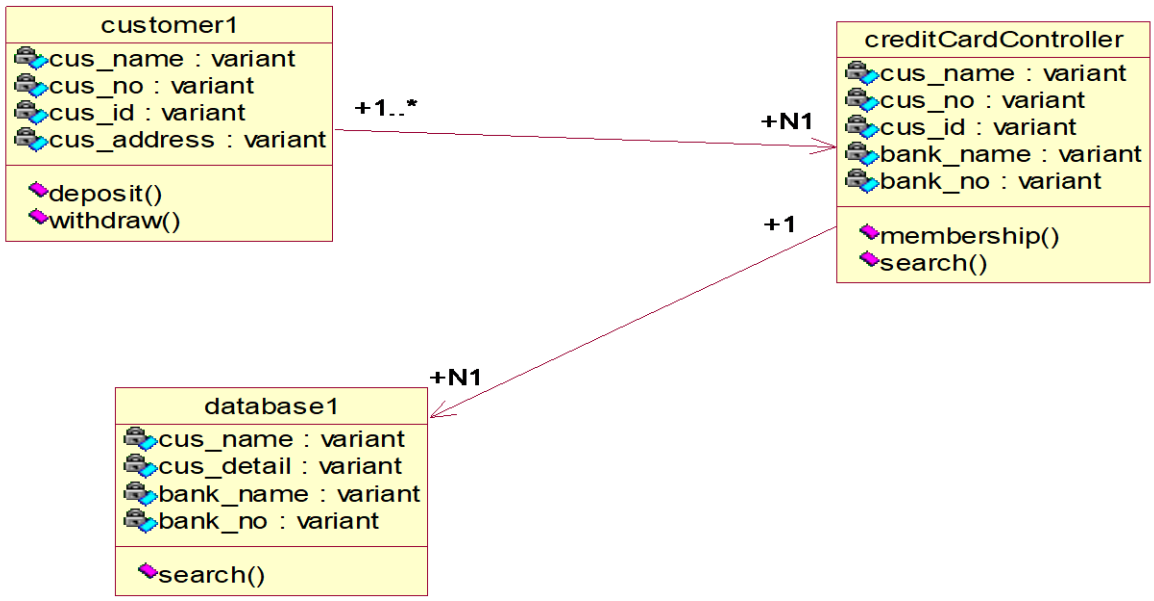
Statediagram:



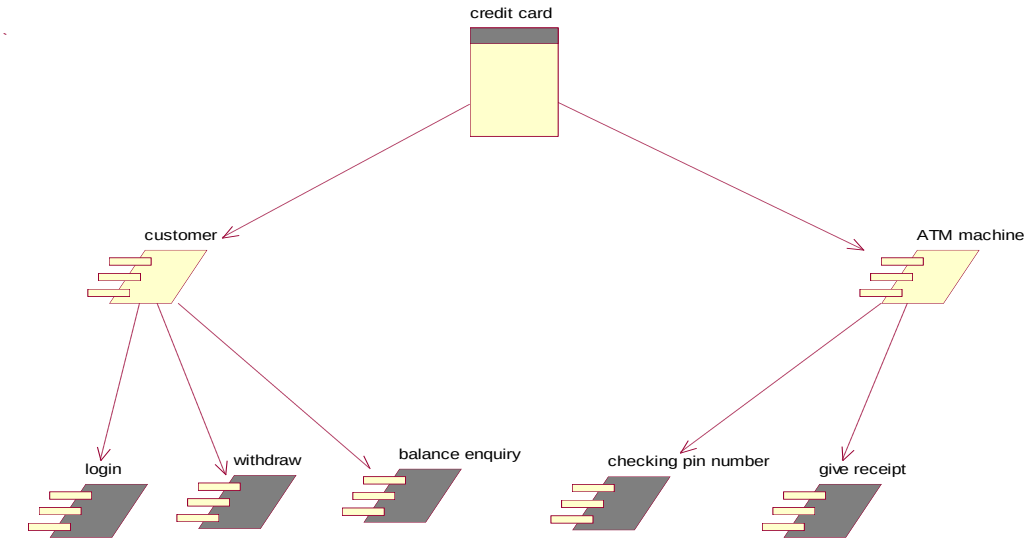
Deployment Diagram:



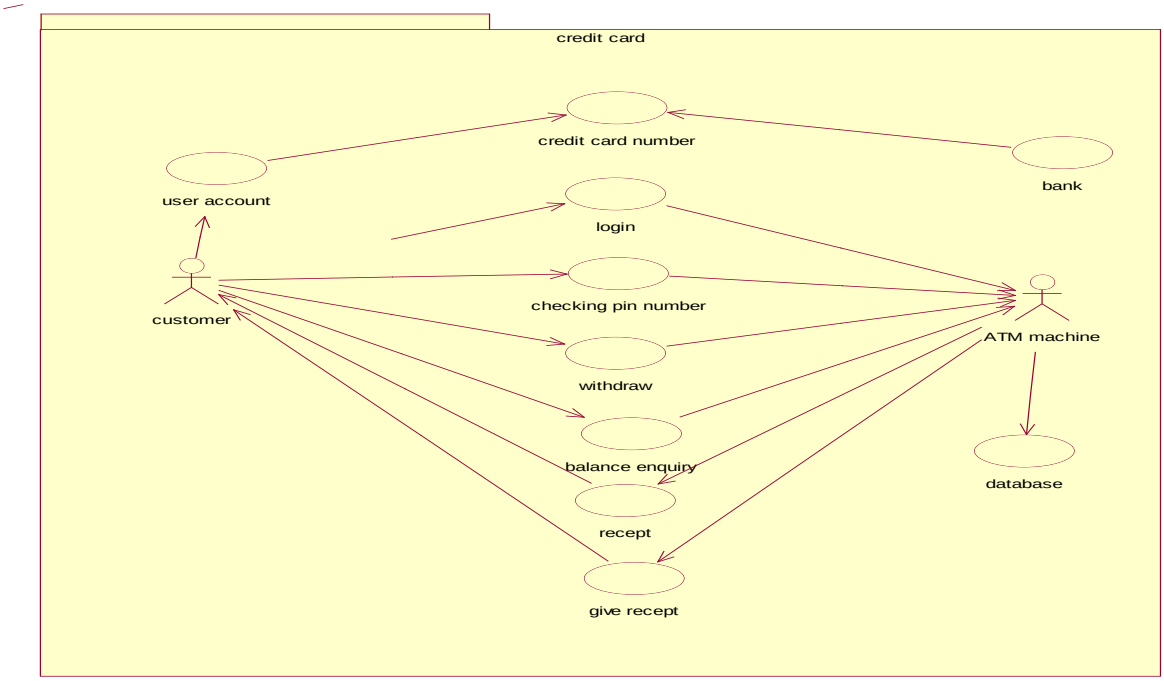
Class diagram:



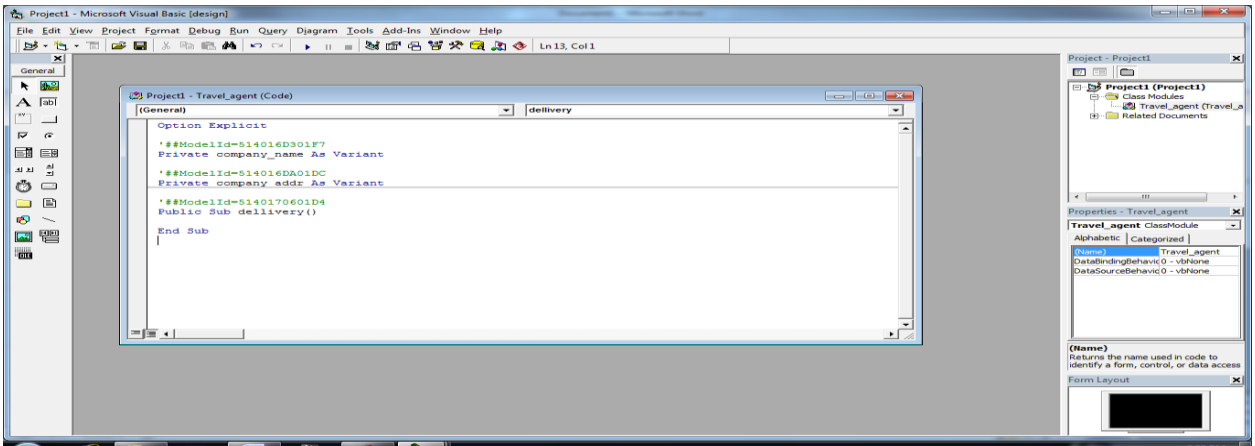
Component diagram:

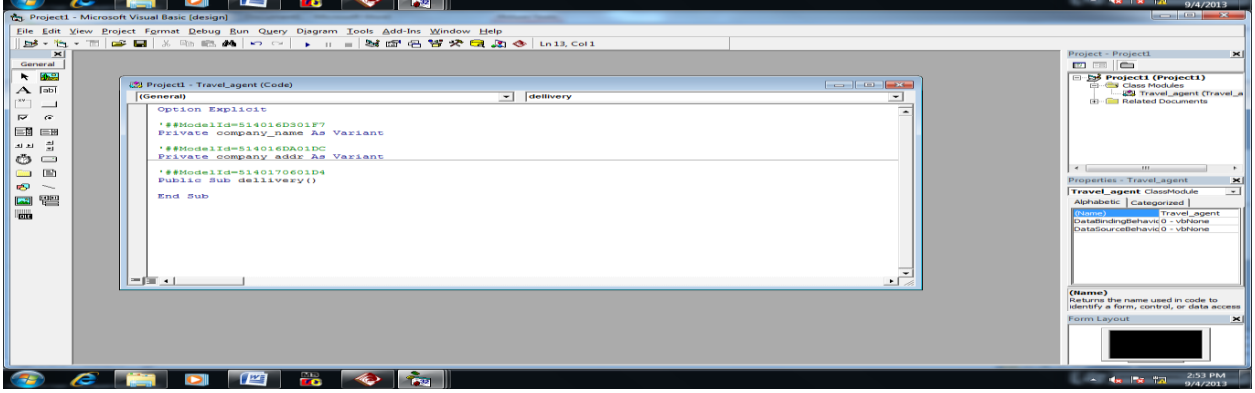


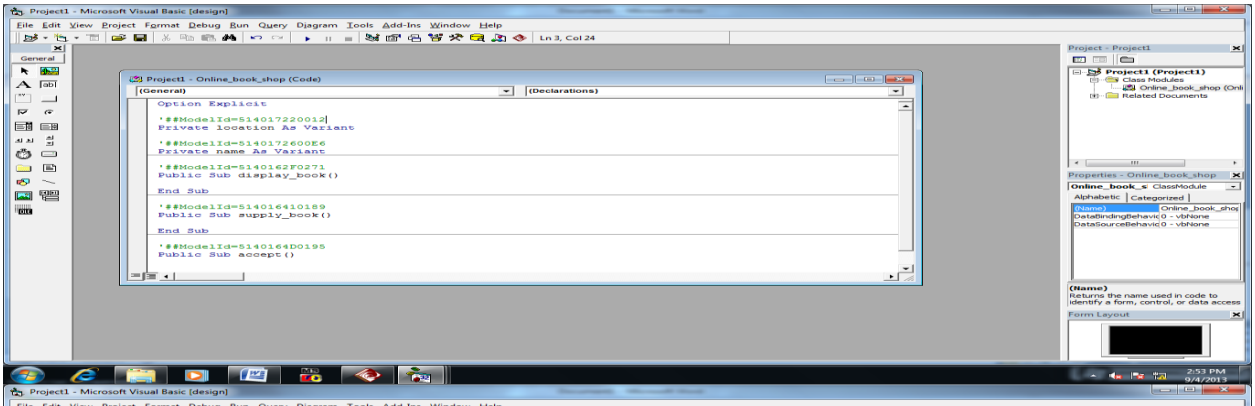
Package diagram:

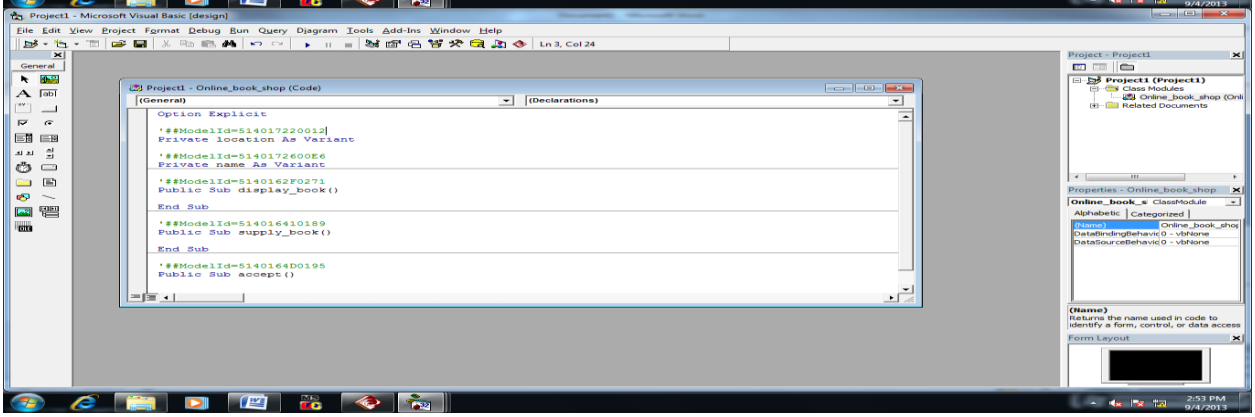


Output:









Result:

Thus the credit card processing system was designed and code is generated successfully.

EX.NO:10 E-BOOK MANAGEMENT SYSTEM

DATE:

Aim

To analyze and design E-book management system using Umbrello software.

Procedure

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

E-book includes the following functions:entering the user details,searching for the books,selecting the books for purchase,paying the bill issued and delivery of books.The actors of this system include the students,online bookshopping center and travel agent.The usecase includes login,searching the list of books,selecting the books to be purchased,paying the bill and delivery of books.

First the user logins using his name and password which is stored in the database,if it does not exist previously.Otherwise, the id is accepted and the books are displayed.Now the user searches for hi requirements from the books displayed.Then selecting the books is being done by the user.The selected books are displayed along with their amount.The user orders and pays for the books.The books are delivered to the user by the online bookshopping center through a travel agent.

Project scope

The e-book deals with purchase and payment of books.The online bookshopping center allows the user to search and purchase books available at that time.We can search the books in an efficient manner.

Project objectives

The aim of this e-book system is to search and purchase books for the user's requirements.The e-book system should provide all the facilities required.

Software requirement analysis

The e-book system involves the following modules:

- Login
- Searching the books
- Selecting the books
- Purchasing the books
- Delivering the books

Login

This module is used to accept the user name and password if it is valid.

Searching the books

This module is used to search the books according to the user's needs.

Selecting the books

This module is used to select a particular set of books from a large set.

Purchasing the books

This module is used to pay and purchase the books required.

Delivering the books

This module is used to deliver the books to the user through a travel agent.

SOFTWARE RESOURCE SPECIFICATION

1.0 INTRODUCTION

E-Book is the interface between the students and Librarian. It aims at improving the efficiency in the Issue of books or magazines and reduces the complexities involved in it to the maximum possible extent.

1.1 PURPOSE

If the entire process of 'Issue of Books or Magazines' is done in a manual manner then it would take several months for the books or magazines to reach the applicant. Considering the fact that the number of students for Book Bank is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process. The system has been carefully verified and validated in order to satisfy it.

1.2 SCOPE

The System provides an online interface to the user where they can fill in their personal details and submit the necessary documents (may be by scanning). The authority concerned with the issue of books can use this system to reduce his workload.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

- **E-book manager**
  Refers to the super user who is the Central Authority who has been vested with the privilege to manage the entire system.

- **User**
  One who wishes to obtain the Books or Magazines.
- **Visitor**
  One who visits to obtain Books or Magazines.
- **Administrator**
  One who manages and maintain Books or Magazines.

1.4 REFERENCES

IEEE Software Requirement Specification format

1.5 TECHNOLOGIES TO BE USED

- ✓ **HTML** - Markup Language used for creating web pages.
- ✓ **J2EE** - Java 2 Enterprise Edition is a programming platform and it is the part of the java platform for developing and running distributed java applications.
- ✓ **HTTP** -Hyper Text Transfer Protocol
- ✓ **TCP/IP** - Transmission Control Protocol/Internet Protocol is the communication protocol used to connect hosts on the Internet.

1.6 TOOLS TO BE USED

- ✓ Eclipse IDE (Integrated Development Environment
- ✓ Umbrello tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements.
Overall description will describe major role of the system components and inter-connections.
Specific requirements will describe roles & functions of the actors.

2.0 OVERALL DESCRIPTION

It will describe major role of the system components and inter-connections.

2.1 PRODUCT PERSPECTIVE

The ORS acts as an interface between the user and the 'e-book manager'. This system tries to make the interface as simple as possible and at the same time not risking the security of data stored in. This minimizes the time duration in which the user receives the books or magazines.

2.2 SOFTWARE INTERFACE

Front End Client

The Student and Librarian online interface is built using JSP and HTML. The Librarians local interface is built using Java.

Web Server

Apache Tomcat application server (Oracle Corporation).

Back End

Oracle 11g database

2.3 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

2.4 SYSTEM FUNCTIONS

Secure Registration of information by the Students.

Librarian can generate reports from the information and is the only authorized personnel to add the eligible application information to the database.

2.5 USER CHARACTERISTICS

User

They are the people who desire to obtain the books and submit the information to the database.

Visitor

They are the person who visits the E-book system

Administrator

He has the certain privileges to add the books and to approval of the reservation of books.

2.6 CONSTRAINTS

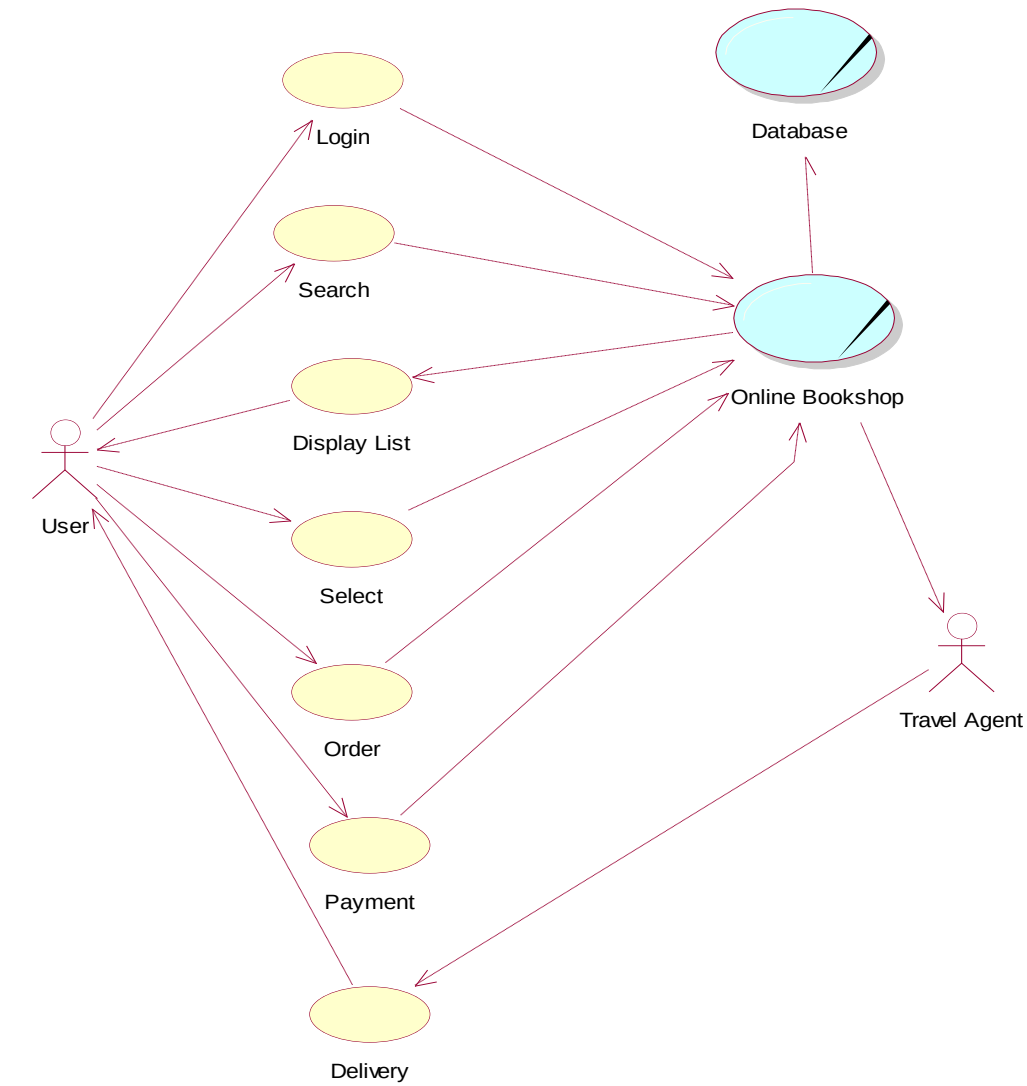
- ✓ The Students require a computer to submit their information.
- ✓ Although the security is given high importance, there is always a chance of intrusion in the web world which requires constant monitoring.
- ✓ The user has to be careful while submitting the information. Much care is required.

2.7 ASSUMPTIONS AND DEPENDENCIES

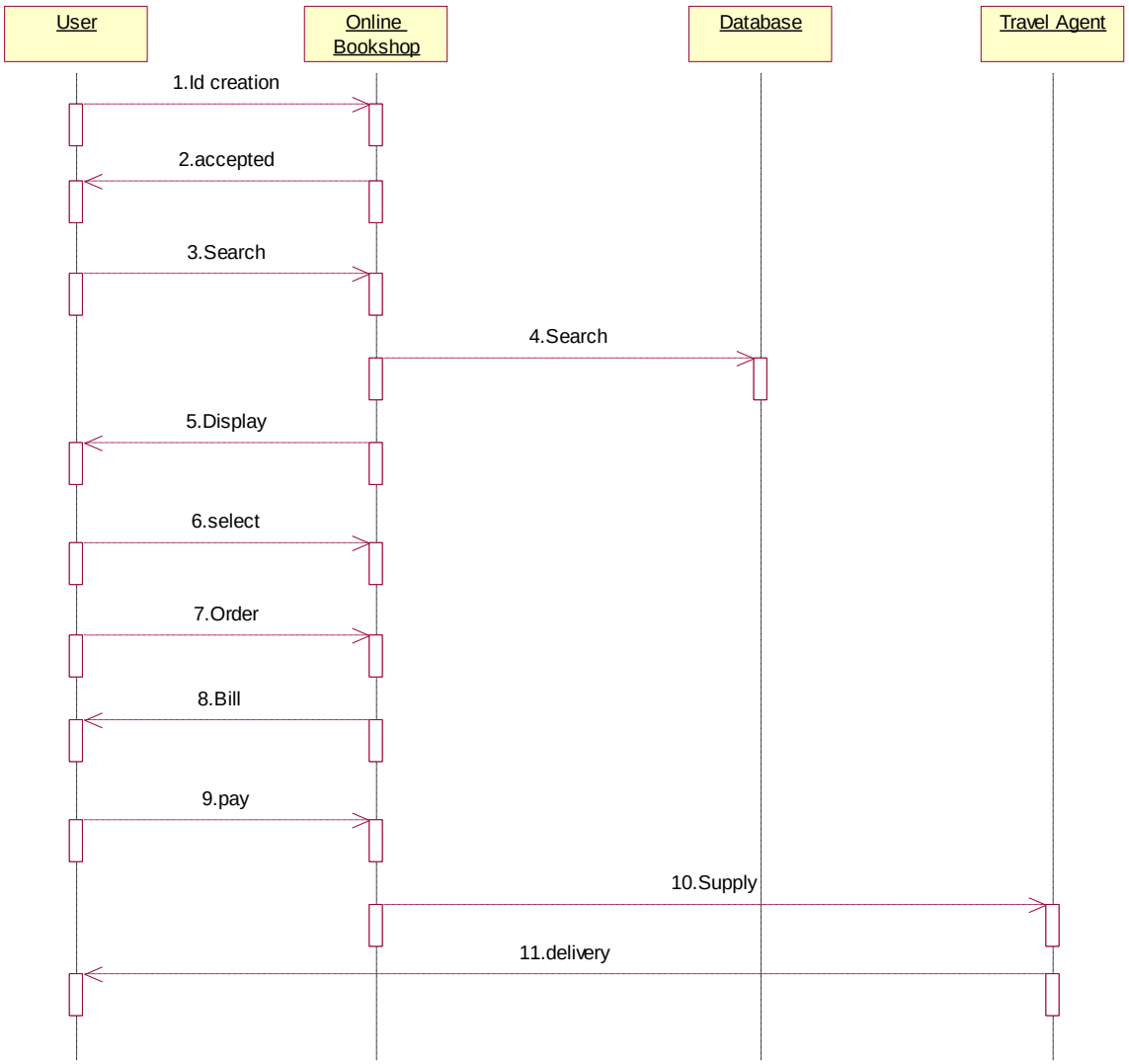
The user and e-book manager must have basic knowledge of computers and English Language. The user may be required to scan the documents and send.

UML Diagrams

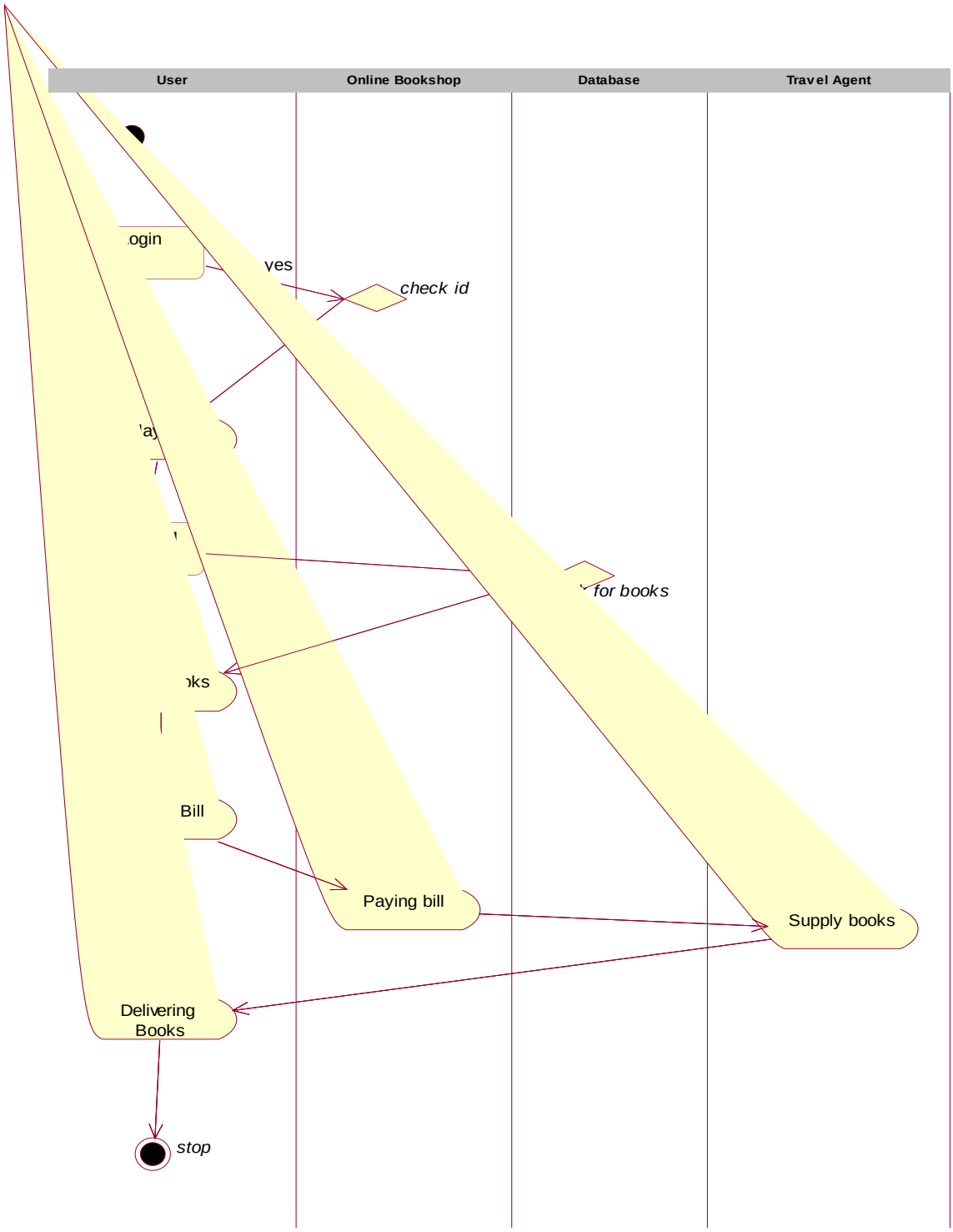
Usecase diagram



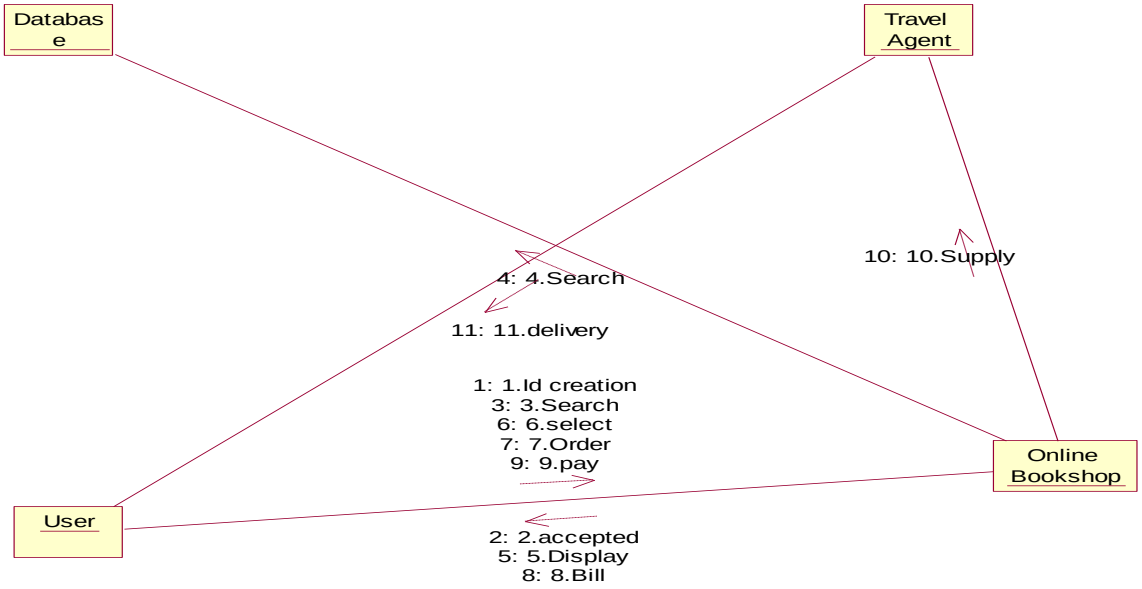
Sequence diagram



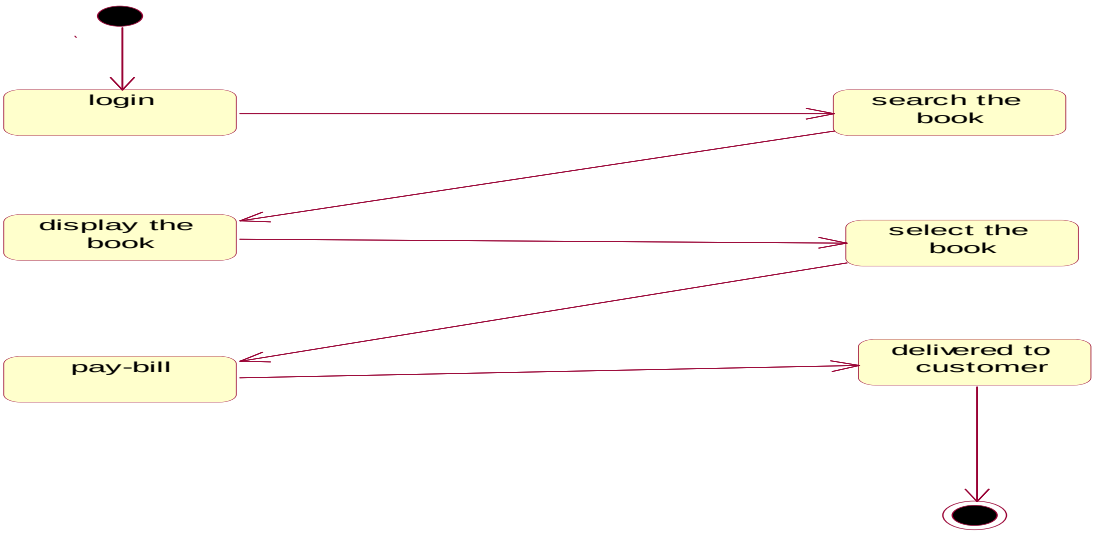
Activity diagram



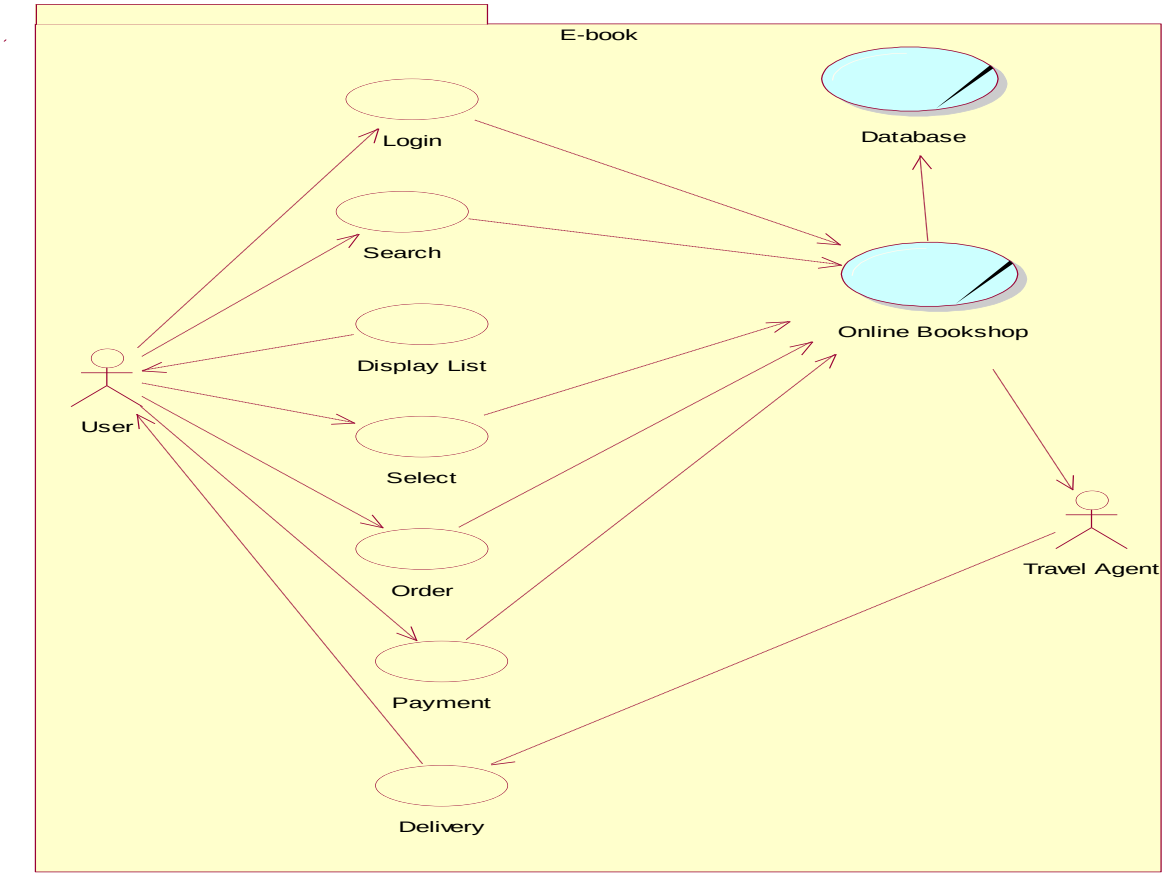
Collaboration diagram



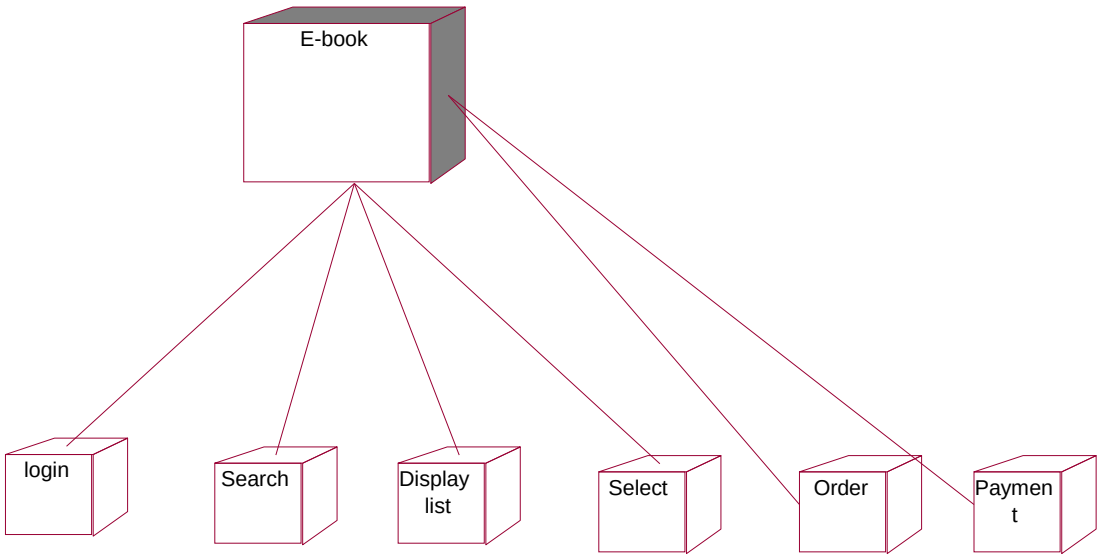
State diagram



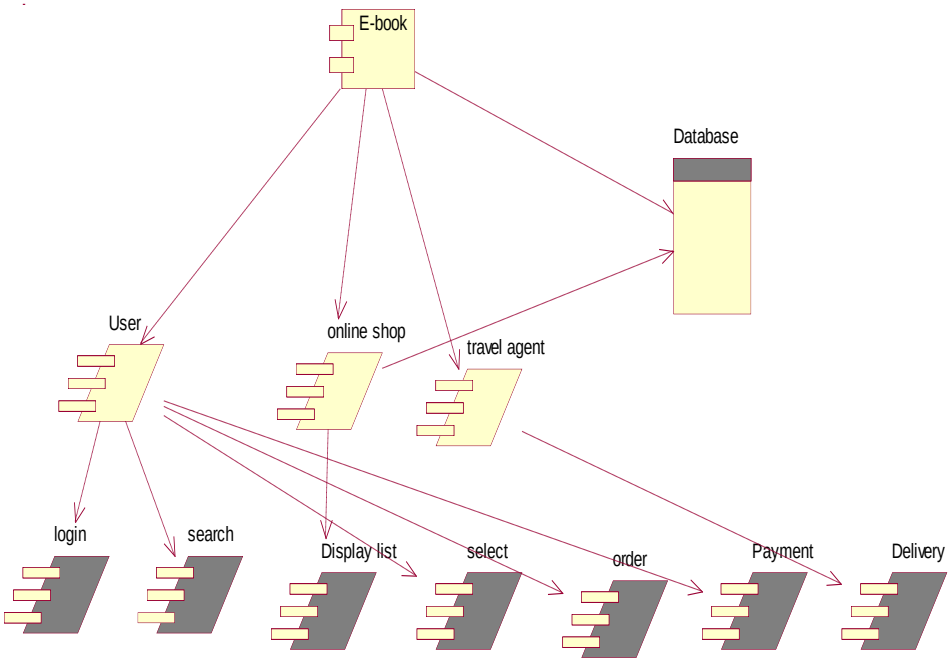
Package diagram



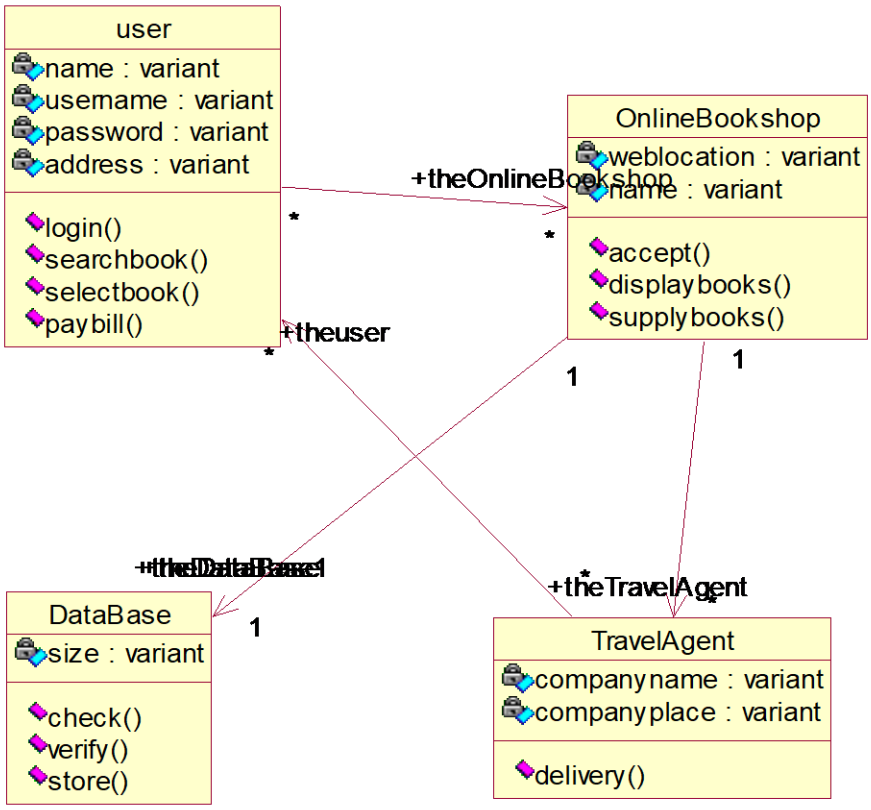
Deployment diagram



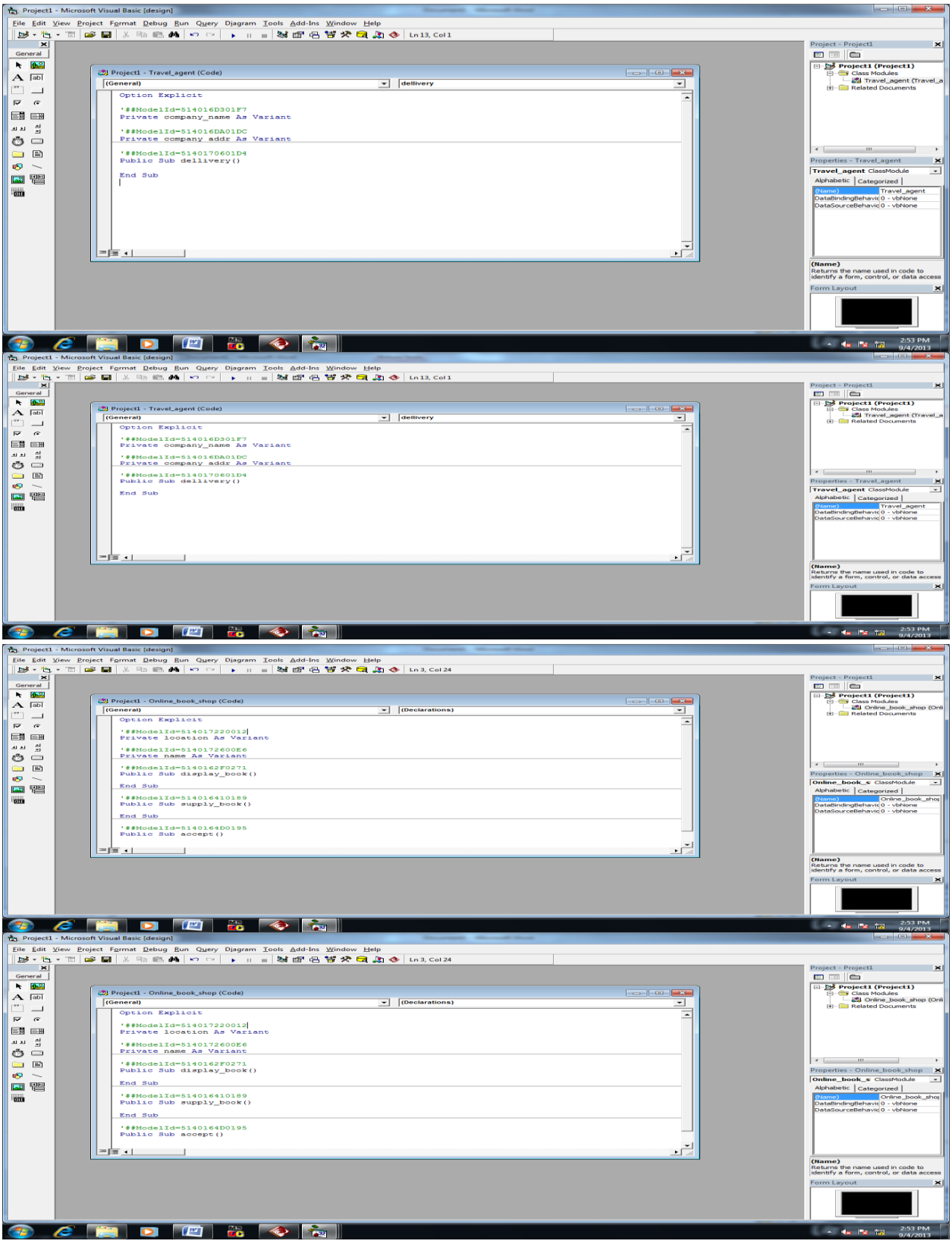
Component diagram



Class Diagram



Output



Result

Thus the E-Book Management system was designed and the code was generated.

EX.NO:11 RECRUITMENT SYSTEM

DATE:

Aim:

To analyze and design recruitment system using Umbrello Software.

Procedure:

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

Recruitment system includes the following functions submitting the resume, conducting aptitudes test, conducting group discussion display short list, conducting technical and personal interview, display selected candidates. The actors of this system include job seek, clerk, company team and HR Management. The use of the system include submit the resume, conduct aptitude test, group discussion etc. They will display a short list. Then the HR Management will conduct personal interview and they will display the selected candidates.

Project scope

The recruitment system deals with applying jobs through Online the company team will conduct aptitude test, group discussion etc. They will display a short list. Then the HR Management will conduct personal interview and they will display the selected candidates.

Project objectives

The objective of these websites is to serve as common meeting ground for jobseekers and employers, both logically and globally, where the candidates find their dream job and recruiter find the right candidate to full fill their needs.

Software requirement analysis

The recruitment system includes the following modules,

- Submit Resume
- Verify Resume
- Conduct Aptitude Test
- Conduct Group Discussion
- Display Short List
- Conduct Technical and Personal Interview

Submit Resume :

First the user will submit the resume to the clerk.

->Verify Resume :

The clerk will verify the user's resumed and store it in his database.

->conduct aptitude test:

This company team will conduct aptitude test to the user.

->conduct group discussion:

This module is used to conduct group discussion to the user by the company team.

->display shortlist:

This module is used to display the selected in the test conducted by the company team.

->conduct technical and personal interview:

This module is used to conduct final interview to the user by the HR management.

SOFTWARE REQUIREMENT SPECIFICATION

1.0 INTRODUCTION

Recruitment System is an interface between the Applicant and the Organization responsible for the Recruitment. It aims at improving the efficiency in the Recruitment process and reduces the complexities involved in it to the maximum possible extent.

1.1 PURPOSE

If the entire process of 'Recruitment' is done in a manual manner then it would takes several days for the recruitment. Considering the fact that the number of applicants for recruitment is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process.

1.2 SCOPE

- The System provides an online interface to the user where they can fill in their personal details and apply for the job.
- The Organization (HR-Department) concerned with the recruitment process can make use of this system to reduce their workload and process the application in a speedy manner.
- Provide a communication platform between the Applicant and the Organization.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

- **Organization**
  Refers to the super user who is the Central Authority with the privilege to manage the entire system. It can be any higher official in the HR department.
- **Applicant**
  One who wishes to apply for the job.
- **RS**
  Refers to this Recruitment System.
- **HTML**
  Hypertext Markup Language used for creating web pages.

- **J2EE**

Java 2 Enterprise Edition is a programming platform java platform for developing and running distributed java applications.

- **HTTP**

Hyper Text Transfer Protocol.

- **TCP/IP**

Transmission Control Protocol/Internet Protocol is the communication protocol used to connect hosts on the Internet.

1.4 REFERENCES

IEEE Software Requirement Specification format.

1.5 TECHNOLOGIES TO BE USED

• HTML
• JSP
• Javascript
• Java

1.6 TOOLS TO BE USED

• Eclipse IDE (Integrated Development Environment)
• Umbrello tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements
Overall Description will describe major role of the system components and inter-connections.
Specific Requirements will describe roles & functions of the actors.

2.0 OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

The RS acts as an interface between the "Applicant and the 'Organization'. This system tries to make the interface as simple as possible and at the same time not risking the security of data stored in. This minimizes the time duration for recruitment process.

2.2 SOFTWARE INTERFACE

• **Front End Client** –The Applicants and Organization online interface is built using JSP and HTML. The Administrators' local interface is built using Java.
• **Web Server** - Glassfish application server (SQL Corporation).
• **Back End** - SQL database.

2.3 HARDWARE INTERFACE

The server is directly connected to the client systems. The client systems have access to the database in the server.

2.4 SYSTEM FUNCTIONS

- The applicant views the jobs through Advertisement.
- Applicants apply for the job.
- Test and Interview are conducted.
- Recruited Applicants are informed.
- HR Manager can generate reports from the information and he/she is the only authorized personnel to add the eligible application information to the database.

2.5 USER CHARACTERISTICS

- **Applicant**

  These are the persons who desire to apply for the job.

- **Organization**

  These are the person with certain privileges to announce recruitment depending upon the organization need. He/ She may contain a group of persons under him/her to publish advertisement and give suggestion whether or not to approve the recruitment.

- **HR**

  He/ She is the person who upon receiving intimation from the RS, perform a personal verification of the applicants and see if he/she has eligibility for the advertised job through a process of Test and Interview.

2.6 CONSTRAINTS

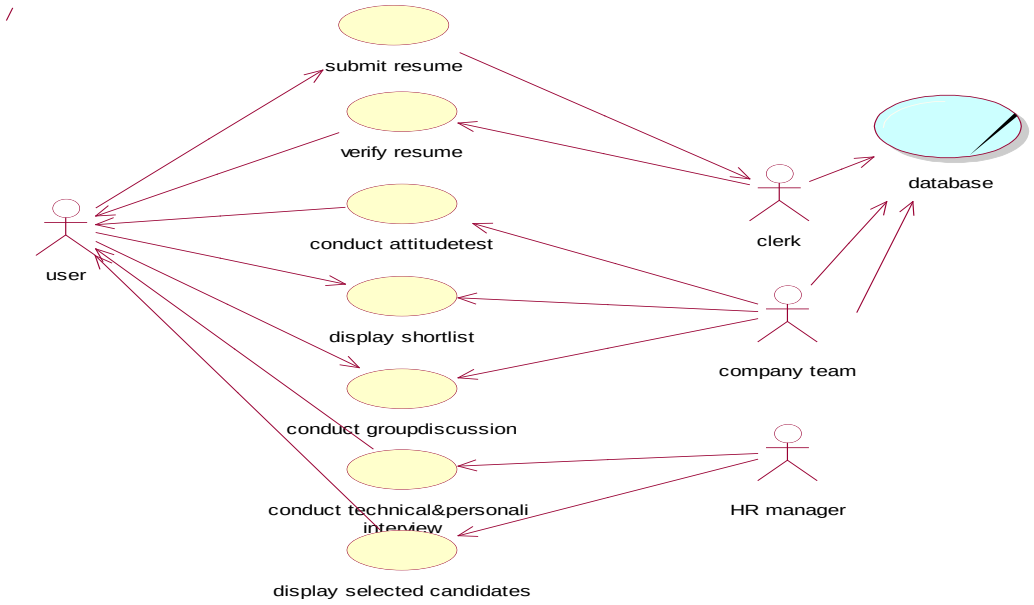
The Applicants require a computer to submit their information.

2.7 ASSUMPTIONS AND DEPENDENCIES

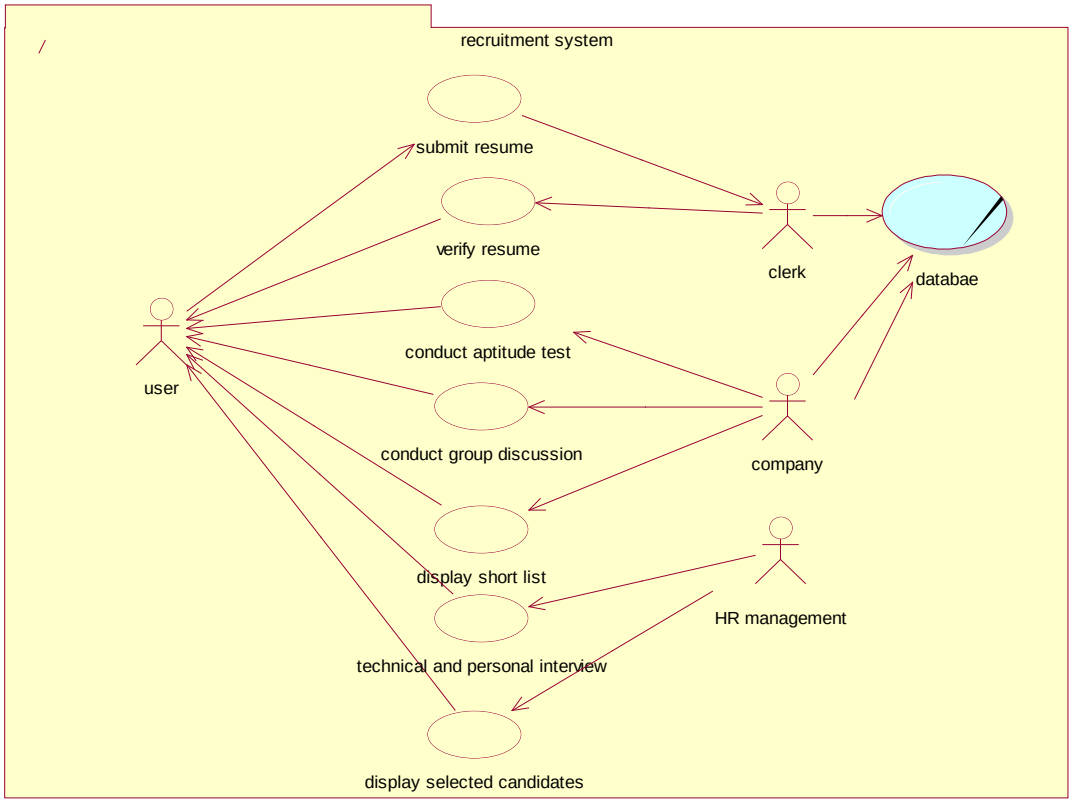
The Applicants and HR must have basic knowledge of computers and English Language.

UML Diagrams:

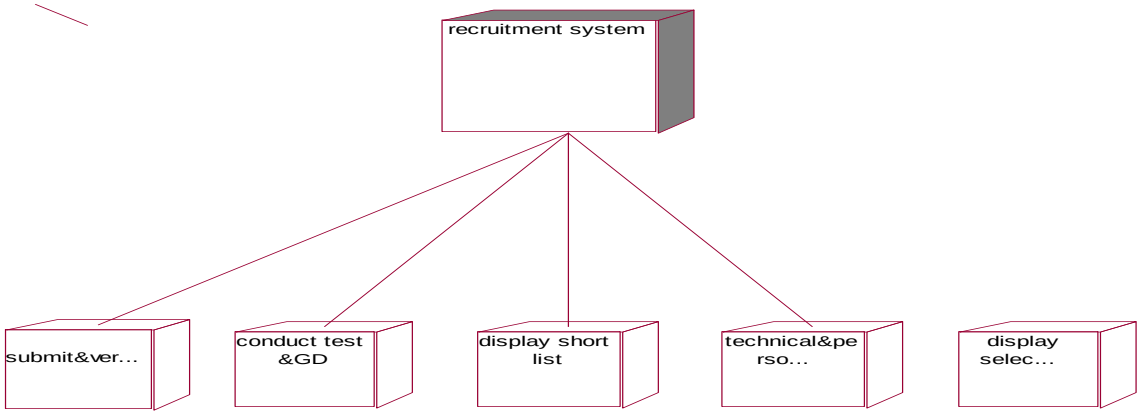
Use case diagram:



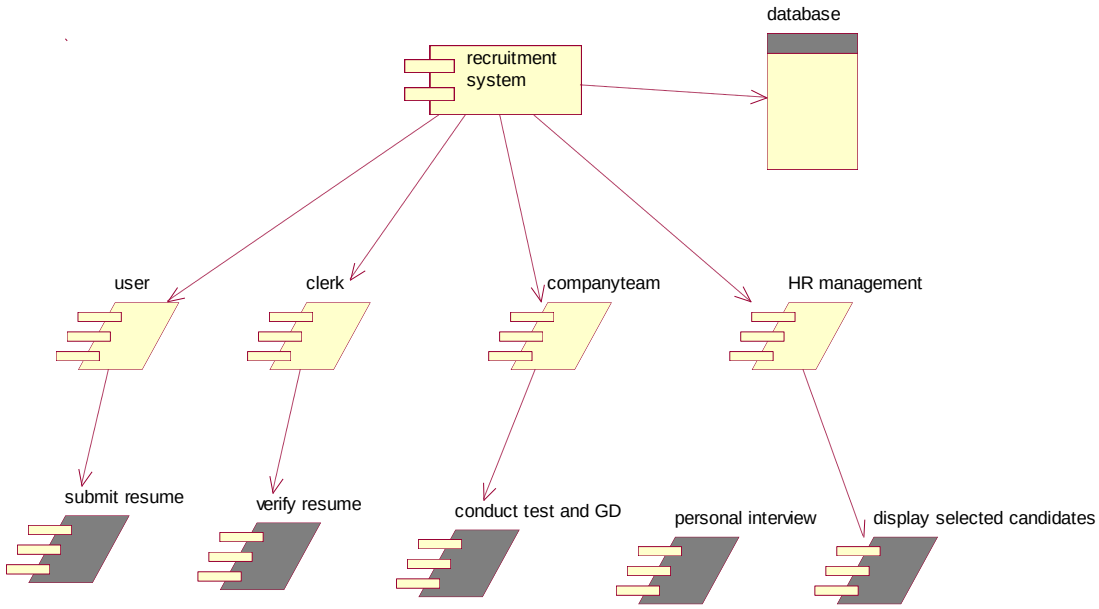
Packagediagram



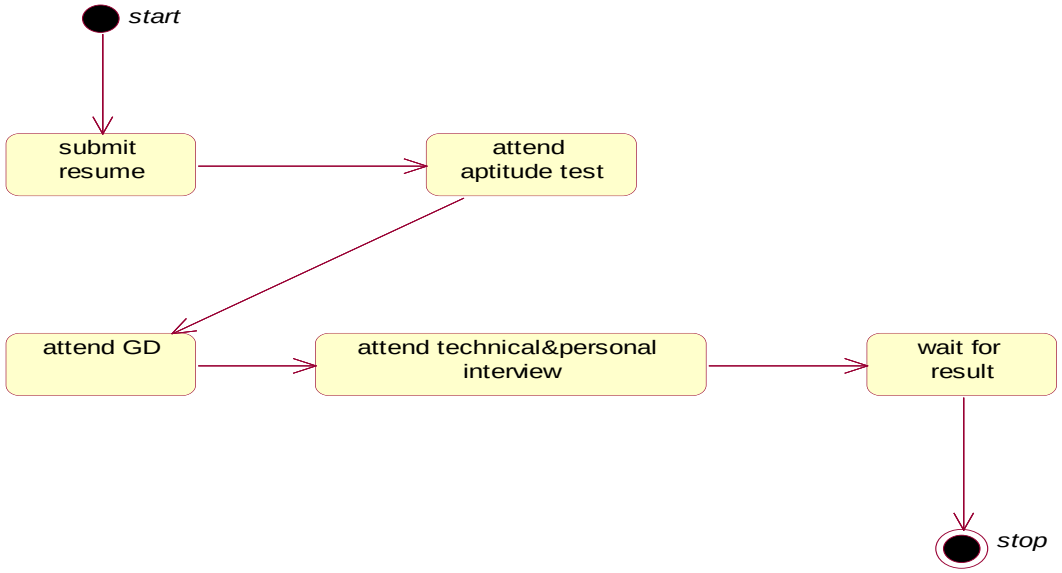
Deployment diagram:



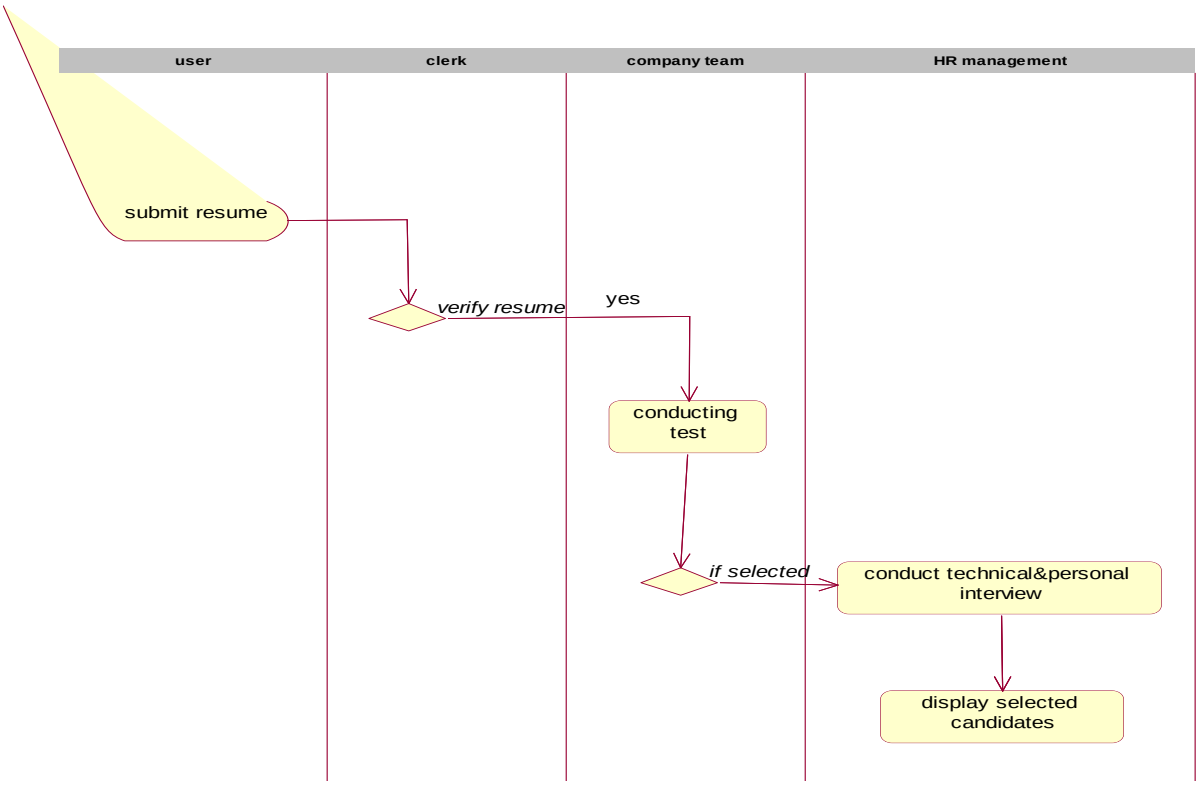
Component diagram



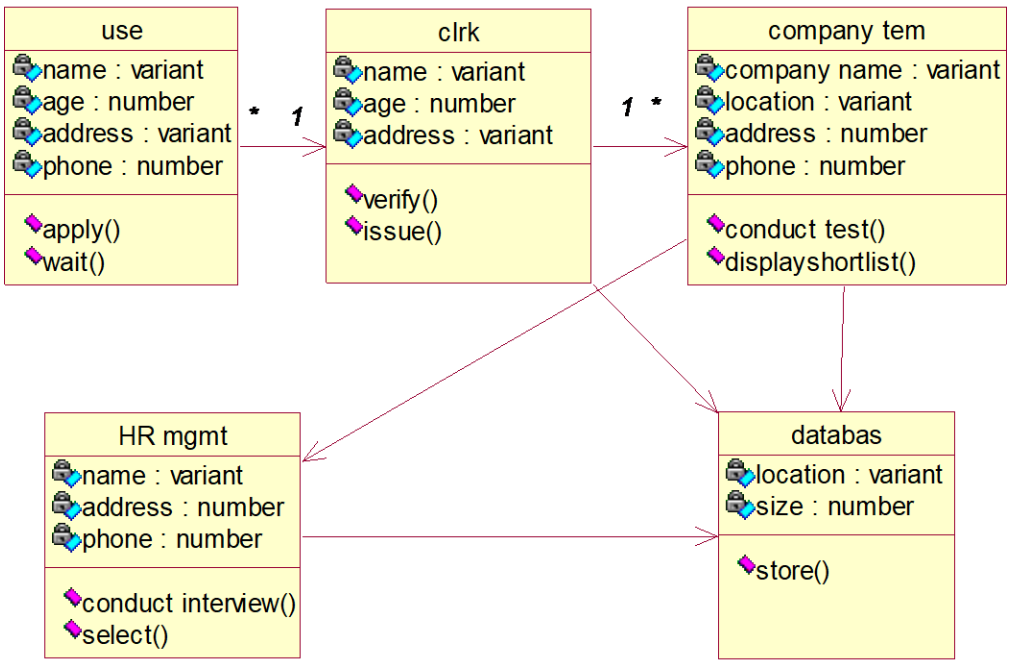
State Diagram:



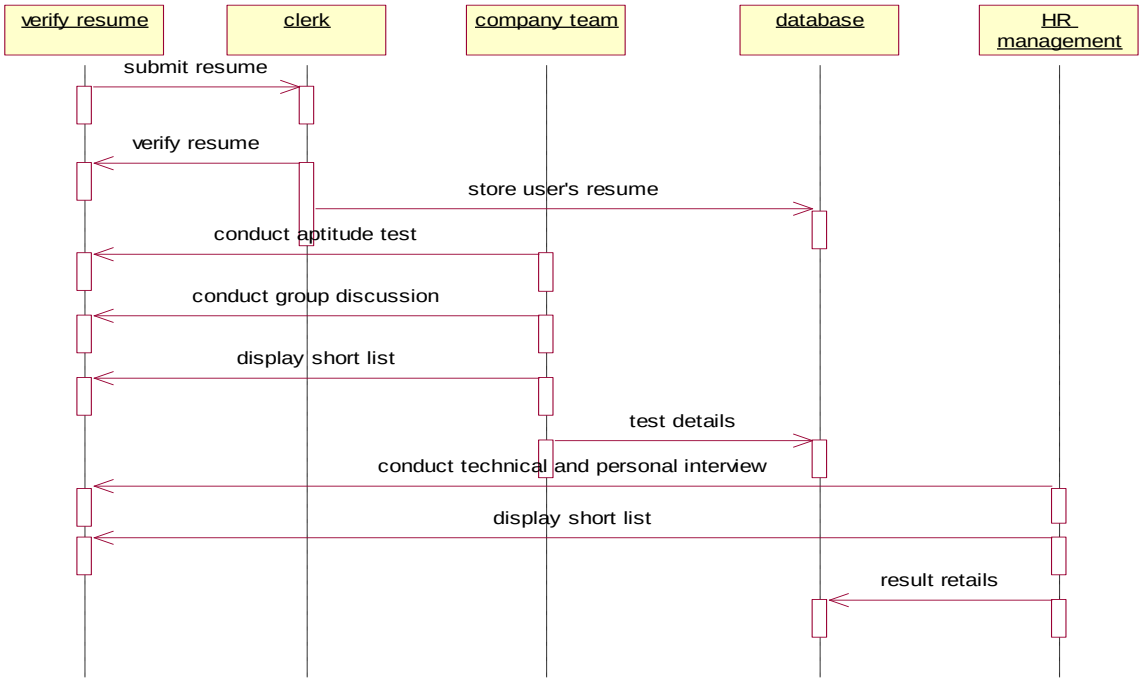
Activity Diagram:



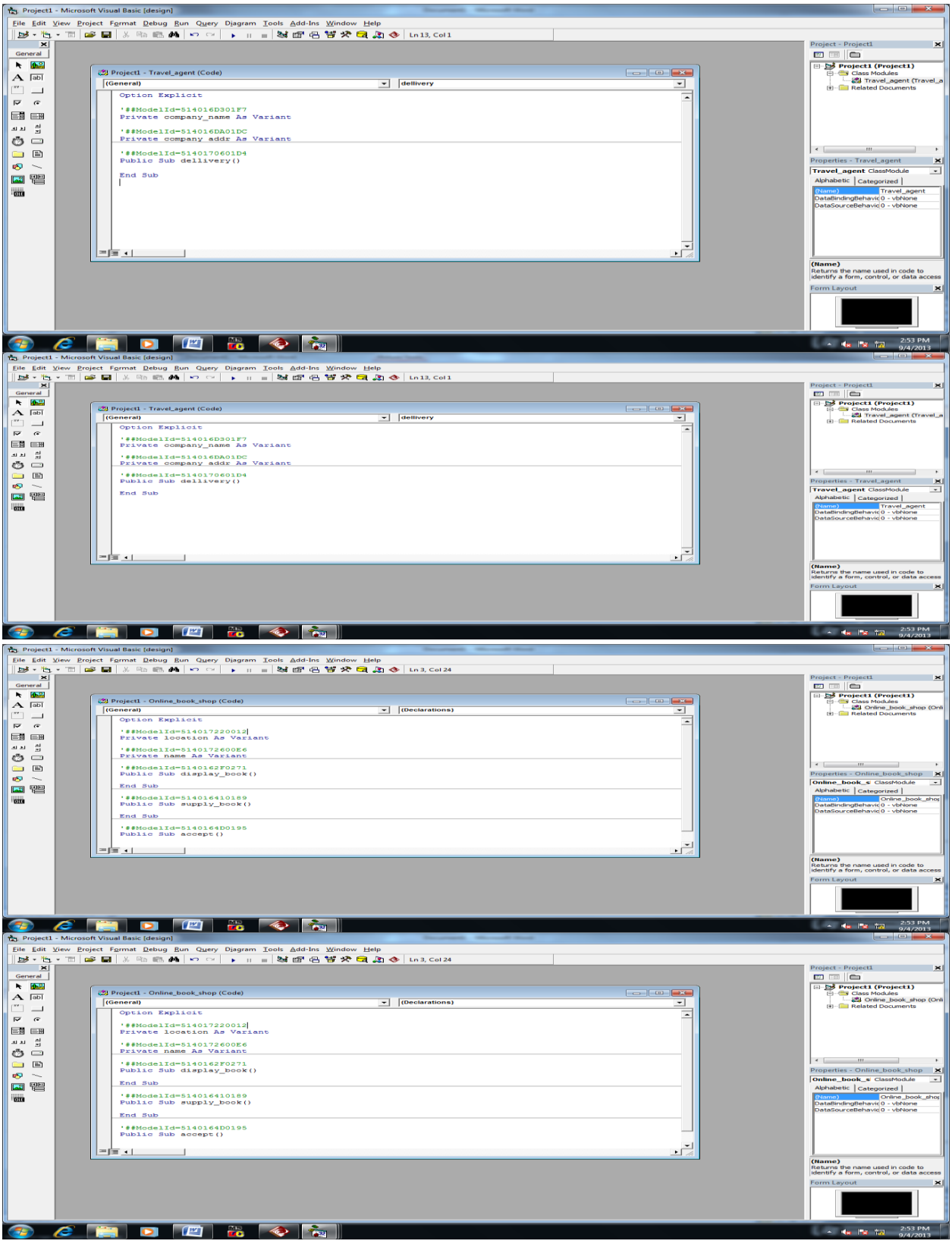
Class Diagram:



Sequence Diagram:



Output:



Result:

Thus the design for recruitment system and the code has been generated successfully.

EX.NO:12 FOREIGN TRADING SYSTEM

DATE:

Aim:

To analyze and design online foreign trading system using Umbrello Software.

Procedure:

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

The foreign trading system is used for import and export the product in one country to another one country. First both importer and exporter agreement with the company. After that the importer sends the order for the material.

The importer sends the money for the particular product then the exporter release the product. After that the importer sale the retailer to the retailer and the retailer sale the product and get the feedback about the product.

Project scope

The main scope of this project is to import and the materials from one country to another one country.

Project Objectives

The aim of the foreign trading system is to import and export products one country to another.

Software Requirement Analysis (Modules)

The modules of the foreign trading system are:

- Agreement
- Order
- Send the money
- Export the material
- Sell the material
- Send the feedback

Agreement

Both importer and the exporter agreement with each other

Order

After the agreement the importer orders for the product

Send the money

The importer sends money to the exporter which depends upon the products

Export the material

After getting the money exporter exports the material**.**

Sell the material

Getting a product the importer sells the product to retailer.

Feedback

The customer purchases the product and use it after that the customer tells some feedback about the product.

Send the Feedback

The retailer gets the feedback from the customer after that send the feedback to the importer.

Get the Feedback

The importer sends the feedback to exporter. After that the exporter modifying the product depends upon the customer need (or) feedback**.**

SOFTWARE REQUIREMENTS SPECIFICATION

1.0 INTRODUCTION

Foreign trading system is the interface between the exporter and buyer. It aims at improving the efficiency in the production, export process and reduce the complexities involved in it to the maximum possible extent.

1.1 PURPOSE

Considering the fact that the number of buyer is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in this process. The system has been carefully verified and validated in order to satisfy it.

1.2 SCOPE

The System provides an online interface to the buyer where they can fill in their personal details and submit the necessary documents (may be by scanning). The authority concerned with the production and shipment of goods can use this system to reduce his workload and process the application in a speedy manner.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

- **Exporter** – One who wishes to export the goods
- **Importer** - One who wishes to obtain the goods
- **Regional authorities –** Who provide service authority to individual and business firms intending to export and/or import goods
- **IEC Number –** Importer-Exporter Number
- **QC –** Quality Control in order to ensure the quality of products

1.4 REFERENCES

IEEE Software Requirement Specification format.

1.5 TECHNOLOGIES TO BE USED

- ✓ HTML
- ✓ JSP
- ✓ Java script
- ✓ Java

1.6 TOOLS TO BE USED

- Eclipse IDE (Integrated Development Environment)
- Rational Rose tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements.
Overall description will describe major role of the system components and inter-connections.
Specific requirements will describe roles & functions of the actors.

2.0 OVERALL DESCRIPTION:

It will describe major role of the system components and inter-connections.

2.1 PRODUCT PERSPECTIVE

This system tries to make the interface as simple as possible and at the same time not risking the security of data stored in. This minimizes the time.

2.2 SOFTWARE INTERFACE

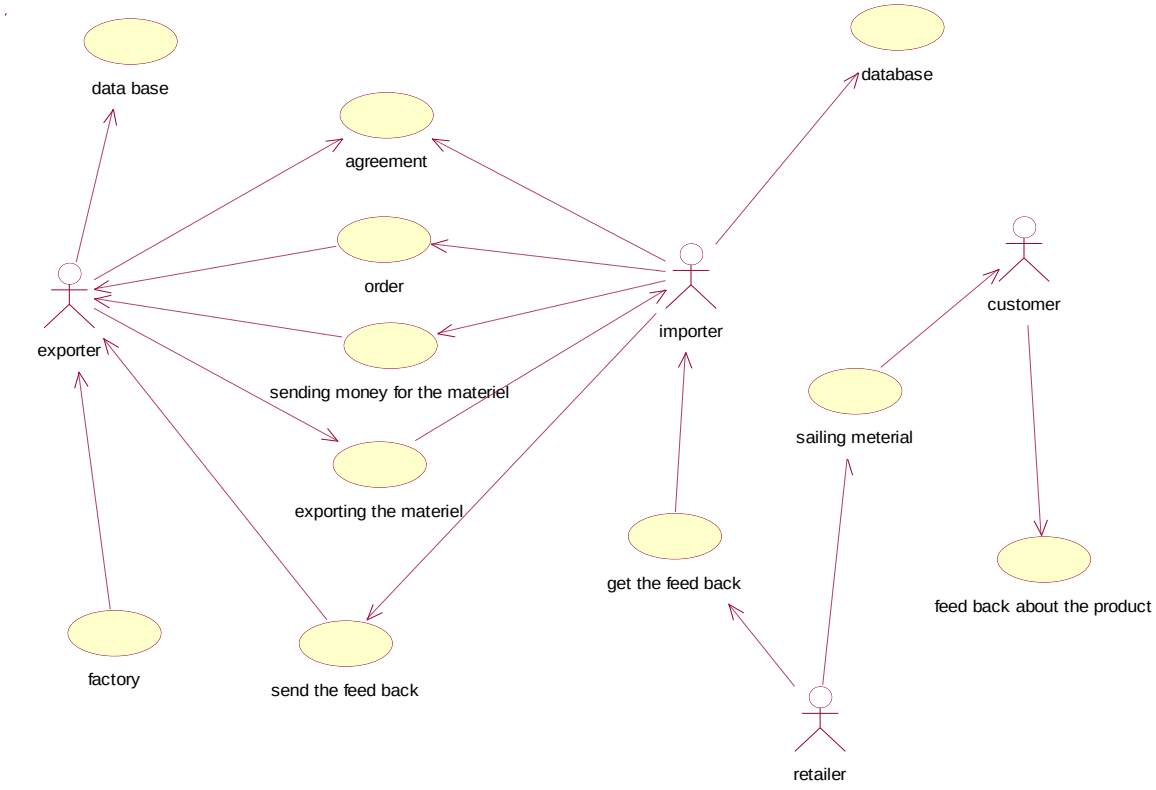
- **Front End Client -** The exporter online interface is built using JSP and HTML.
- **Web Server** – Apache Tomcat Server (Oracle Corporation)
- **Back End -** Oracle 11g database

2.3 HARDWARE INTERFACE

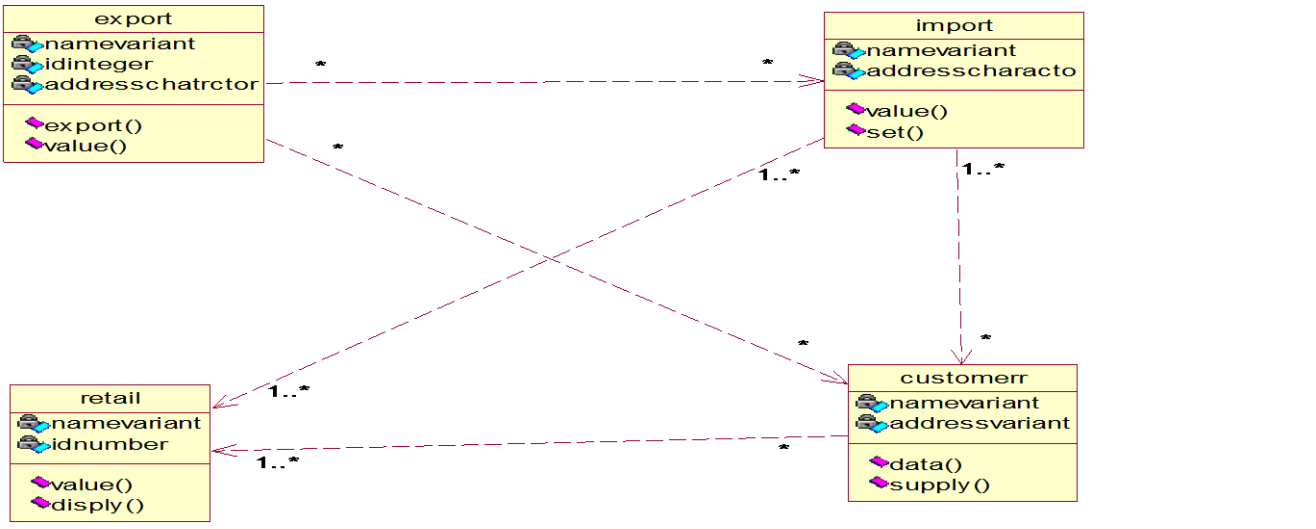
The server is directly connected to the client systems. The client systems have access to the database in the server.

UML Diagrams:

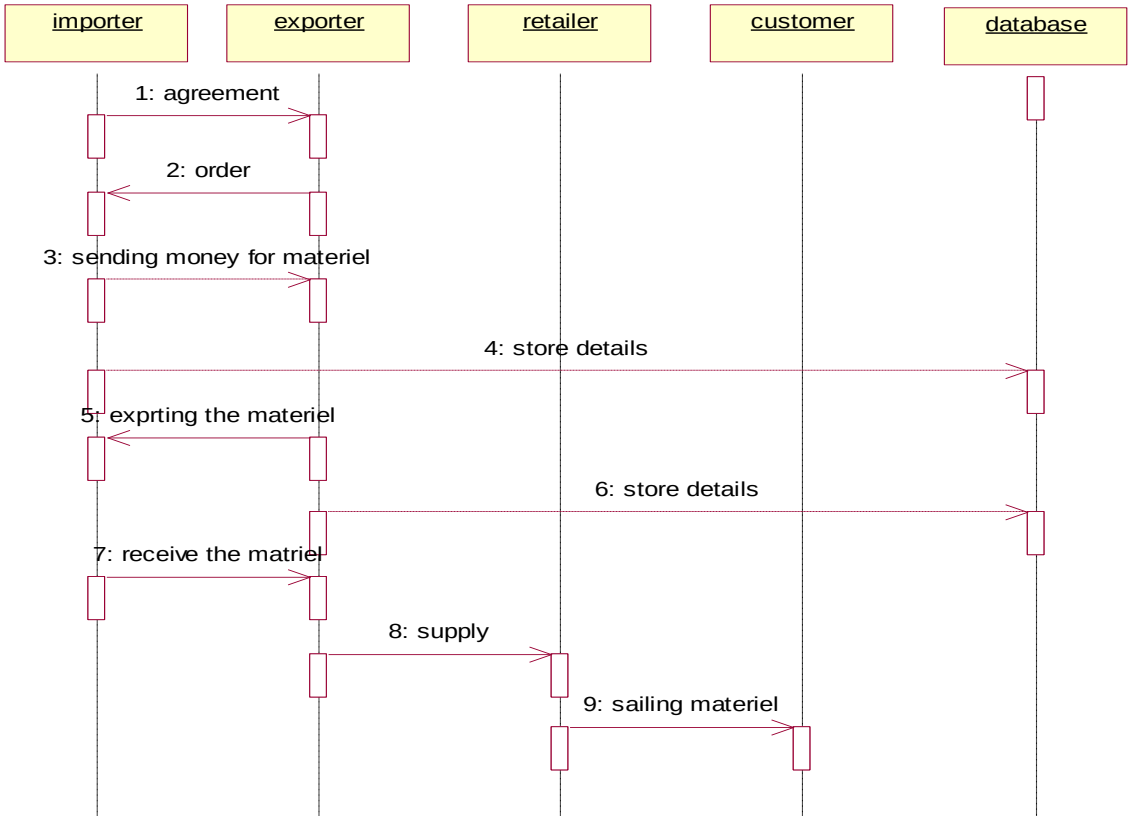
Use case Diagram:



Class diagram



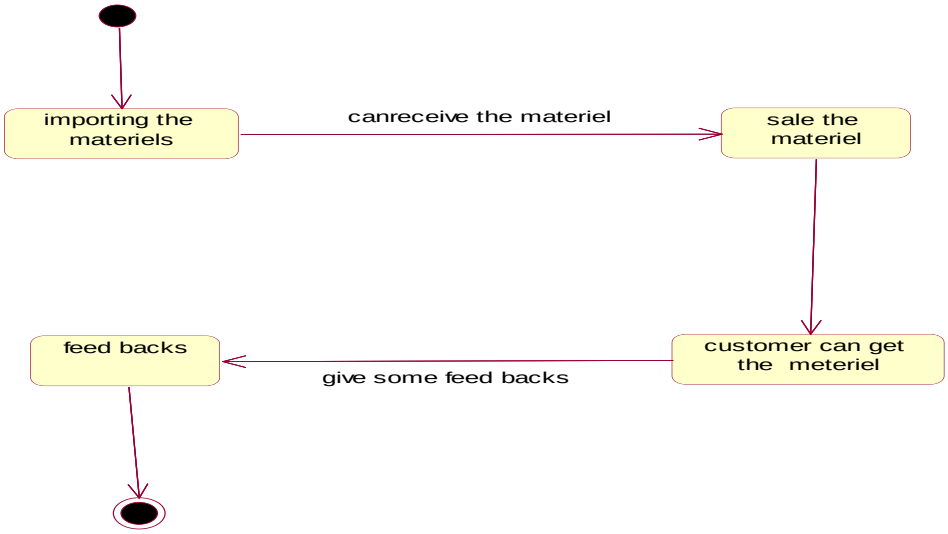
Sequence diagram



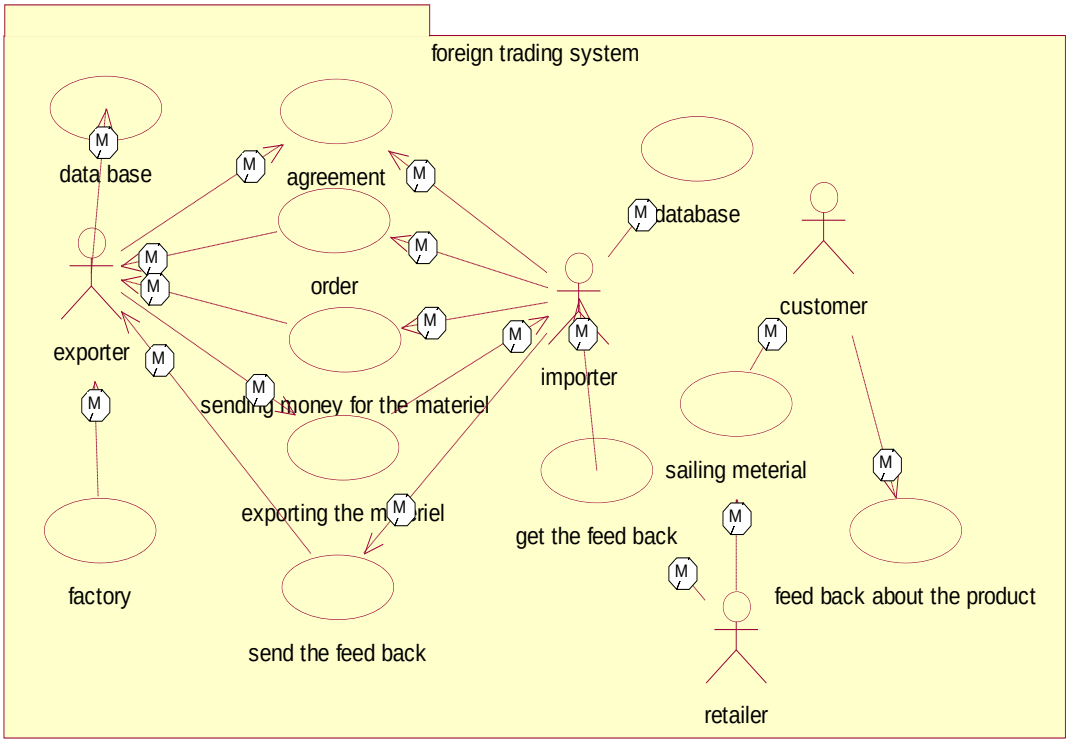
Collaboration Diagram



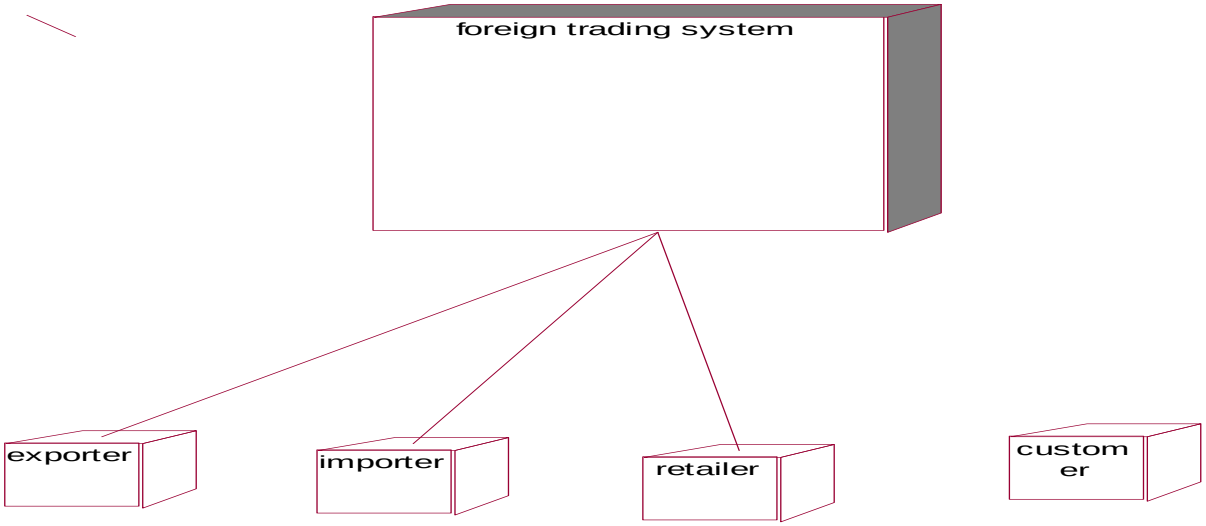
State Diagram



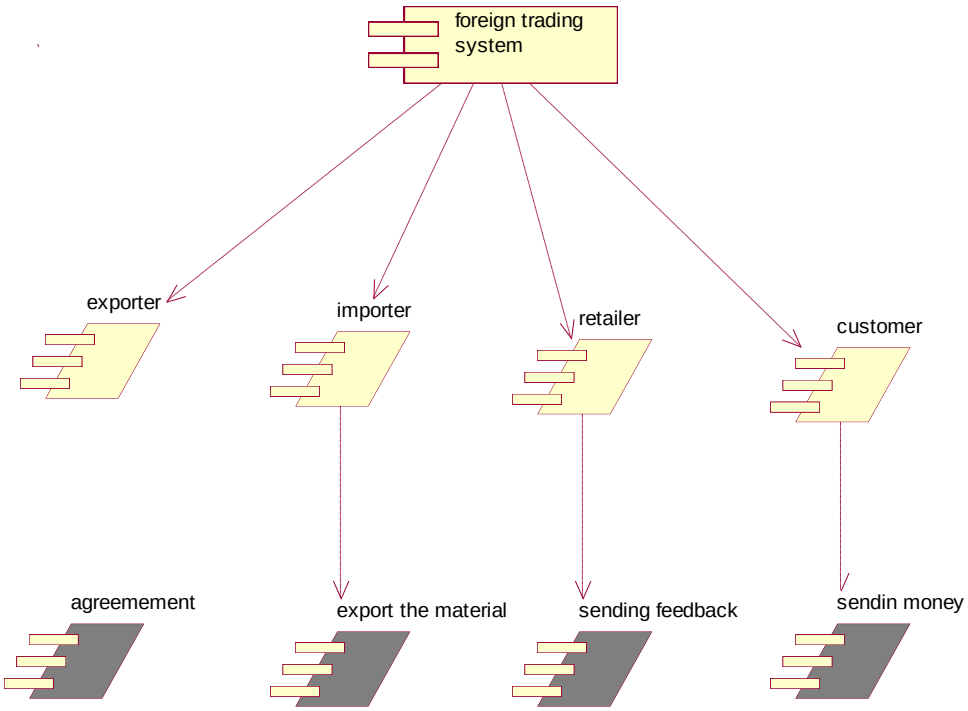
Package Diagram:



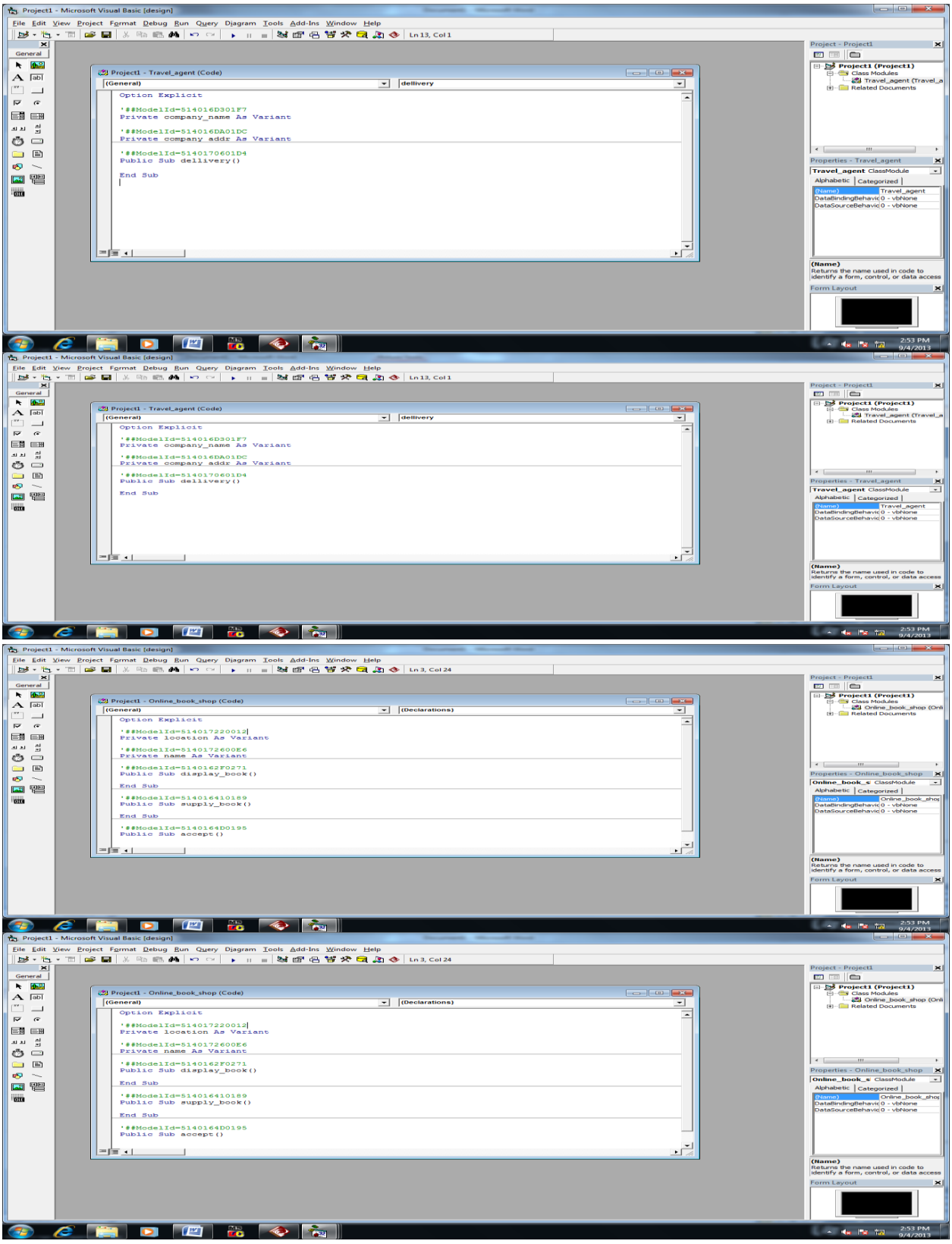
Deployment Diagram



Component Diagram:



Output:



Result:

Thus the Foreign Trading system was designed and the code was generated.

EX.NO:13 CONFERENCE MANAGEMENT SYSTEM

DATE:

Aim:

To analyze and design conference management system using Umbrello Software.

Procedure:

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

The process of the candidates is to login the conference system and submit the paper through online. Then the reviewer reviews the paper and sends the acknowledgement to the candidate either paper selected or rejected. This process of on conference management system are described sequentially through following steps,

• The candidate login to the conference management system.

• The paper title is submitted.

• The paper is been reviewed by the reviewer.

• The reviewer sends acknowledgement to the candidate.

• Based on the selection, the best candidate is selected.

• Finally the candidate registers all details.

Project scope

The scope of this conference management process is to select the best candidate from the list of candidates based on their performance in the process.

Project objectives

The aim of the conference management system is that the system can easily review the process. The main process in this document is the submission of paper by the candidate, reviewing process by the reviewer and sending of acknowledgement to the candidates whose paper is selected

.**Software requirement analysis**

CANDIDATE

The candidate can login and submit the paper to the reviewer. After getting acknowledgement the candidate will submit the revised and camera ready paper then registration process will be carried out.

REVIEWER

Reviewer will reviews the paper and sending acknowledgement to the candidate

DATABASE

Database is used to verify login and store the details of selected candidates.

SOFTWARE REQUIREMENT SPECIFICATION

1.0 INTRODUCTION

This software specification document consist full set of features and function for online conference management system. In this we give specification about the system requirements that are apart from the functionality of the system to perform the candidate paper valuation. It tells the usability, reliability defined in use case specification.

1.1 PURPOSE

The purpose of the conference management system is that the system can easily review the process. The main process in this document is the submission of paper by the candidate, reviewing process by the reviewer and sending of acknowledgement to the candidates whose paper is selected.

1.2 SCOPE

The scope of this conference management process is to select the best candidate from the list of candidates based on their performance in the process.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

- **CANDIDATE** - The candidate can login and submit the paper to the reviewer. After getting acknowledgement the candidate will submit the revised and camera ready paper then registration process will be carried out.
- **REVIEWER** - Reviewer will reviews the paper and sending acknowledgement to the candidate
- **DATABASE** - Database is used to verify login and store the details of selected candidates.
- **HTML** - Markup Language used for creating web pages.
- **J2EE** – Java 2 Enterprise Edition is a programming platform java platform for developing and running distributed java applications.
- **HTTP** - Hyper Text Transfer Protocol.
- **TCP/IP** – Transmission Control Protocol/Internet Protocol is the communication protocol used to connect hosts on the Internet.

1.4 REFERENCES

IEEE Software Requirement Specification format.

1.5 TECHNOLOGIES TO BE USED

• HTML
• JSP
• Java

1.6 TOOLS TO BE USED

• Eclipse IDE (Integrated Development Environment)
• Umbrello tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements –

Overall Description will describe major role of the system components and inter-connections.
Specific Requirements will describe roles & functions of the actors.

2.0 OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

The process of the candidates is to login the conference system and submit the paper through online. Then the reviewer reviews the paper and sends the acknowledgement to the candidate either paper selected or rejected.

2.2 SOFTWARE INTERFACE

- ✓ **Front End Client** - The exporter online interface is built using JSP and HTML.
- ✓ **Web Server** – Apache Tomcat Server (Oracle Corporation)
- ✓ **Back End** - Oracle 11g database

2.3 HARDWARE INTERFACE

The BPO system's server is directly connected to the client systems via ftp. The client systems have access to the database in the server.

2.4 SYSTEM FUNCTIONS

This process of on conference management system are described sequentially through following steps,
• The candidate login to the conference management system.
• The paper title is submitted.
• The paper is been reviewed by the reviewer.
• The reviewer sends acknowledgement to the candidate.
• Based on the selection, the best candidate is selected.
• Finally the candidate registers all details.

2.5 USER CHARACTERISTICS

- **Candidate** - Logins the conference system and submits the paper then do the registration process.
- **Reviewer** – Review the paper, select best candidate and send acknowledgement to them.

2.6 CONSTRAINTS

- Although the security is given high importance, there is always a chance of intrusion in the web world which requires constant monitoring.

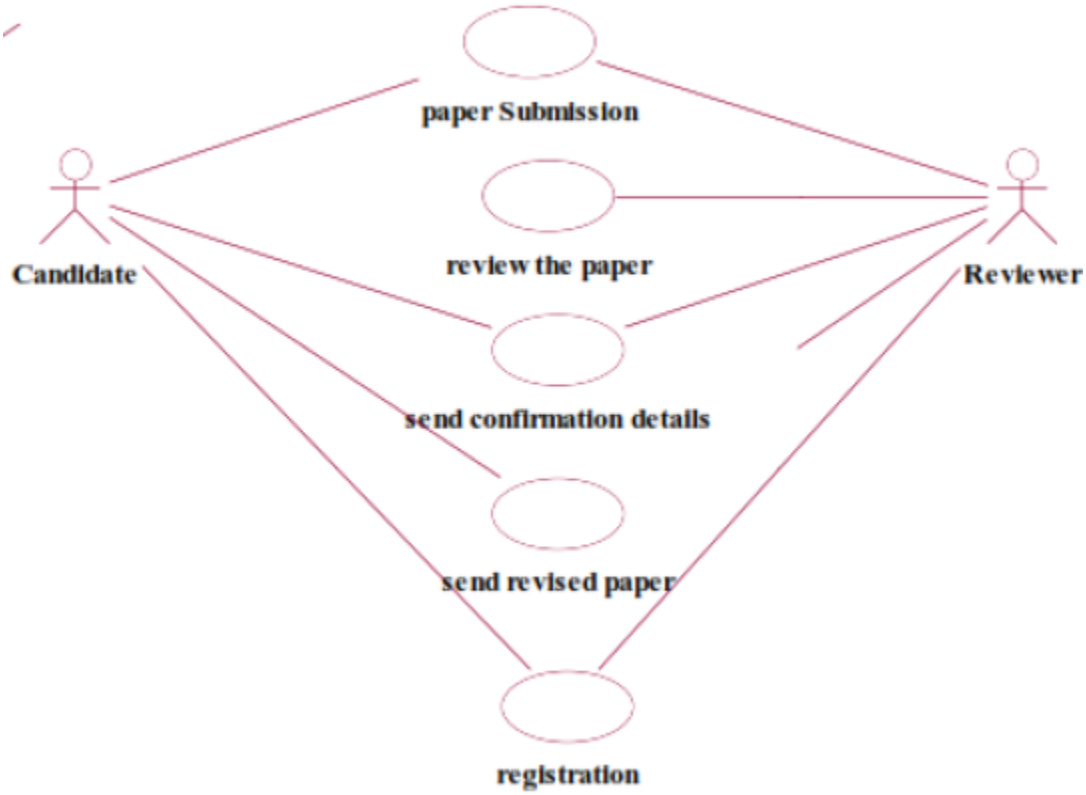
- The user has to be careful while submitting the information. Much care is required.

2.7 ASSUMPTIONS AND DEPENDENCIES

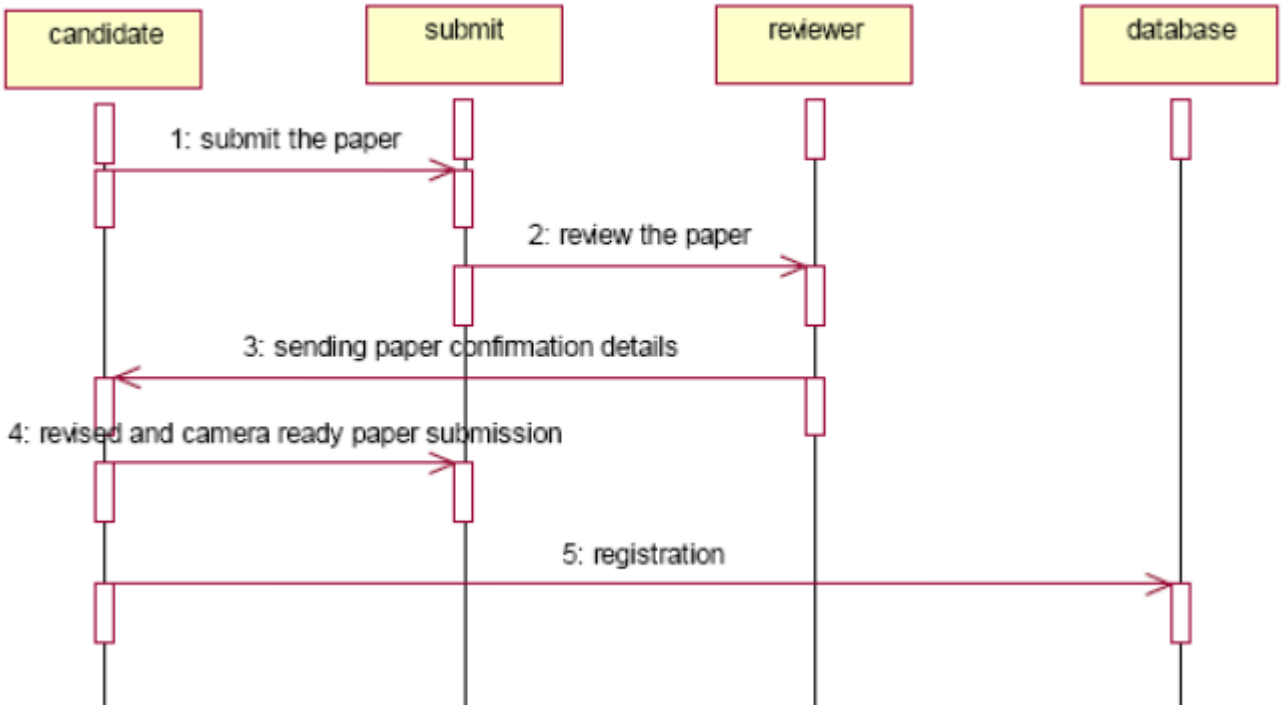
- The candidate and reviewer must have basic knowledge of computers and English Language.
- Provide privacy and security for the documents and candidate information

UML diagrams:

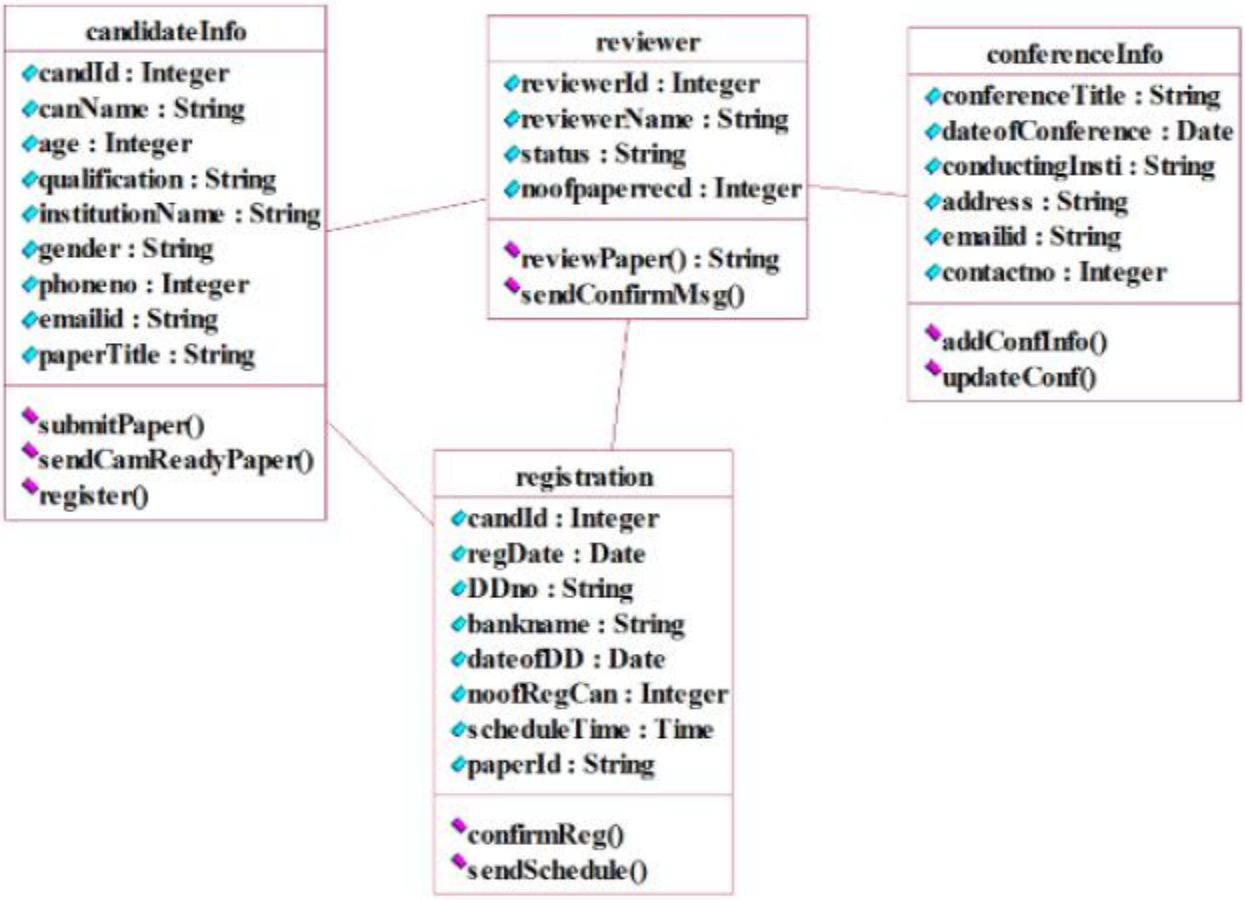
Usecase diagram:



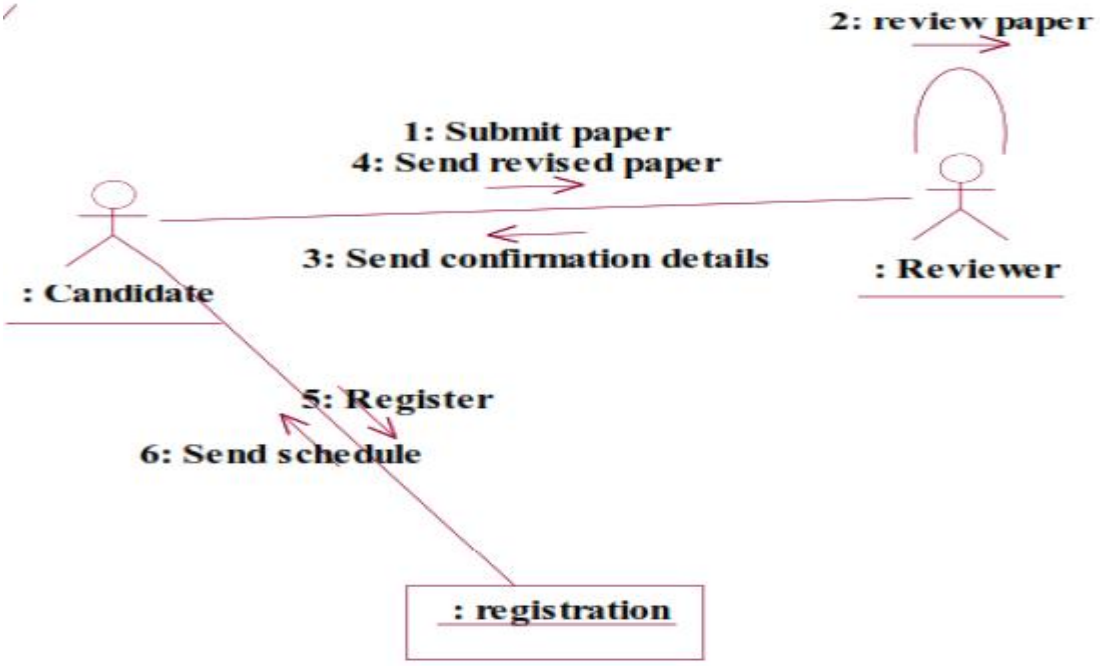
Sequence diagram:



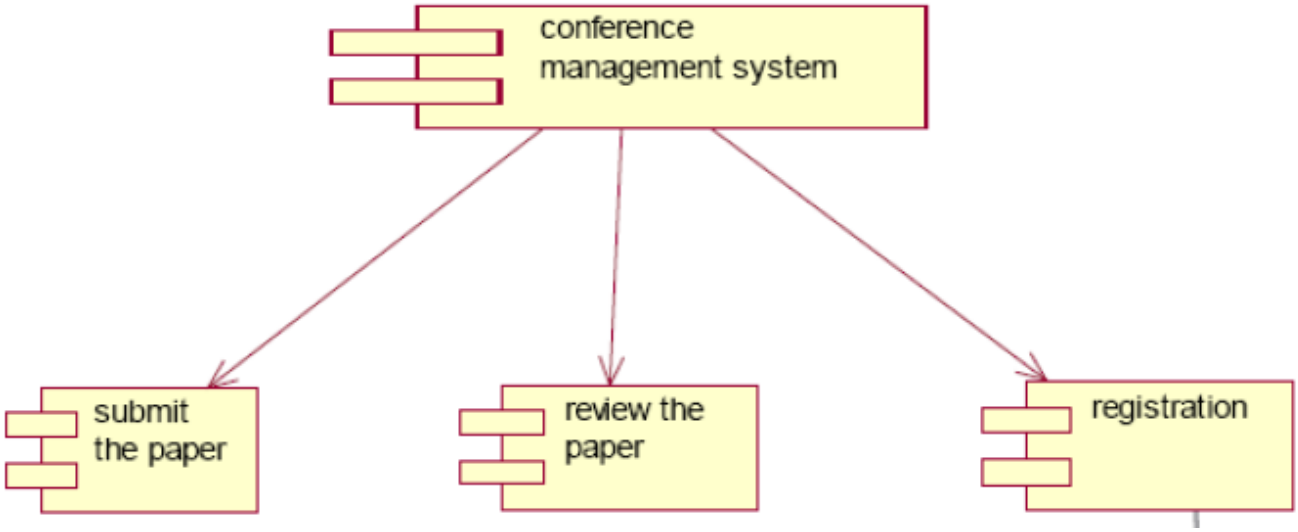
Class diagram:



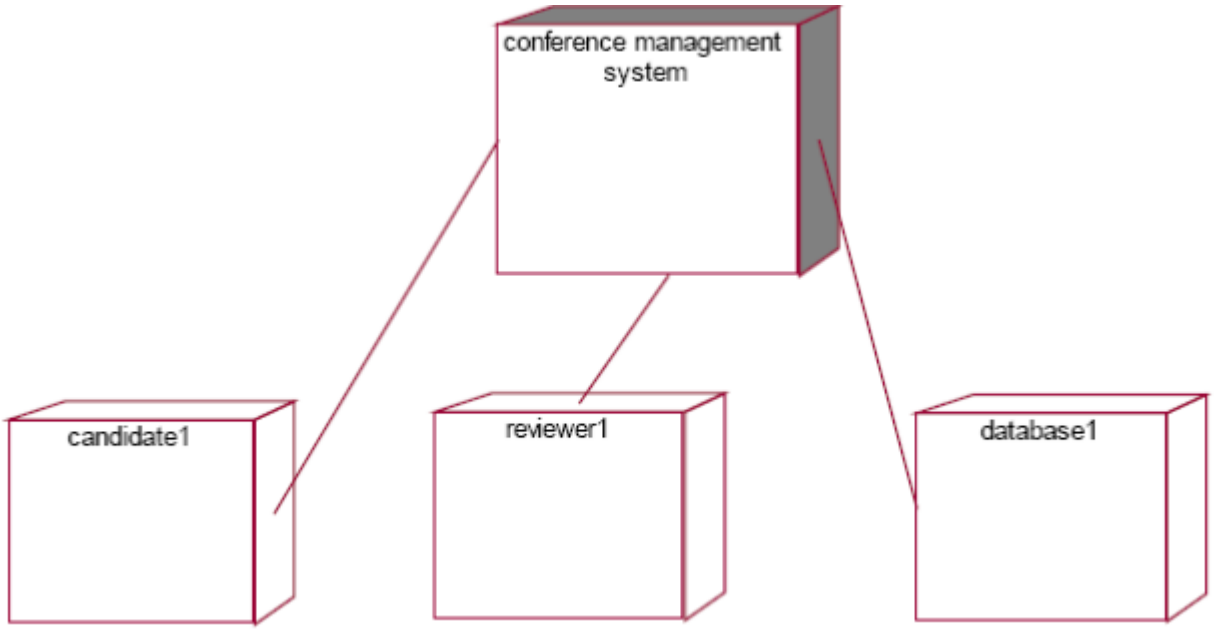
Collaboration Diagram:



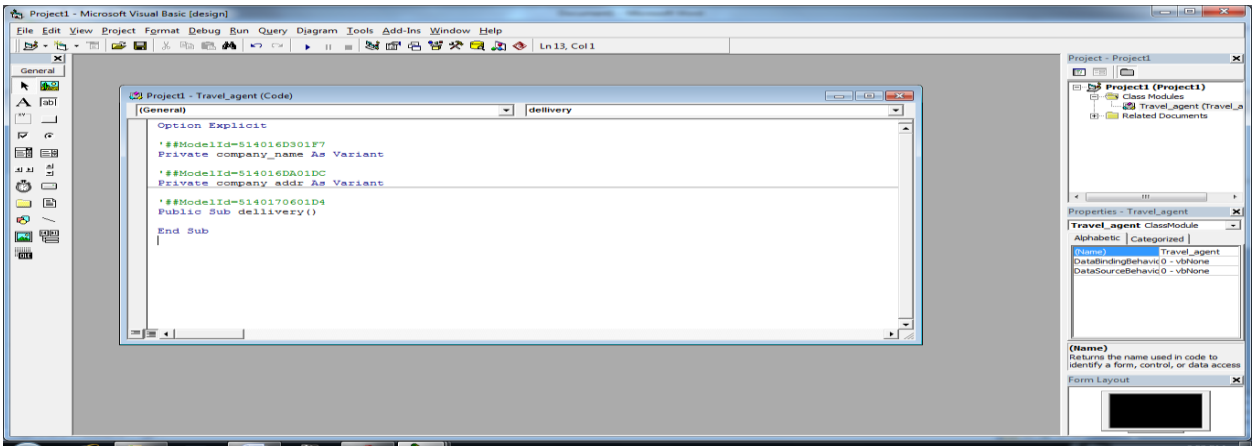
Component diagram:

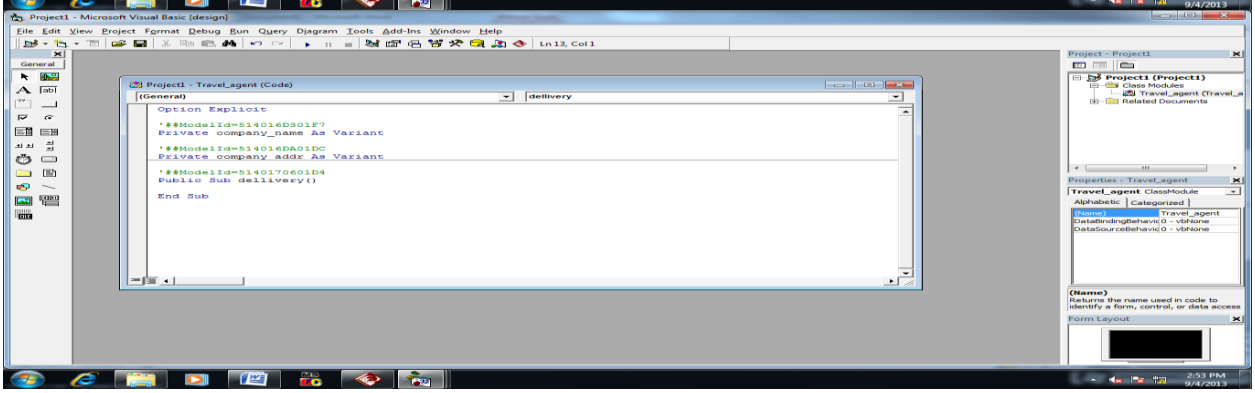


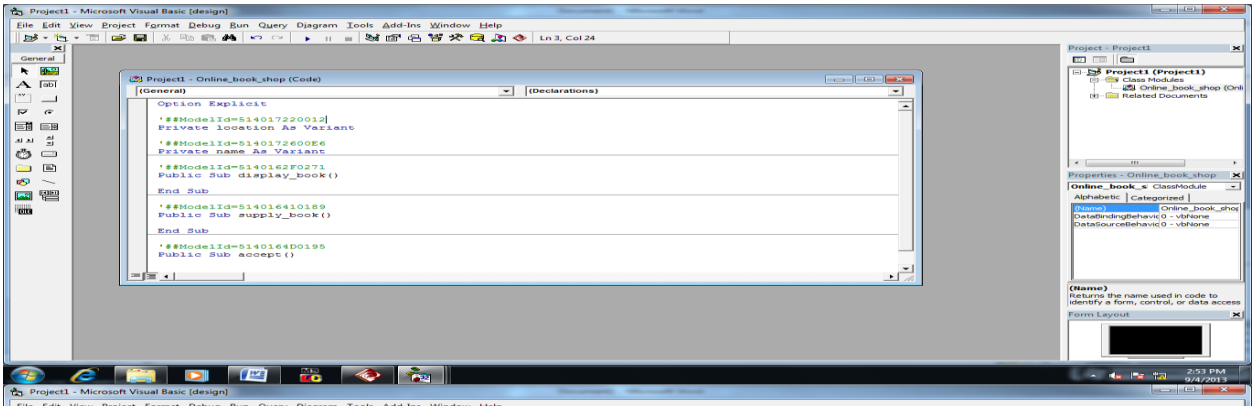
Deploymentdiagram:

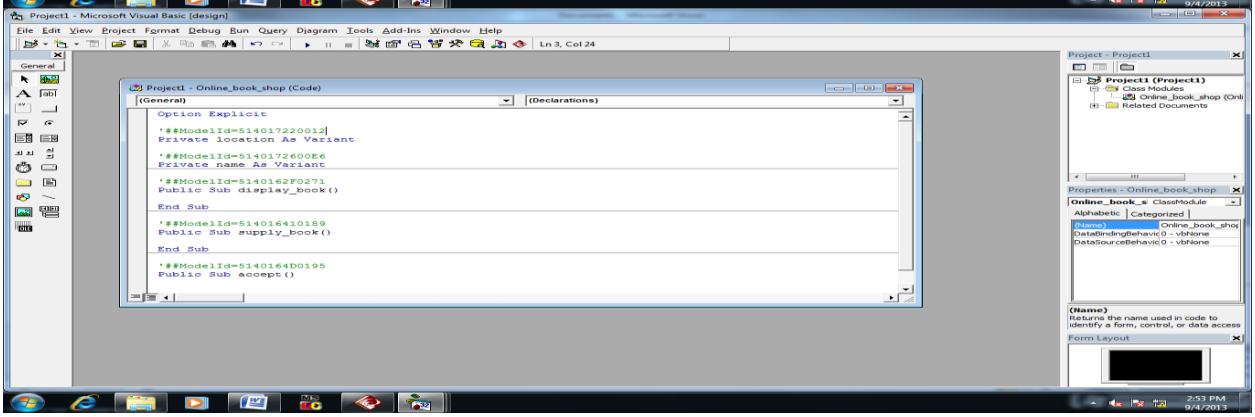


Output:









Result:

Thus, the design for conference management system and code has been generated successfully.

EX.NO:14 BPO MANAGEMENT SYSTEM

DATE:

Aim:

To analyze and design online course registration system using Umbrello Software.

Procedure:

1. Develop a problem statement.
2. Identify Use cases and develop the Use case Model.
3. Design activity diagram and sequence diagram for use case.
4. Draw state chart diagram.
5. Draw package, component and deployment diagrams.
6. Draw class diagram and generate code.

Problem statement

BPO is the process of hiring another company to handle business activity program. The activity typically includes finance and administration HR. The main aim of this is a subset of out sourcing that involves contracting of the operations and responsibilities of specific business functions to a third party service provider.BPO is typically categorized into back office outsourcing which include internal business functions such as human resource or finance and accounting a front office outsourcing which includes customer related service such as contact counter.

Project scope

The course registration system deals with the registration of course through websites or directly. In this students can search the different courses available and know details about their registration.

Project objectives

The course registration system deals with the registration of course through websites or directly. In this students can search the different courses available and know details about their registration.

Software requirement analysis

The BPO Management analyzing system includes the following modules:

- Make a call.
- Accept.
- Asking information.
- Verify the problem.
- Giving solution.

Make a call:

If the user have any doubt, first the user makes a call to the call centre executer.

Accept:

The call centre executer accepts the call and then registered it.

Asking information:

Here, first for clarifying the doubts the user asks for some information and the call centre executer ask information for the verification purposes.

Verify the problem:

As the executer receiving the data from the user, he verifies the problems.

Giving solutions:

After the verifications, he gave solutions for the corresponding problems.

SOFTWARE REQUIREMENT SPECIFICATION

1.0 INTRODUCTION

Business process outsourcing (**BPO**) is a subset of outsourcing that involves the contracting of the operations and responsibilities of specific business functions (or processes) to a third-party service provider. Originally, this was associated with manufacturing firms, such as Coca Cola that outsourced large segments of its supply chain.[1] In the contemporary context, it is primarily used to refer to the outsourcing of business processing services to an outside firm, replacing in-house services with labor from an outside firm.

BPO is typically categorized into back office outsourcing - which includes *internal business functions* such as human resources or finance and accounting, and **front office outsourcing** - which includes *customer-related services* such as contact center services.

BPO that is contracted outside a company's country is called offshore outsourcing. BPO that is contracted to a company's neighboring (or nearby) country is called nearshore outsourcing.

1.1 PURPOSE

- ✓ Huge reduction in printing, dispatch costs.
- ✓ Seamless process that is fully integrated ensuring better quality of service to customers.
- ✓ Cost reductions by automation of upload processes from clients; automatic routing of documents to operators using OCR.
- ✓ Documents as well as the status of process is accessible quickly and from anywhere to BPO management as well as customers.
- ✓ Security of documents as they are stored in digital form.

.

1.2 SCOPE

- As part of BPO, documents need to be managed between the outsourcing company and the offshore company.
- Multiple clients need to be managed by the BPO Company.
- Security of the documents has to be ensured so that there is no unauthorized access of the documents to other organizations.
- Quick turnaround times have to be managed.
- Appropriate process flow of the documents has to be present in the system to check the status of the documents at any point of time.

1.3 DEFINITIONS, ACRONYMS AND THE ABBREVIATIONS

• **Client**- One who wishes to outsource their job

• **BPOS**- Refers to this Business Process Outsourcing System

• **BPO Organization**- Refers to this Business Process Outsourcing Organization

• **HTML** - Markup Language used for creating web pages.

• **J2EE** – Java 2 Enterprise Edition is a programming platform java platform for developing and running distributed java applications.

• **HTTP** - Hyper Text Transfer Protocol.

•**TCP/IP** – Transmission Control Protocol/Internet Protocol is the communication protocol used to connect hosts on the Internet.

1.4 REFERENCES

IEEE Software Requirement Specification format.

1.5 TECHNOLOGIES TO BE USED

• HTML
• JSP
• Javascript
• Java

1.6 TOOLS TO BE USED

• Eclipse IDE (Integrated Development Environment)
• Umbrello tool (for developing UML Patterns)

1.7 OVERVIEW

SRS includes two sections overall description and specific requirements –
Overall Description will describe major role of the system components and inter-connections.
Specific Requirements will describe roles & functions of the actors.

2.0 OVERALL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

The BPOS acts as an interface between the 'client' and the 'BPO organization'. This system tries to make the interface as simple as possible and at the same time not risking the security of data stored in. This minimizes the time duration in which the user receives the documents.

2.2 SOFTWARE INTERFACE

- **Front End Client** - The exporter online interface is built using JSP and HTML.
- **Web Server** – Apache Tomcat Server (Oracle Corporation)
- **Back End** - Oracle 11g database

2.3 HARDWARE INTERFACE

The BPO system's server is directly connected to the client systems via ftp. The client systems have access to the database in the server.

2.4 USER CHARACTERISTICS

• **BPO Organization** - They are the people who desire to obtain the outsourcing job from various clients and submit the information to the database.
• **Client** - He has the certain privileges to outsource their jobs and to approve the issue of document. He may contain a group of persons under him to verify the documents and give suggestion whether or not to approve the dispatch of job.

2.5 CONSTRAINTS

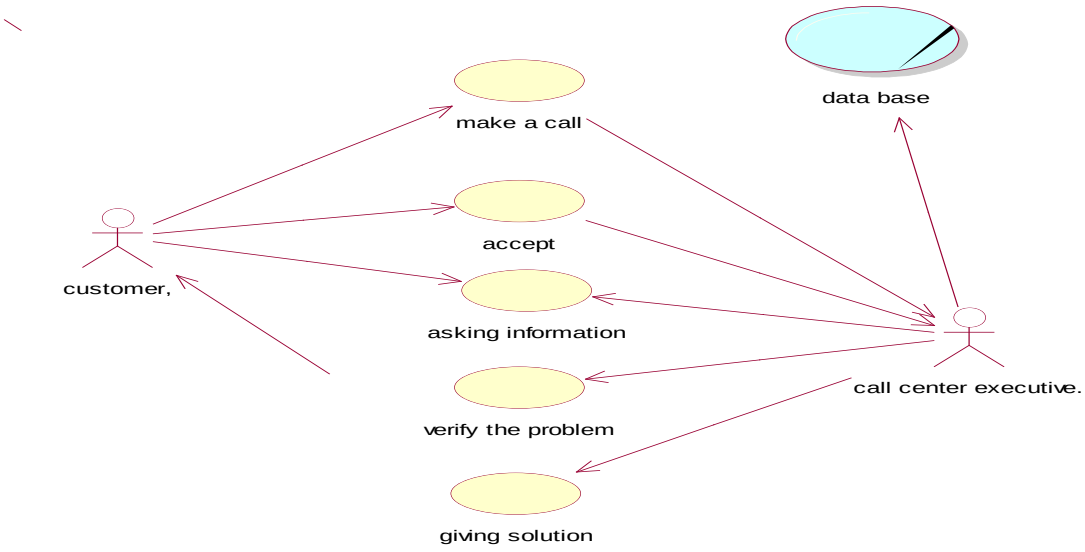
• The BPO require a computer (FTP) to submit their information.
• Although the security is given high importance, there is always a chance of intrusion in the web world which requires constant monitoring.
• The user has to be careful while submitting the information. Much care is required.

2.6 ASSUMPTIONS AND DEPENDENCIES

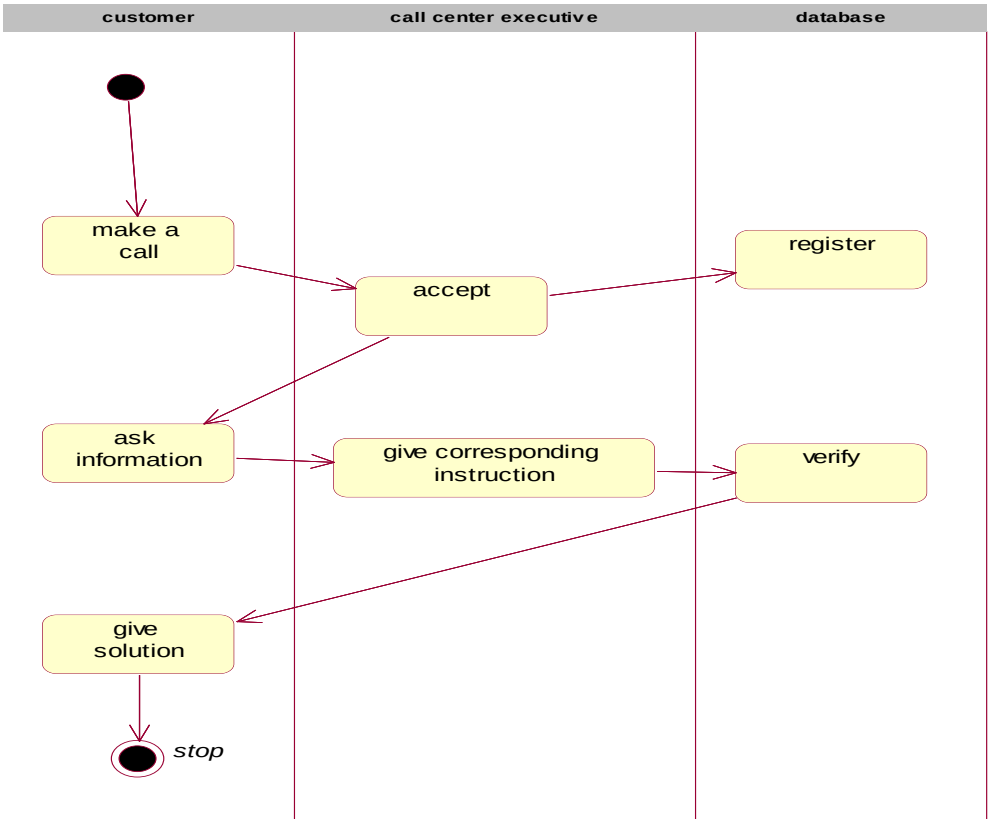
• The organization and client must have basic knowledge of computers and English Language.
• Provide privacy and security for the documents and client information

UML diagrams:

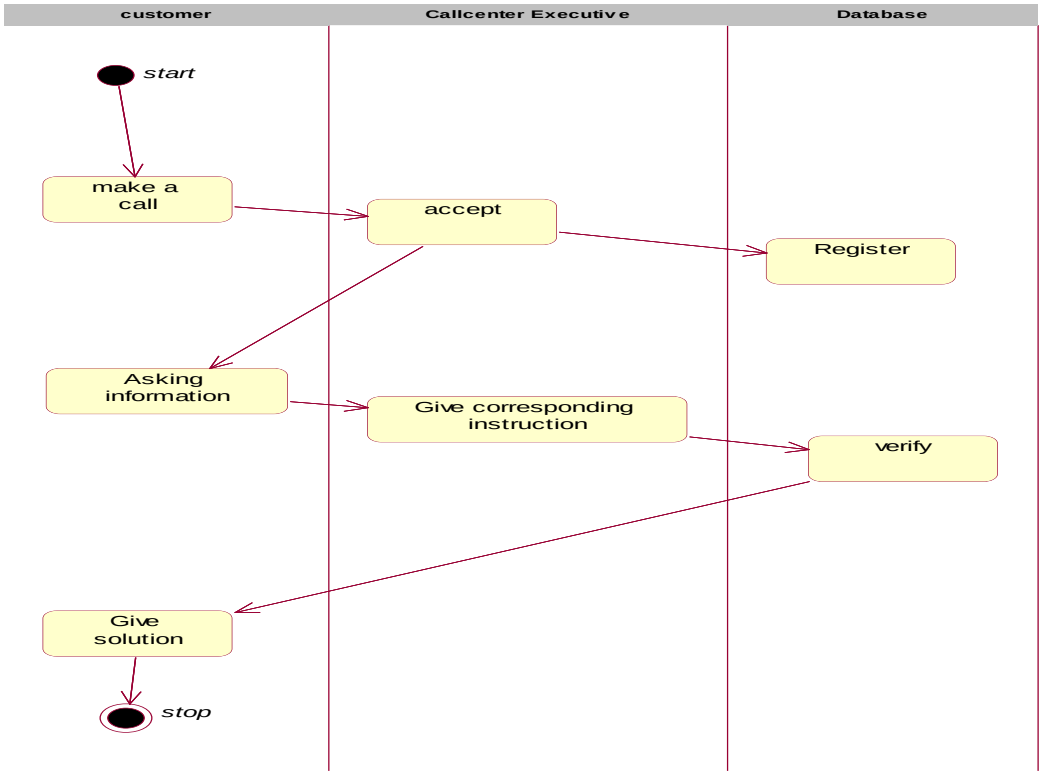
Usecase diagram:



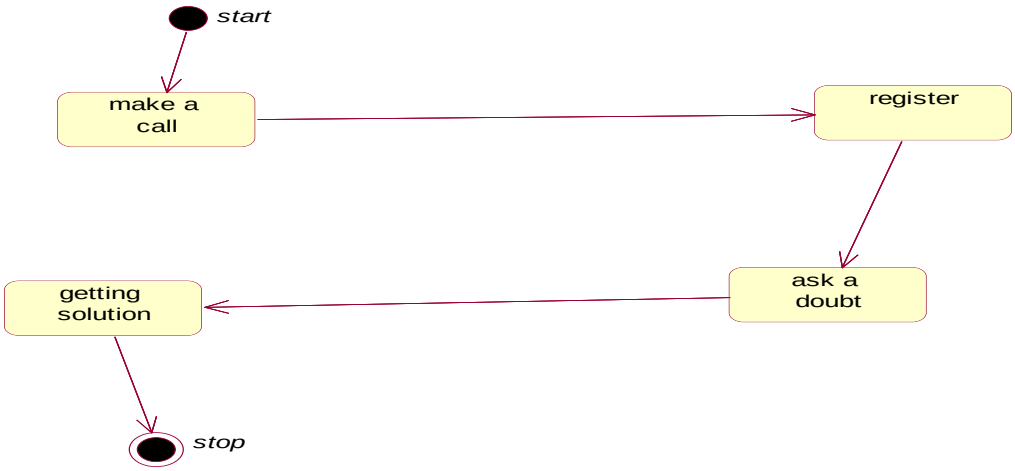
Sequence diagram:



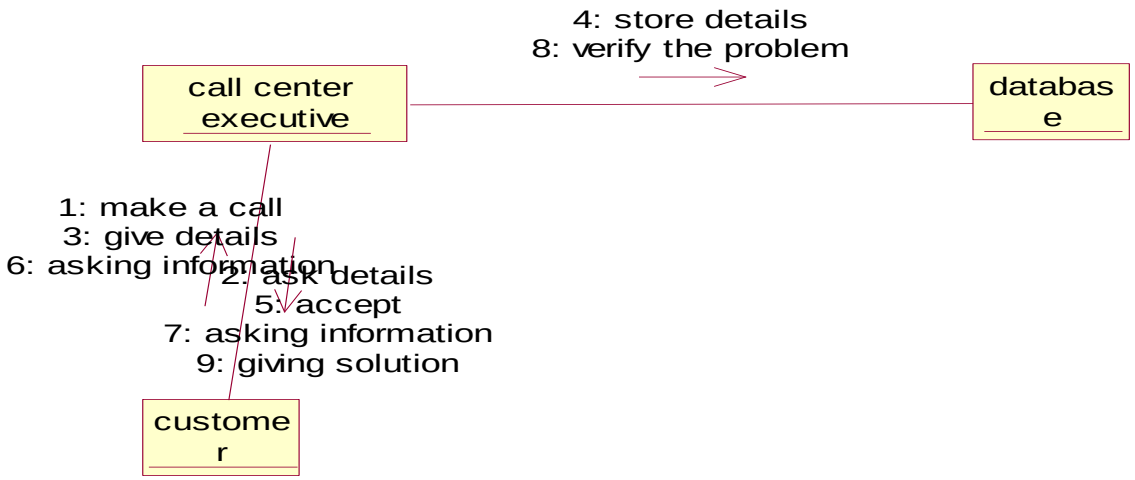
Activity diagram:



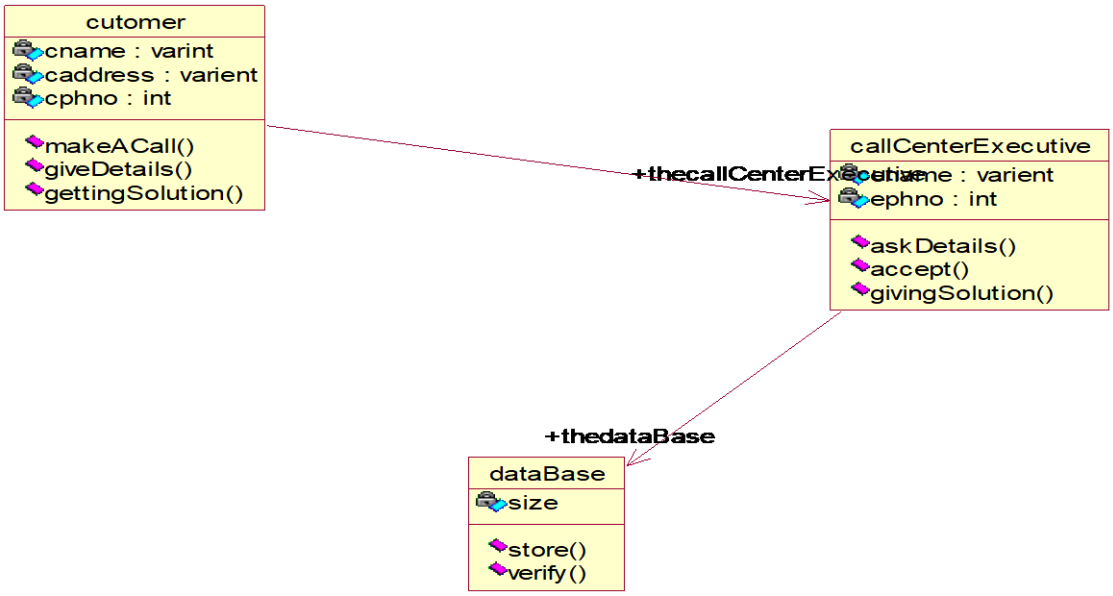
state diagram:



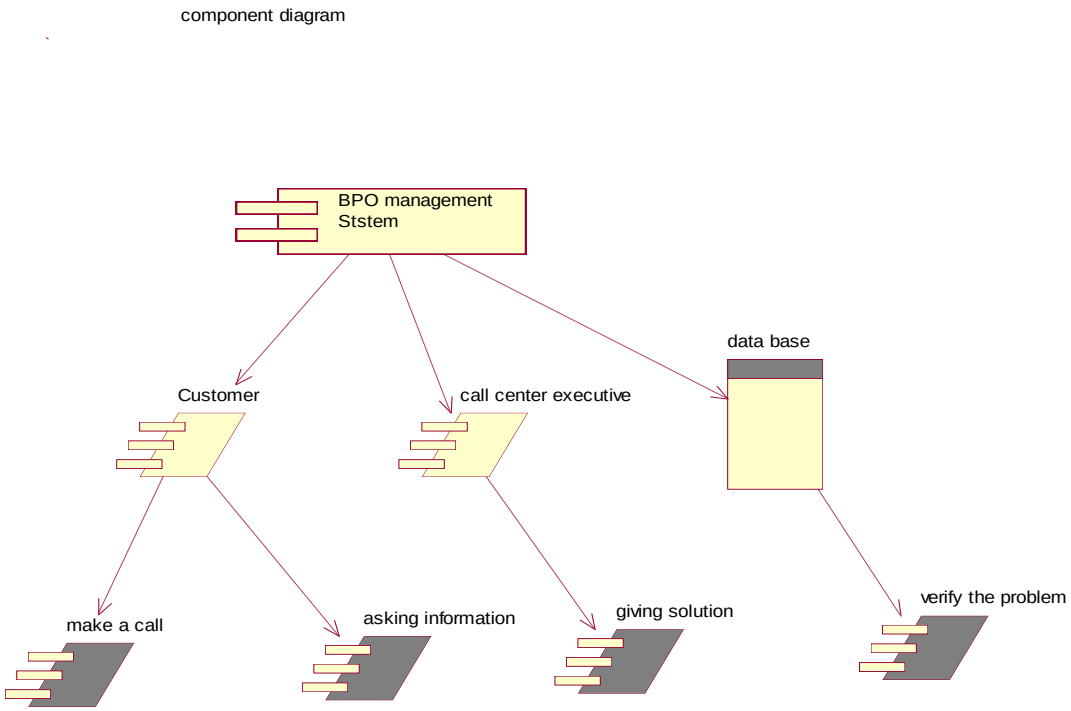
Collaboration diagram:



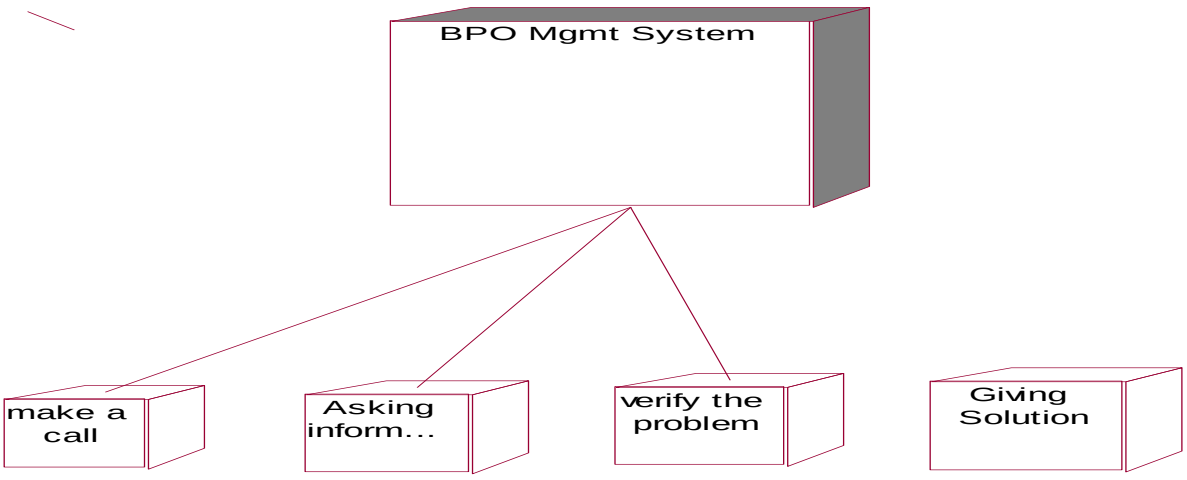
Class diagram:



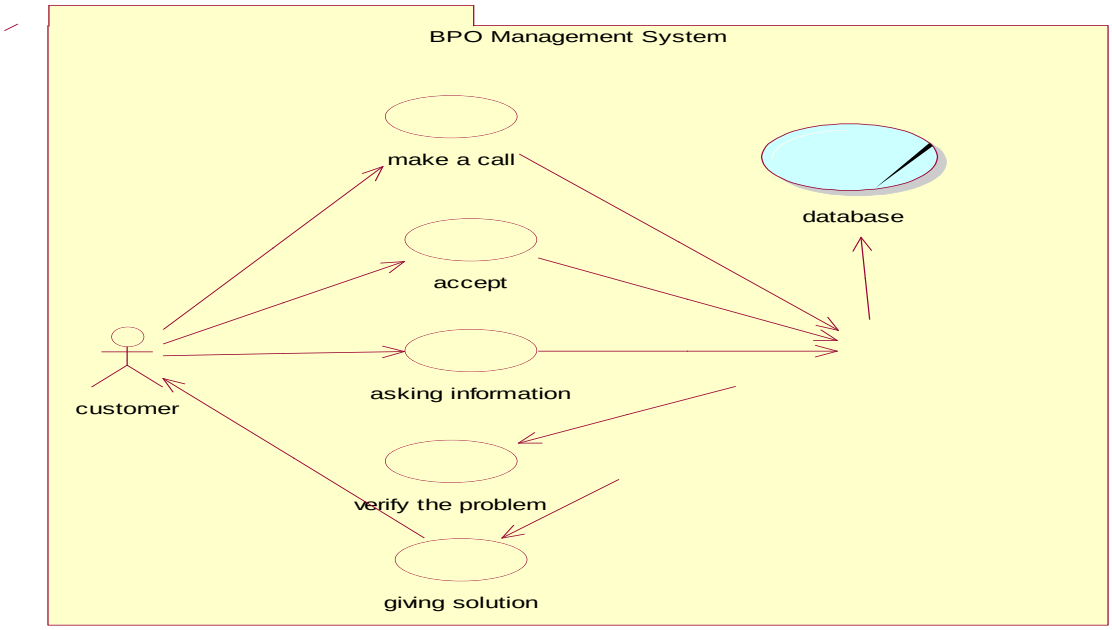
Component diagram:



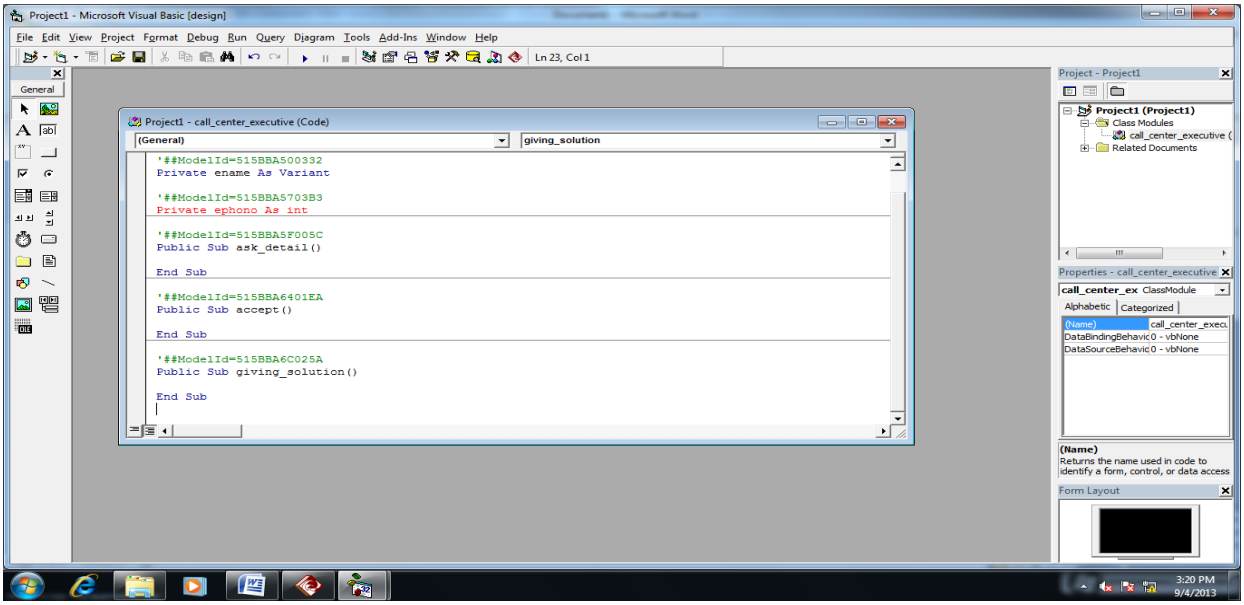
Deployment diagram:

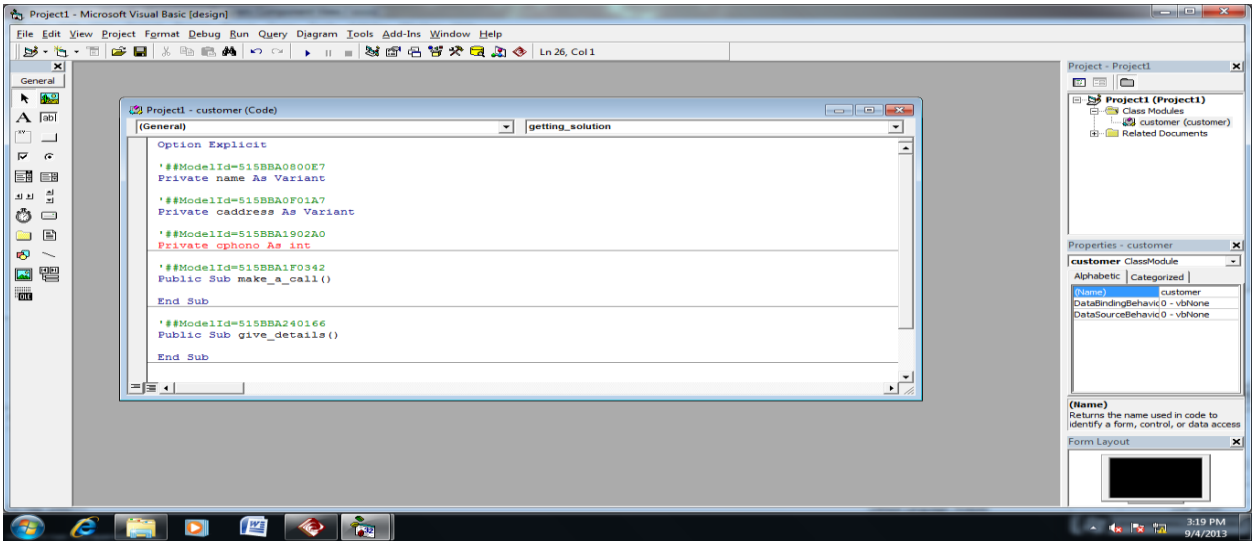


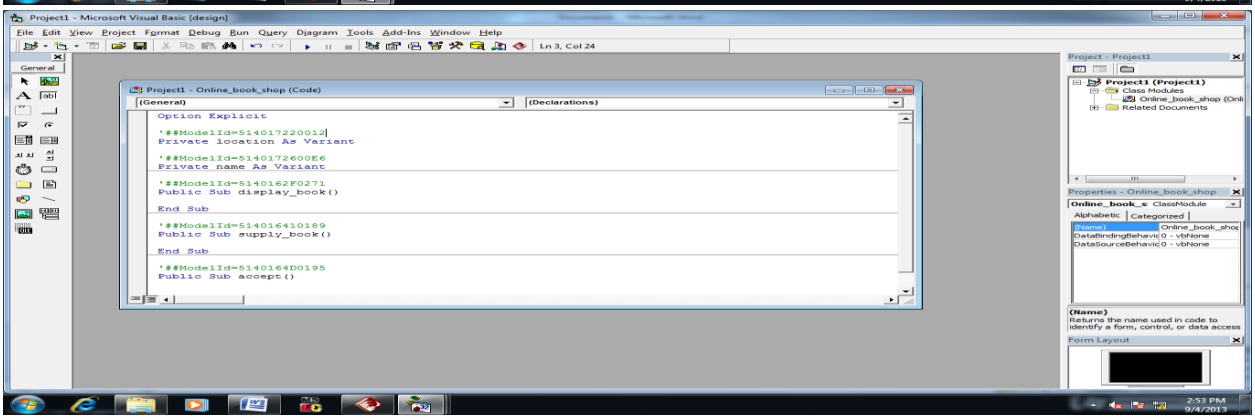
Package diagram:



OUTPUT:







Result:

Thus the BPO Management system was done and the code is generated successfully

EX.NO:15 LIBRARY MANAGEMENT SYSTEM

DATE:

AIM

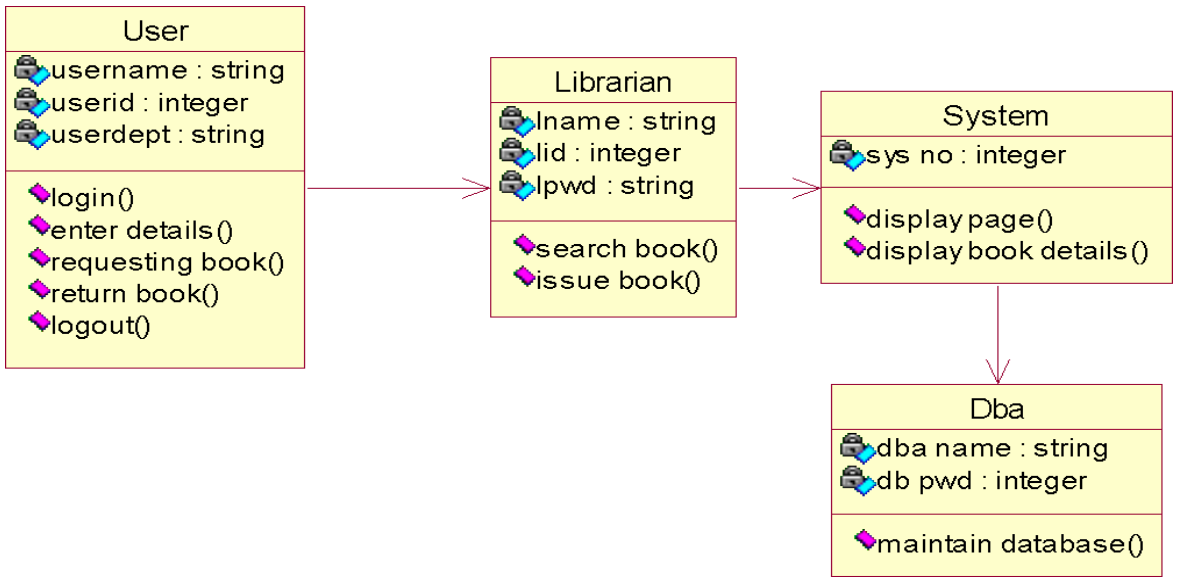
To design an object oriented model for Library Management System using Umbrello software and to implement it using Java.

PROBLEM STATEMENT

The library management system is a software system that issues books and magazines to registered students only. The student has to login after getting registered to the system. The borrower of the book can perform various functions such as searching for desired book, get the issued book and return the book.

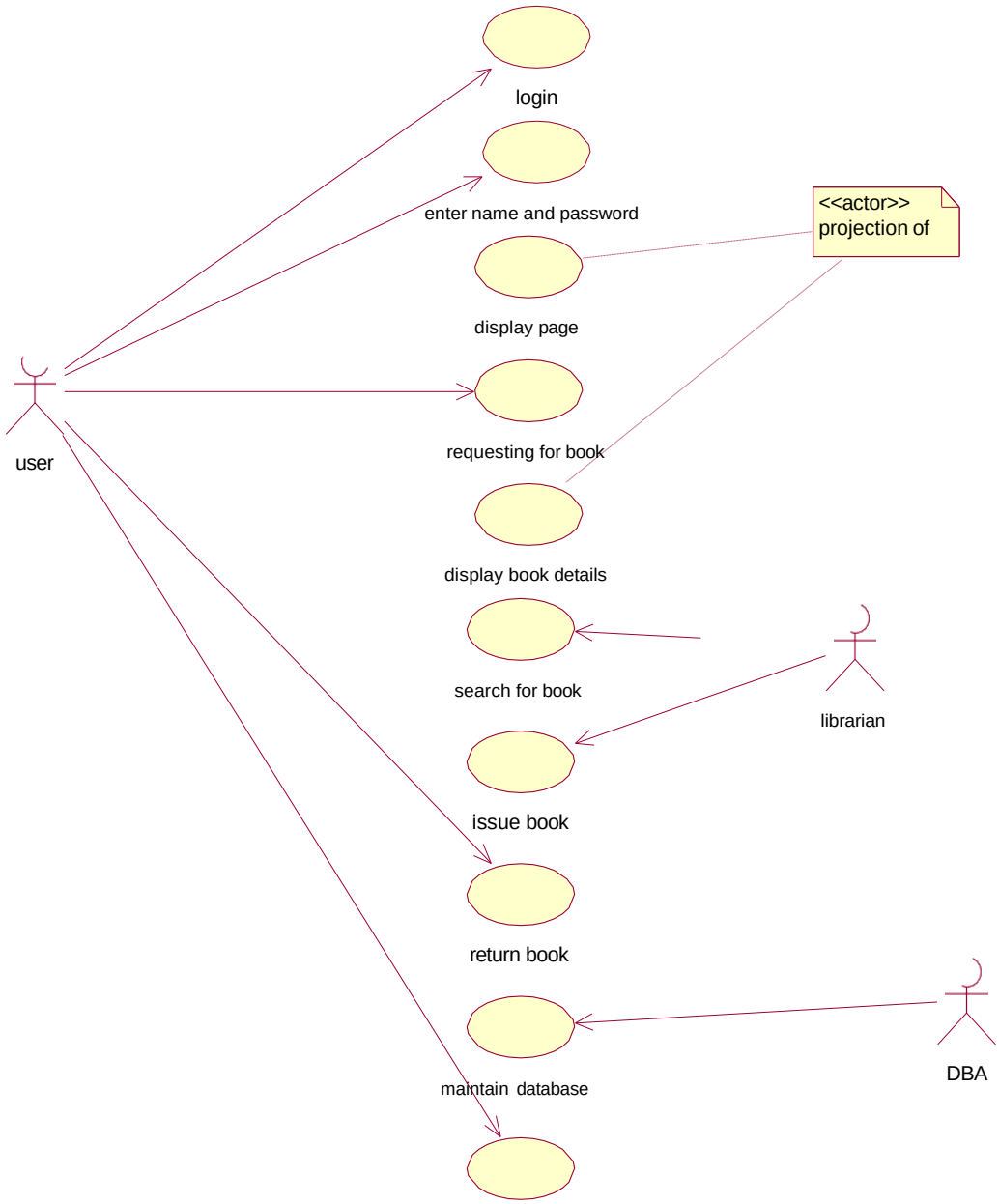
CLASS DIAGRAM

A class diagram in the unified modeling language is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations and the relationships among objects. The library management system makes use of the following classes user, librarian, system and DBA.



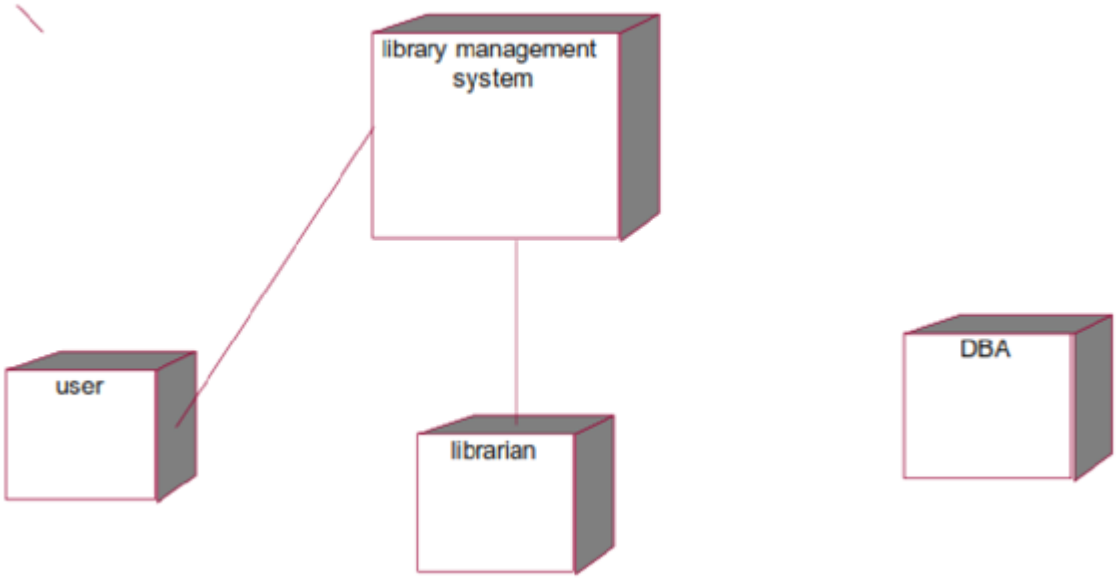
USE CASE DIAGRAM

Use case is a list of actions or events. Steps typically defining the interactions between a role and a system to achieve a goal. The use case diagram consists of various functionality performed by actors like user, librarian, system and DBA.



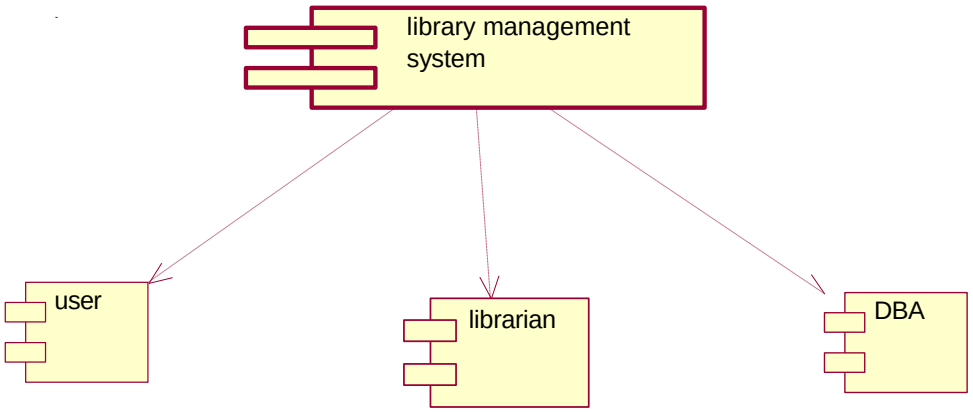
DEPLOYMENT DIAGRAM

Deployment diagram is a structure diagram which shows architecture of the system as deployment of software artifacts to deployment target. It is the graph of nodes connected by communication association. It is represented by three dimensional box. The device node is library management system and execution environment nodes are user, librarian, system and DBA.



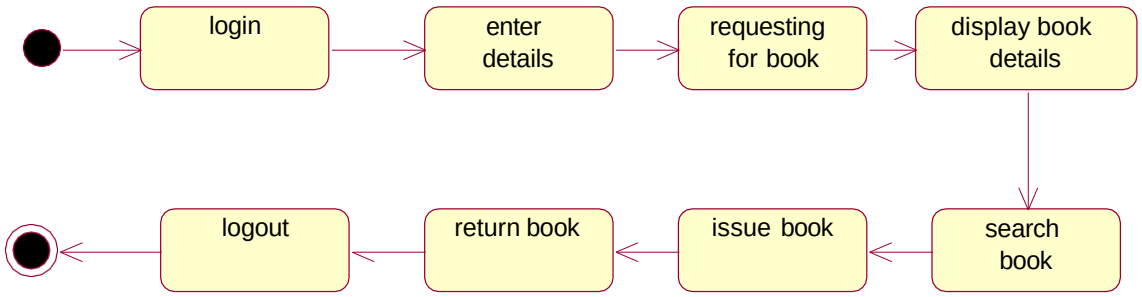
COMPONENT DIAGRAM

Component diagram shows the dependencies and interactions between software components. Component diagram carries the most important living actors of the system i.e, user, librarian and DBA.



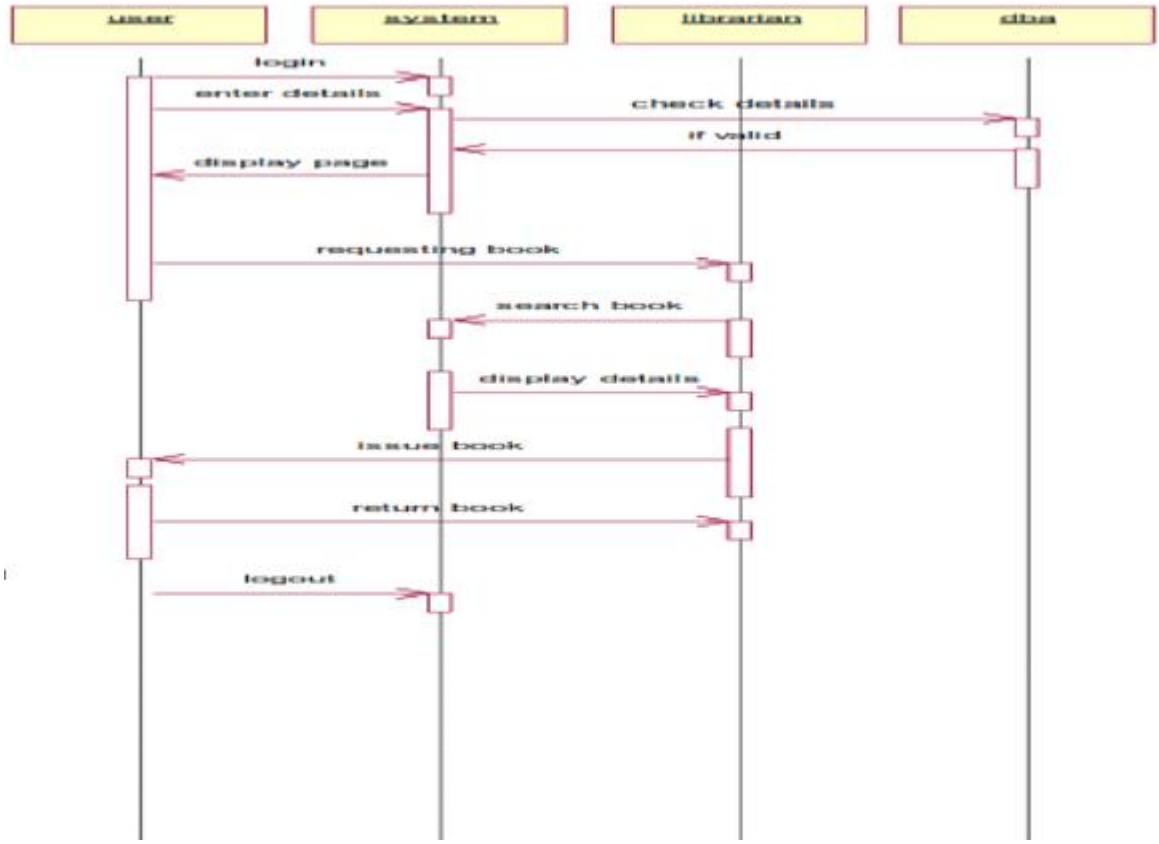
STATECHART DIAGRAM

State chart diagram is also called as state machine diagram. The state chart diagram contains the states in the rectangular boxes and the states are indicated by the dot enclosed. The state chart diagram describes the behavior of the system. The state chart diagram involves eight stages such as login, enter details, requesting for book, display book details, search book, issue book, return book and logout.



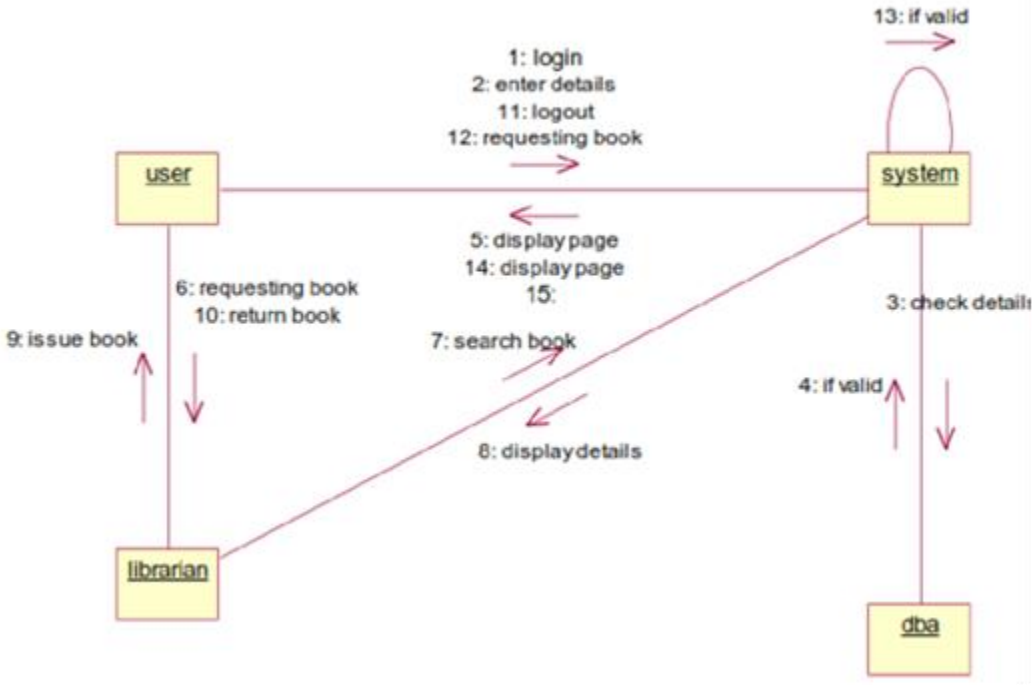
SEQUENCE DIAGRAM

A sequence diagram represent the sequence and interactions of a given use case or scenario. Sequence diagram capture most of the information about the system. It is also represent in order by which they occur and have the object in the system send message to one another. Here the sequence starts with interaction between user and the system followed by database. Once the book have been selected the next half of sequence starts between librarian and user followed by database.

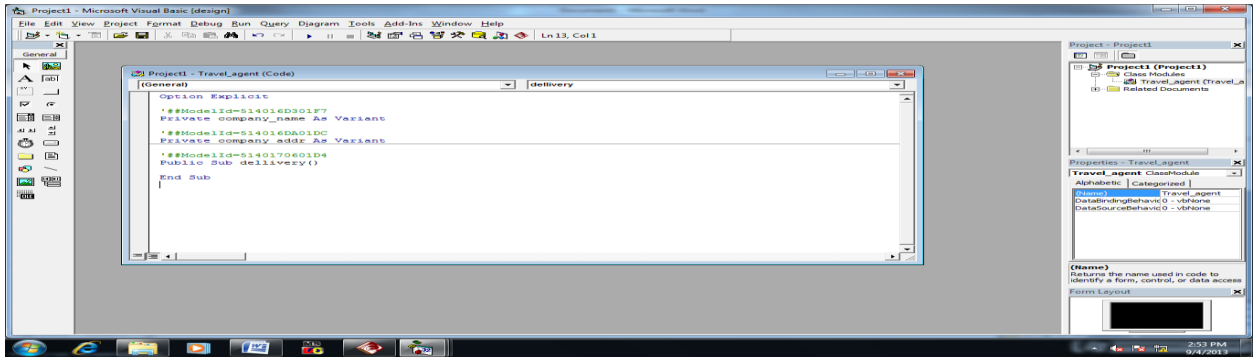


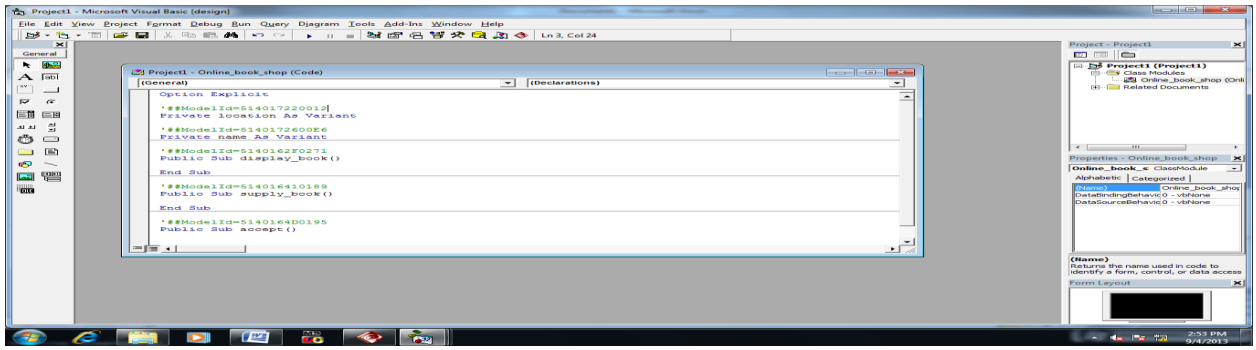
COLLABORATION DIAGRAM

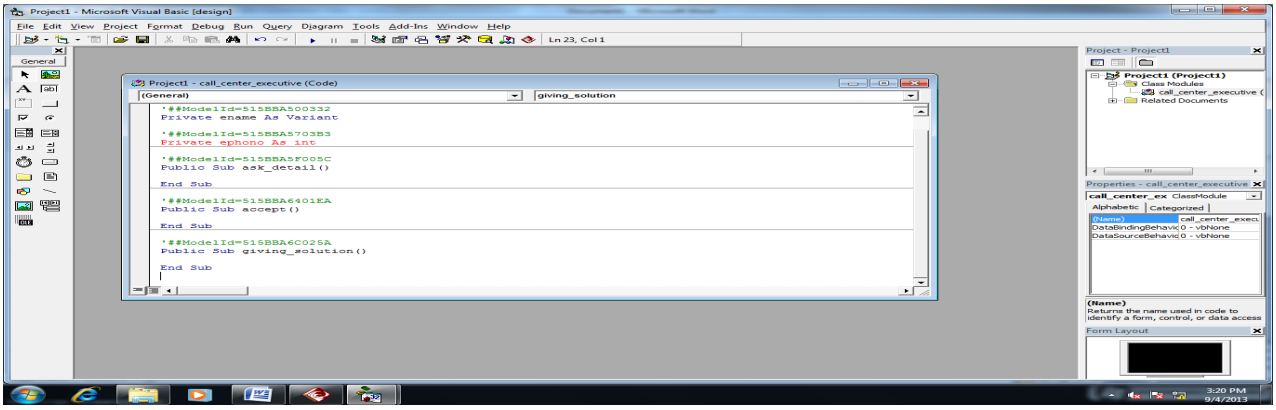
Like sequence diagram collaboration diagrams are also called as interaction diagram. Collaboration diagram convey the same informations as sequence diagram but focus on the object roles instead of the times that messages are sent. Here the actions between various classes are represented by number format for the case of identification.

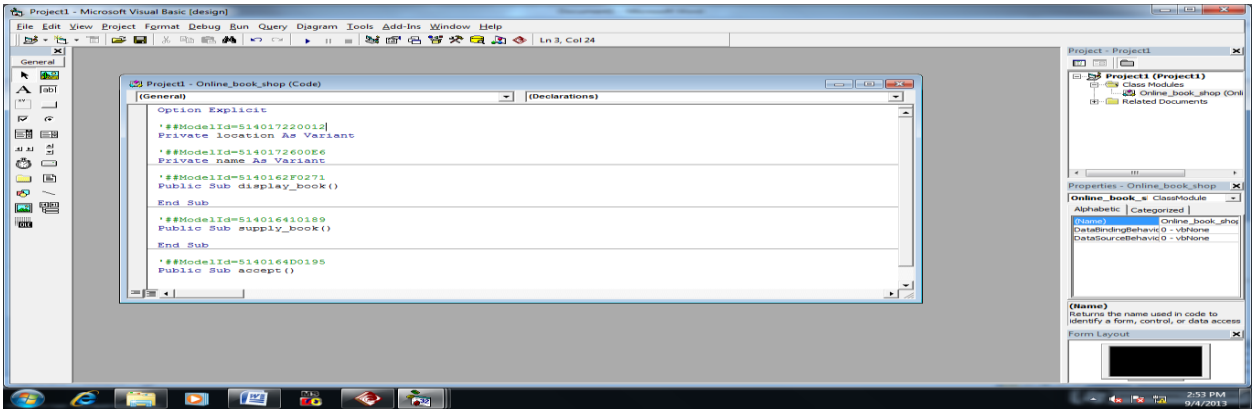


OUTPUT:









RESULT

Thus the various UML diagrams for library management system was drawn and the code was generated successfully.

EX.NO:16 STUDENT INFORMATION SYSTEM

DATE:

AIM

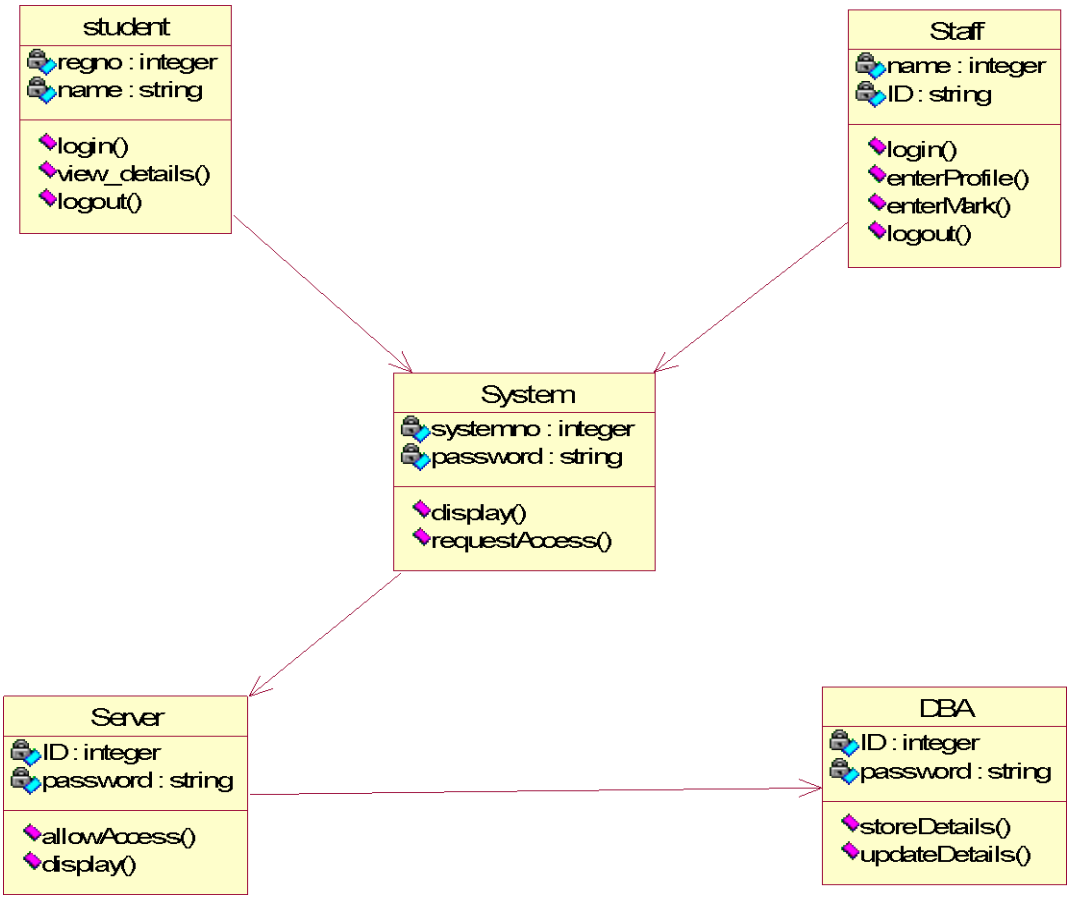
To design an object oriented model for Student information system using Umbrello software.

PROBLEM STATEMENT

The student must register by entering the name and password to login the form. The admin select the particular student to view the details about that student and maintaining the student details. This process of student information system is described sequentially through following steps. The student registers the system. The admin login to the student information system. He/she search for the list of students. Then select the particular student. Then view the details of that student. After displaying the student details then logout.

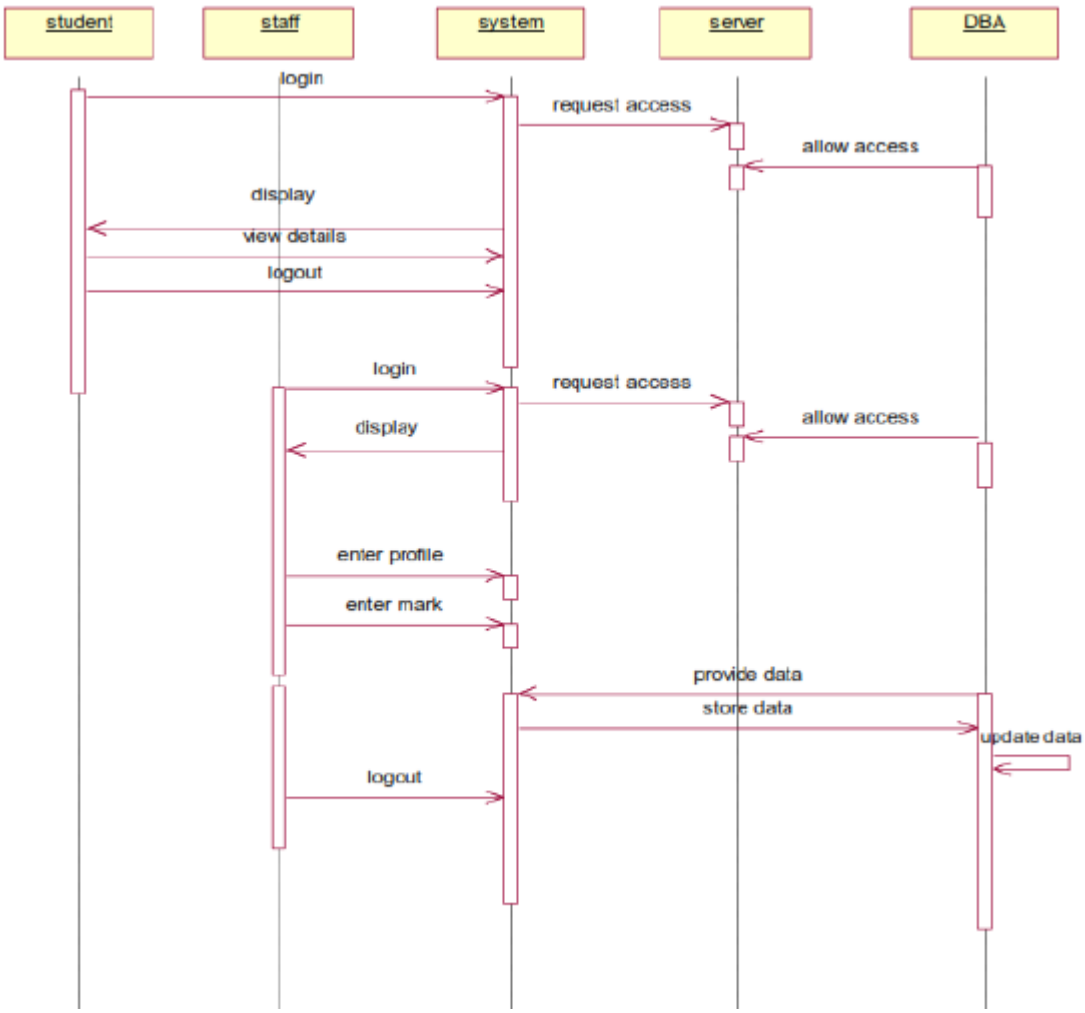
CLASS DIAGRAM

The class diagram is the graphical representation of all classes used in the system. The class diagram is drawn as rectangular box with three components or compartments like class name, attributes and operations. The student information system makes use of the following classes like student, staff, system, DBA and server.



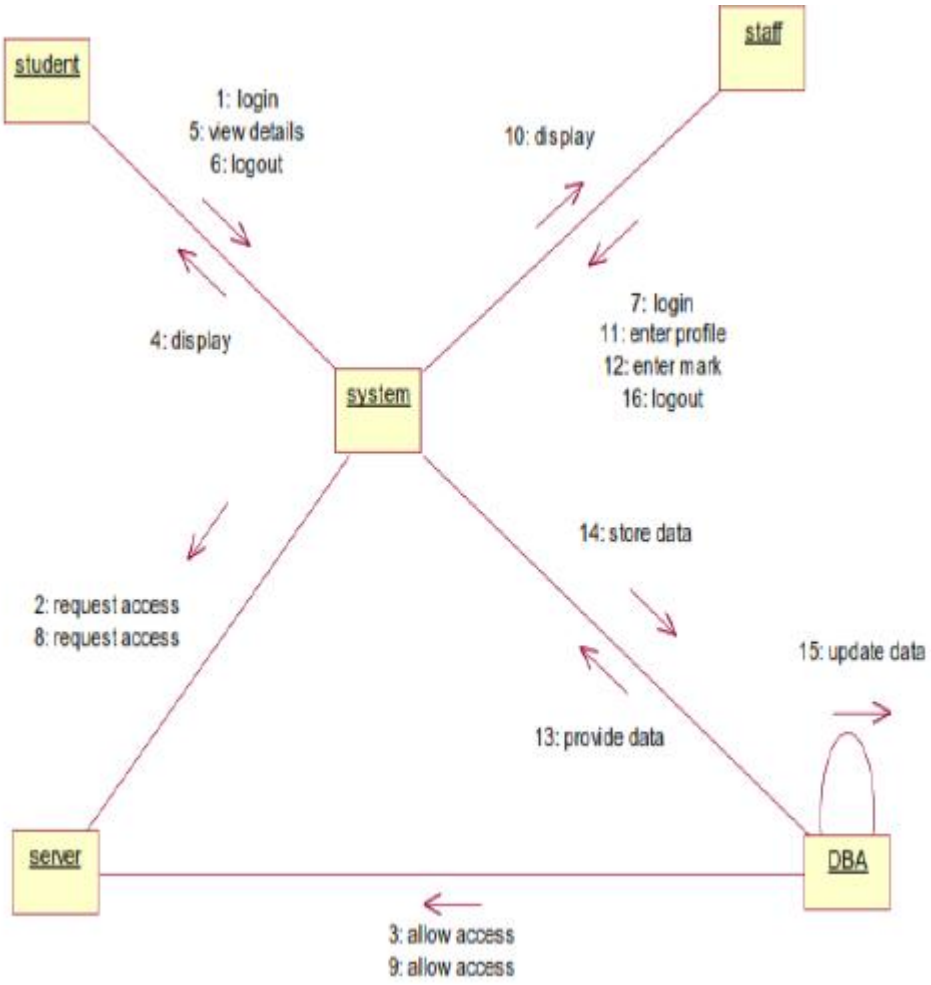
SEQUENCE DIAGRAM

A Sequence diagram represent the sequence and interaction of a given usecase or scenario. Sequence diagram capture most of the information about the system. Here the sequence starts between the student and the system. The second half of interaction takes place between staff and system then by police and followed by database. The student first login to the system and then view the details of the details. Staff login to the system enter mark and enter the details of the student. DBA store and update the details of the student.



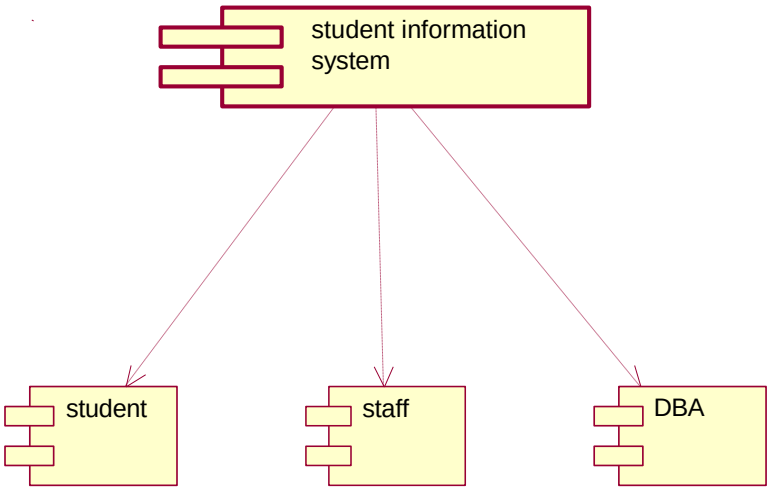
COLLABORATION DIAGRAM

A Collaboration diagram represents the collaboration in which is a set of objects related to achieve a desired outcome. In collaboration, the sequence is indicated by numbering the message several numbering schemes are available. Login, request access, allow access, display, view details, logout, login, request access, allow access, display, enter profile, enter mark, provide data, logout, store data, update data.



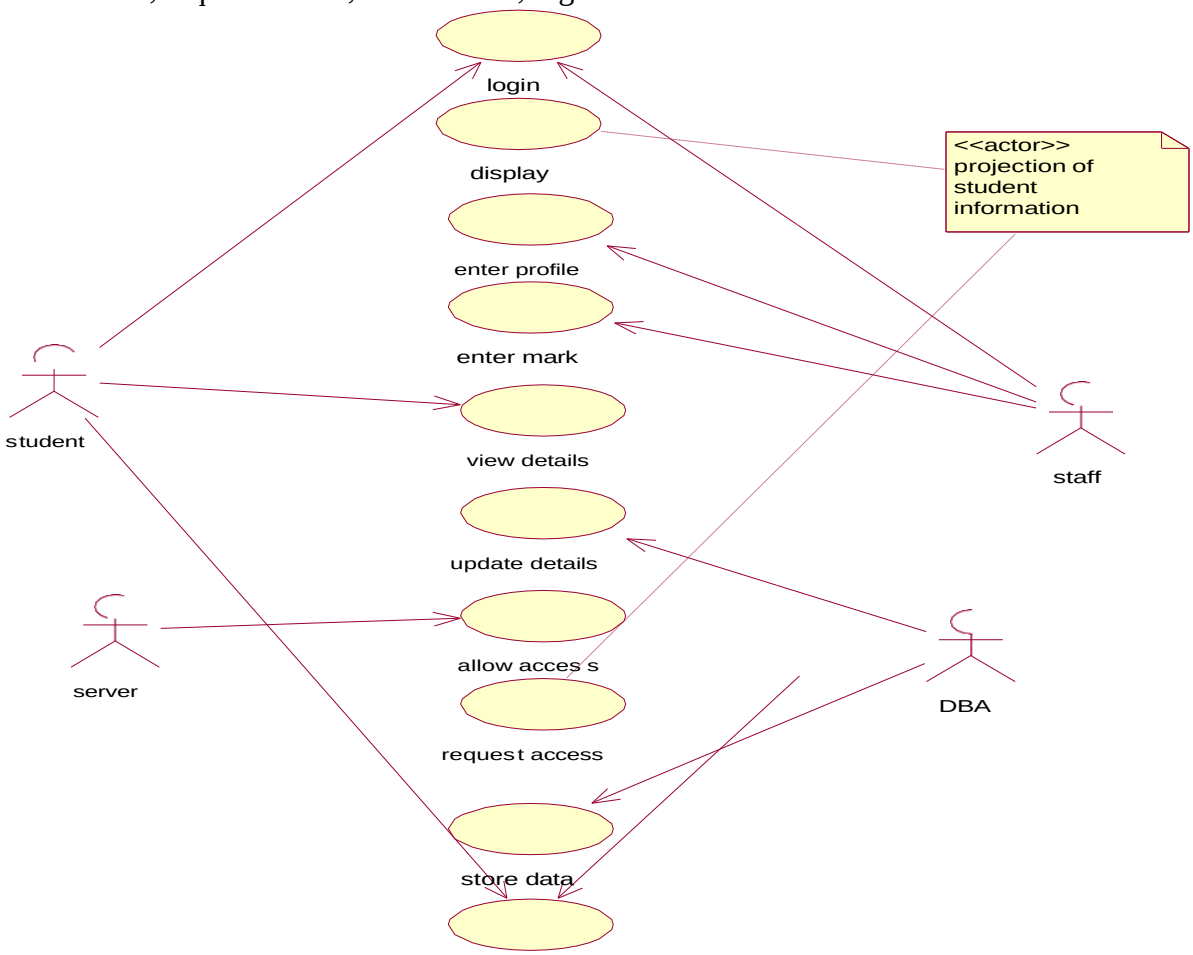
COMPONENT DIAGRAM

Component diagram carries the major living actors of the system. The component diagram main purpose is to show the structural relationship between components of the system.The main component of the system is student information system and the other components of the system are student, staff and DBA.



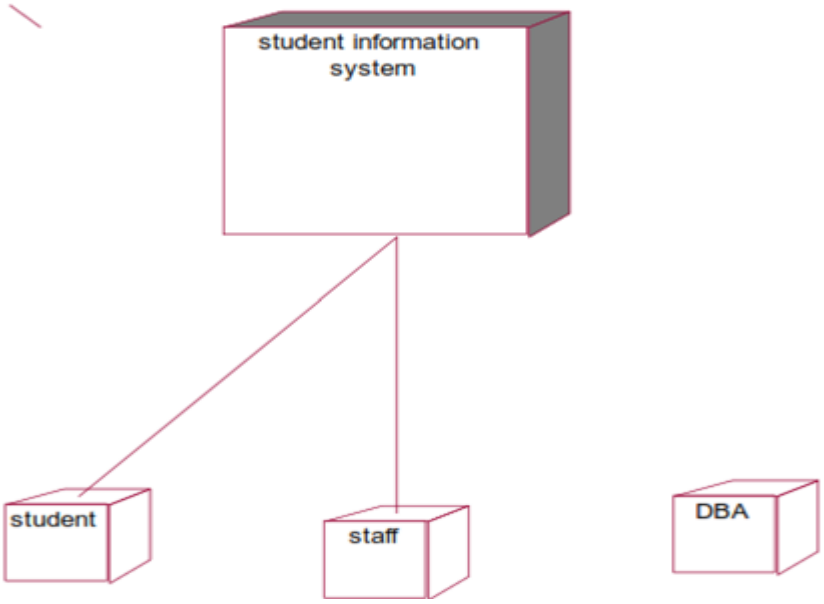
USE CASE DIAGRAM

Use case diagram is a graph of actors, a set of use cases, association between the actors and the use cases and generalization among the cases. Use case diagram is a list of actions or events. Use case diagram was drawn to represent the static design view of the system. Steps typically defined the interactions between a role and a system to achieve a goal. The use case diagram consists of various functionality performed by the actors like student, staff, system, DBA and server. The use case diagram consists of various functionality like login, display, enter profile, enter mark, view details, update details, allow access, request access, store details, logout.



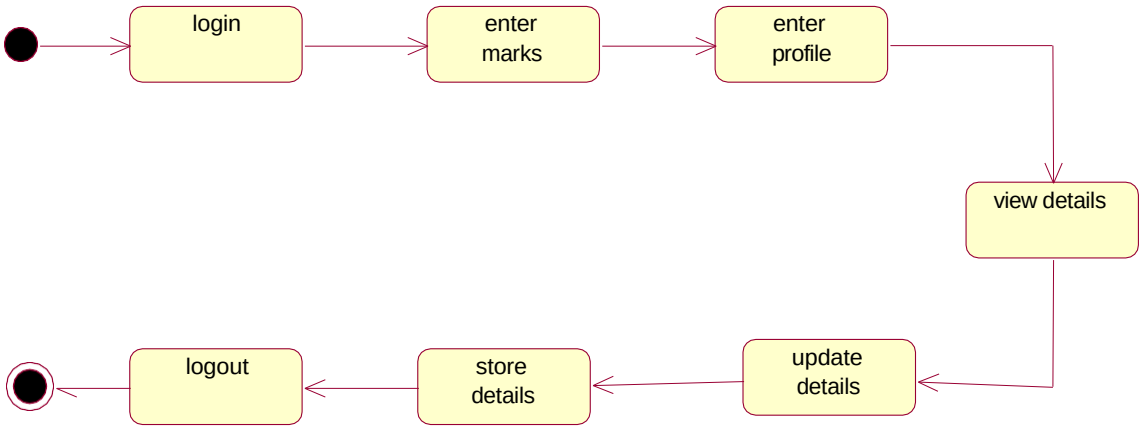
DEPLOYMENT DIAGRAM

Deployment diagram shows the configuration of runtime processing elements and the software components processes and objects that live in them. Component diagram are used in conjunction with deployment diagram to show how physical modules code are distributed on various hardware platform. The processor node in the system is student information system and the execution environment nodes or device nodes are student, staff and DBA.



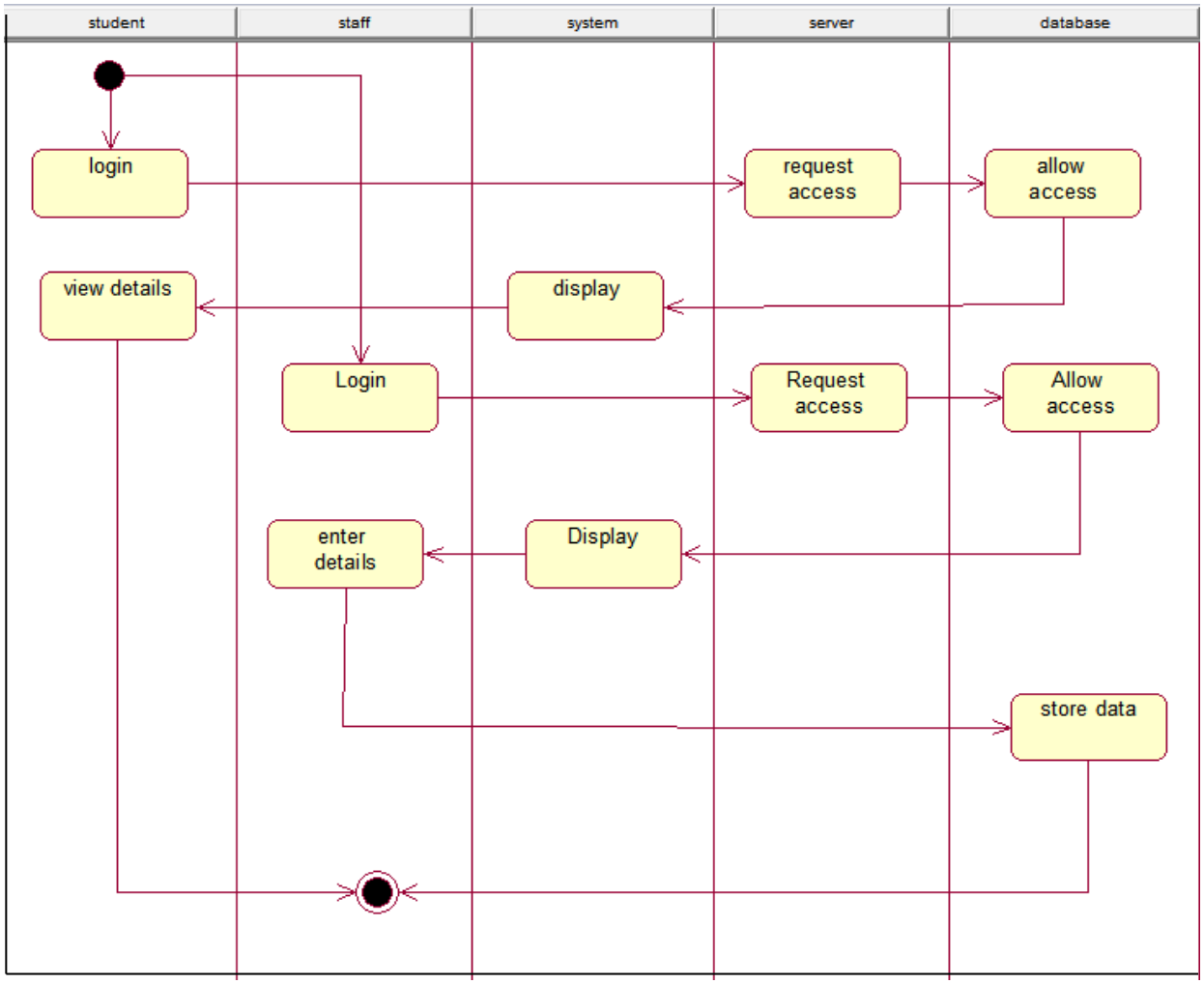
STATE CHART DIAGRAM

A State chart diagram is also called as state machine diagram. The state chart contains the states in the rectangular boxes and the states are indicated by the dot enclosed. The state chart diagram describes the behavior of the system. The state chart involves six stages such as login, enter mark, enter profile, view details, provide details, update details, store details and logout.

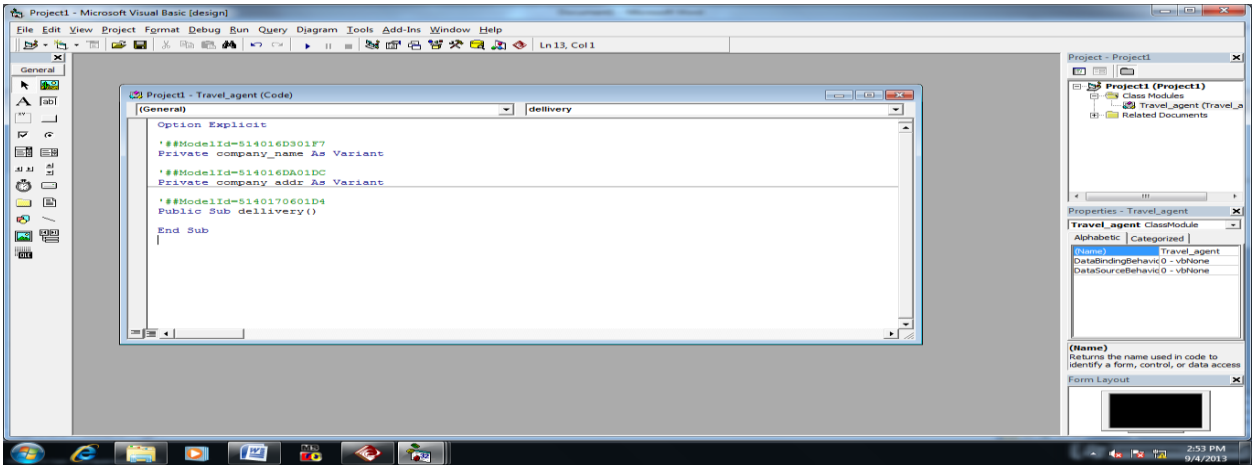


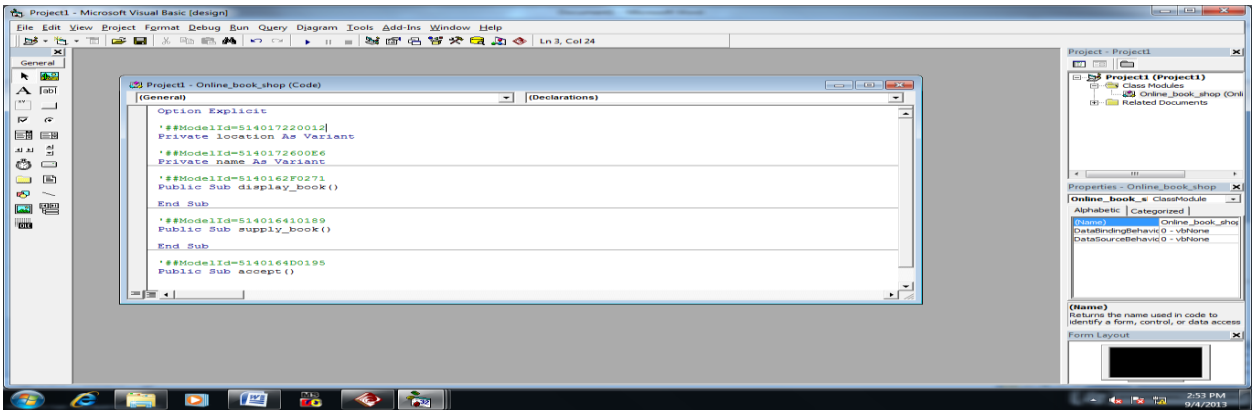
ACTIVITY DIAGRAM

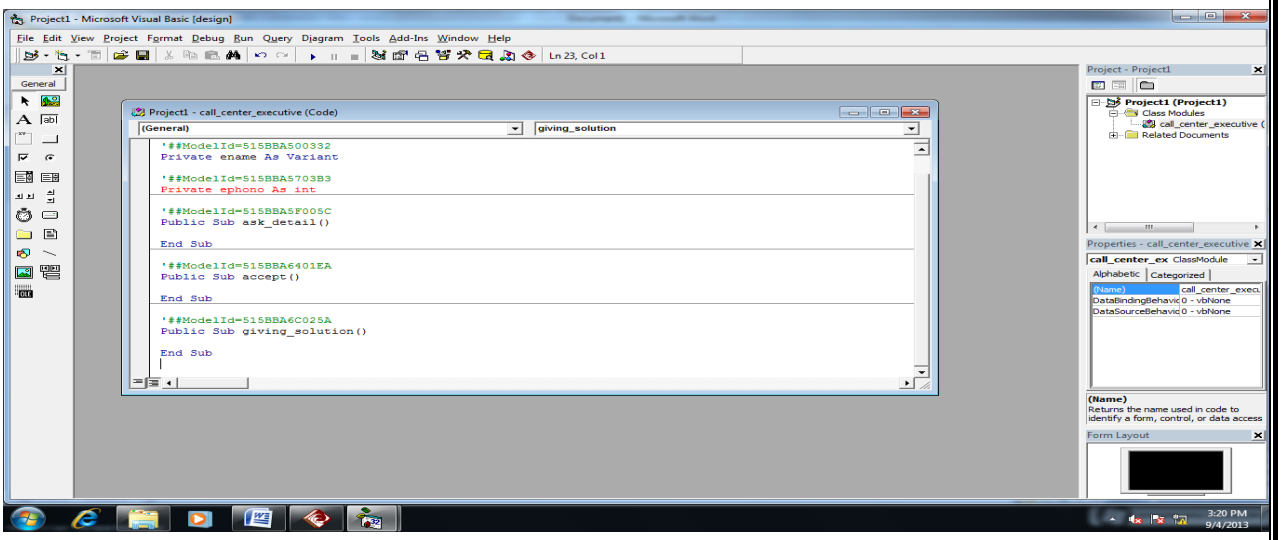
Activity diagram are graphical representation of stepwise activities and actions with support for choice, interaction and concurrency. Here in the activity diagram the student login to the system and view the details of the student. The staff login to the system for entering the student details and update the details in the database. The final interaction is the DBA store the details of the student.

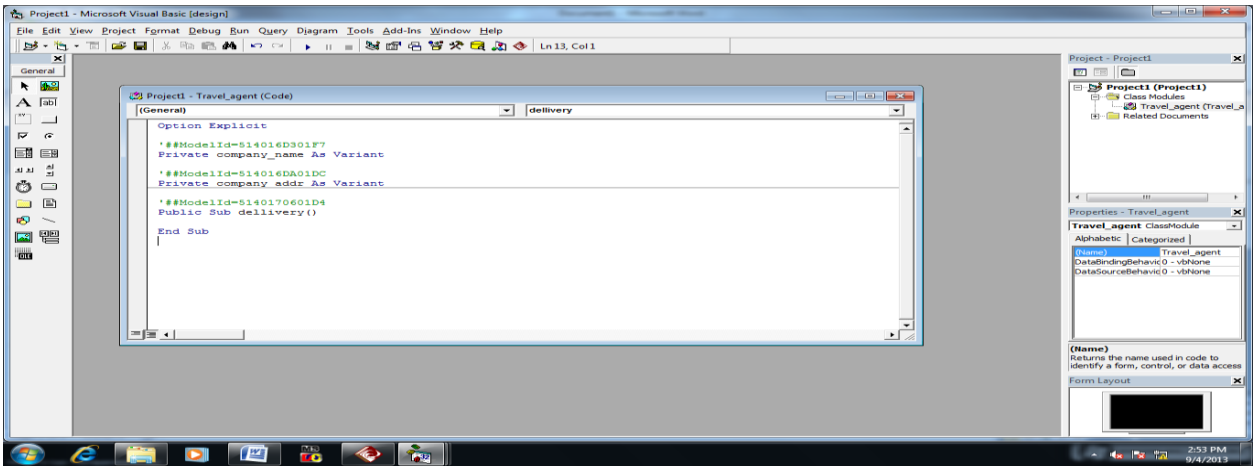

student
staff
system
server
database
login
request access
allow access
view details
display
Login
Request access
Allow access
enter details
Display
store data

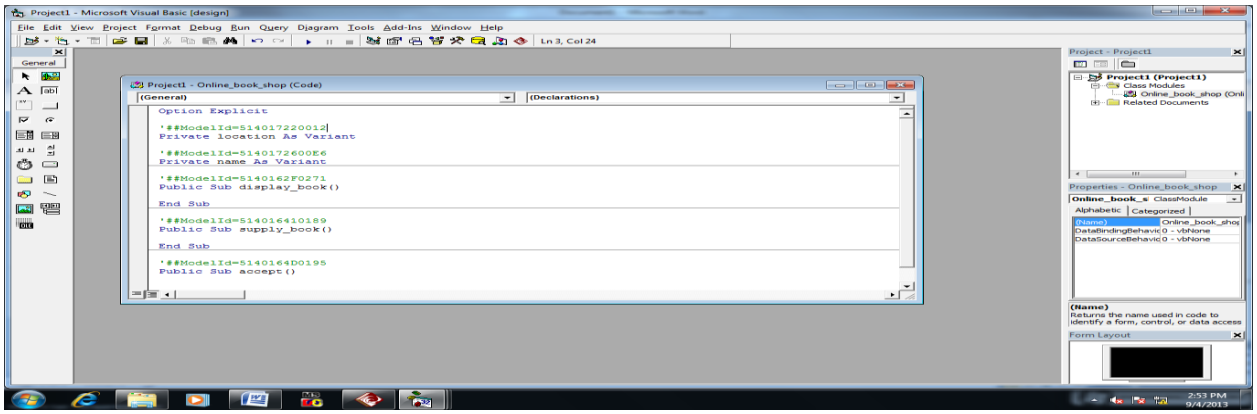
OUTPUT:


Project1 - Microsoft Visual Basic [design]
Project1 - Travel_agent (Code)
(General)
dellivery
Option Explicit
'##ModelId=514016D301F7
Private company_name As Variant
'##ModelId=514016DA01DC
Private company_addr As Variant
'##ModelId=5140170601D4
Public Sub dellivery()
End Sub


Project1 - Microsoft Visual Basic [design]
Project1 - Online_book_shop (Code)
(General)
(Declarations)
Option Explicit
'##ModelId=514017220012
Private location As Variant
'##ModelId=5140172600E6
Private name As Variant
'##ModelId=5140162F0271
Public Sub display_book()
End Sub
'##ModelId=514016410189
Public Sub supply_book()
End Sub
'##ModelId=5140164D0195
Public Sub accept()


Project1 - Microsoft Visual Basic [design]
Project1 - call_center_executive (Code)
(General)
giving_solution
'##ModelId=515BBA500332
Private ename As Variant
'##ModelId=515BBA5703B3
Private ephono As int
'##ModelId=515BBA5F005C
Public Sub ask_detail()
End Sub
'##ModelId=515BBA6401EA
Public Sub accept()
End Sub
'##ModelId=515BBA6C025A
Public Sub giving_solution()
End Sub





RESULT

Thus the various UML diagrams for student information system were drawn and code was generated successfully.

CONTENT BEYOND THE SYLLABUS

Ex. No: 4 HOSTEL MANAGEMENT SYSTEM

Date:

Aim:

To develop a Hostel Management using Rational Rose Software and to implement the software in Java.

Problem Analysis:

Hostel Management System is a Customize and user-friendly software for Hostel. It has been designed to automate, manage and look after the over-all processing of even very large hostel. It is capable of managing Enquiry details, Student Details, Payment Details etc. Hostel Management System is a Customize and user-friendly software for Hostel which provide hostel information, hostel room information, hostel accounts information.Hostel Management Software Systemis offering a maximum of stability, cost-effectiveness and usability. It provides the most flexible and adaptable standards management system software solutions for hostel.

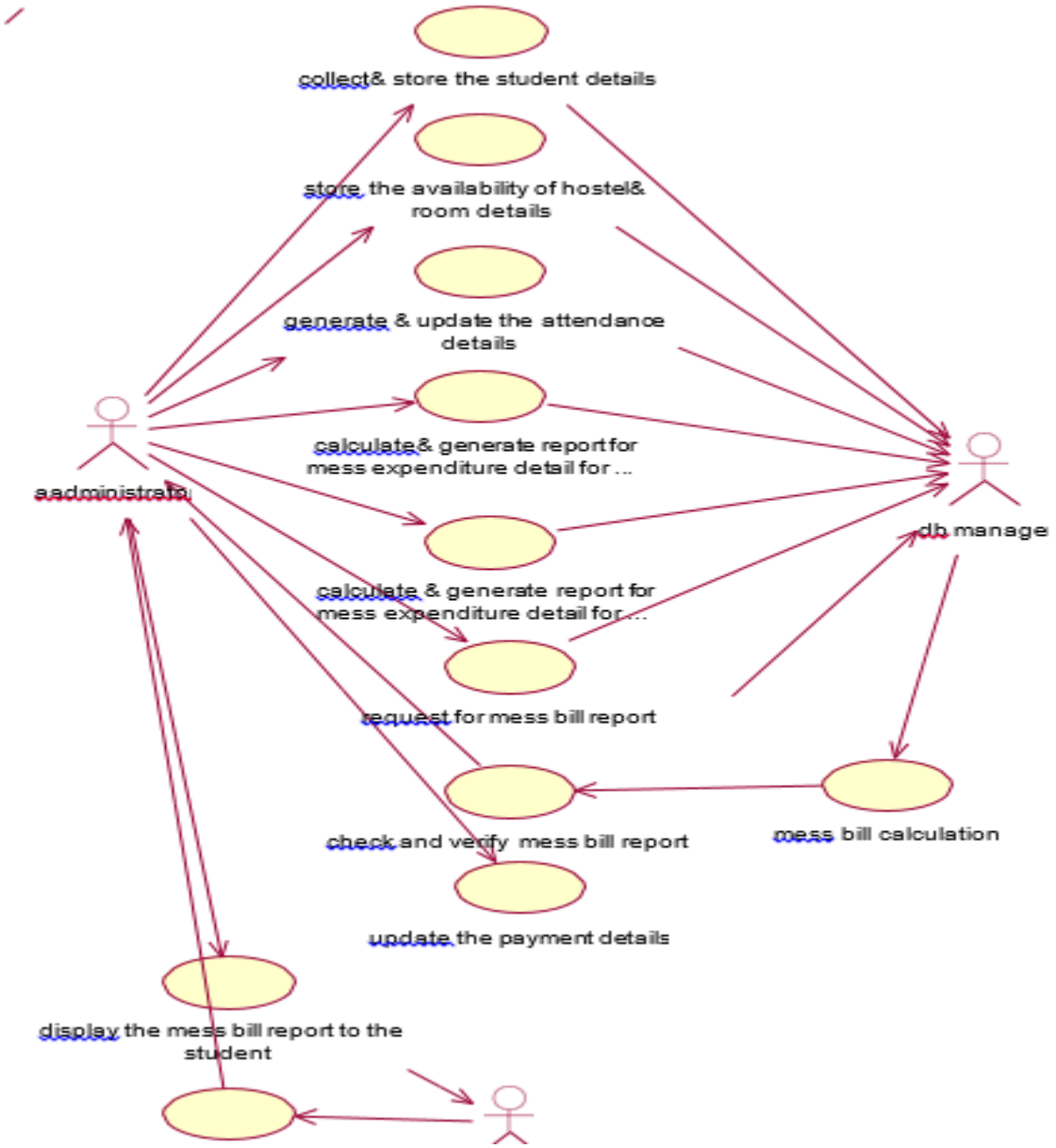
Problem statement

a. Effective for Administration Purpose
b. Cheap
c. Better Service

UML DIAGRAMS:

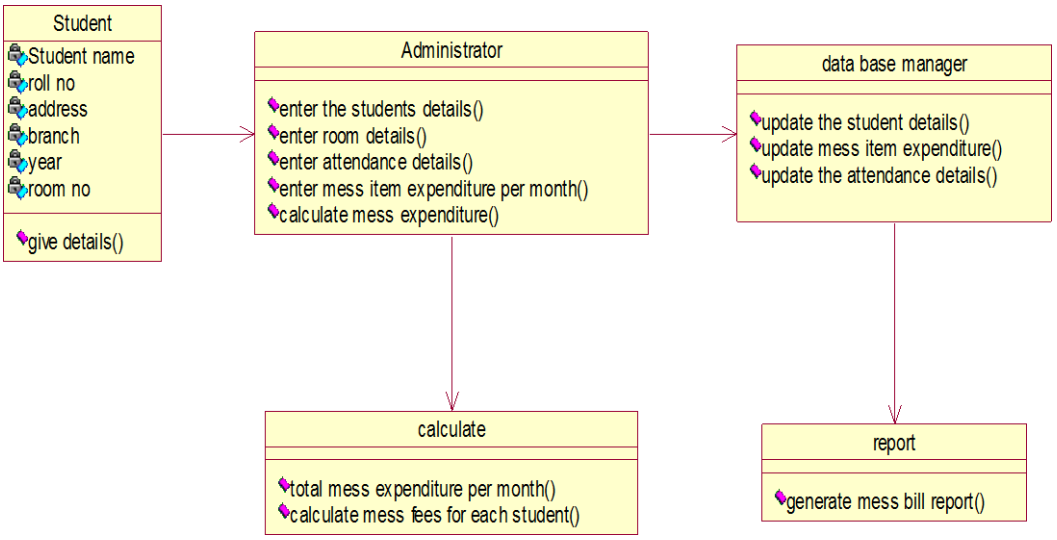
USE CASE DIAGRAM:

A use case is a methodology used in system analysis to identify, clarify, and organize system requirements. The use case is made up of a set of possible sequences of interactions between systems and users in a particular environment and related to a particular goal. It is represented using ellipse. Actor is any external entity that makes use of the system being modeled. It is represented using stick figure.



CLASS DIAGRAM:

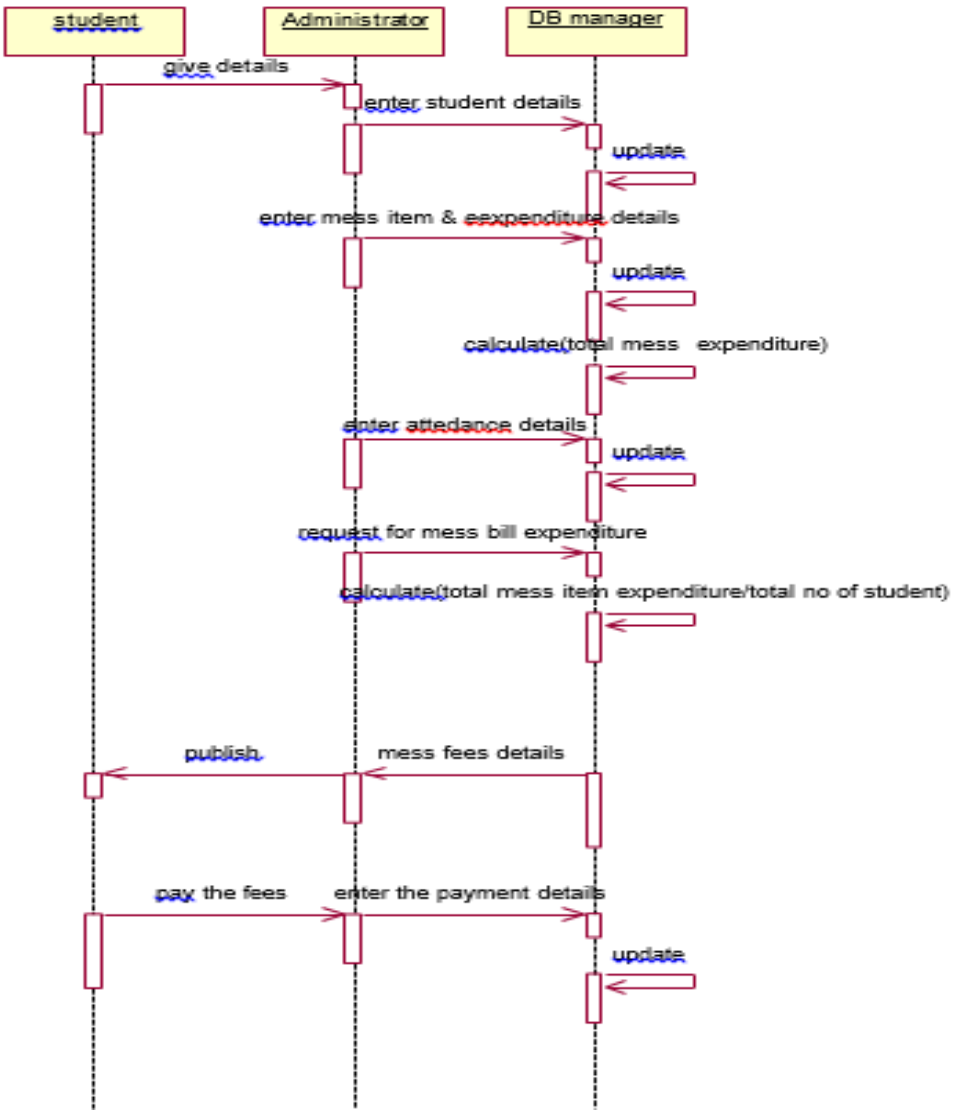
A class diagram in the unified modeling language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, and the relationships between the classes. It is represented using a rectangle with three compartments. Top compartment have the class name, middle compartment the attributes and the bottom compartment with operations.



SEQUENCE DIAGRAM:

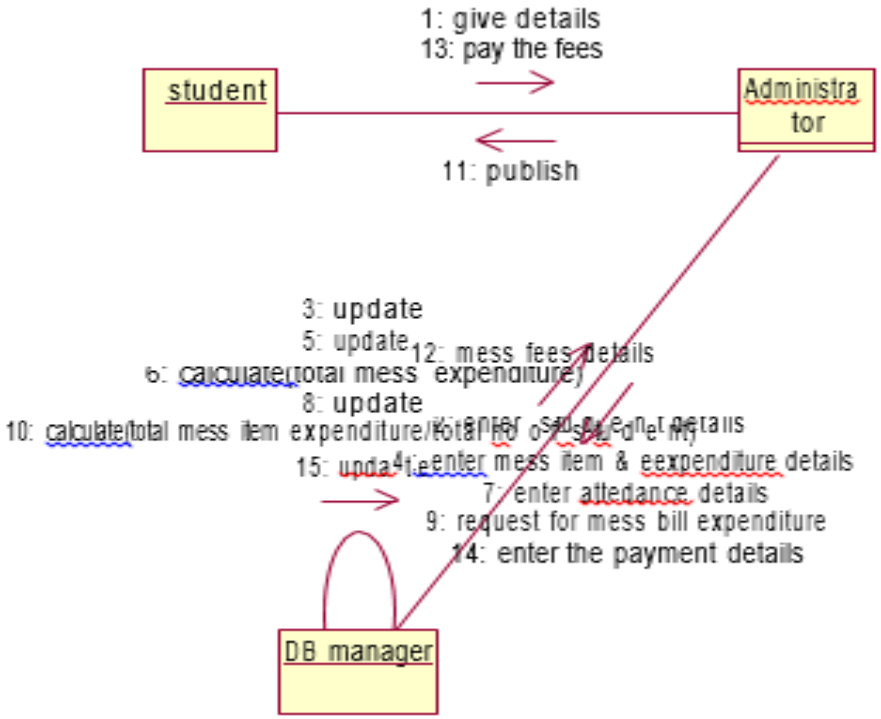
A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. There are two dimensions.

1. Vertical dimension-represent time.
2. Horizontal dimension-represent different objects



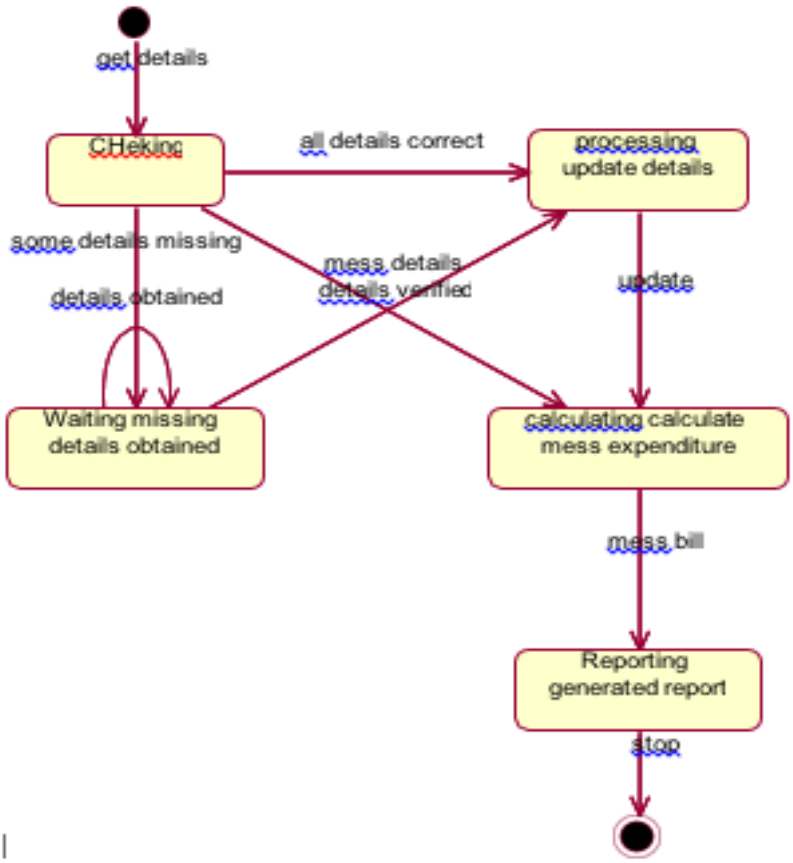
COLLABORATION DIAGRAM:

A collaboration diagram, also called a communication diagram or interaction diagram. A sophisticated modeling tool can easily convert a collaboration diagram into a sequence diagram and the vice. A collaboration diagram resembles a flowchart that portrays the roles, functionality and behavior of individual objects as well as the overall operation of the system in real time

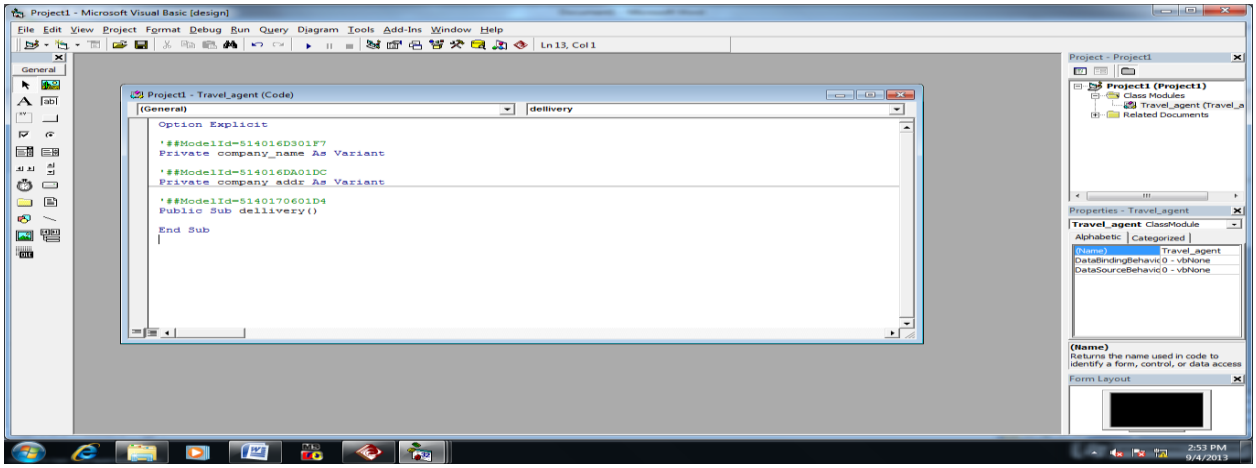


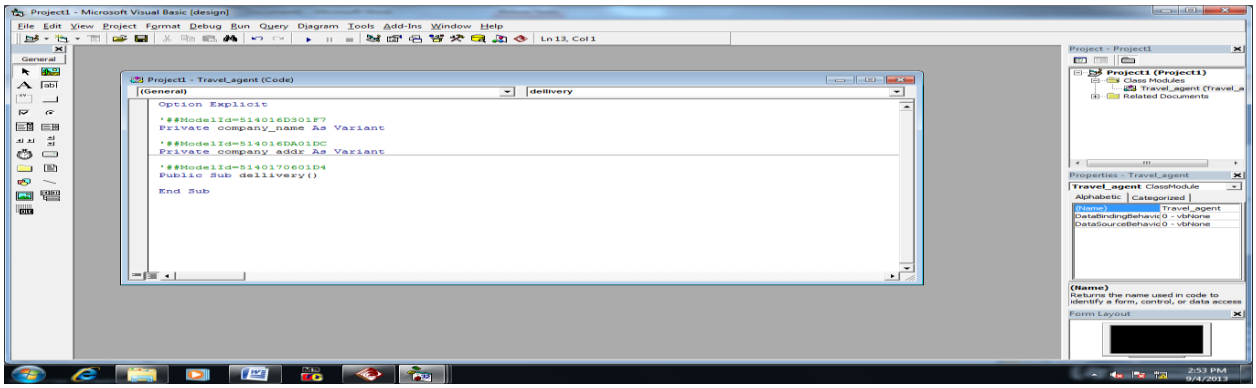
State Diagram:

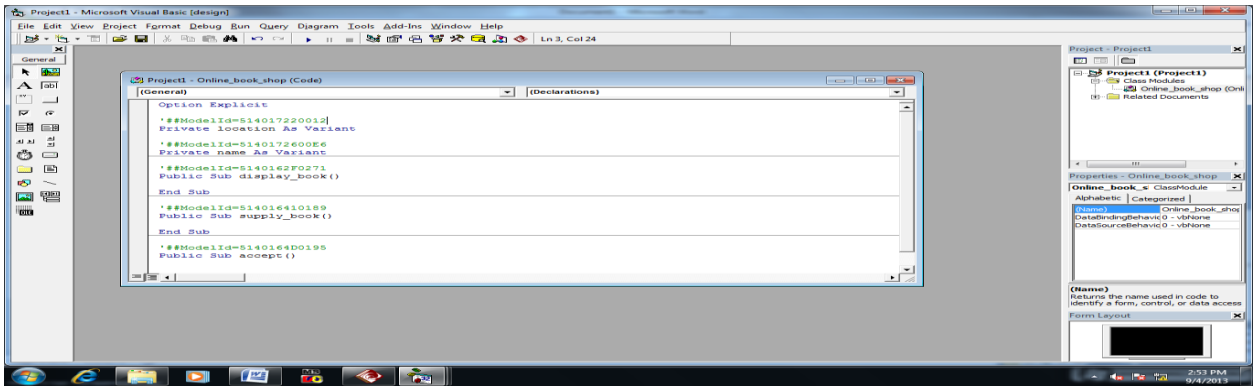
State diagram are a familiar technique to describe the behavior of a system. They describe all of the possible states that a particular object can get into and hoe the objects state changes as a result of events that reach the object.

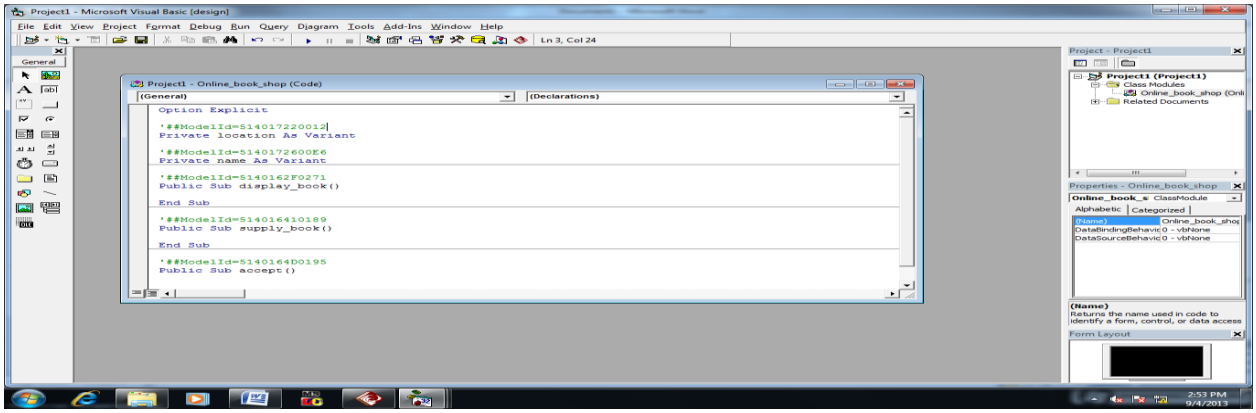

get details
CHeking
all details correct
processing
update details
some details missing
details obtained
mess details
details verified
update
Waiting missing
details obtained
calculating calculate
mess expenditure
mess bill
Reporting
generated report
stop

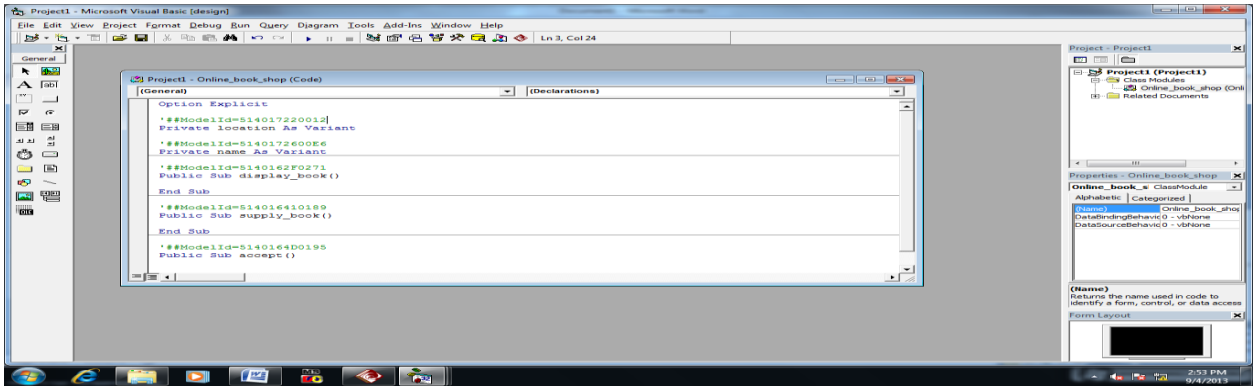
OUTPUT:


Project1 - Microsoft Visual Basic [design]
Project1 - Travel_agent (Code)
(General)
dellivery
Option Explicit
'##ModelId=514016D301F7
Private company_name As Variant
'##ModelId=514016DA01DC
Private company_addr As Variant
'##ModelId=5140170601D4
Public Sub dellivery()
End Sub


Project1 - Microsoft Visual Basic [design]
Project1 - Travel_agent (Code)
(General)
dellivery
Option Explicit
'##ModelId=514016D301F7
Private company_name As Variant
'##ModelId=514016DA01DC
Private company_addr As Variant
'##ModelId=5140170601D4
Public Sub dellivery()
End Sub


Project1 - Microsoft Visual Basic [design]
Project1 - Online_book_shop (Code)
(General)
(Declarations)
Option Explicit
'##ModelId=514017220012
Private location As Variant
'##ModelId=5140172600E6
Private name As Variant
'##ModelId=5140162F0271
Public Sub display_book()
End Sub
'##ModelId=514016410189
Public Sub supply_book()
End Sub
'##ModelId=5140164D0195
Public Sub accept()





RESULT:

Thus the hostel management was designed and the code was generated successfully.

VIVA QUESTIONS

1. Define Object Oriented Analysis?
2. Define Modular Design**.**
3. Write the characteristics of an object.
4. What is a class?
5. Name two types of object diagram.
6. What is an attribute? Give example.
7. What is multiple inheritance?
8. What is dynamic binding?
9. What is static binding?
10. Write the four quality measures for software development?
11. What is object persistence?
12. What is polymorphism? Give an example.
13. What is cardinality?
14. What is a formal class or abstract class?
15. What is a meta-class?
16. Define Encapsulation?
17. What is the need of an Object diagram?
18. What is state of an object?
19. Write some applications of object model?
20. Define Concurrency.
21. Name the three general approaches for classification?
22. Name the five levels of process maturity in OOD?
23. Name the two process used by Grady BOOCH in his OO software development?
24. Name the four steps in Micro development process?
25. What are the steps followed in macro development process?
26. Short notes on OMT functional model.
27. Names the diagrams of Booch Methodology.
28. Name the models in objectory.
29. What is unified modeling language?
30. Name the available layers of the three layered approach to software development.
31. Write the two responsibilities of access layer?
32. Write any two advantages of modeling?
33. What is Objectory?
34. Define Static model?
35. Define Dynamic model?
36. What is an association? Give one example.
37. What is a qualifier? Give one example.
38. What is a method?
39. Define Component Diagram
40. Define Deployment Diagram

41. What is meant by implementation diagram?
42. What is Collaboration Diagram?
43. What is interaction diagram? Mention the types of interaction diagram.
44. Define Class Diagram
45. Define Activity Diagram.
46. What is a use case?
47. Define usecase diagram.
48. What are the three kinds of Actors and three common use case formats?
49. What are the artifacts in inception phase
50. What Tests Can Help Find Useful Use Cases?
51. What is UML?
52. What are the three ways and perspectives to Apply UML?
53. What are the primary goals in the design of UML?
54. Name the three types of relationships in a use case diagram.
55. Write the two types of Implementation diagram?
56. What is an activity?
57. Name the types of relationships among the objects.
58. Write the guidelines for identifying the associations
59. Name the two properties of a part of relationship.
60. Write the Guidelines for identifying part of relationship.
61. Define Prototype?
62. Name the types of attributes.
63. Write the syntax for presenting the attribute that was suggested by UML.
64. Write the syntax for presenting the operation that was suggested by UML
65. What is a Façade?
66. What is the need of an Interaction diagram?
67. What is the need of a Class diagram?
68. What is Behavior of an object?
69. Define forward engineering and revere engineering.
70. What are traditional development methodologies?
71. What is Object Oriented development methodology?
72. How messages are different from methods?
73. What is the distinction between traditional development methodologies and object oriented development methodologies?
74. What are the advantages of object oriented approach?
75. What is Unified Approach(UA)?
76. What is Package diagram?
77. Explain the include relationship.
78. Explain the extend relationship.
79. What is the generalization relationship?
80. What is meant by POS system?
81. What do you mean by the term elaboration?

82. Enlist various tasks that are involved in elaboration.
83. What is domain model?
84. Why domain model is referred as visual dictionary?
85. How to create domain model?
86. Define conceptual classes.
87. What are description classes?
88. What is attributes?
89. Define aggregation and composition.
90. Differentiate single inheritances.
91. Distinguish between method and message in object.
92. When to create a subclass of superclass?
93. How to create instance?
94. Give a sample category list for identifying conceptual classes.
95. What are steps for mapping design to coding?
96. What are the issues in OO Testing?
97. List the four level of OO testing.
98. Define GUI Testing.
99. What is regression testing?
100. Define System Events and the System Boundary